Interest Rate Derivatives Intensive

EMEA Programme 2022

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Goldman Sachs

LondonFinancialStudies
capital markets learning

Interest Rate Derivatives Intensive

EMEA 2022

Course Notes and Material

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Introduction

The Course

This course is an intensive four-day version of the original Goldman Sachs Derivatives University program. The course has established a reputation for excellence over many years and has been kept up to date by the regular addition of new topics.

The program is designed to provide more experienced participants with a practical understanding of a wide range of interest rate derivatives. It is divided into cumulative half-day modules, each section using material from previous parts of the course. All modules contain practical exercises which, will be done in small groups organised in the virtual classroom.

Access to the LFS Live virtual classroom, course readings, material, quizzes and further support are available at gs.londonfs.com.

LFS Live

This Program will be delivered via the LFS Live learning platform. This is a virtual classroom that has been running for the last five years as an extension to the London Financial Studies physical teaching facility in London. It is designed to allow participants to take part in a program of learning as if they were in in the physical space.

There will be a program producer on-line at all times so that participants can deal with technical or administrative questions without interrupting the flow of the class.

The virtual classroom can be delivered through all of the standard video conferencing platforms as well as PC based applications such as Webex and Zoom. This course will be delivered via Zoom and a set of technical instructions will be sent out several days before the course.

The Teacher

David Cox has wide practical experience of the financial markets and an international reputation as a teacher of high-level short courses. His career includes ten years in banking, primarily with Bank of America Capital Markets. After leaving The City he joined the staff of London Business School, where he set up the financial markets seminar program, did research and maintained an active external teaching and consultancy practice.

Dr. Cox has been teaching Goldman Sachs professionals and clients for over ten years and has a thorough grasp of the firm's culture. He specialises in quantitative techniques, risk management and derivative products and is the founding director of London Financial Studies.

London Financial Studies

LFS has been working with Goldman Sachs over many years to provide the highest standard of teaching to clients and staff. The company ran the first Derivatives University programs and was instrumental in developing the brand.

London Financial Studies concentrates exclusively on capital markets. LFS teachers are prominent practitioners and academics all of whom have a clear and thorough grasp of their subjects and wide practical experience. LFS offers GS professionals and their clients a unique and expert teaching resource that combines theoretical understanding with practical experience.

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System Instructions (Zoom)

Introduction

LFS Live is a virtual classroom that has been running for the last five years as an extension to the London Financial Studies physical teaching facility in London. It is designed to allow participants to take part in a programme of learning as if they were in in the actual classroom.

The virtual classroom can be delivered through all of the standard video conferencing platforms as well as PC based applications such as Webex and Zoom. These instructions are for delivery via Zoom.

Equipment

In order to get the maximum from each session it is most important to use suitable equipment:

A PC or Mac with a large screen is most important (phones and ipads are not suitable).

You will need an installation of Microsoft Excel. It can be useful, but not essential, to have this installed on a second PC or laptop.

A webcam is required to enable you to participate in the online classes, take part in breakout groups and interact with the teacher and other delegates.

A headset with microphone generally provides the best audio quality.

We strongly suggest using a wired internet connection not Wi-Fi for the best quality picture and sound.

What to expect

Virtual courses tend to run slightly more slowly than those in the classroom and exercises will generally be shorter to allow all participants to participate fully.

There will be a programme producer on-line at all times so that you can deal with technical or administrative questions without interrupting the flow of the class.

It is always best to watch your live session in full-screen view without any other applications running apart from those used during the course such as MS Excel.

To avoid background noise we recommend keeping your microphone muted when not talking to the teacher or the group.

Please keep your video on whenever you are in the virtual classroom so that you can take part fully in the sessions and be seen by the teacher and the group.

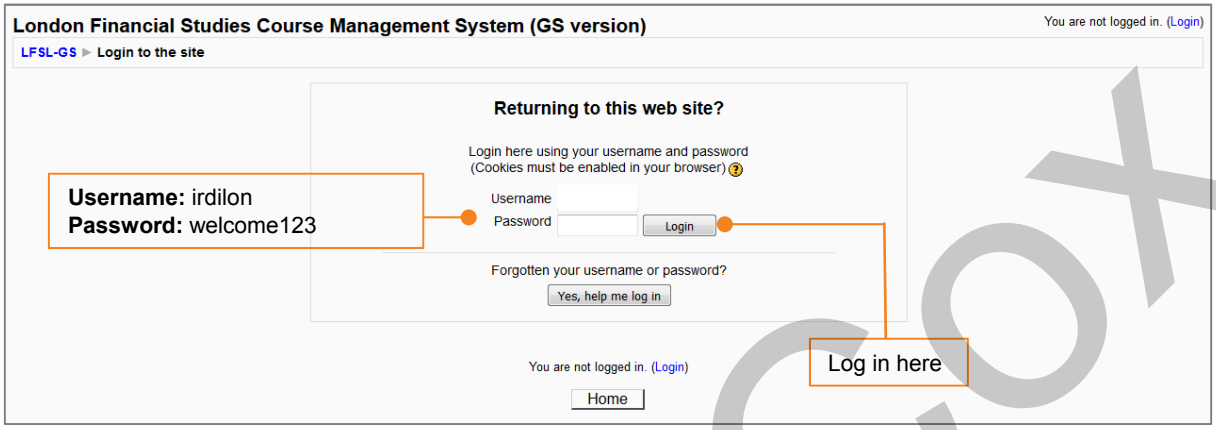
Do not hesitate to request technical support at any time via online chat.

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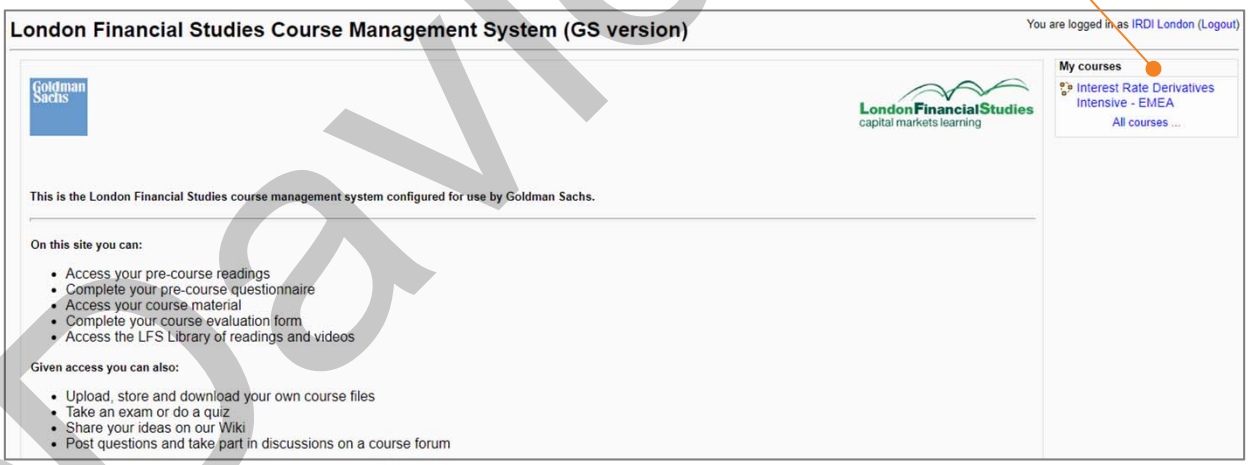
Log in and Access

Open the following page in your browser: http://gs.londonfs.com/
Enter the password and username shown below to enter the LFS Learning platform



LFS Learning is used before, during and after your programme to store material and provide access to the virtual classroom and other resources.

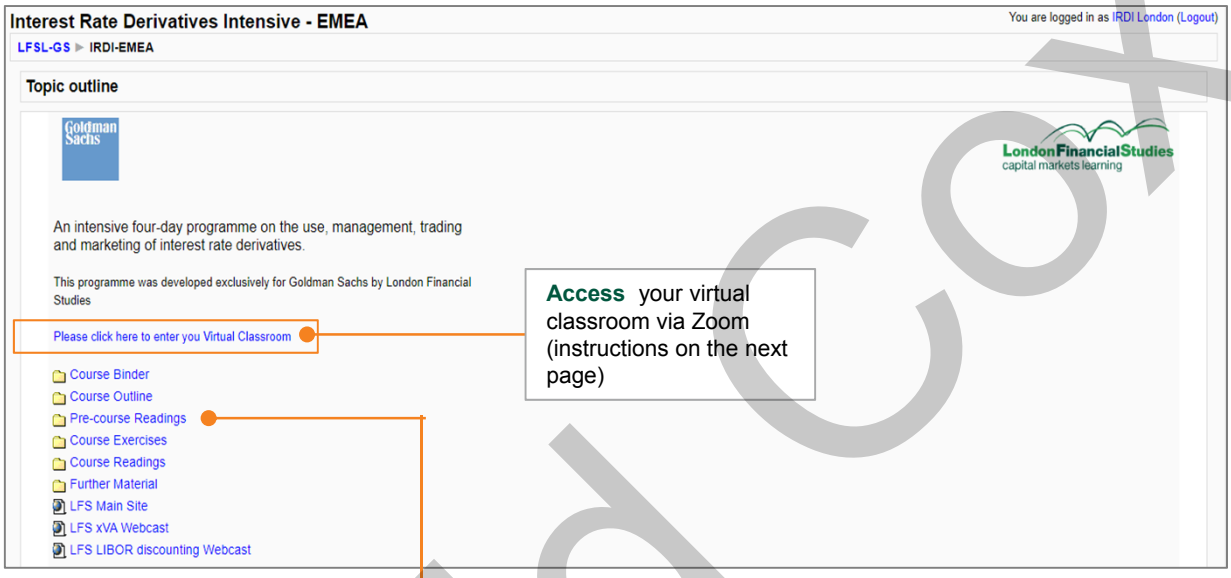
Once logged in, at the top right hand side of the page, click on “Interest Rate Derivatives Intensive” under “My courses”.



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Access to the Classroom & Course Materials

Once on the course page you can access your virtual classroom and download course material as shown below:



Access your virtual classroom via Zoom (instructions on the next page)

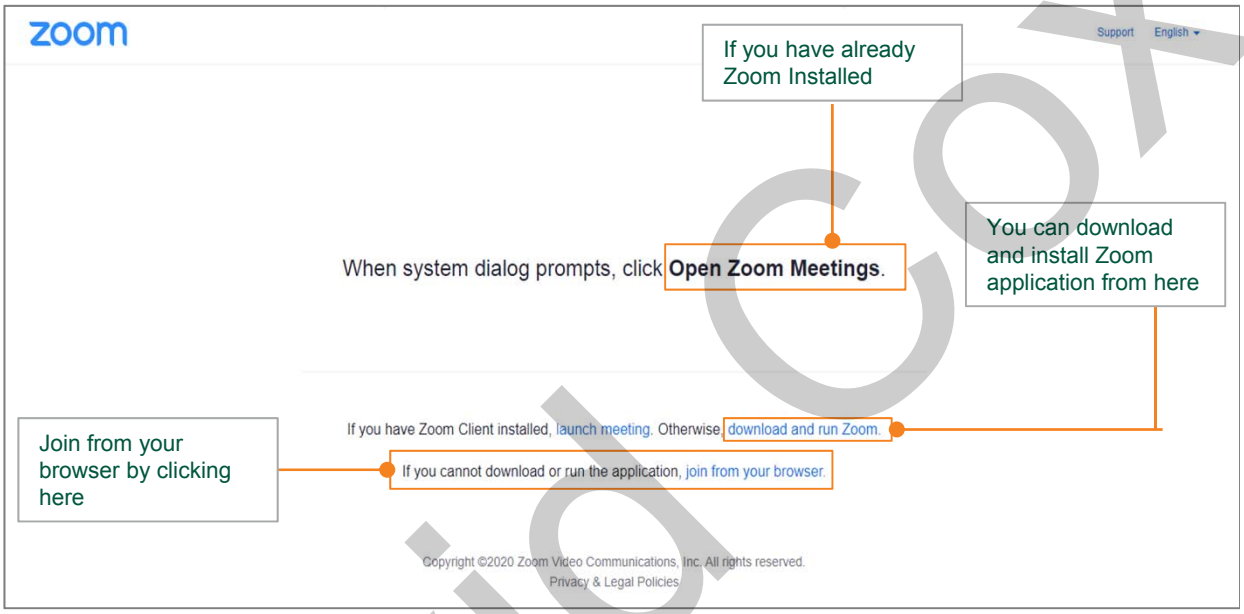
Access / Download your course materials and other resources before, during, and after the course

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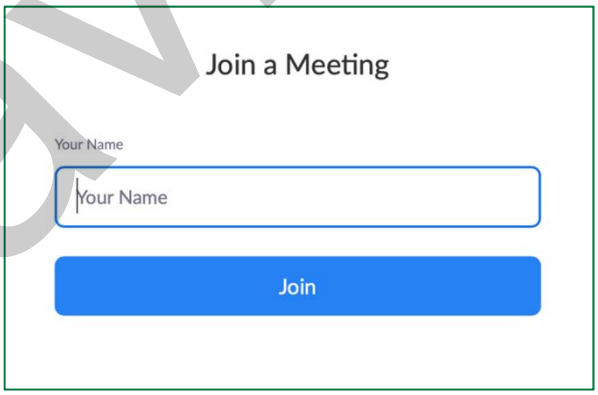
System Instructions

You can join your Virtual Classroom through Zoom by following one of the routes below:

- Open Zoom Meetings: if you have already Zoom application installed
- Download and Run Zoom: if it is not installed and you are able to download the app.
- Join from your browser: If you are not able to download Zoom on your PC or Mac



After having done one of the above, type your name and click “Join” to enter the classroom



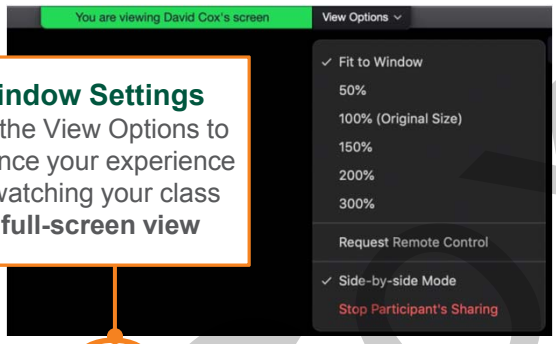
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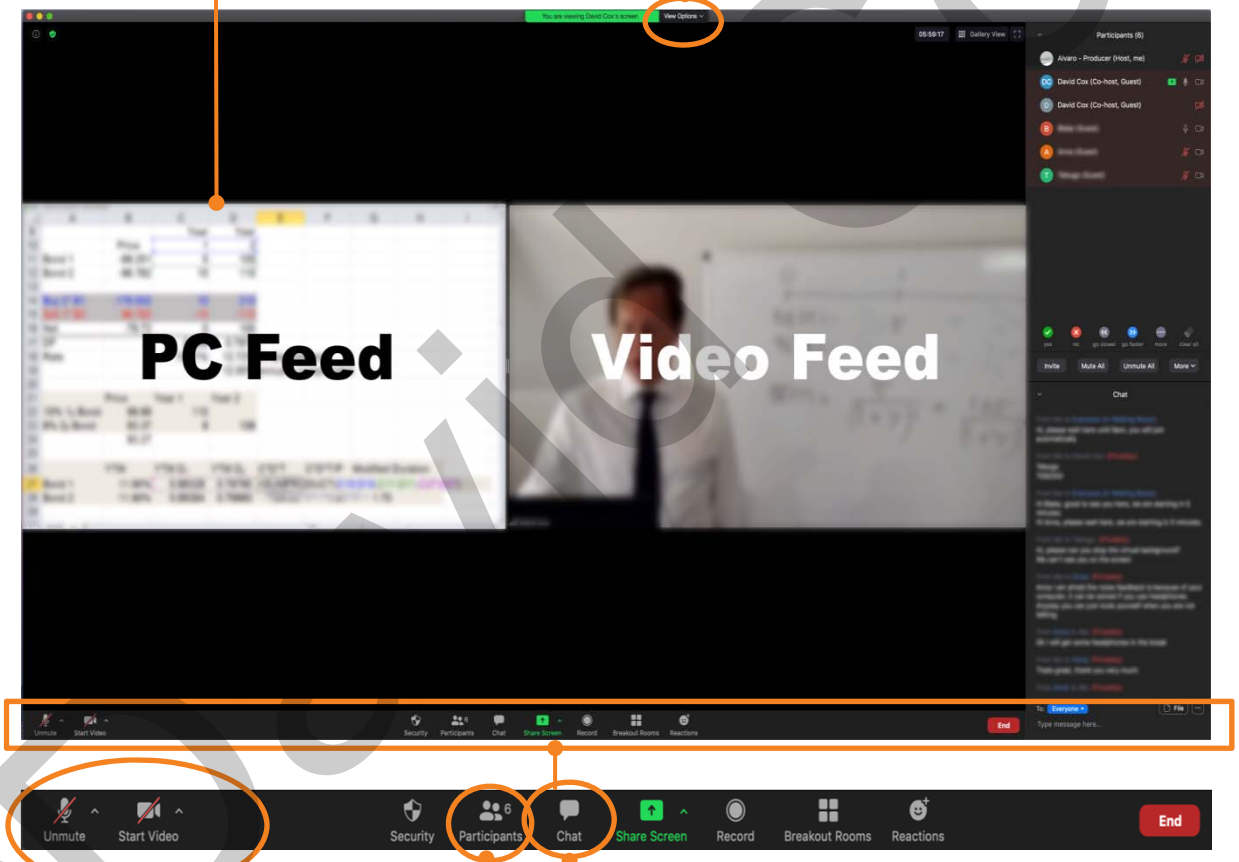
Zoom Control Panel

As LFS Live is being delivered through Zoom you can make use of all of the features of that product:

Live video feeds
All classroom action is captured by our cameras, edited and broadcast live by our production team

Window Settings
Use the View Options to enhance your experience by watching your class in **full-screen view**





Microphone & webcam
Please keep your **microphone MUTED** when you are not addressing the teacher or the group and keep your **webcam ON** at all times

List of online delegates

Online chat
Receive real-time **technical support** from the LFS Live production team

© Document

Contents

Day One Programme

Fundamentals	Lecture slides
	Readings available at gs.londonfs.com:
	Extracts from: Brealey, R. A and Myers, S. C. (1988) Principals of Corporate Finance, chapters two and three, McGraw-Hill.
	Extracts from ISDA definitions.
Futures and FRAs	Lecture slides
	Readings available at gs.londonfs.com:
	CME Contract Specifications for Interest Rate Products
	CME Span Overview
	Eurodollar Futures Strips, Bundles and Packs, CME Publications.
	Burghart, G. and Hoskins, W. (1994) The Convexity Bias in Eurodollar Futures. Dean Witter Institutional Futures Research Note.

Day Two Programme

Yield Curve Building	Lecture Slides
	Readings available at gs.londonfs.com:
	Schaefer, S.M. (1977) The Problem With Redemption Yields. Financial Analysts Journal.
	Cubic Spline Interpolation. Extract from Numerical Recipes Text.
Interest Rate Swaps	Lecture Slides
	Readings available at gs.londonfs.com:
	Extending USD OIS Curves Using FED Funds Basis Swap Quotes. Bloomberg Paper, 2011.
	Mercurio, F. (2011) The New Swap Math. OTC Derivatives and Structured Notes. Bloomberg Paper.

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Day Three Programme

Cross Currency and Asset Swaps — Lecture Slides

Readings available at gs.londonfs.com:

Basis for Change. Risk Magazine Vol 8, No.10, 1995

Option Essentials for the Practitioner — Lecture Slides

Readings available at gs.londonfs.com:

Boyle, P.P., Broadie, M. Glasserman, P. (1997) Monte Carlo Methods for Security Pricing. Journal of Economic Dynamics and Control. 21 1267-1321.

Amin, K. (1995) Option Pricing Trees. The Journal of Derivatives 2(4) 34-46.

Extract from: Chapter 20 Brealey, R.A. and Myers, S.C. (1992) Principals of Corporate Finance, McGraw-Hill Inc.

Rubinstein, M. (1998) Edgeworth Binomial Trees Journal of Derivatives Spring 1998

Hull, J.C. and White, A. (1993) Efficient procedures for valuing European and American Path Dependent Options. The Journal of Derivatives (Fall) 21-31

Day Four Programme

Interest Rate Options: Caps, Floors and Collars — Lecture Slides

Interest Rate Options: Swaptions — Lecture Slides

Readings available at gs.londonfs.com:

Neuberger, A.J. Pricing Swap Options Using the Forward Swap Market. IFA Working Paper 139-91. London Business School.

Hagan, P. S.; Kumar, D.; Lesniewski, A. S.; Woodward, D. E. (2002) Managing Smile Risk. Working Paper.

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Day One

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Day One

8.15 – 8.30	**Sign-in, Welcome and Course Introduction**
8.30 – 10.30	**Fundamentals and Rate Benchmarks** • Old reference rates: "ibors" • Changes to Euribor • New reference rates: SOFR, ESTA • Cashflows, conventions and compounding intervals • Discount factors and their meaning • Internal rate of return v zero coupon rates • Forward rates and discount factors **Worked Examples: Basic calculations**
10.30 – 10.45	Break
10.45 – 13.00	**Fundamentals Continued** • Bond mechanics • The limitations of yield to maturity • Introduction to duration and convexity • Some practical implications of convexity **Further worked examples**
13.00 – 14.00	Lunch
14.00 – 15.30	**Managing Short Term Interest Rate Risk** • Examples of short term interest rate risk Libor v overnight • FRA contract details (Euribor) • Constructing hedges with FRAs • Basic market jargon: rally, sell-off, bull steepener, bear flattener etc. • Revaluation and DV01 • Revaluation and Roll down • Euribor Futures • SOFR Futures • Constructing hedges with SOFR futures • Futures execution and clearing • Initial and variation margin • Credit risk
15.30 – 15.45	Break
15.45 – 17.15	**Exercise: Hedging strategies**
17.15 – 17.30	**Online Quiz**

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The Basics



You sell me your old car

I agree to pay

Either £2,000 today

Or £4,000 in one year

Which would you take?

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Old Reference Rates: "ibors"

Libor: London Interbank Offered Rate

Unsecured interbank funding rate

- Quoted in GBP, USD, JPY, CHF and EUR, at 11.00am London time, published at about 11.30am

Maturities:

- Overnight, spot next, 1 Week, 1,2,3,6 and 12 Months
- Gaps by linear interpolation

Calculated for ICE (IBA) by Thompson-Reuters.

- Deposit rates obtained from 11 - 18 banks depending on currency
- Top and bottom 25% of quotes discarded
- Average of the middle two quartiles taken
- Usefulness depends on liquid money markets and accurate submissions
- Will continue until end 2021. Various replacements

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Old Reference Rates: "ibors"

Euribor: Euro Interbank Offered Rate

Updated version will continue beyond end 2021

Wholesale unsecured funding rate

- Calculated by European Money Markets Institute ("EMMI")

Quoted in EUR at 11.00am Frankfurt time (CET)

- Maturities: 1 week, 1 month, 3 months, 6 months and 12 months.

Hierarchical three levels of submission from panel

- Eliminate highest and lowest 15%, arithmetic average:
- 1) Volume-weighted rate for prior TARGET day transactions, or if not available:
- 2) Wider spectrum of transactions from recent TARGET days (historical and linearly interpolated), or if not available:
- 3) Range of closely related markets, modelling and/or judgment

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New Reference Rates

New benchmarks to replace Libor are required as:

- Volume of cash transactions much reduced
- Panel of banks unrepresentative of market
- "Expert judgment" used for submissions
- Regulatory and reputational risk to banks in panel

Many legal and contractual hurdles in making the transfer to a transaction based benchmark!

- UK: Modified SONIA (unsecured)
- US: SOFR (Median overnight treasury repo rate)
- Euro: €STA (Volume weighted trimmed mean of unsecured o/n)
- Switzerland: SARON (interbank overnight repo)
- Japan: TONA (Tokyo Over Night Average unsecured)

How can longer tenors be based on OIS rates?

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New Reference Rates

ESTA

- Replaces EONIA (Euro Overnight Index Average)
- Calculated by the ECB as a volume weighted trimmed mean of unsecured transactions reported under MMSR Regulation regulation (No 1333/2014)
- Transactions ordered by volume, top and bottom 25% removed, weighted average of remaining published before 9am CET

SOFR

- Replacing Fed Funds Effective Rate
- Volume-weighted median of Repo rates
- A broad measure of the overnight cost of financing Treasuries
- Published daily at around 8am New York time the following day.

SONIA (Sterling Overnight Index Average)

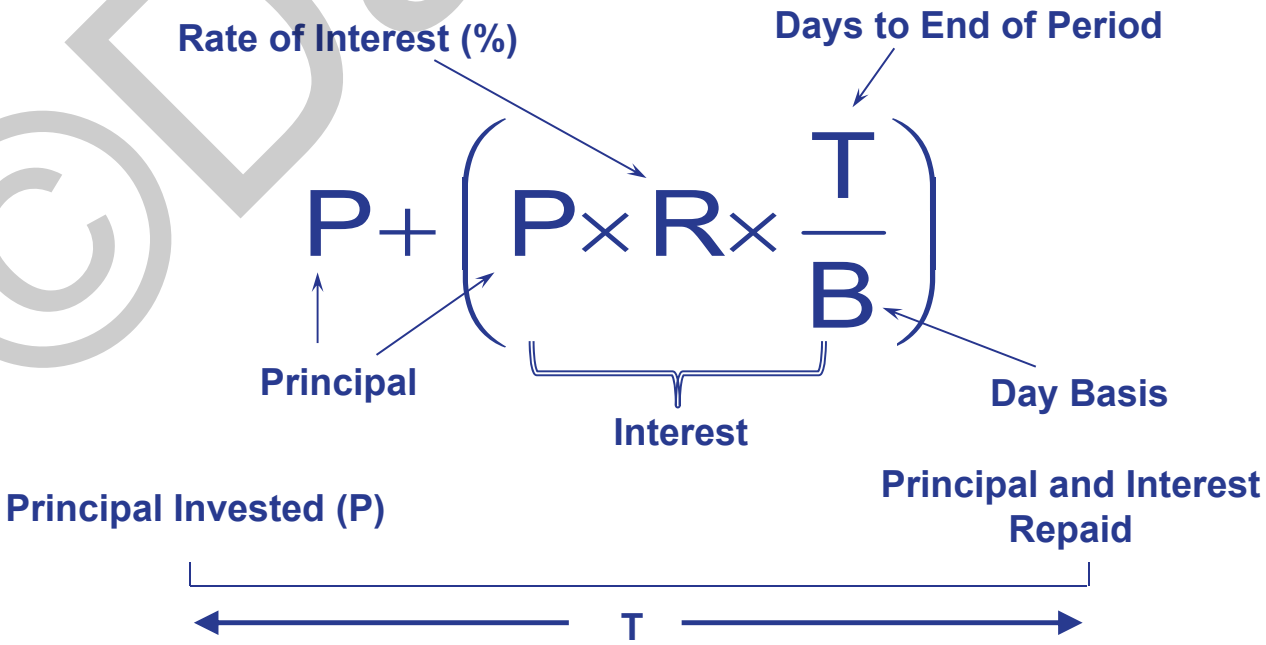
- Calculated by the Bank of England and published at 9am the following day. Stirling Libor replacement

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These are averages over a day not fixings at a specific time

Simple Interest on Cash Deposit

Rate set at start, interest paid at end of the period:



How to calculate T?

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Interest Calculation Conventions

Modified following business day

If a day falls on a weekend or holiday then move to the following business day unless this takes you into the next month. If so move to the previous business day

One month from 20 January would normally be 20 February
If 20 February is a Saturday then the date becomes *22 February*

One month from 28 January would normally be 28 February
If 28 February is a Sunday then the date becomes *26 February*

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Interest Calculation Conventions

Day Basis

Money Market Basis: Act/360

Actual number of days in the period divided by 360
This applies to US Dollars, Euros and Swiss Francs

Money Market Basis: Act/365

This applies to Sterling (and old Commonwealth countries), Yen, Zloty, Rouble, and many other currencies

Bond Basis: 30/360

The year is treated as if it consists of twelve thirty day months
This applies to Bonds of many types
Unadjusted 30/360 takes no account of payment days being moved to avoid weekends and holidays
Adjusted 30/360 provides extra interest if the payment day is moved

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Present and Future Value

Single Period

£2,000 — T = 365 Days — £2,100.00

R = 5%

Future Value of £2,000: $£2{,}000 \times \left(1 + \left(5\% \times \frac{365}{365}\right)\right) = £2{,}100.00$

$2,000 — T = 365 Days — $2,101.39

R = 5%

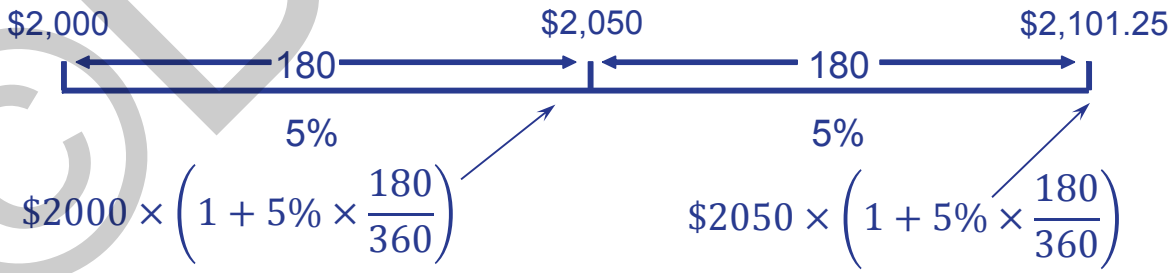
Future Value of $2,000: $\$2{,}000 \times \left(1 + \left(5\% \times \frac{365}{360}\right)\right) = \$2{,}101.39$

Market conventions make a difference!

© David Cox

Present and Future Value

More than one period, same rate for all



Interest is reinvested at the end of 6 Months

In this case Bond (30/360) day count conventions are used

The final payment is known at the start

$$\$2{,}000 \times \left(1 + \left(5\% \times \frac{180}{360}\right)\right)^2 = 2{,}101.25$$

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Present and Future Value

More than one period, same rate for all



Invest \$100 — Receive after T days

Compounding Interval		Rate = 5% T = 360
Annual (Rate R_a)	$P\times\left(1+R_a\right)^{\frac{T}{360}}$	\$105
Semi-annual (Rate R_s)	$P\times\left(1+\frac{R_s}{2}\right)^{\frac{T}{180}}$	\$105.06
Quarterly (Rate R_q)	$P\times\left(1+\frac{R_q}{4}\right)^{\frac{T}{90}}$	\$105.09
Daily (Rate R_d)	$P\times\left(1+\frac{R_d}{360}\right)^{T}$	\$105.13

Assuming a constant reinvestment rate an annual interest rate can be converted to semi-annual: $R_s = 2\times\left(\sqrt{1+R_a}-1\right)$

In this case Bond (30/360) day count conventions are used

© David Cox

Present and Future Value

Continuous Compounding

In general the amount received after investment for t years at R is:

$$P\times\left(1+\frac{R}{m}\right)^{mt}$$

m = number of times compounded per year

R = interest rate compounded with frequency m (expressed as a decimal)

As the interval tends to zero and m goes to infinity:

Amount received = $P\times e^{R_c t}$ $\quad \$100\times e^{0.05}=\$105.13$

(R_c = continuously compounded rate of interest, t = years)

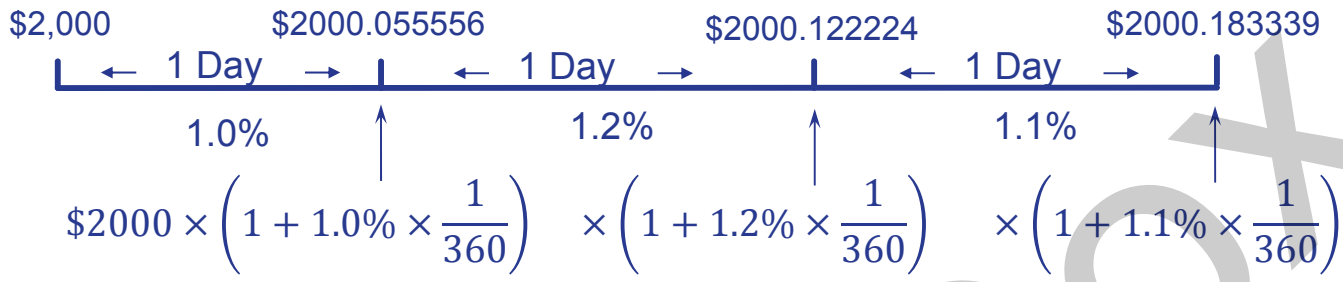
$$e\equiv\lim_{m\to\infty}\left(1+\frac{1}{m}\right)^{m}=2.7182819.....$$

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Present and Future Value

More than one period, different overnight rates



Principal and interest are reinvested at the end of each day at different overnight rates

The final payment is not known at the start

This is what we have with SOFR instead of Libor

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Discount Factor

Ratio of value between two points in time

$2,000 —— R= 5% T = 360 ——→ $2,100

R is a simple interest rate (known at the start of the period)

$$\text{Discount Factor} = \frac{2{,}000}{2{,}100} = 0.95238 = \frac{1}{\left(1 + R \times \frac{T}{360}\right)}$$

$2,000 —— R_s= 5% T = 360 ——→ $2,101.25

R_s is a semi-annually compounded interest rate

$$\text{Discount Factor} = \frac{2{,}000}{2{,}101.25} = 0.951814 = \frac{1}{\left(1 + R_s \times \frac{1}{2}\right)^{\frac{T}{180}}}$$

Present Value = Future Cashflow × Discount Factor

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Discount Factor

The price of a zero coupon bond that pays one unit of currency at maturity T

$$D = \frac{1}{\left(1 + R \times \frac{T}{360}\right)}$$ (R: Simple)

$$D = \frac{1}{\left(1 + R_a\right)^{\frac{T}{360}}}$$ (R_a: Annual)

$$D = e^{-R_c t}$$ (R_c: Continuous)

Zero Coupon Interest Rate

The yield on a zero coupon bond with maturity T

Makes no assumptions about reinvestment rates

Sometimes known as the *Spot Interest Rate*

$$R = \left(\frac{1}{D} - 1\right) \times \frac{360}{T}$$

$$R_a = \left(\frac{1}{D}\right)^{\frac{360}{T}} - 1$$

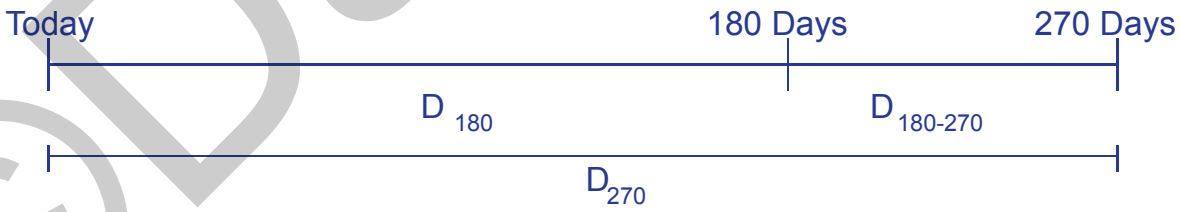
$$R_c = \frac{-\mathrm{Ln}(D)}{t}$$

R = Interest Rate D = Discount Factor t = Time (Years) T = Time (Days)

© David Cox

Forward Discount Factor

Relates value between future points in time



Discounting can be done in one or two stages

For no arbitrage there must only be one price for an asset so:

$$D_{270} = D_{180} \times D_{180\text{-}270}$$

Forward discount factor: $$D_{180-270} = \frac{D_{270}}{D_{180}}$$

Forward rate: $$R_{180\text{-}270} = \left(\frac{1}{D_{180\text{-}270}} - 1\right)\frac{360}{90}$$

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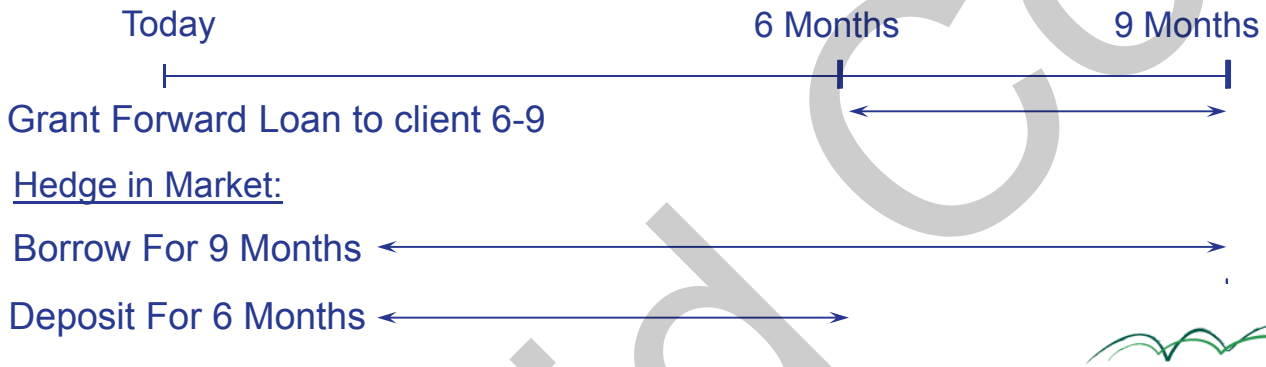
Forward Rate from Cash

How would you lock in a future rate of interest with no derivatives?

Cash Rates (%)	
3 Month	8.5 - 8.3
6 Month	8.9 - 8.7
9 Month	9.3 - 9.1

$$R_f = \left(\frac{1}{D_f} - 1\right)\frac{360}{90}$$

$$D_{Forward} = \frac{D_{Long\ period}}{D_{Short\ period}}$$



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Bond Fundamentals

Fixed rate of interest (the bond's "coupon") is paid until maturity

- Annual, semi-annual or quarterly payment intervals

Nominal value is repaid at maturity



Nominal = 100, price = 103, coupon = 5% (annual), maturity = 4y

- Coupon cashflow is calculated using bond conventions (often 30/360)
- If you buy a bond between coupon dates you receive all of the next coupon and must pay a proportion of this to the seller in the form of an adjustment to the price called "accrued interest"
- Accrued interest calculations often use day count conventions other than 30/360

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Bond Fundamentals

For fair value, a bond's price is the sum of discounted cash flows

Four year bond paying annual coupons:

Year:	0	1	2	3	4
Cashflow: -103		+5	+5	+5	+105
Discount Factor:		D_1	D_2	D_3	D_4

$$\text{Value} = 5 \times D_1 + 5 \times D_2 + 5 \times D_3 + 105 \times D_4$$

If a bond's market price is not equal to fair value then we need to know why

- It doesn't necessarily mean a market opportunity, but it may do

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Yield to Maturity

Also known as "redemption yield" or "internal rate of return"

A single rate of interest (Y) used to discount a bond's cashflows to equal its price

Year:	0	1	2	3	4
Cashflow: -103		+5	+5	+5	+105

$$103 = \frac{5}{(1+Y)^1} + \frac{5}{(1+Y)^2} + \frac{5}{(1+Y)^3} + \frac{105}{(1+Y)^4}$$

Y = 4.17%

Not a zero coupon rate

- A complex average of zero coupon rates
- Not directly related to the discount factor for any one period
- For many purposes much less use than a zero coupon rate

Makes the assumption that the yield curve is flat!

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Interest Rate Sensitivity

What happens to a bond's price if interest rates change?

Can be approximated by what happens to price if the yield to maturity changes

Take the four year bond as an example:

Year:	0	1	2	3	4
Cashflow:	-103	+5	+5	+5	+105

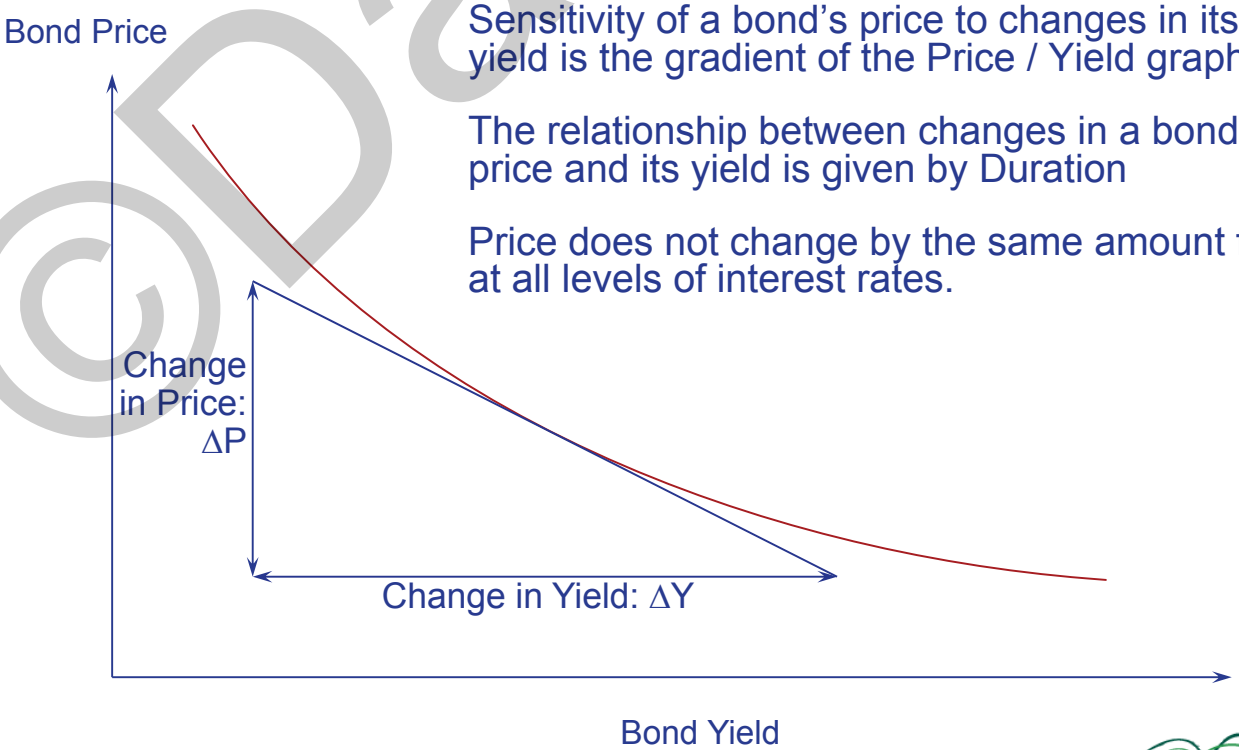
$$103 = \frac{5}{(1+Y)^1} + \frac{5}{(1+Y)^2} + \frac{5}{(1+Y)^3} + \frac{105}{(1+Y)^4}$$

Price = 103 and yield to maturity (Y) = 4.17%

How does the price change if the yield increases by 10bp?

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Interest Rate Sensitivity



Sensitivity of a bond's price to changes in its yield is the gradient of the Price / Yield graph

The relationship between changes in a bond's price and its yield is given by Duration

Price does not change by the same amount for at all levels of interest rates.

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Interest Rate Sensitivity

Duration

Year:	0	1	2	3	4
Cashflow: -103		+5	+5	+5	+105
Discount Factor:		D_1	D_2	D_3	D_4

$$\text{Price} = \frac{5}{(1+Y)^1} + \frac{5}{(1+Y)^2} + \frac{5}{(1+Y)^3} + \frac{105}{(1+Y)^4}$$

Gradient of price / yield graph is first derivative:

$$\frac{\Delta P}{\Delta Y} = -\frac{1}{(1+Y)}\left[\frac{1\times 5}{(1+Y)} + \frac{2\times 5}{(1+Y)^2} + \frac{3\times 5}{(1+Y)^3} + \frac{4\times 105}{(1+Y)^4}\right]$$

When divided by price this is weighted average of cashflow timing

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Interest Rate Sensitivity

Macaulay Duration

Concept developed in 1930s

The weighted average of the timing of bond's cashflows where the weights are the discounted cashflows

For the 4 year bond with coupon of 5%:

$$\text{Macaulay Duration} = \frac{1}{\text{Price}}\left[\frac{1\times 5}{(1+Y)} + \frac{2\times 5}{(1+Y)^2} + \frac{3\times 5}{(1+Y)^3} + \frac{4\times 105}{(1+Y)^4}\right]$$

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Calculating Duration

Four Year Bond

Coupon: 5% pa Price = 103 YTM = 4.17%

Year (T)	Cashflow	DF from YTM	C*DF	C*DF*T
1	5	0.959967	4.80	4.80
2	5	0.921537	4.61	9.22
3	5	0.884646	4.42	13.27
4	105	0.849231	89.17	356.68
Sum:			103.00	383.96

$$\frac{1}{(1+Y)^T}$$

$$\text{Duration} = \frac{383.96}{103} = 3.73 \text{ years}$$

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Interest Rate Sensitivity

Modified Duration

Percentage price change per change in yield: $\frac{1}{P}\frac{\Delta P}{\Delta Y}$

$$\text{Modified Duration} = -\frac{1}{(1+Y)} \times \text{Macaulay Duration}$$

$$-\frac{1}{(1+4.17\%)} \times 3.73 = -3.58$$

If yield increased by 10bp then the price will drop to:

$$103 - (3.58 \times 0.001 \times 103) = 102.631$$

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Duration

A measure of the sensitivity of a bond's price to changes in its yield to maturity

- The weighted average life of a bond where the weights are discounted cashflows
- Duration changes with yield and is only accurate for small changes in yield
- Duration reflects changes in yield to maturity which assumes that that the yield curve is flat.
- It is only accurate for parallel shifts in the yield curve

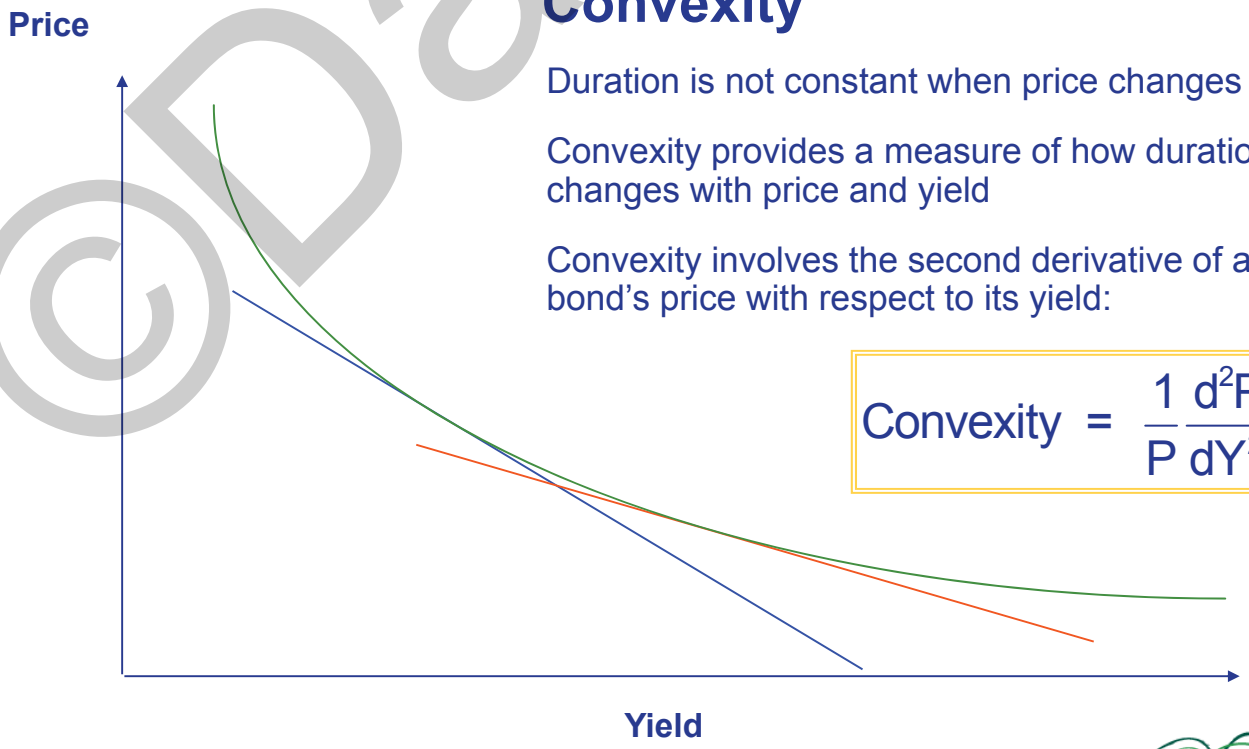
A measure of the sensitivity of cashflow NPV to small parallel yield curve shifts

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Interest Rate Sensitivity

Convexity



Duration is not constant when price changes

Convexity provides a measure of how duration changes with price and yield

Convexity involves the second derivative of a bond's price with respect to its yield:

$$\text{Convexity} = \frac{1}{P}\frac{d^2P}{dY^2}$$

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Interest Rate Sensitivity

Convexity

The relationship between yield and price is not linear

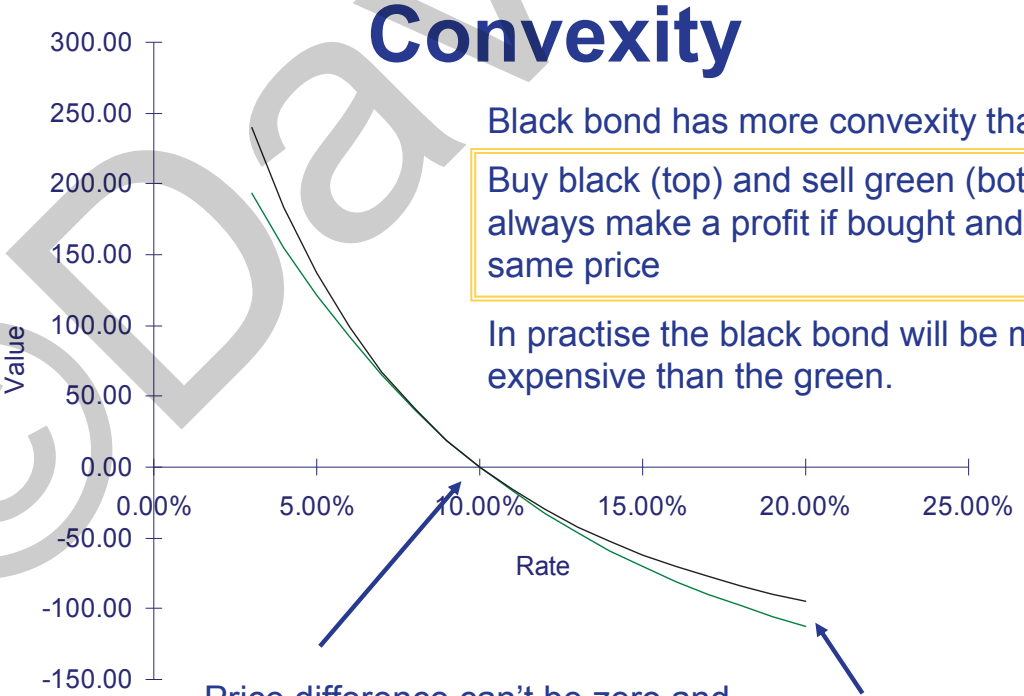
Two bonds with different convexities, but the same price and yield, will give rise to arbitrage profits

- All else being equal, the more convex bond must have a higher price than the less convex bond
- The price difference is a function of how far the market expects interest rates to move up or down
- Interest rate volatility affects relative value in the bond market

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Convexity



Black bond has more convexity than green.

Buy black (top) and sell green (bottom) would always make a profit if bought and sold at the same price

In practise the black bond will be more expensive than the green.

Price difference can't be zero and must reflect the range of likely future rate moves (volatility)

Profit if rates move up or down

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Interest Rate Sensitivity
Convexity

Price as a function of yield can be expanded using a Taylor series:

$$dP = \frac{dP}{dY}dY + \frac{1}{2}\frac{d^2P}{dY^2}dY^2 + \text{Error}$$

So:

$$\frac{dP}{P} = \frac{1}{P}\frac{dP}{dY}dY + \frac{1}{2}\left(\frac{1}{P}\frac{d^2P}{dY^2}dY^2\right) + \frac{\text{Error}}{P}$$

Percentage Price Change

Modified Duration

Convexity

© David Cox

Convexity
Second derivative of price:

$$\frac{d^2P}{dY^2} = \sum_{t=1}^{n}\frac{t(t+1)C}{(1+Y)^{t+2}} + \frac{n(n+1)M}{(1+Y)^{n+2}}$$

t = payment period
Y = yield to maturity (quoted per period)
C = coupon payment per period
M = principal repayment at maturity (after n periods)
P = bond price

$$\text{Convexity} = \frac{1}{P}\frac{d^2P}{dY^2}$$

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Discount Factors from Bonds

Quick Question

Derive discount factors and spot rates from these two bonds

Price	Annual Coupon	Maturity
88.251	5	2 Years
96.782	10	2 Years

Use the discount factors to price the following:

1 year bond with an annual 10% coupon

2 year bond with an annual 8% coupon

If you have time calculate the Yield to Maturity, Duration and Convexity of the 5% and 10% bonds

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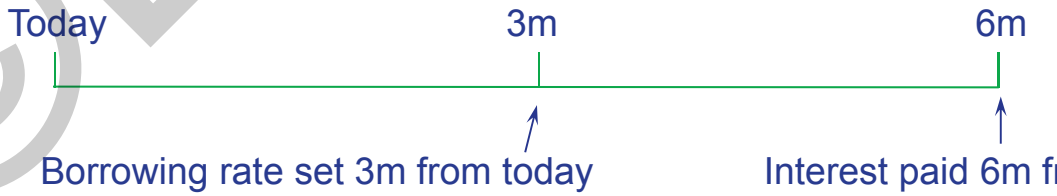
Short Term Risk

€100,000,000 Revolving credit facility @ Euribor

Paid every three months

Rate reset every three months

Consider payment at month six



What is your risk?

How would you deal with it?

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Forward Rate Agreement

FRA

Provides the ability to lock in future rates of interest

Forward contract

- Nothing paid when the agreement is made
- Payments are made on the (forward) settlement date
- Seller pays and buyer receives if interest rates rise

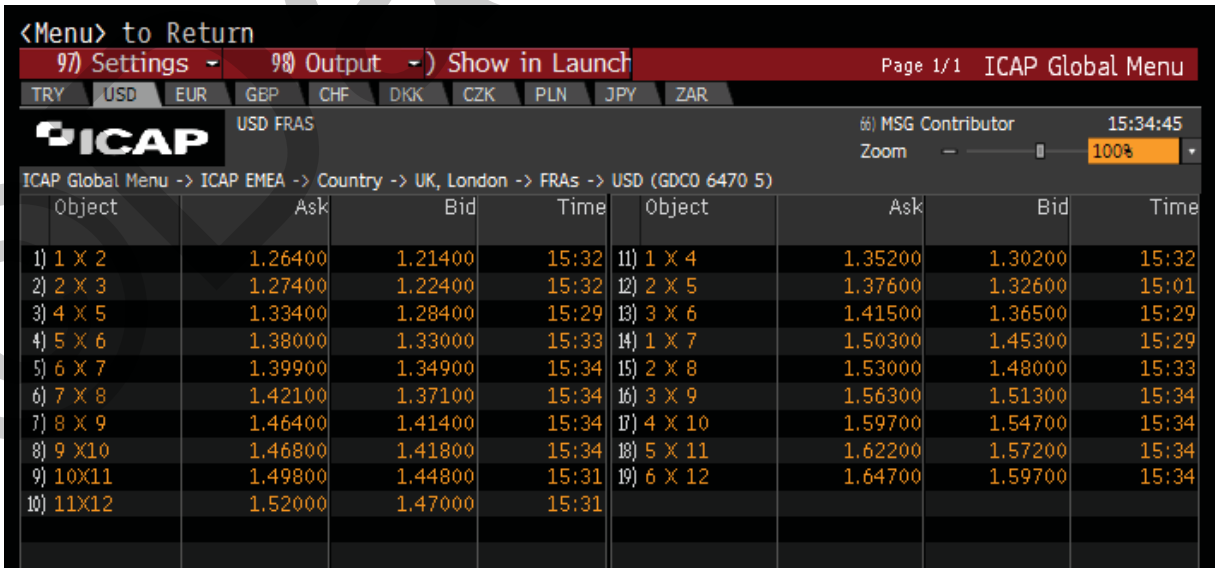
Works as a hedge for Euribor reset risk

- No good for overnight risk

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FRA Rates (ICAP)



<Menu> to Return

97) Settings 98) Output) Show in Launch Page 1/1 ICAP Global Menu

TRY | USD | EUR | GBP | CHF | DKK | CZK | PLN | JPY | ZAR

ICAP USD FRAS 66) MSG Contributor 15:34:45 Zoom 100%

ICAP Global Menu -> ICAP EMEA -> Country -> UK, London -> FRAs -> USD (GDCO 6470 5)

	Object	Ask	Bid	Time		Object	Ask	Bid	Time
1)	1 X 2	1.26400	1.21400	15:32	11)	1 X 4	1.35200	1.30200	15:32
2)	2 X 3	1.27400	1.22400	15:32	12)	2 X 5	1.37600	1.32600	15:01
3)	4 X 5	1.33400	1.28400	15:29	13)	3 X 6	1.41500	1.36500	15:29
4)	5 X 6	1.38000	1.33000	15:33	14)	1 X 7	1.50300	1.45300	15:29
5)	6 X 7	1.39900	1.34900	15:34	15)	2 X 8	1.53000	1.48000	15:33
6)	7 X 8	1.42100	1.37100	15:34	16)	3 X 9	1.56300	1.51300	15:34
7)	8 X 9	1.46400	1.41400	15:34	17)	4 X 10	1.59700	1.54700	15:34
8)	9 X10	1.46800	1.41800	15:34	18)	5 X 11	1.62200	1.57200	15:34
9)	10X11	1.49800	1.44800	15:31	19)	6 X 12	1.64700	1.59700	15:34
10)	11X12	1.52000	1.47000	15:31					

Source: Bloomberg

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Forward Rate Agreement

Agreement between two parties based on:

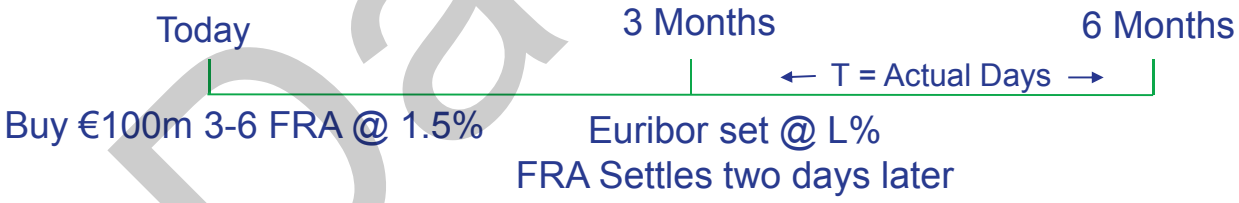
- Notional principal (P)
- Future Euribor rate (L)
- Rate agreed today (FRA rate "F")
- Tenor of future Euribor in days (T)
- Day-count basis (B)

Buyer receives cash if Euribor is over FRA rate, pays if below

$$\text{Cash Payment} = P \times [(L - F) \times T/B] \times \frac{1}{(1 + (L \times T/B))}$$

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FRA Settlement



$$\text{Settlement} = €100\text{m} \times (L - 1.5\%) \times \frac{T}{360} \times D_{3-6}$$

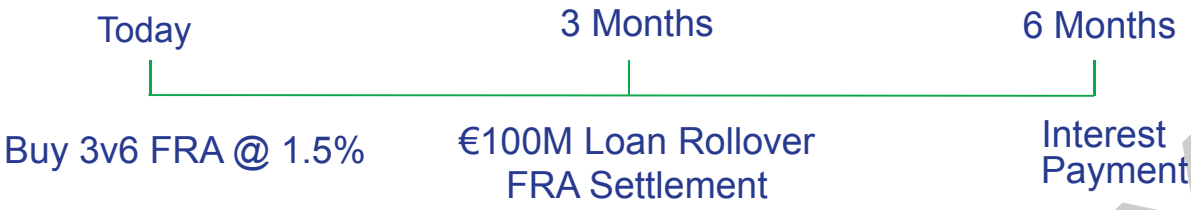
$$D_{3-6} = \frac{1}{\left(1 + L \times \frac{T}{360}\right)}$$

Buyer receives/pays if amount is positive/negative

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Borrowing Hedged with FRA



3 Month Libor in 3 Months	FRA Settlement	Value @ Month 6	Borrowing Cost	Cost of Funds	Rate
2.5%	248,447.20	250,000.00	(625,000.00)	(375,000.00)	1.5%
0.5%	(249,687.89)	(250,000.00)	(125,000.00)	(375,000.00)	1.5%

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Some Market Jargon

Rate "rally" = fall in interest rates
- Bond prices go up

Rate "sell off" = rise in interest rates
- Bond prices fall

Long the market
- Make money when rates rally
- Buy a bond or bond futures
- Buy Eurodollar futures
- Sell FRA
- Receive fixed on an interest rate swap

Short the market
- Make money when rates sell off
- Sell a bond or bond futures
- Sell Eurodollar futures
- Buy FRA
- Pay fixed on an interest rate swap

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More Market Jargon

Curve (yield curve) “flattening”

- Difference between long and short term rates decreases

Curve “steepening”

- Long term rates rise in relation to short term rates

Curve “inversion”

- Long term rates fall below short term rates

“Bear flattener”

- A trade that makes money when rates sell off (rise) and the yield curve flattens

“Bull steepener”

- A trade that makes money when rates rally (fall) and the yield curve steepens

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PV of FRA Position

At the point of trading the value of a FRA position will be zero (Why?)

- Subject to the bid-ask spread and transactions costs.

After trading the value may differ from zero as rates change and time moves on.

- Closing out a position may result in a profit or loss

The present value of an old FRA position is the P/L on closing the position in today’s terms.

This is calculated by using the FRA rate for the forward period and discounting the future cashflow based upon the initial and current rates.

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FRA P/L and DV01

Today | 3 Months ← T = Actual Days → | 6 Months

Libor set @ L%, FRA Settlement:

Buy €100m 3-6 FRA @ 150bp: $€100\text{m} \times (\text{L} - 1.5\%) \times \frac{\text{T}}{360} \times \text{D}_{3-6}$

Sell €100m 3-6 FRA @ 151bp: $€100\text{m} \times (1.51\% - \text{L}) \times \frac{\text{T}}{360} \times \text{D}_{3-6}$

PV of Net Position: $€100\text{m} \times 0.01\% \times \frac{\text{T}}{360} \times \text{D}_{3-6} \times \text{D}_3$

FRA DV01: $€100\text{m} \times 0.01\% \times \frac{\text{T}}{360} \times \text{D}_6$

The DV01 is how much you make or lose in today's terms from a 1bp move in the forward rate.

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FRA Convexity

Today | 3 Months ← T = Actual Days → | 6 Months

Libor set @ L%, FRA Settlement:

FRA DV01: $€100\text{m} \times 0.01\% \times \frac{\text{T}}{360} \times \text{D}_6$

As rates rally DV01 gets bigger because the six month discount factor (a ZCB price) increases

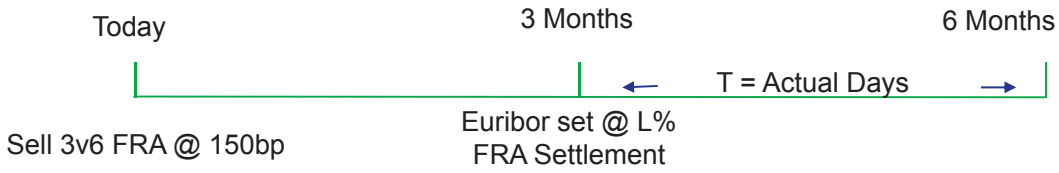
The FRA present value is convex (non linear function of rates):

Yield Curve Flat	D_3	D_6	FRA Settlement	Value Today	FRA DV01
2.50%	0.99378882	0.987654321	248,456.79	246,913.58	2,469.14
1.51%	0.996239197	0.992506575	2,490.63	2,481.27	2,481.27
1.50%	0.99626401	0.992555831	0	0	2,481.39
1.49%	0.996288824	0.992605092	- 2,490.76	- 2,481.51	2,481.51
0.50%	0.998751561	0.997506234	- 249,688.28	- 249,376.56	2,493.77

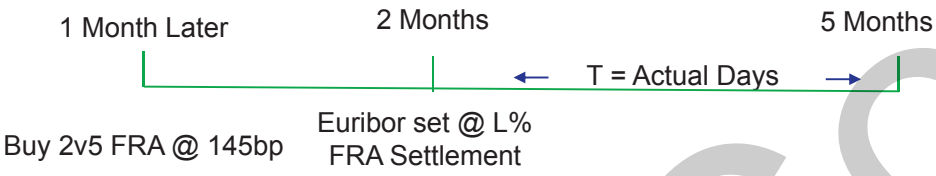
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FRA Roll Down



In one month the position will be a 2v5 FRA:



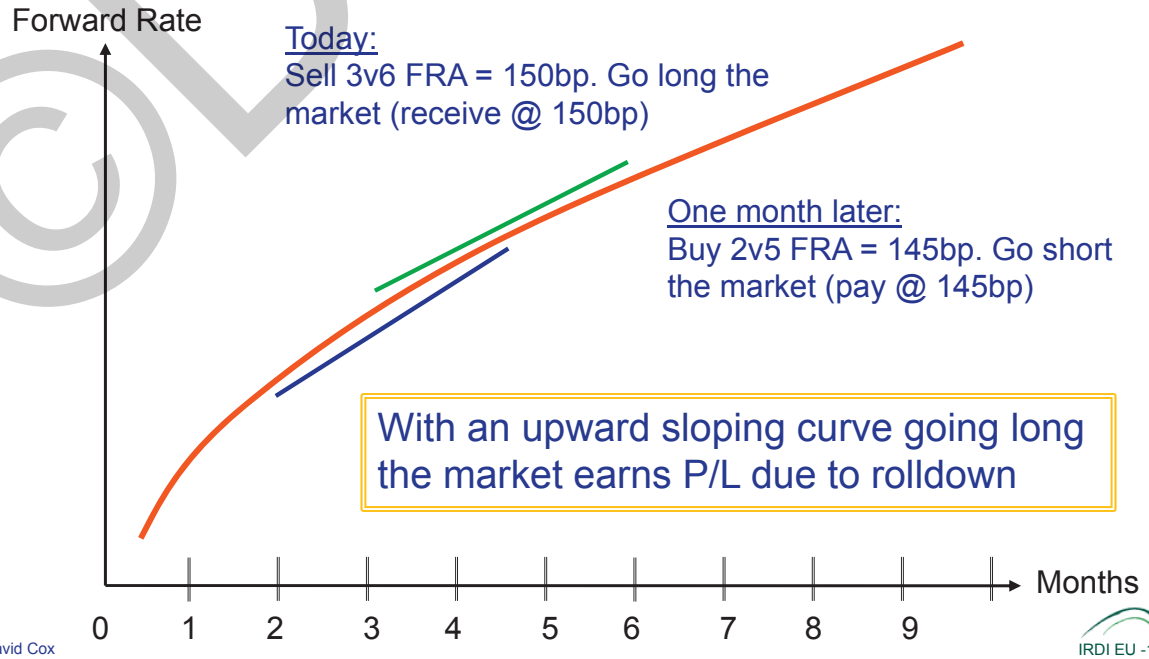
- The PV of an FRA position changes with the level of the forward rate and discount factor, but also the passage of time
- The passing of time gives rise to "roll down" as the open position becomes sensitive to changes in a part of the yield curve nearer to today
- Even if the curve does not change it's level or shape roll down can generate P/L

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FRA Roll Down

3v6 FRA position becomes 2v5 in one month

The yield curve remains exactly the same but "roll down" has reduced the FRA rate by 5bp generating P/L



With an upward sloping curve going long the market earns P/L due to rolldown

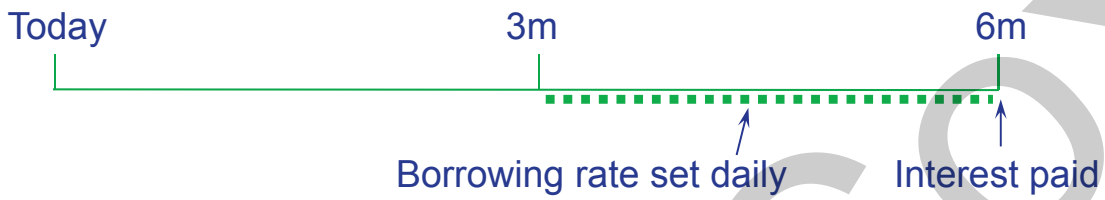
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Short Term Risk

$100,000,000 Revolving credit facility @ SOFR

- Paid every three months
- Rate reset every day
- Consider payment at month six



What is your risk?
How would you deal with it now?

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3 Month Euribor Futures

Contract for future "delivery" of a financial instrument

Specific financial instrument

- 3 month Euro deposit

Standard contract

- Quantity: €1,000,000 contract size
- Dates: IMM dates and 4 serial months

Exchange traded (ICE also Eurex, CurveGlobal)

- Marked to market daily (provision for intraday)
- Traded on margin

Short term rate futures are cash settled

Liquid in the near months

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3 Month Euribor Futures



Expiry price (EDSP) = 100 - Euribor

- Before expiry: if price = 99.500 implied rate for the forward period = 50bp
- Price will be over 100 for negative rates (100.500: -50bp)
- This is not exactly the forward rate as Futures and FRAs have different convexities

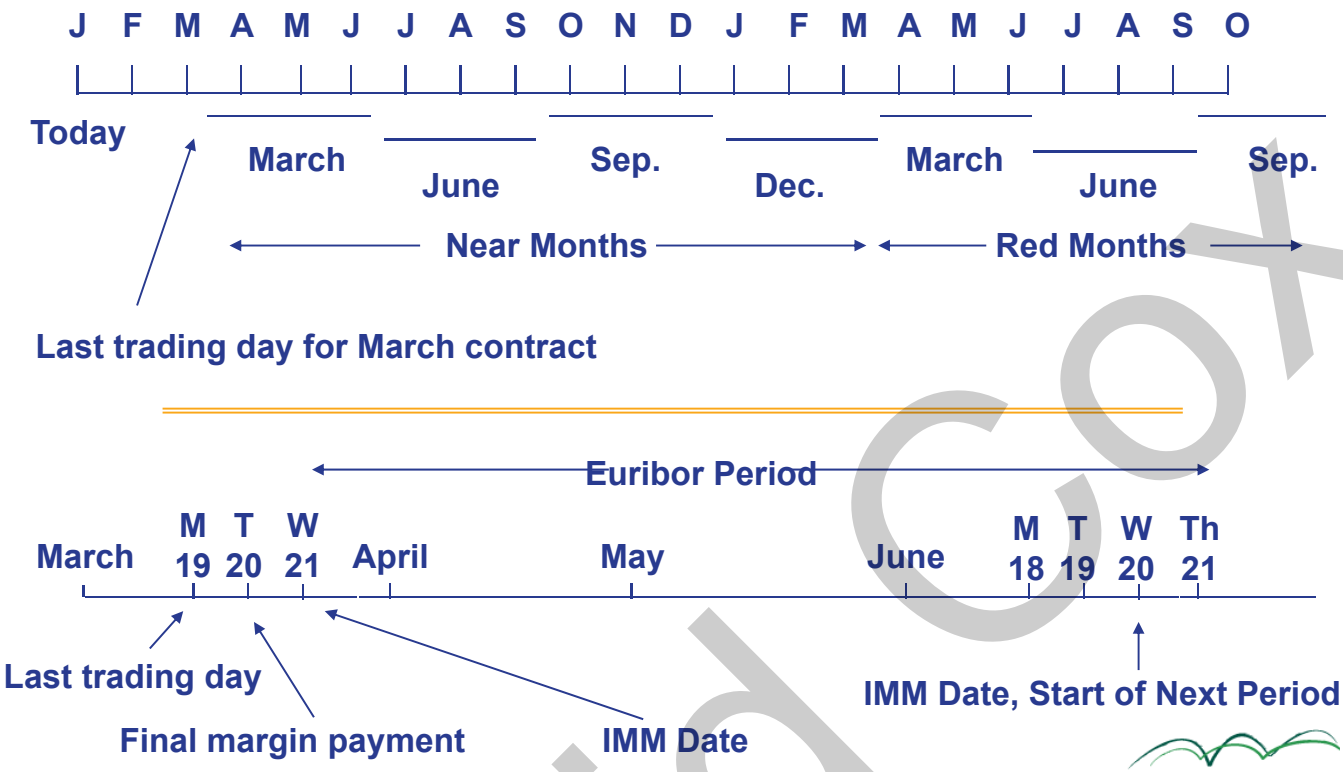
Minimum price move:

- 0.005 (half tick)
- On one contract a move of one tick represents a profit or loss of €25.00
- (0.01% * €1,000,000 * 90/360 = €25)
- This is 90 days, but the contract settles to Euribor

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3 Month Euribor Futures



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Euribor Futures Hedge



Sell 100 Euribor Futures @ 98.50

€100M Loan Rollover

Close Futures Position: P/L in Margin Account

Interest Payment

3 Month Euribor in 3 months	Futures Price	Futures P/L	Loan Interest	Approx. Total Cost of Funds	Approx. Effective Rate
2.5%	97.50	250,000	625,000	375,000	1.5%
0.5%	99.50	-250,000	125,000	375,000	1.5%

$$\text{Number of contracts} = \frac{\text{Change in NPV of Loan for 1 bp Rate Move}}{\text{Futures DV01 (Tick Value: €25)}}$$

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Eurodollar Futures

United States | 97) Export To Exce | Delayed | 07/31/2017

	Start Date	Days	Ticker	Last
	08/02/2017	49	Front	98.7535
10)	09/20/2017	91	EDU7	98.6700
11)	12/20/2017	91	EDZ7	98.5550
12)	03/21/2018	91	EDH8	98.4750
13)	06/20/2018	91	EDM8	98.4100
14)	09/19/2018	91	EDU8	98.3450
15)	12/19/2018	91	EDZ8	98.2650
16)	03/20/2019	91	EDH9	98.2100
17)	06/19/2019	91	EDM9	98.1600
18)	09/18/2019	91	EDU9	98.1100
19)	12/18/2019	91	EDZ9	98.0400
20)	03/18/2020	91	EDH0	97.9950
21)	06/17/2020	91	EDM0	97.9500
22)	09/16/2020	91	EDU0	97.9000

United States | 97) Export To Exce | Delayed | 07/31/2017

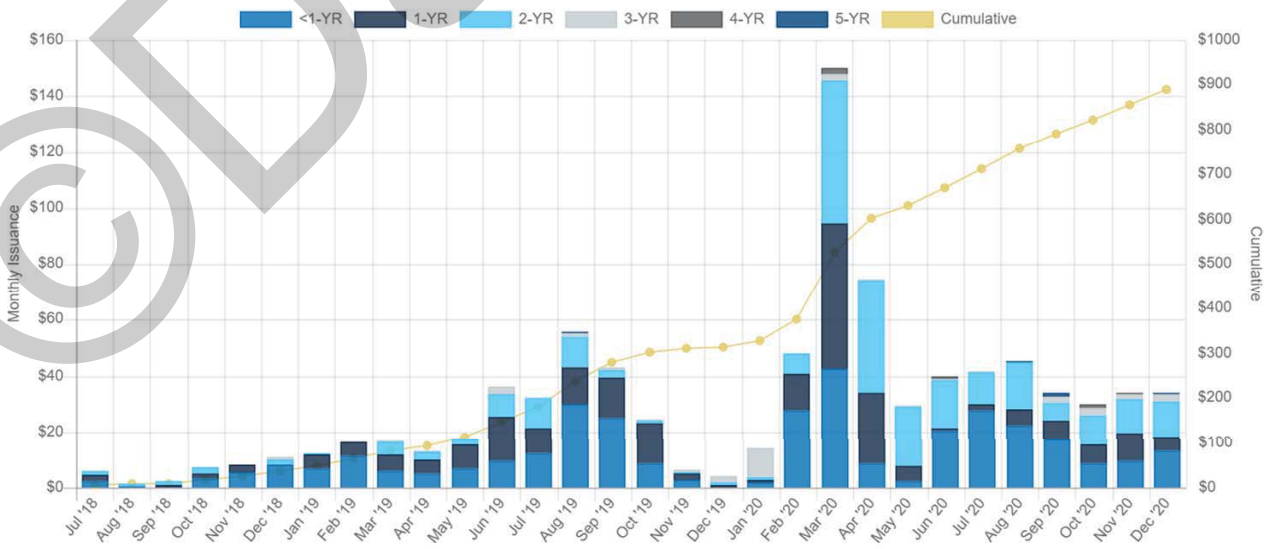
	Start Date	Days	Ticker	Last
23)	12/16/2020	91	EDZ0	97.8400
24)	03/17/2021	91	EDH1	97.7950
25)	06/16/2021	91	EDM1	97.7450
26)	09/15/2021	91	EDU1	97.6950
27)	12/15/2021	91	EDZ1	97.6400
28)	03/16/2022	91	EDH2	97.5950
29)	06/15/2022	98	EDM2	97.5500
30)	09/21/2022	91	EDU2	97.5050
31)	12/21/2022	84	EDZ2	97.4500
32)	03/15/2023	98	EDH3	97.4200
33)	06/21/2023	91	EDM3	97.3800
34)	09/20/2023	91	EDU3	97.3500
35)	12/20/2023	91	EDZ3	97.3200
36)	03/20/2024	91	EDH4	97.3100

Source: Bloomberg

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SOFR Issuance

By Tenor ($Bn)



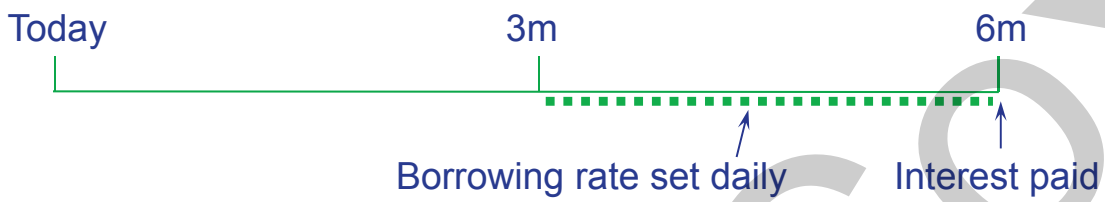
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Short Term Risk

$100,000,000 Revolving credit facility @ SOFR

- Paid every three months
- Rate reset every day
- Consider payment at month six



What is your risk?
How would you deal with it now?

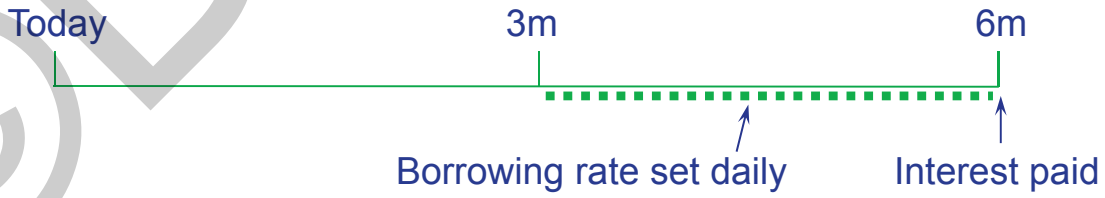
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Short Term Risk

$100,000,000 Revolving credit facility @ SOFR

- Paid every three months
- Rate reset every day
- Consider payment at month six



$$\underbrace{\left(1+r_1\times\frac{1}{360}\right)}_{\text{Wednesday}}\times\underbrace{\left(1+r_2\times\frac{1}{360}\right)}_{\text{Thursday}}\underbrace{\times\left(1+r_3\times\frac{3}{360}\right)}_{Friday\ and\ Weekend}\ldots.\ \underbrace{\times\left(1+r_n\times\frac{1}{360}\right)}_{\text{Tuesday}}=\left(1+R\times\frac{91}{360}\right)$$

Once at the end of the period it is possible to calculate (R), a simple interest rate for the whole period needed to generate the interest paid at the end. This is only known at the end of the period

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3 Month SOFR Futures

Contract for future "delivery" of a financial instrument

Specific financial instrument

- Daily Compounded USD deposit

Standard contract

- Deposit size defined such that one basis point of daily compounded rate = $25
- IMM dates
- Compounding period starts on the contract IMM date
- Ceases trading on the day before the delivery IMM date (the end of the compounding period).
- March contract final price is known only on the June IMM date

Exchange traded (CME also ICE)

- Marked to market daily
- Traded on margin

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3 Month SOFR Futures

March SOFR Contract



EDSP = 100 – R

- R = Simple interest rate derived from daily compounded overnight interest between IMM dates
- This is a backward looking rate (unlike ibor)
- Equivalent to the floating leg of a SOFR Swap
- Forward period is usually 91 days (can be ± 7 days)

Minimum price move:

- 0.0025 for <4m to expiry (quarter tick), 0.005 for the rest
- On one contract a move in rate of one tick represents a profit or loss of $25.00

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3 Month SOFR Futures

EDSP = 100 – R

$$R = \frac{360}{D} \times \left(\prod_{t=1}^{n} \left(1 + \frac{r_t d_t}{360} \right) - 1 \right)$$

D = number of days between IMM dates (normally 91)

n = number of business days in period between IMM dates

r_t = overnight rate for day t (Friday's rate used for the weekend)

d_t= number of days in the overnight period (1 for weekdays, 3 for Friday and the weekend)

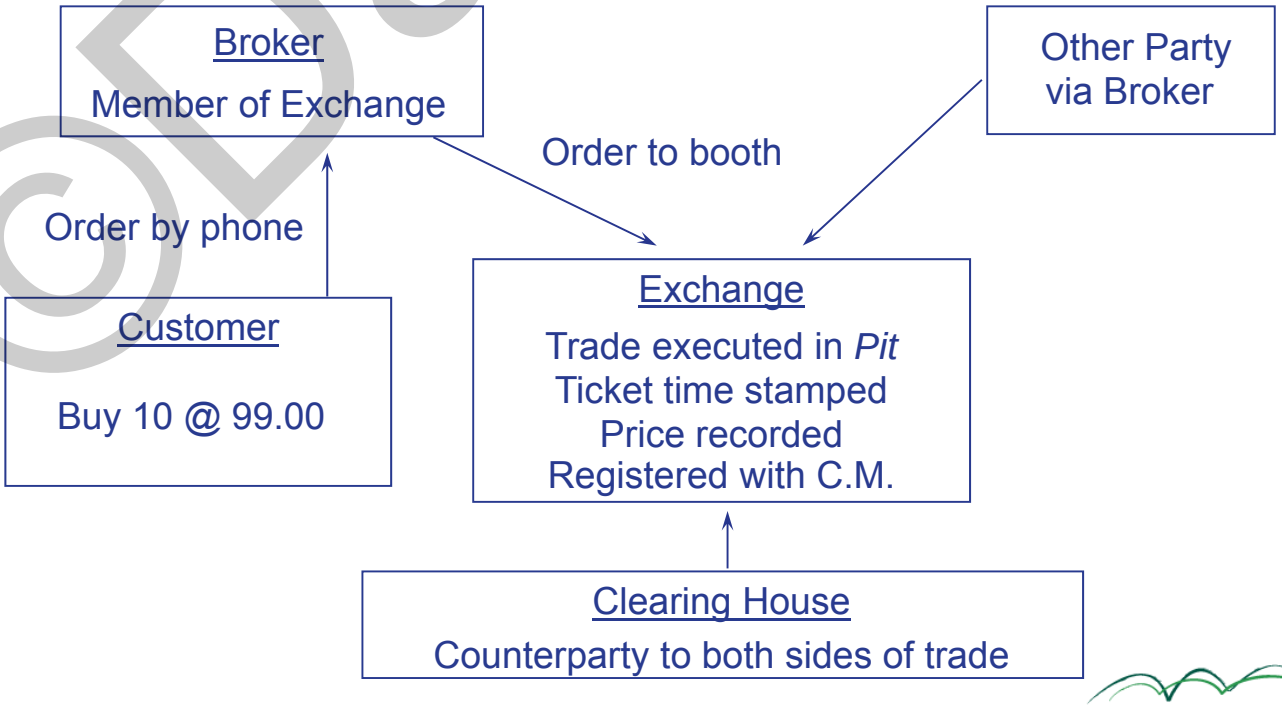
$$\underbrace{\left(1 + r_1 \times \frac{1}{360}\right)}_{\text{Wednesday}} \times \underbrace{\left(1 + r_2 \times \frac{1}{360}\right)}_{\text{Thursday}} \underbrace{\times \left(1 + r_3 \times \frac{3}{360}\right)}_{Friday\ and\ Weekend} \ldots \underbrace{\times \left(1 + r_n \times \frac{1}{360}\right)}_{\text{Tuesday}} = \left(1 + R \times \frac{91}{360}\right)$$

N.B. The one month SOFR contract uses averaging, like Fed Funds futures, not compounding

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Interest Rate Futures

Mechanics of Old Pit Trading



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CME GLOBEX Trading Platform

June 1992	Launched based on Reuters technology.
Aug 1992	Eurodollar launched. 2063 contracts in FX futures traded. Trading from Chicago, New York London and Paris.
Oct 2002	2nd generation platform implemented based on NSC trading system developed by MATIF (now Euronext)
Sept 2003	Total volumes reach 1 million (all contracts)
2004	Additional telecommunication hubs established in Frankfurt, Dublin, Amsterdam, Gibraltar, Milan and Singapore. Eurodollar volume on Globex reaches 72% of total CME Eurodollar volume.
2006	Average daily volume nears 3.2 million contracts 69% of total CME volume.
2008	85% of Eurodollar volume on Globex, volumes decreased at end of 2008 due to low interest rates, started to pick up Jan 2010
Mar 2014	Daily ED volume 2.8 million up 47% from Mar 2013

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Electronic Execution

Establish account

National Futures Association (NFA) registered broker or Futures Commission Merchant

Order entry to Globex

Phone to broker or directly via a computer using a Globex trading front-end application. (From brokers, CME or Independent Software Vendors)

Execution takes less than one millisecond

Electronic trading available around the clock

CME Globex Electronic Markets: 5:00 p.m. – 4:00 p.m. CT, Sunday to Friday

Electronic trading carries on in parallel with trading by open outcry on the CME floor 7:20 a.m. – 2:00 p.m. CT.

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Order Matching

CME has predefined sets of matching rules (algorithms) for the products traded. One is applied to each product :

First-in, First-out

Price and time only criteria for filling order.

All orders at same price level filled according to time priority.

Allocation (Pro-Rata)

Top order filled first, then rest allocated depending on size of orders placed (orders all at same price)

* Used for the Eurodollar contract

Lead Market Maker (LMM)

Allocation to increase liquidity in certain markets, individuals selected by CME given LMM status, must always make 2 way prices, in return get a guaranteed allocation percentage of incoming orders

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Exchange Clearing Member

Transactions are executed through CM and cleared by CME Clearing

Highly capitalized,

Closely monitored

CM Stands behind all trades made through CME

Assume full financial and performance responsibility.

CMs are responsible and accountable for every position they carry

Including for the account of a member, member firm, non-member customer or own account.

Customers, members and member firms must all have a relationship with a CM to be able to trade

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CME Clearing

Division of CME through which:

- All trades are confirmed, matched and settled daily until offset or delivered
- All financial settlements are made.

Takes one side and a clearing member the opposite side of each transaction

Conducts business only with clearing members

- No transaction with customers, individual or corporate members

Held accountable to clearing members for performance on all open positions

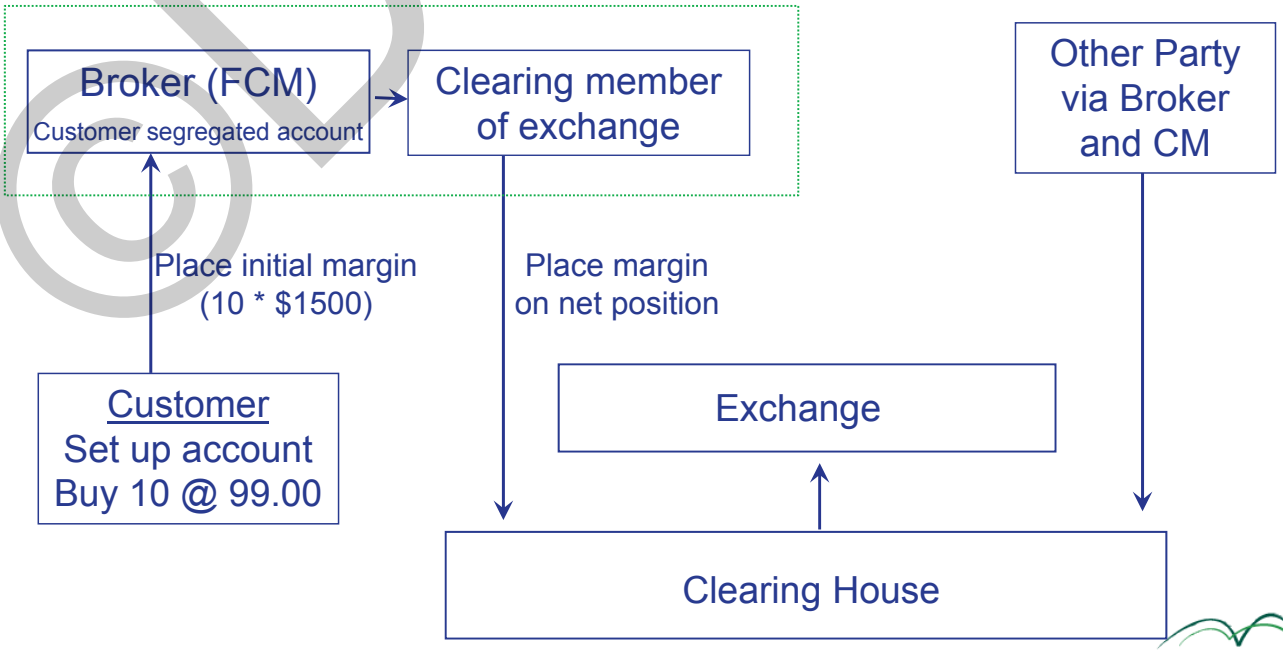
Guarantees performance on each contract

- Protects buyers and sellers from loss

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Interest Rate Futures

Initial Margin

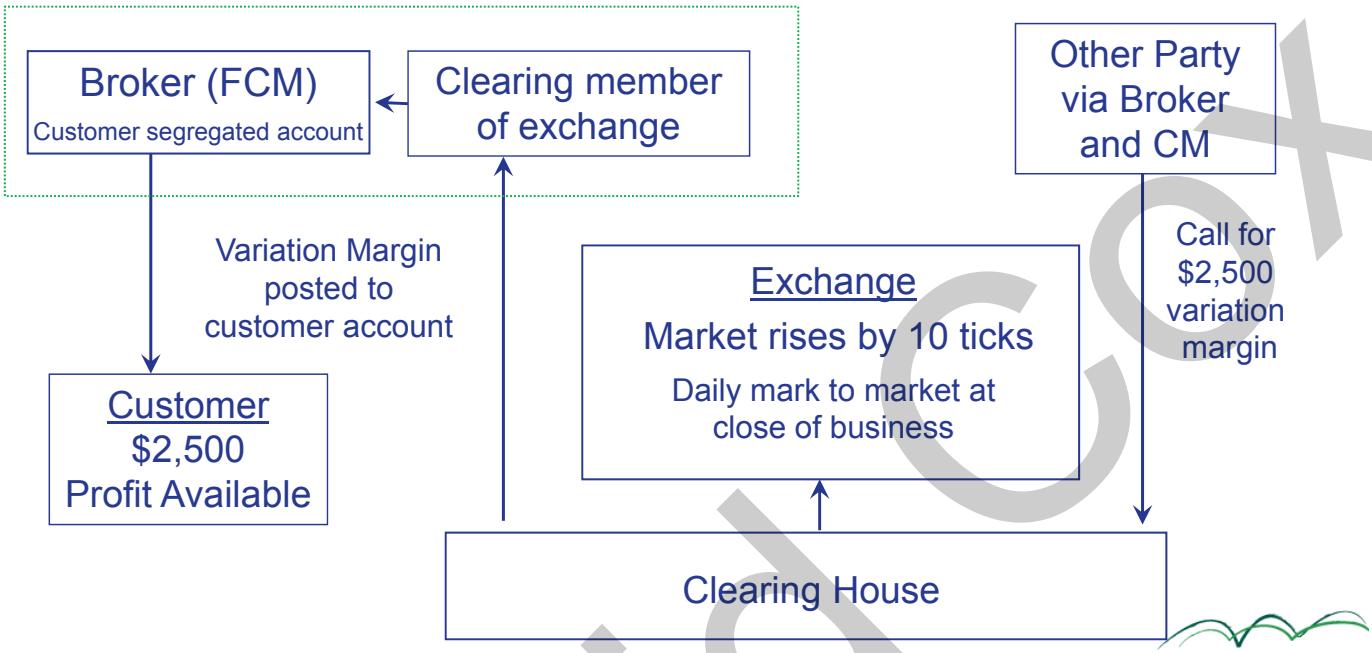


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Interest Rate Futures

Variation Margin



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Medium Term Instruments

FRAs	Cash	Futures
Can be OTC	Cash	Always exchange traded
Clearing available	OTC	Standard periods
Any maturity	Any maturity	Fixed contract size
Any principal	Any principal	High liquidity
Counterparty risk	Counterparty risk	Clearing house counterparty
Collateral posting	Liquidity a problem	Margin administration
“Off” balance sheet	Fully on balance sheet	“Off” balance sheet

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Day Two

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Day Two

8.30 – 10.30 **Overnight Index Swaps**

- Mechanics
- Valuation
- DV01
- Forward OIS and futures
- Convexity correction

Yield Curve Building

- Why build a yield curve
- Bootstrapping an OIS curve
- Three month v six month Euribor basis and multiple forward curves
- The short end of the curve in today's market
- What happens when rates go negative
- Interpolation methods and their impact

10.30 – 10.45 Break

10.45 – 13.00 **Exercise: Curve building**

13.00 – 14.00 Lunch

14.00 – 15.30 **Interest Rate Swaps in General**

- The background to swaps
- Today's market and jargon
- Market conventions and standard swap configurations
- The importance of basis swaps
- Multiple swap trades: Curve and Fly
- Rules of thumb and quick pricing formulae
- Pricing for zero NPV and execution
- Revaluation
- Carry and roll-down
- Hedging a swap book's market risk
- Spot and forward rate sensitivities and risk measures

15.30 – 15.45 Break

15.45 – 17.15 **Exercise: Pricing, hedging and revaluing Swaps**

17.15 – 17.30 **Online Quiz**

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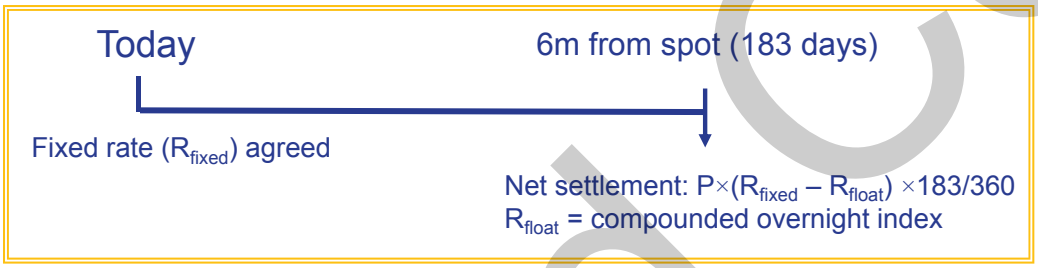
Overnight Index Swap (OIS)

Agreement between parties to pay:

- Fixed rate of interest in exchange for a rate derived from a compounded overnight index
- Money market conventions used for cashflow calculations
- Euro: ESTA, GBP: SONIA, USD: SOFA

No exchange of principal

- Maturities under one year generally have a single payment at maturity
- Swaps with maturity over one year often have annual payments



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Overnight Index Swap (OIS)

Floating Payments



$$\left(1 + r_1 \times \frac{1}{360}\right) \times \left(1 + r_2 \times \frac{1}{360}\right) \times \left(1 + r_3 \times \frac{3}{360}\right) \times \left(1 + r_4 \times \frac{1}{360}\right) \cdots = \left(1 + R_{float} \times \frac{183}{360}\right)$$

Floating rate (R_{float}) is a simple interest rate giving a cashflow equivalent to that generated by overnight compounding throughout the period

Calculated in the same way as for SOFR futures

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Overnight Index Swap (OIS)

DV01

Today	6m from spot (183 days)
9:00 am receive Fixed @ 100bp on 100m 6m OIS	100m×(100bp - R_{float})×183/360
16:00 pm Close by agreeing to pay Fixed @ 99bp	100m×(R_{float} - 99bp)×183/360
PV: 100m×0.01%×183/360×D_{183}	Net: 100m×0.01%×183/360

Swap DV01 = 100m×0.01%×183/360×D_{183}

For swaps with multiple periods add up the DV01s of all the individual cashflows

$$DV01 = P \times 0.01\% \times \sum_{t=1}^{M} \tau_t \times D_t$$

P = Swap notional

τ_t= Day count fraction for each cashflow at time t (Days/360)

D_t = Discount factor for each time t

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Forward OIS

Fixed rate agreed today, casflows on both sides based on a forward period

Forward period can be linked to IMM dates, but not necessarily

Today 26 Jan 2021 Execute Forward Swap (0)	Dec 2025 IMM Date Compounding Starts (1786)	March 2026 IMM Date Compounding Ends (1877)
Fixed rate (R_{fixed}) agreed		Swap settles
		Net settlement: P×(R_{fixed} – R_{float}) ×91/360 R_{float} is derived from compounded overnight rates between IMM dates

$$\text{Forward OIS DV01} = P \times 0.01\% \times \frac{91}{360} \times D_{1877}$$

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Convexity

Today	Dec 25	March 26
26 Jan 21	IMM Date	IMM Date
(0)	(1786)	(1877)
Execute Forward Swap		Swap settles
Futures Hedge		Futures expire

Receive fixed on $100m 57v60 Forward IMM OIS @1.00%

Hedge by Selling Dec 25 SOFR Futures @ 99.00

Yield curve is flat @ 1% daily compounded

$D_{1877} = 0.949197712$

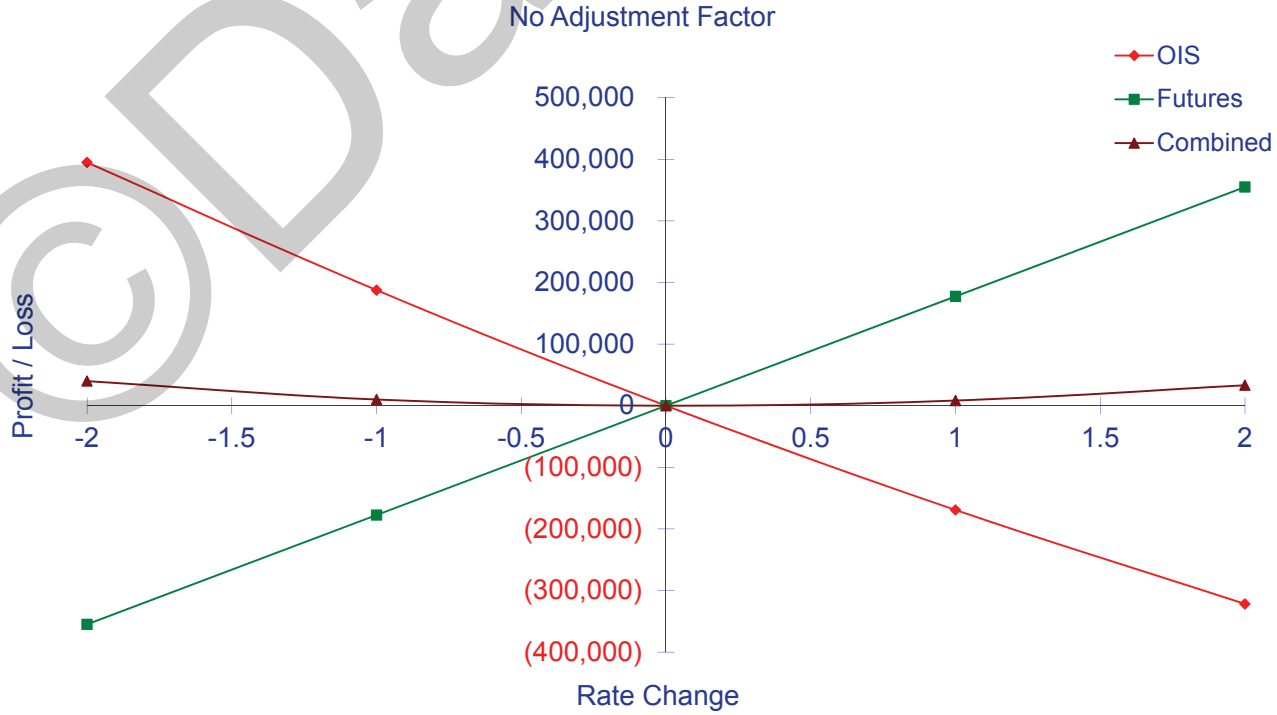
Swap DV01 (Value for 1bp parallel shift in YC):

$$100\text{m} \times 0.01\% \times \frac{91}{360} \times D_{1877} = 2{,}399.36$$

Sell $\frac{2{,}399.36}{25} = 96$ Contracts to hedge

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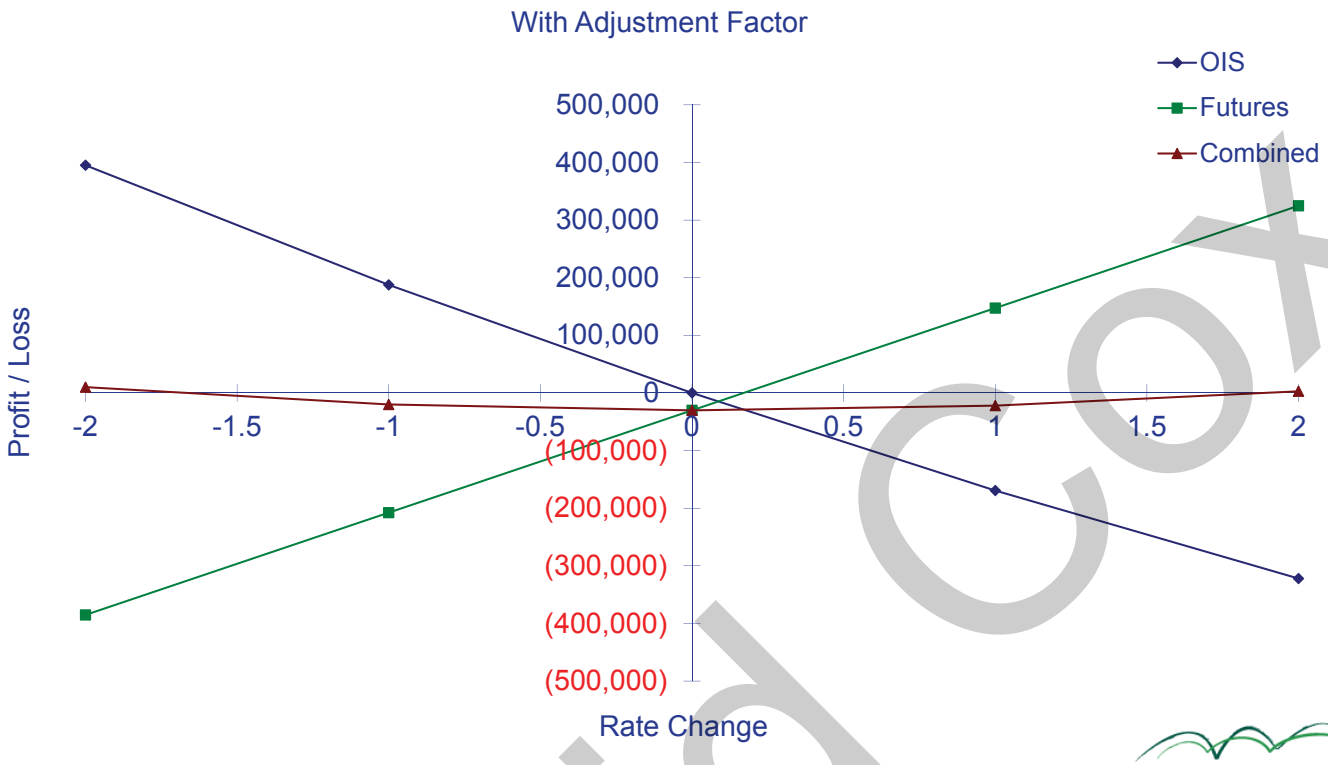
Receive Fixed: $100m 57 v 60 IMM Dated SOFR OIS (Long) Sell SOFR Futures (Short)



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Receive Fixed: $100m 57 v 60 IMM Dated SOFR OIS (Long) Sell SOFR Futures (Short)



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Futures and Forwards

Often Assumed that 100-Futures price = forward rate

This is not exact:

- Interest differentials on margin surplus and funding
  - Short the market with futures: sell off gives profit and a higher margin investment rate, reverse for a rally
- Forward contract DV01 depends on Discount Factor so has convexity from today's perspective
- Futures P/L is linear in rates as DV01 does not depend on discounting ($25 per tick)

"Convexity" Stochastic interest rates give rise to differences:

- Cox, Ingersoll and Ross (1981) JFE

The Convexity Bias in Eurodollar Futures:

- Burghart and Hoskins (1994) DW Research Note

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Convexity

Forward and Spot Rates are Highly Correlated

Today a Long position in OIS has positive convexity

Discount Factor gets bigger as market rallies

Futures have zero convexity

$25 per tick remains constant regardless of expiry date

Difference must be paid for:

100 - futures price is greater than the forward OIS rate:

100 - P = (F+x)

Adjustment factor (x) takes into account the volatility of the forward and spot rates and their correlation

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Adjustment Factor

Typical Values

Vol = 90bp Corr = 0.95

Six Months:	0.16bp
One Year:	0.49bp
Two Years:	1.7bp
Five Years:	10bp

Important beyond 12 months

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Convexity Correction

Eurodollar Futures

31 July 2017

Adjustment not ticked

United States | 97) Export To Excel | 98

Delayed | 07/31/2017 | Adju

	Start Date	Days	Ticker	Last	Rate
23)	12/16/2020	91	EDZ0	97.8400	2.1600
24)	03/17/2021	91	EDH1	97.8000	2.2000
25)	06/16/2021	91	EDM1	97.7500	2.2500
26)	09/15/2021	91	EDU1	97.6950	2.3050
27)	12/15/2021	91	EDZ1	97.6400	2.3600
28)	03/16/2022	91	EDH2	97.5950	2.4050
29)	06/15/2022	98	EDM2	97.5500	2.4500
30)	09/21/2022	91	EDU2	97.5050	2.4950
31)	12/21/2022	84	EDZ2	97.4500	2.5500
32)	03/15/2023	98	EDH3	97.4200	2.5800
33)	06/21/2023	91	EDM3	97.3800	2.6200
34)	09/20/2023	91	EDU3	97.3500	2.6500
35)	12/20/2023	91	EDZ3	97.3200	2.6800
36)	03/20/2024	91	EDH4	97.3100 y	2.6900

Adjustment ticked

United States | 97) Export To Excel | 98

Delayed | 07/31/2017 | Adju

	Start Date	Days	Ticker	Last	Rate
23)	12/16/2020	91	EDZ0	97.8400	2.1269
24)	03/17/2021	91	EDH1	97.7950	2.1674
25)	06/16/2021	91	EDM1	97.7450	2.2127
26)	09/15/2021	91	EDU1	97.6950	2.2578
27)	12/15/2021	91	EDZ1	97.6400	2.3076
28)	03/16/2022	91	EDH2	97.5950	2.3472
29)	06/15/2022	98	EDM2	97.5500	2.3861
30)	09/21/2022	91	EDU2	97.5050	2.4253
31)	12/21/2022	84	EDZ2	97.4500	2.4748
32)	03/15/2023	98	EDH3	97.4200	2.4980
33)	06/21/2023	91	EDM3	97.3800	2.5315
34)	09/20/2023	91	EDU3	97.3500	2.5549
35)	12/20/2023	91	EDZ3	97.3200	2.5780
36)	03/20/2024	91	EDH4	97.3100 y	2.5810

Source: Bloomberg

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Yield Curves

Basic Types

Bond Yield Curve

- YTM against bond tenor
- YTM of bonds priced at par
- These do not show an accurate term structure of rates

"Zero Coupon" Yield Curve

- Spot interest rates against time
- Directly related to discount factor curve
- Ibor and OIS curves are not the same
- Forward Ibor curves of different tenors are not the same
- Correct curves are essential for valuing interest rate derivatives
- Essential for correct valuation of all cash flows over time

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Yield Curve Building

Risk Management Issues

Assumptions and Purpose

Choose appropriate techniques

Interpolation

Techniques contain a range of different assumptions

Multiple Curves

Incomplete markets lead to many possible yield curves from the same data

Liquidity and credit effects give different curves for different underlying intervals of rate (6m Ibor v 3m Ibor v OIS)

Spread risk

Measures of Confidence

Regression techniques for markets with limited data provide yield curves measured within confidence limits

External effects like taxes

Certain regimes create tax specific yield curves

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Discount Factor

Principal Constraints

Always one or less?

Money today is worth more than the same money in the future

Only true when interest rates are positive!

Always greater than zero:

Money is always worth having however far in the future

Always decreasing?

Money gets less valuable the further away it is in time.

Not so if forward rates are negative!

No discontinuities in the curve

Forward interest rates are always finite

Zero coupon bonds only have one price

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Zero Coupon Yield Curve

Ingredients for Basic Bootstrapping

Cash Rates

FRA Rates

Futures Prices

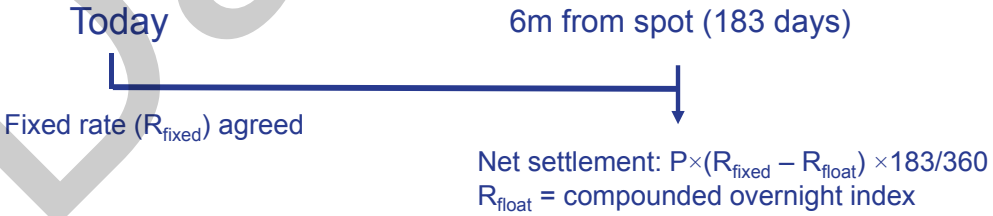
Swap Rates

These days we need more than just a Libor curve

© David Cox

Building an OIS Curve

6m OIS



All cashflows discounted at OIS rates so:

- Floating leg has a value of par If we include a principal payment at maturity (discounting and earning at the same rate)
- Fixed and floating legs have the same value as no payment is made at execution

OIS can be treated as a par value bond with a coupon equal to the fixed rate

- Principal needs to be included in final cashflows even though not settled

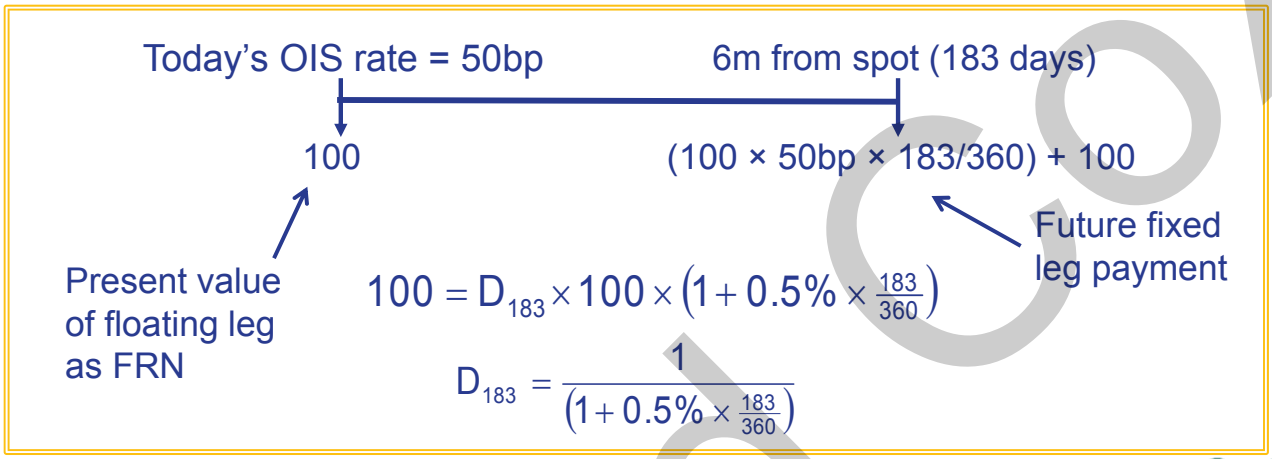
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Building an OIS Curve

Under one year OIS makes a single payment

- Treat as floating rate note (FRN) and zero coupon bond (ZCB)
- Floating leg treated as an FRN paying the overnight rate and discounted at the same rate will have par value
- Fixed leg is treated as a ZCB paying the swap rate plus par
- Swap legs have equal NPV



$$100 = D_{183} \times 100 \times \left(1 + 0.5\% \times \frac{183}{360}\right)$$

$$D_{183} = \frac{1}{\left(1 + 0.5\% \times \frac{183}{360}\right)}$$

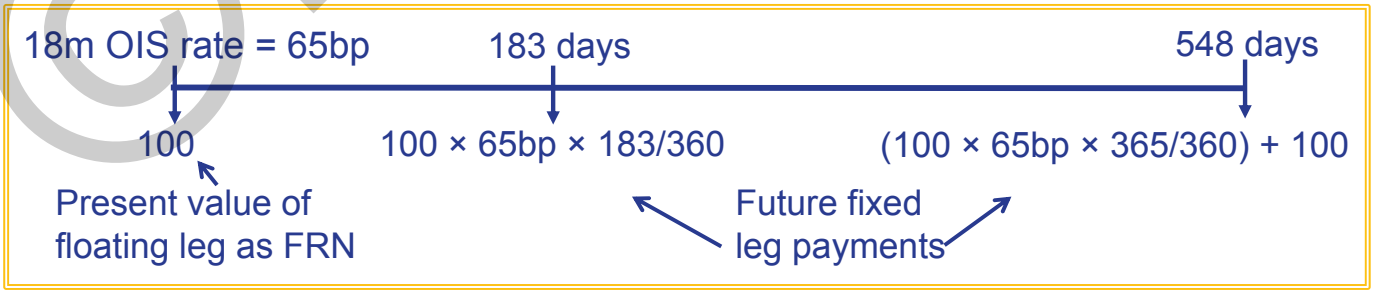
Below one year maturity OIS swap rates are zero coupon

© David Cox

Building an OIS Curve

Beyond one year OIS makes annual payments with a short stub at the beginning if necessary

- Treat floating leg as par value FRN
- Treat fixed leg as bond paying coupon equal to swap rate
- Swap legs have equal NPV



$$100 = D_{183} \times 100 \times 0.65\% \times \frac{183}{360} + D_{548} \times 100 \times \left(1 + 0.65\% \times \frac{365}{360}\right)$$

$$D_{548} = \frac{1 - D_{183} \times 0.65\% \times \frac{183}{360}}{\left(1 + 0.65\% \times \frac{365}{360}\right)}$$

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Building an OIS Curve

OIS rates (money market basis):

1 Year: 0.5%	367 Days
2 Years: 0.55%	365 Days
3 Years: 0.6%	366 Days

$$D_1 = \frac{1}{\left(1 + 0.5\% * \frac{367}{360}\right)} = 0.994929 \qquad D_2 = \frac{1 - 0.994929 \times 0.55\% * \frac{367}{360}}{\left(1 + 0.55\% \times \frac{365}{360}\right)} = 0.988907$$

$$100 = D_1 \times 100 \times 0.6\% \times \frac{367}{360} + D_2 \times 100 \times 0.6\% \times \frac{365}{360} + D_3 \times \left(100 \times 0.6\% * \frac{366}{360} + 100\right)$$

$$D_3 = \frac{1 - \left(0.994929 \times 0.6\% \times \frac{367}{360}\right) - \left(0.988907 \times 0.6\% \times \frac{365}{360}\right)}{1 + 0.6\% \times \frac{366}{360}} = 0.981909$$

Continue to the longest maturity OIS
Beyond this use Ibor v OIS basis swaps

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Fed Funds v Libor Basis Swap

Term	Mid	Bid	Ask
03M	29.055	28.055	30.055
06M	29.500	28.500	30.500
09M	30.750	29.750	31.750
01Y	30.250	29.250	31.250
18M	31.500	30.500	32.500
02Y	32.750	31.750	33.750
03Y	33.000	32.000	34.000
04Y	33.000	32.000	34.000
05Y	32.750	31.750	33.750
07Y	32.125	31.125	33.125
10Y	30.375	29.375	31.375
12Y	29.000	28.000	30.000
15Y	27.500	26.500	28.500
20Y	25.500	24.500	26.500
25Y	24.125	23.125	25.125
30Y	23.000	22.000	24.000

Subtract spread from Libor to get fixed swap v Fed Funds

Source: Bloomberg page BGUS: BGC Partners

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USD Basis Swap



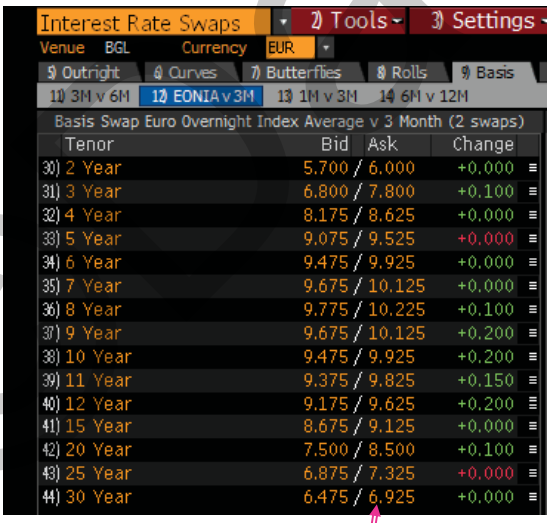
Source: Bloomberg

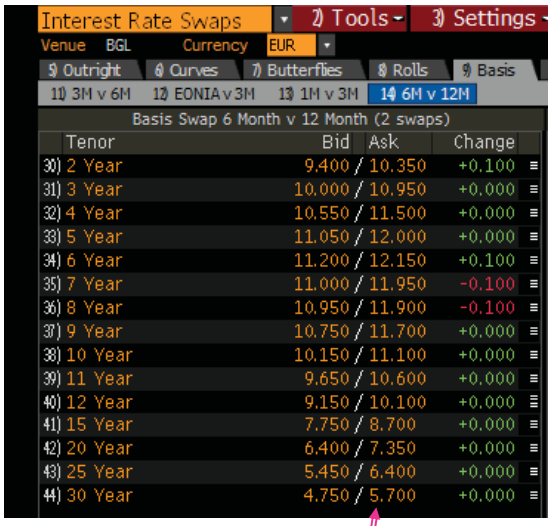
40y price taker pays 1m Libor + 9.15bp, receives 3m Libor Flat

30y price taker pays 3m Libor + 12.801bp, receives 6m Libor Flat

© David Cox

Euro Basis Swaps





Source: Bloomberg

30y price taker pays Eonia+ 6.925bp, receives 3m Euribor flat

30y price taker pays 6m Euribor + 5.7bp, receives 12m Euribor flat

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Yield Curve Building Summary

Need to build several curves

OIS curve

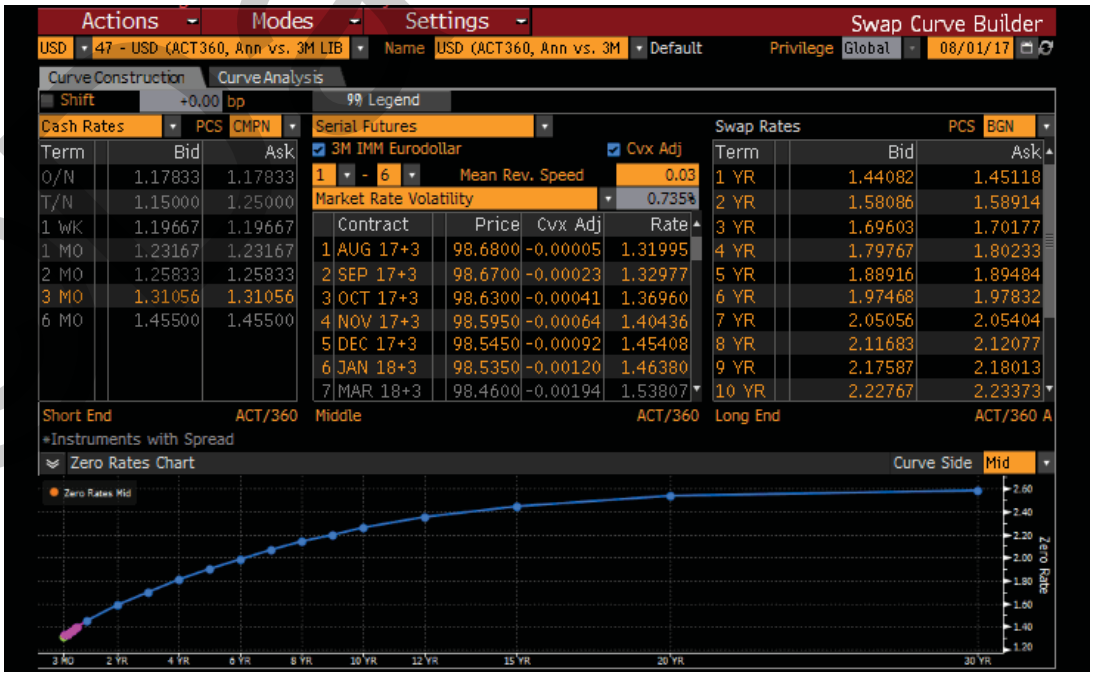
- Used for discounting cashflows when derivatives are collateralised.
- Ibor swaps can no longer be treated as having par value on the floating side

Multiple Libor/Euribor forward curves

- Future cashflows based upon Euribor and can be hedged with forward contracts (Futures, FRAs Swaps etc.)
- Due to liquidity and credit differences sequences of forward rates with different tenors are not equivalent
- Liquid basis swaps allow for conversion between Euribor periods

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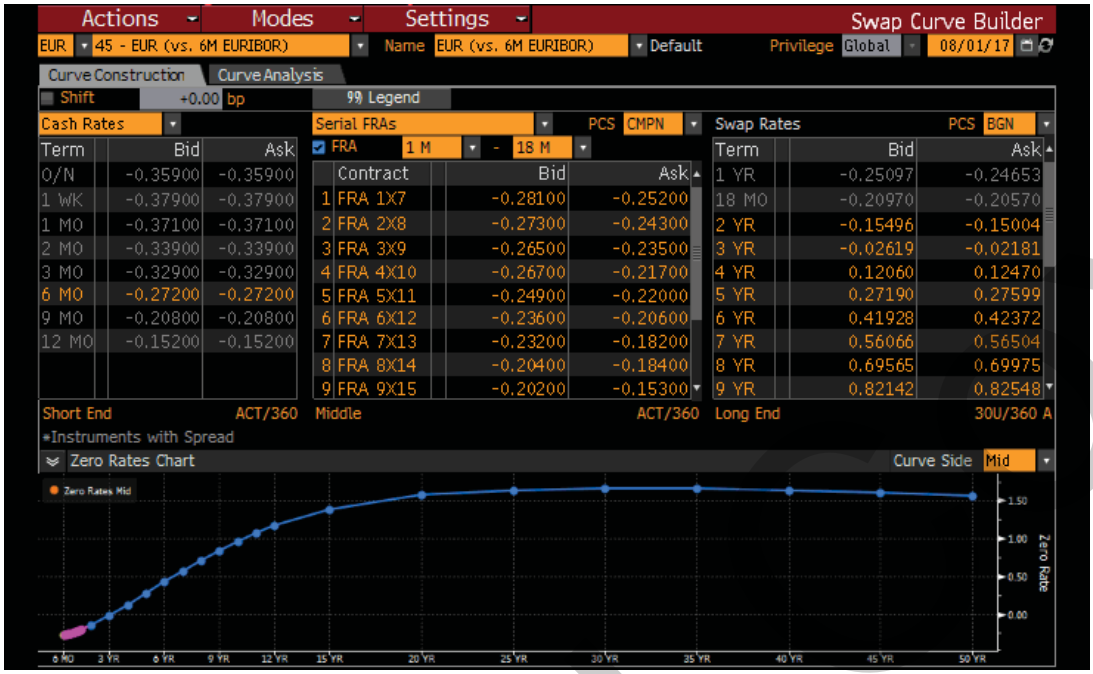
3m USD Libor Curve



Source: Bloomberg

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6m Euribor Curve



Source: Bloomberg

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Interpolation

Finding New Points On a Yield Curve

- Straight Line
- Polynomial
  - Single high order polynomial
  - Unstable between points and at ends
- Splined Polynomial
  - Chain of low order polynomials
- Weighted sum of other functions
  - Basis Splines
  - Bernstein Polynomials

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Interpolation

Straight Line

Easy to Calculate

Not Accurate For Sharp Bends

Cannot use accurately on discount factors

Discontinuous Forward Rates

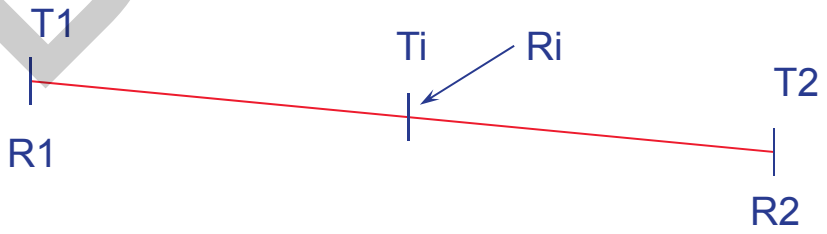
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Interpolation

Straight Line

Intermediate values lie on a line between the two nearest data points.



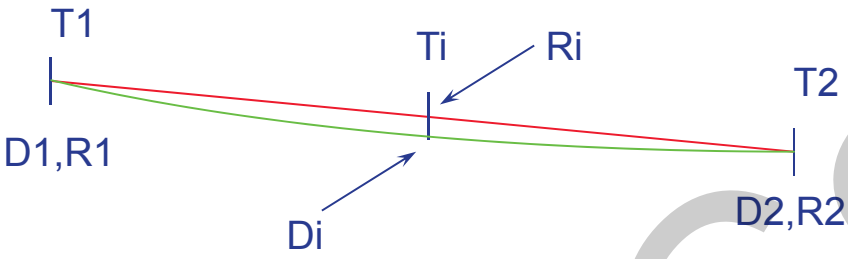
$$R_i = R_1(1-\lambda) + R_2(\lambda)$$

$$\lambda = \frac{(T_i - T_1)}{(T_2 - T_1)}$$

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Interpolation

If interest rates follow a straight line discount factors do not



Using Rates
R1 = R2 = 6.00%
T1 = 180
T2 = 360
Ti = 270
Ri = 6.00%

Using Discount Factor
D1 = 0.970874
D2 = 0.943396
Di = 0.957135
Ri = 5.9713%

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Interpolation

The problem of forward rates

Accurate forward rates are essential for the pricing of a wide range of instruments e.g. forward start swaps and caps.

Straight line interpolation can give discontinuous forward rates if the slope of the curve changes

Cubic splines will avoid this because of the smoothness conditions

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Interpolation

Cubic Splines

A different cubic polynomial is fitted between each pair of data points.

At each point the linked polynomials are twice differentiable

The slope of the total curve is smooth as is the rate of change of the slope

Because cubic splines can reflect the shape of the discount curve between data points they can be used on both rates and discount factors without significant loss of accuracy.

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Interpolation

Cubic Spline

Smooth curve

Twice differentiable at every data point

Not so easy to calculate

Good for bendy curves

Continuous forward rates

Can suffer from oscillation

Can be used with discount factors

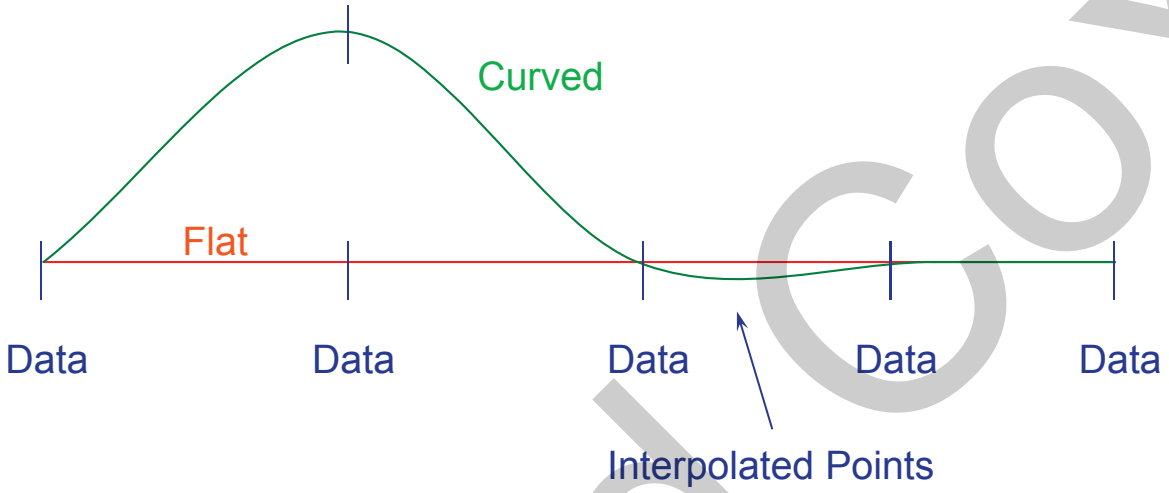
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Cubic Splines

Oscillation

Interpolation techniques produce estimates not observations and each has its limitation.



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Interest Rate Swap

Agreement between two parties

One party could be a clearing house

Exchange of interest payments

No exchange of principal on vanilla IRS, unlike CCS

Payments based on:

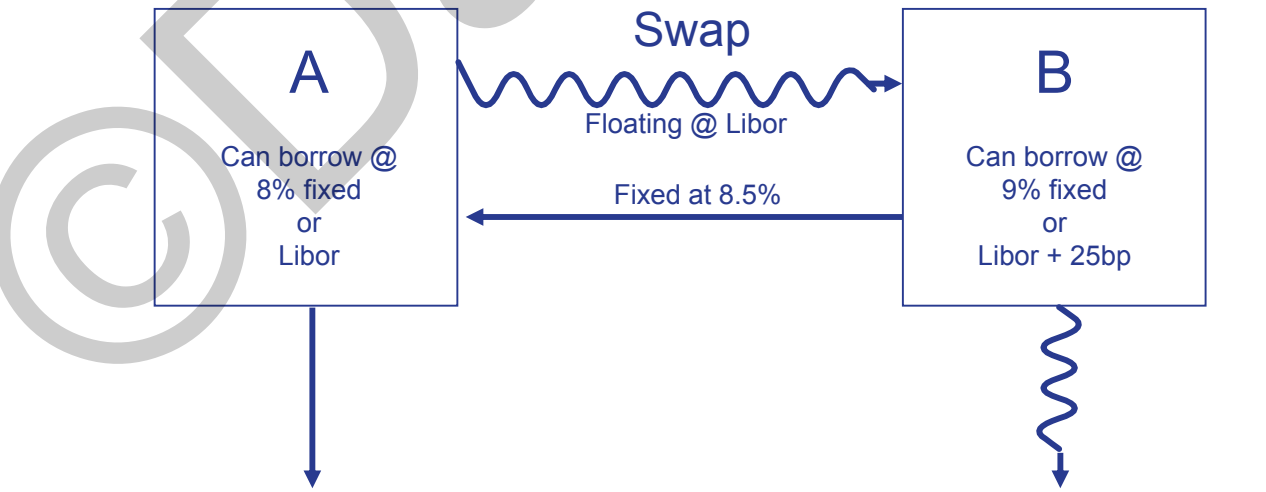
- Notional principal
- Fixed rate of interest
- Floating rate (Ibor or overnight)
- Spread over floating rate
- Payment interval
- Start date and tenor
- Day count fractions
- Holiday calendar
- Reset delay

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Interest Rate Swaps

How the market began in the early 1980s



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Interest Rate Swaps
Today's Market

Very active and transparent 2 way market
- Liquid in major currencies
- Major institutions run large books
- Quoted fixed against Ibor or overnight
- Minimum market size: $50 million
- Maximum size depends on DV01, larger amounts for shorter dates
- Central clearing
- Electronic execution

Liquidity concentrates around specific maturities:
- 1-5y, 7y, 10y, 15y, 20y ….
- Other dates trade regularly

Short (money market) swaps most liquid
- Arbitraged and hedged with futures

Term (longer dated) swaps bond related

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Interest Rate Swaps

Market participants and purposes

Intermediaries

Brokers: arrangement only in illiquid markets

Banks and other market makers: OTC counterparty, manage mismatched cash flows, take credit risk, run book

SEFs: electronic execution

CCPs: central counterparty and clearing

Participants

Banks, Funds

Local authorities (municipalities)

Corporations

Supranational institutions

Sovereign borrowers

Purposes

Trading and arbitrage

Hedging and risk management

Debt structuring

Asset creation

Cash flow transformation

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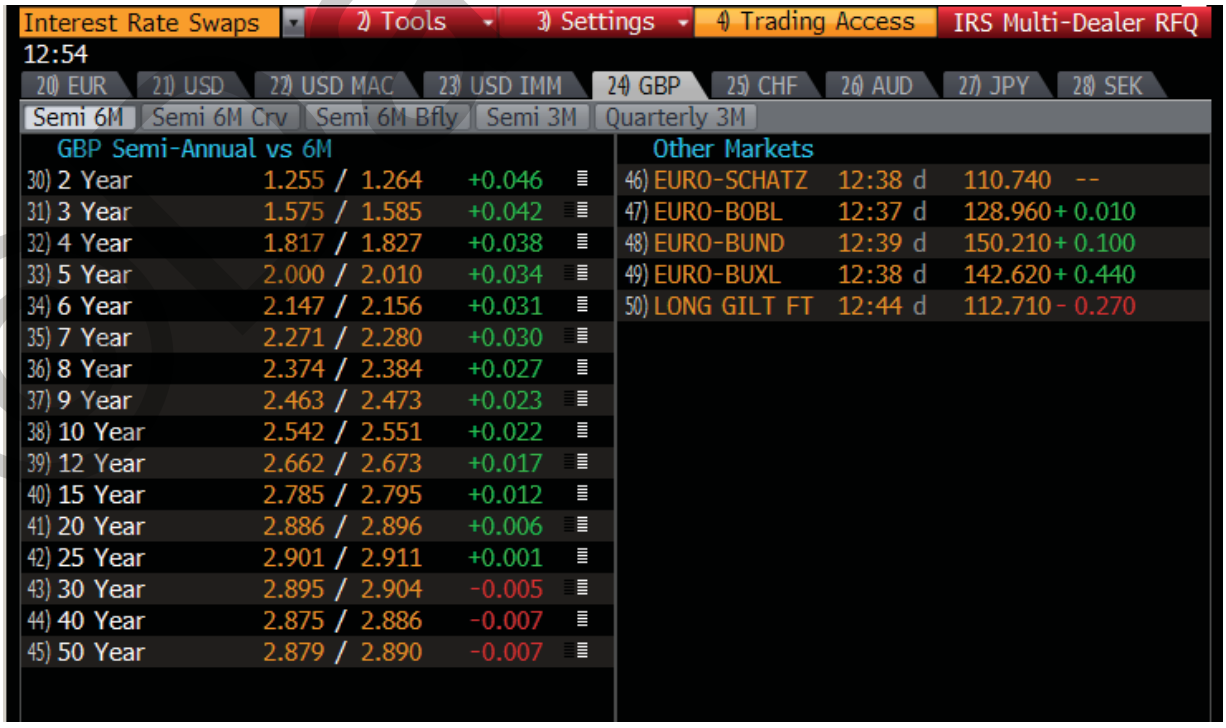
Ibor Swap Conventions

	USD	EUR	JPY	GBP
Fixed Side:				
Payment Period	Semi-Annual	Annual	Semi-Annual	Semi-Annual
Day Count Fraction	30/360	30/360	Actual/365	Actual/365
Payment Holidays	New York,London	TARGET	Tokyo,London	London
Business Day Convention	Modified Following	Modified Following	Modified Following	Modified Following
Floating Side:				
Floating Rate	LIBOR	EURIBOR	LIBOR	LIBOR
Designated Maturity	3m	6m	6m	6m
Payment Period	Quarterly	Semi-Annual	Semi-Annual	Semi-Annual
Reset Period	Quarterly	Semi-Annual	Semi-Annual	Semi-Annual
Day Count Fraction	Actual/360	Actual/360	Actual/360	Actual/365
Payment Holidays	New York,London	TARGET	Tokyo,London	London
Business Day Convention	Modified Following	Modified Following	Modified Following	Modified Following
Reset Holidays	London	TARGET	London	London
Reset Delay (t+X days)	2	2	2	0

Cross border payment system for the Euro

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GBP Libor IRS



Source: Bloomberg

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Euribor IRS

Interest Rate Swaps | 2) Tools | 3) Settings | IRS Trading Portal

Venue BGL Currency EUR

5) Outright | 6) Curves | 7) Butterflies | 8) Rolls | 9) Basis | 10) Cross-asset

11) 6M | 12) MAC v 6M | 13) 3M | 14) 12M | 15) 1M | 16) OIS | 17) IMM v 6M | 18) IMM v 3M

Annual v 6 Month Euribor

	Tenor	Bid	Ask	Change
30)	1 Year	-0.250	-0.247	+0.001
31)	18 Month	-0.208	-0.205	+0.003
32)	2 Year	-0.153	-0.150	+0.003
33)	3 Year	-0.022	-0.019	+0.004
34)	4 Year	0.126	0.130	+0.007
35)	5 Year	0.282	0.286	+0.012
36)	6 Year	0.432	0.435	+0.013
37)	7 Year	0.577	0.581	+0.016
38)	8 Year	0.716	0.720	+0.018
39)	9 Year	0.845	0.849	+0.018
40)	10 Year	0.962	0.966	+0.018
41)	11 Year	1.067	1.071	+0.019
42)	12 Year	1.159	1.162	+0.019
43)	13 Year	1.237	1.241	+0.018
44)	14 Year	1.307	1.311	+0.019
45)	15 Year	1.365	1.369	+0.017
46)	16 Year	1.414	1.419	+0.018
47)	17 Year	1.456	1.461	+0.017
48)	18 Year	1.490	1.495	+0.017

Annual v 6 Month Euribor

	Tenor	Bid	Ask	Change
49)	19 Year	1.520	1.525	+0.017
50)	20 Year	1.544	1.548	+0.016
51)	25 Year	1.610	1.615	+0.014
52)	30 Year	1.637	1.642	+0.012
53)	40 Year	1.637	1.642	+0.010
54)	50 Year	1.586	1.593	+0.008

Other Markets

	Tenor	Price	Change	Time	Delayed
100)	EURO-SCHATZ	112.075	--	13:50	d
101)	EURO-BOBL	132.080	-0.030	13:50	d
102)	EURO-BUND	161.850	-0.130	13:50	d
103)	EURO-BUXL	161.460	-0.080	13:50	d

Source: Bloomberg

© David Cox

MXN IRS v TIIE

Mexico | Browse | 98) Settings | 15:08:26 | Markets Monitor

1) Foreign Exchange | 2) Fixed Income | 3) Swaps | 4) Equity | 5) Commodities

6) TIIE | 7) RR Derivatives

Bloomberg

	Bid	Ask
TIIE		
3x1	7.87	7.91
6x1	7.97	8.01
9x1	8.04	8.08
13x1	8.04	8.10
26x1	7.95	7.99
39x1	7.86	7.90
52x1	7.83	7.87
65x1	7.83	7.87
91x1	7.92	7.97
130x1	8.05	8.10
195x1	8.23	8.28
260x1	8.32	8.36
390x1	8.51	8.57

Prebon

	Bid	Ask
TIIE		
3x1	7.87	7.91
6x1	7.94	8.04
9x1	8.01	8.11
13x1	8.02	8.12
26x1	7.92	8.02
39x1	7.83	7.93
52x1	7.80	7.90
65x1	7.80	7.90
91x1	7.89	7.99
130x1	8.02	8.12
195x1	8.27	8.38
260x1	8.37	8.47
390x1	8.54	8.64

Bloomberg

	Bid	Ask
Basis		
3x1	19.00	29.00
6x1	18.00	28.00
9x1	10.00	20.00
13x1	3.00	13.00
26x1	-2.00	4.00
39x1	5.00	5.00
52x1	-5.00	5.00
65x1	-5.00	5.00
91x1	-5.00	5.00
130x1	-7.00	3.00
195x1	-9.00	1.00
260x1	-12.00	-2.00
390x1	-8.00	2.00

Tullett

	Bid	Ask
Basis		
3x1		
6x1		
9x1		
13x1		
26x1		
39x1		
52x1		
65x1		
91x1		
130x1		
195x1		
260x1		
390x1		

	Last	Chg
Official Rates		
Target	7.50	.25
Bank	7.57	.01
Govt	7.58	.02
TIIE		
TIIE 28D	7.8250	-.01
TIIE 91D	7.8850	.00
UDI	5.992379	.00
LIBOR		
1M	1.59375	.00
6M	2.10613	.01

MXN SWAP (TIIE... Daily Line



8) Mexico Interest Rates

9)	Fidelity Says Bond Bears Are Getting It Wrong on Mexican Debt	BN	01:01
10)	Rand Jumps on Zuma Exit; China Boosts U.S. Debt Share: EM...	BN	02/18
11)	It's a First: S. African Bonds Getting More Love Than Mexico...	BN	02/16
12)	SocGen Lifts South Africa Overweight Exposure in Bond Portfolio	BFW	13:23
13)	Fidelity Says Bond Bears Are Getting It Wrong on Mexican Debt	BN	01:01
14)	Rand Jumps on Zuma Exit; China Boosts U.S. Debt Share: EM...	BN	02/18
15)	Artemis: Mexico 7.2 quake rattles new World Bank cat bond	BLG	02/17

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Source: Bloomberg

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USD Libor IRS

Interest Rate Swaps | 2) Tools | 3) Settings | IRS Trading Portal

Venue BGL Currency USD

5) Outright | 6) Curves | 7) Butterflies | 8) Rolls

9) S/A v 3M | 10) S/A v 1M | 11) Ann v 3M | 12) MACS/A | 13) IMM S/A | 14) IMM Ann | 15) OIS | 16) Basis

Semi-annual v 3 Month Libor

	Tenor	Bid	Ask	Change
30)	6 Months	1.381	1.384	+0.002
31)	12 Months	1.457	1.460	+0.005
32)	18 Months	1.537	1.540	+0.007
33)	2 Year	1.598	1.601	+0.008
34)	3 Year	1.715	1.718	+0.010
35)	4 Year	1.816	1.819	+0.007
36)	5 Year	1.909	1.912	+0.007
37)	6 Year	1.995	1.998	+0.008
38)	7 Year	2.072	2.075	+0.008
39)	8 Year	2.140	2.143	+0.008
40)	9 Year	2.201	2.204	+0.009
41)	10 Year	2.254	2.257	+0.008
42)	12 Year	2.344	2.347	+0.010
43)	15 Year	2.433	2.436	+0.010
44)	20 Year	2.516	2.519	+0.009
45)	25 Year	2.552	2.555	+0.011
46)	30 Year	2.566	2.569	+0.011
47)	40 Year	2.554	2.558	+0.010
48)	50 Year	2.526	2.529	+0.011

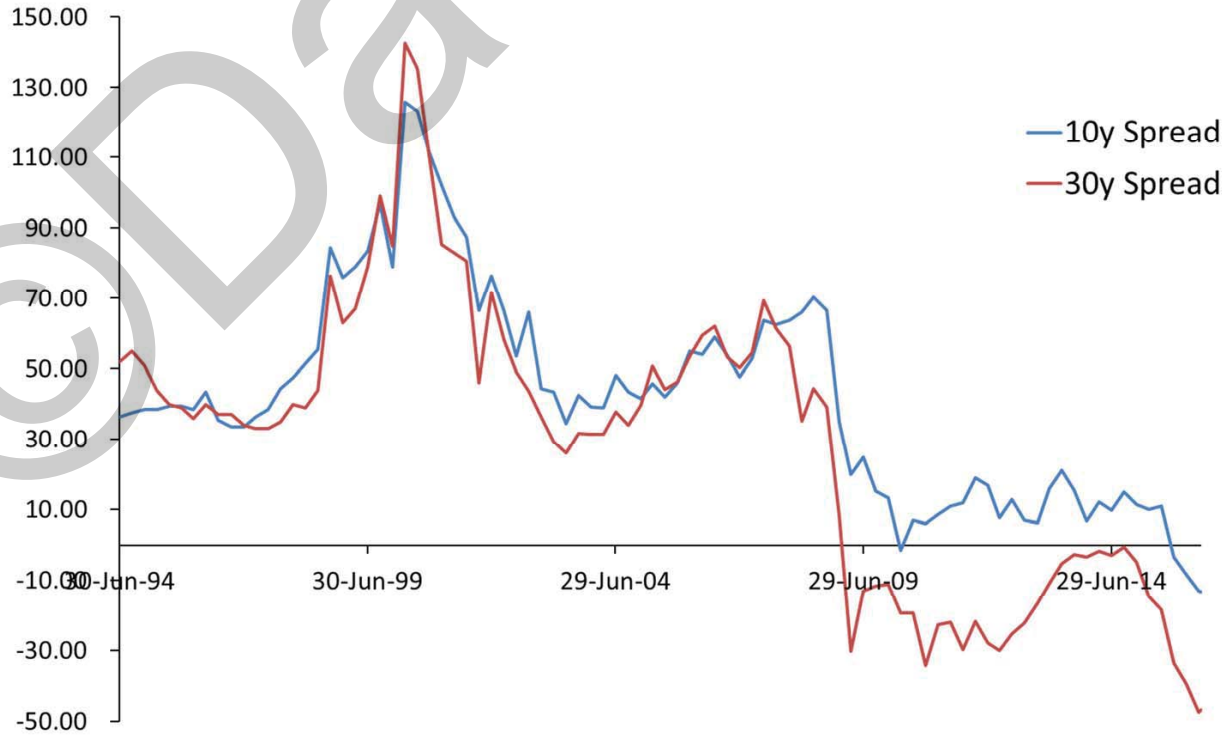
Spreads v Treasuries

	Tenor	Bid	Ask	Change
	1 Year	24.134	25.225	+0.175
70)	2 Year	24.414	24.822	+0.424
71)	3 Year	21.039	21.373	+0.891
	4 Year	14.460	15.000	+0.640
72)	5 Year	7.108	7.442	+0.369
	6 Year	1.962	2.294	+0.106
73)	7 Year	-3.928	-3.594	-0.116
	8 Year	-3.503	-3.171	-0.212
	9 Year	-3.848	-3.515	-0.432
74)	10 Year	-4.825	-4.450	-0.324
	12 Year	4.148	4.481	-0.186
	15 Year	-2.196	-1.863	-0.355
	20 Year	-9.200	-8.707	-0.503
	25 Year	-20.865	-20.55	-0.533
75)	30 Year	-34.870	-34.53	-0.766

Source: Bloomberg

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Treasury Bond v Swap



Swap Rate - Treasury Yield

Source: Bloomberg

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18 Month Quarterly Quarterly

Quarterly fixed against three month Euribor

Euribor set for second period (3 months from today)

Fixed and floating payments for period 2*



Net settlement of period 1 fixed and floating payments*

Fixed rate and first period Euribor set when swap is dealt

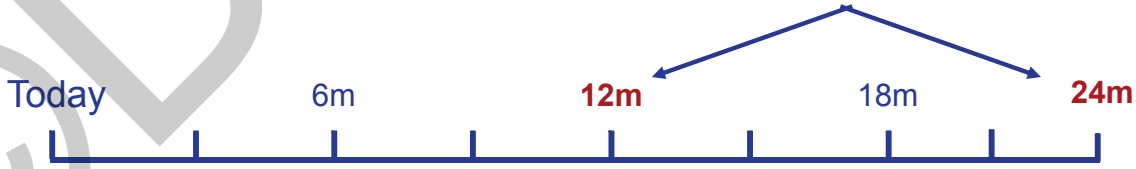
* $ and € swaps have payment dates calculated from spot
£ swaps have payment dates calculated from today

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2 Year Annual Threes

Annual fixed against three month Euribor

Annual fixed payments @ 12, 24 months*



Quarterly floating payments (@ 3, 6, 9 …… months*)

Euribor set quarterly (3, 6, 9 ….. months from today)

Fixed rate and first floating rate set (Euribor)

* $ and € swaps have payment dates calculated from spot
£ swaps have payment dates calculated from today

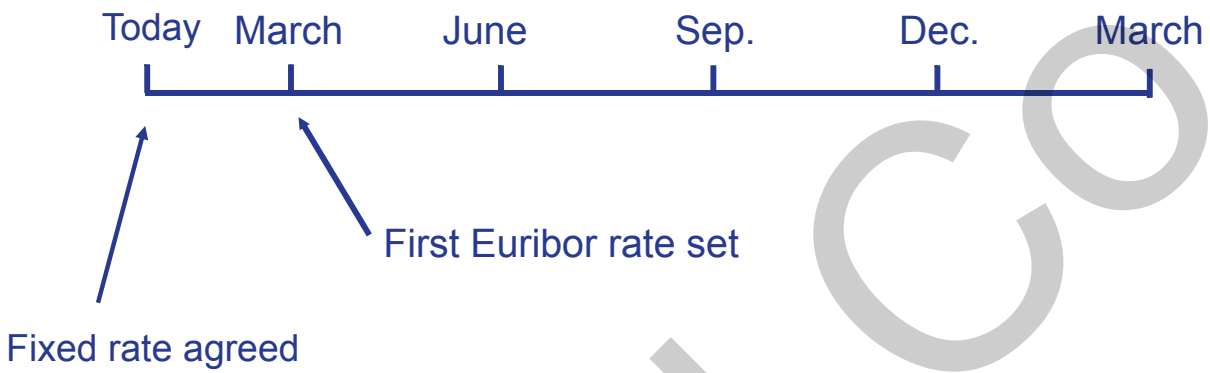
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March - March (IMM)

Forward starting swap linked to IMM Dates

Fixed and floating payments made on each IMM date
(third Wednesday in March, June, September or December)



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IMM Swaps

Interest Rate Swaps | 2) Tools | 3) Settings | 4) Trading Access | IRS Multi-Dealer RFQ

13:19

20) EUR | 21) USD | 22) USD MAC | 23) USD IMM | 24) GBP | 25) CHF | 26) AUD | 27) JPY | 28) SEK

IMM S/A | S/A Crv1 | S/A Crv2 | S/A Roll | S/A Bfly | IMM Ann | Ann Crv1 | Ann Crv2 | Ann Roll | Ann Bfly

USD Semi-Annual vs 3M IMM

	Sep14	Dec14	Mar15	Jun15
1Y	0.333 / 0.337	0.456 / 0.460	/	0.851 / 0.854
2Y	0.714 / 0.718	0.897 / 0.902	1.100 / 1.105	1.335 / 1.339
3Y	1.150 / 1.155	1.336 / 1.341	1.530 / 1.535	1.731 / 1.735
4Y	1.517 / 1.521	1.681 / 1.686	1.847 / 1.853	2.017 / 2.022
5Y	1.799 / 1.803	1.941 / 1.945	2.086 / 2.091	2.235 / 2.239
6Y	2.020 / 2.024	2.149 / 2.153	2.276 / 2.281	2.408 / 2.412
7Y	2.201 / 2.205	2.317 / 2.321	2.434 / 2.439	2.551 / 2.554
8Y	2.353 / 2.358	2.459 / 2.464	2.563 / 2.568	2.673 / 2.677
9Y	2.481 / 2.486	2.577 / 2.582	2.677 / 2.682	2.775 / 2.779
10Y	2.590 / 2.594	2.681 / 2.686	2.772 / 2.777	2.863 / 2.867
12Y	2.764 / 2.769	2.844 / 2.848	/	3.001 / 3.005
15Y	2.943 / 2.948	3.008 / 3.013	/	3.139 / 3.143
20Y	3.103 / 3.106	3.154 / 3.159	/	3.256 / 3.261
25Y	3.175 / 3.180	3.219 / 3.222	/	3.306 / 3.309
30Y	3.212 / 3.215	3.250 / 3.254	/	3.326 / 3.330

Source: Bloomberg

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Swap "Curve" Trade

Two swaps executed together as a package

- Different maturities
- Opposite directions
- Notionals chosen so that net DV01 is zero

Value not affected by small parallel shifts in YC

P/L when the YC steepens or flattens

5v10 Curve:

- Pay fixed for 5Y on 260m notional
- Receive fixed for 10Y on 142m notional.
- DV01 on each leg of $125,000

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Swap "Curve" Trade



	2Y	3Y	4Y	5Y	6Y	7Y	8Y	9Y
1Y	35.1 / 35.8	78.1 / 78.9	115.5 / 116.3	144.2 / 144.9	166.9 / 167.7	185.4 / 186.2	200.6 / 201.3	213.7 / 214.4
2Y		42.7 / 43.4	80.1 / 80.8	108.8 / 109.4	131.5 / 132.2	150.0 / 150.7	165.2 / 165.8	178.3 / 178.9
3Y			37.0 / 37.8	65.7 / 66.4	88.4 / 89.2	106.9 / 107.7	122.1 / 122.8	135.2 / 135.9
4Y				28.3 / 29.0	51.0 / 51.8	69.5 / 70.3	84.7 / 85.4	97.8 / 98.5
5Y					22.4 / 23.1	40.9 / 41.6	56.1 / 56.7	69.2 / 69.8
6Y						18.1 / 18.9	33.3 / 34.0	46.4 / 47.1
7Y							14.8 / 15.5	27.9 / 28.6
8Y								12.8 / 13.4
9Y								
10Y								
12Y								
15Y								
20Y								
25Y								
30Y								
40Y								

Source: Bloomberg

Quoted: rate for back leg – front leg
Positive for upward sloping curve

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Swap Butterfly "Fly"

3 swaps executed together as a package

- Different maturities
- Longest and shortest maturities (wings) in one direction the middle maturity (body) in the other direction
- Wings have equal DV01 and half that of the body so Fly is DV01 neutral in total

Value not affected by small parallel shifts in YC

P/L when the YC curvature changes

5v10v15 "Fly":

- Pay fixed for 5Y on 250m notional (wing)
- Receive fixed for 10Y on 266m notional (body)
- Pay fixed for 15y on 90m notional (wing)
- DV01 on wings of $120,000 each and on body of $240,000

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Swap Butterfly "Fly"

Interest Rate Swaps | 2) Tools | 3) Settings | 4) Trading Access | IRS Multi-Dealer RFQ

13:23

20) EUR | 21) USD | 22) USD MAC | 23) USD IMM | 24) GBP | 25) CHF | 26) AUD | 27) JPY | 28) SEK

Semi 3M | S/A Crv | S/A Bfly | Annual 3M | Ann Crv | Ann Bfly | OIS | Basis

USD Semi vs 3M Butterfly				USD Semi vs 3M Butterfly		
31) 2v3v4	4.9 / 6.2	-0.520	31) 5v10v30	16.3 / 17.7	-0.502
32) 2v3v5	-23.8 / -22.4	-0.247	32) 5v15v30	Rcv / Pay	
33) 2v3v10	-104.5 / -103.2	+0.397	33) 6v7v8	2.5 / 4.1	-0.084
34) 2v3v30	-168.3 / -166.9	+0.538	34) 7v8v9	1.4 / 3.0	-0.061
35) 2v4v6	Rcv / Pay		35) 7v10v12	20.9 / 22.2	-0.232
36) 2v5v10	27.5 / 29.0	-0.307	36) 7v10v15	2.7 / 4.0	-0.140
37) 2v5v30	-36.3 / -34.7	-0.166	37) 8v9v10	1.0 / 2.5	-0.010
38) 2v10v30	125.3 / 126.7	-1.452	38) 10v12v15	-0.9 / 0.3	+0.066
39) 3v4v5	8.1 / 9.4	+0.194	39) 10v15v20	19.2 / 20.5	-0.070
40) 3v5v7	24.0 / 25.5	+0.034	40) 10v15v30	7.8 / 9.1	-0.093
41) 3v5v10	-15.4 / -14.0	+0.291	41) 10v20v30	40.2 / 41.7	-0.188
42) 3v5v30	-79.2 / -77.7	+0.432	42) 12v15v20	1.3 / 2.6	-0.044
43) 3v10v30	82.4 / 83.7	-0.854	43) 15v20v30	4.1 / 5.6	-0.070
44) 4v5v6	5.1 / 6.6	-0.102	44) 20v25v30	3.0 / 4.6	+0.029
45) 5v6v7	3.5 / 5.1	+0.043			
46) 5v7v10	1.1 / 2.6	-0.128			
47) 5v8v10	Rcv / Pay				
48) 5v10v15	Rcv / Pay				

Source: Bloomberg

Quoted: 2 × rate for body – both wings
Sign depends on YC curvature and tenors
Only zero if equally spaced on linear YC

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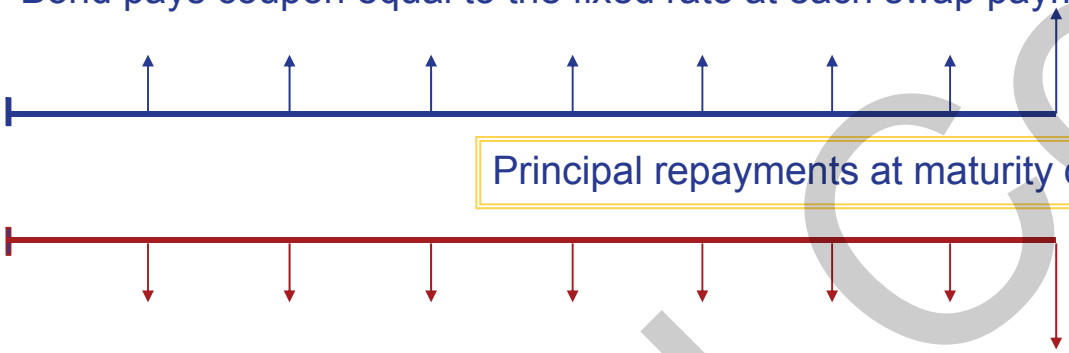
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Euribor Swap as Bond and FRN

An interest rate swap can be decomposed into:

- A floating rate note paying Euribor of the same tenor as the swap
- A fixed rate bond paying the fixed rate on the swap
- Zero NPV swap rate = par value bond yield
- This assumes that discounting is done with the Euribor curve

Bond pays coupon equal to the fixed rate at each swap payment date



Principal repayments at maturity cancel out

FRN pays Euribor at each swap payment date and has a price of par

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Approximate Euribor Swap Rate

Swap fixed rate (S%) is the coupon on a par value bond so:

$$100 = 100 \times \underbrace{\left[S\tau \sum_{t=1}^{T} D_t + D_T \right]}_{\text{Bond}}$$

FRN

Assume 30/360 day count on the fixed leg with year fraction τ

Discount Factors (D_t) are from the appropriate Euribor curve at each payment date

For a spot starting swap:

$$S = \frac{1 - D_T}{\tau \sum_{t=1}^{T} D_t}$$

For a forward starting swap out of date x:

$$S = \frac{D_x - D_T}{\tau \sum_{t=x+1}^{T} D_t}$$

Euribor curve used for both discounting and generating forward rates

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Precise Euribor Swap Pricing

Floating Side

Cashflows valued using the forward curve

- Use forward Euribor to generate “implied” cashflows
- Forward rates used because future Euribor payments can be hedged with forward contracts. Realised or forecast future Euribor is therefore not relevant to pricing.
- The YC used to generate the forward rates has to be built from swaps with the same Euribor frequency as the one being valued

Discount the cashflows to today with ESTA curve

- This gives the NPV of the floating side
- No need to include the notional payment at maturity

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Precise Swap Pricing

Floating Leg

3 year swap, 6m floating: Euribor (Act/360) Principal: € 100,000,000

Today — 1 Year — 2 Years — 3 Years

Days:	181	184	181	184	181	186
ESTA DF:	1.00310477	1.00628939	1.00940688	1.01252131	1.015274679	1.018011186
Forward Euribor:	-0.47%	-0.47%	-0.46%	-0.45%	-0.39%	-0.37%
Hedged Cashflow:	-236,305.56	-240,222.22	-231,277.78	-230,000.00	-196,083.33	-191,166.67
PV Cashflow:	-237,039.23	-241,733.07	-233,453.38	-232,879.90	-199,078.44	-194,609.81

- Discount factors and forward rates are not from the same curve
- Hedging floating payments is done with forward Euribor rates
- Discounting is done with an OIS yield curve
- Hedged cashflow = € 100,000,000 * Forward Rate * Days/360
- PV = OIS Discount factor * Cashflow

Floating NPV = Total PV of hedged cash flows = - €1,338,793.83

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Precise Swap Pricing

Fixed Leg

3 year swap, annual fixed rate: -0.441% (30/360) Principal: € 100,000,000

Today	1 Year	2 Years	3 Years
Days:	**365**	**365**	**367**
ESTA DF:	1.00628939	1.01252131	1.018011186
Fixed Cashflow:	-440,853.59	-440,853.59	-440,853.59
PV Cashflow:	-443,626.29	-446,373.66	-448,793.89

Discount factors from OIS curve

Cashflow = € 100,000,000 * -0.441%* 360/360

PV = Discount factor * Cashflow

Fixed NPV = Total PV of fixed cash flows = - € 1,338,793.83

Total Swap NPV = Fixed NPV - Floating NPV = 0

NB rates and cashflows all negative at the moment!

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Precise Swap Pricing

To value an interest rate swap:

- Build a Euribor zero coupon yield curve from rates with the same floating basis as the swap
- Build an OIS (ESTA) yield curve
- Calculate sequence of dates
- Interpolate discount factors as necessary
- Calculate Euribor forward rates
- Calculate hedged floating cashflows from forward rates
- Calculate NPV of hedged floating cashflows using OIS curve
- Calculate NPV of cashflows based on a fixed rate using OIS discounting

On trade date choose fixed rate so Fixed NPV = Floating NPV

- Swap rate is dependent on all Euribor forward rates up to maturity.

Rate is dependant on choosing the correct yield curves

- Forward rates generated by using a Euribor curve of the correct tenor
- Discounting done by using an OIS (ESTA) curve

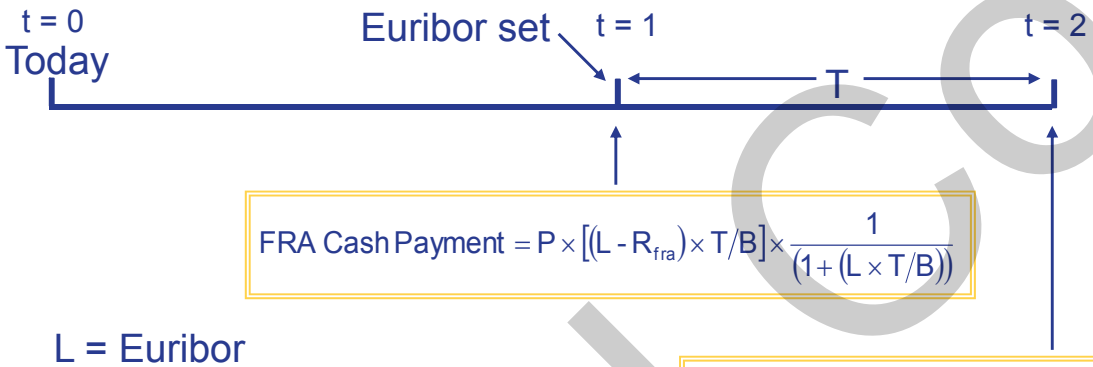
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Swap as Series of FRAs

FRA is equivalent to a single period forward starting Interest Rate Swap with a fixed rate equivalent to the FRA rate

A swap pays at the end of the forward period. An FRA pays at the start, but the payment is discounted at Euribor so that it is equivalent to an undiscounted payment at the end.



$$\text{FRA Cash Payment} = P \times \left[(L - R_{fra}) \times T/B\right] \times \frac{1}{(1 + (L \times T/B))}$$

$$\text{Net Swap Cashflow} = P \times \left[(L - R_{fix}) \times T/B\right]$$

L = Euribor
R_{fra} = FRA rate
R_{fix} = Swap fixed rate

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Quick Swap DV01 Formula

Breaking a swap up into its constituent FRAs means that the swap DV01 is equal to the sum of the FRA DV01s

FRA DV01 = $P \times 0.01\% \times \tau \times D_2$

So for a swap with a start date of t=1 and maturity T:

Swap DV01 = $P \times 0.01\% \times \sum_{t=2}^{T} \tau D_t$

For a vanilla 30/360 swap the DV01 is equal to the sum of discount factors for each payment date times the day count fraction (τ).

If the first cashflow is based upon a reset that has already happened then the net fixed cashflow must be treated separately.

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Swap Carry and Roll Down

Carry: the value of the difference between the first fixed and floating swap payments

- Difference in basis points between the fixed rate on a spot starting swap and a forward swap starting in one reset interval and ending at the same time
- Also the NPV of the forward swap with a fixed rate equal to the spot starting swap divided by the forward swap DV01
- Upward sloping curve provides positive carry to fixed receiver

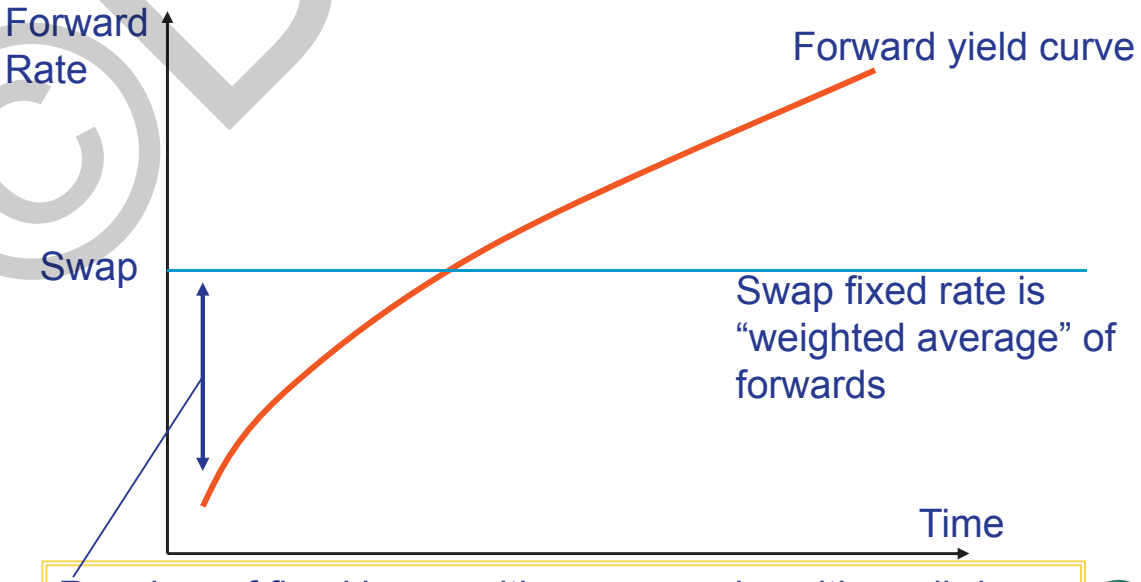
Roll Down: P/L based upon the passing of one reset interval with no change in the YC

- Difference in basis points between the fixed rate on a spot starting swap of a given tenor and that with a tenor one reset interval shorter.
- Also the NPV of the swap with the shorter tenor but a fixed rate equal to the longer swap divided by the shorter swap DV01
- Upward sloping curve provides positive roll-down to fixed receiver

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Market Risk

Unhedged receiver of fixed takes the view that Euribor rates will not rise in line with the forward curve



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Hedging Market Risk

Purpose: Value of swap + hedge constant with changes in Yield Curve

Many Possible Hedges

Fixed Payer/Floating Receiver:

- Buys Bonds
- Buys Eurodollar Futures
- Buys Bond Futures
- Sells FRAs

Most methods work for small parallel changes in the YC, but hedges need adjusting when moves are large and shifts are not parallel

- Bond hedges are also subject to spread risk
- Bond positions need funding

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Hedging with Bond Futures

Bond futures can be used instead of bonds

Fixed payer can effectively "buy" bonds through the futures market. Fixed receiver can just as easily "sell" bonds by using futures.

In calculating the number of contracts need to match swap's duration with that of futures position.

Duration calculations for futures are not as simple as for bonds.

Convexity adjustment can be very different

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Hedging with Bond Futures

With only one CTD bond futures price would closely track forward price of the CTD, divided by the CTD's factor.

Duration of futures contract would be close to the modified duration of the CTD divided by its factor

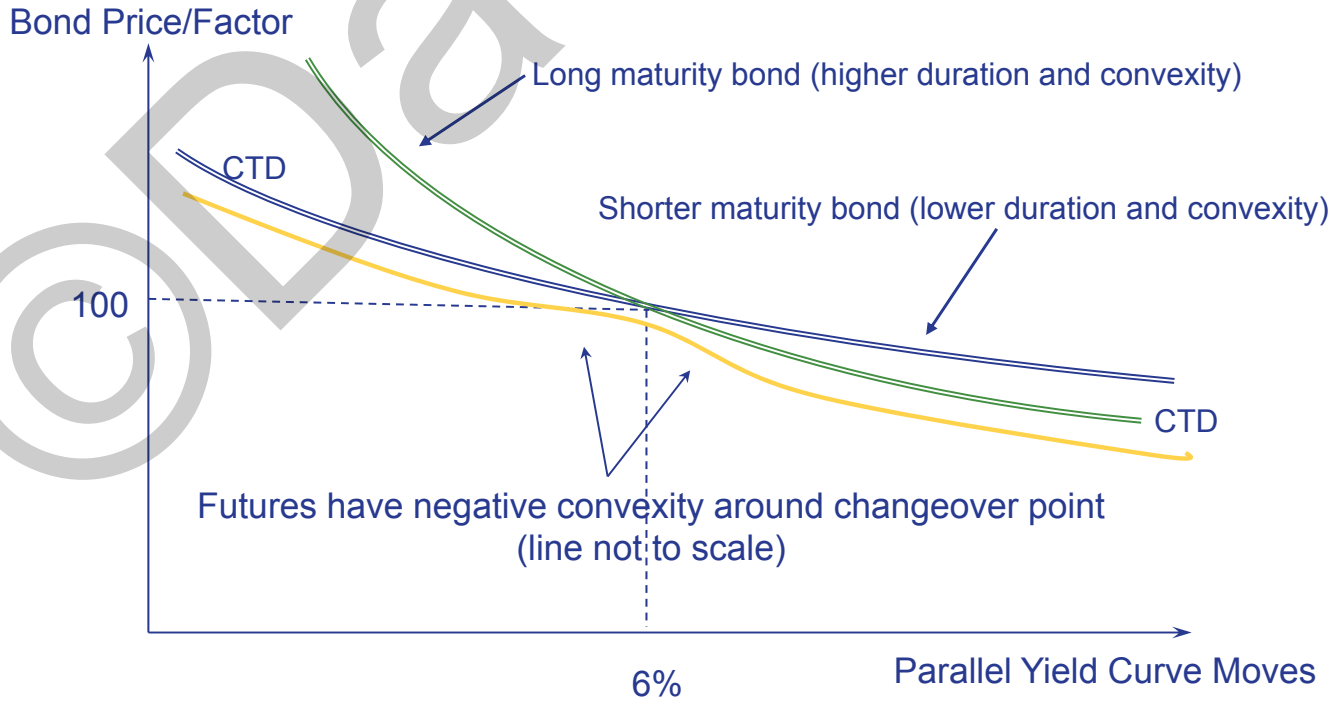
With many possible CTDs big yield moves can change the CTD. This causes uncertainty in the futures contract duration.

Interest rate sensitivity of futures price is a combination of the delivery option delta and the duration of possible CTD bonds

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Hedging with Bond Futures



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Hedging with Bond Futures

To hedge with treasury futures you need to calculate the option-adjusted duration (OAD)

T-Bond futures will have negative convexity if there is uncertainty about the CTD

- As rates rise, the OAD lengthens
- As rates fall, the OAD shortens

This means that the relative convexities of the swap (normally positive) and futures hedge (possibly negative) can be very different

Frequent re-balancing of the hedge may be necessary

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Hedging with Eurodollar Futures

Relate changes in futures prices to change in swap NPV

Calculate zero NPV fixed rate

Change first futures in yield curve by one tick

Calculate new swap NPV for same fixed rate

Repeat for each contract

$$\text{Hedge} = \frac{\text{Change in Swap NPV for a 1 Tick Change in Futures Price}}{\text{Tick Value of Futures Contract}}$$

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Hedging with Eurodollar Futures

YC split into 3m segments starting on IMM dates

DV01 of swap for each curve segment is calculated

This divided by DV01 of futures contract for that segment ($25) gives number of contracts to hedge that period

Each segment of the curve is hedged separately

Protects against non parallel shifts in the curve

Hedge factors will change with large rate moves

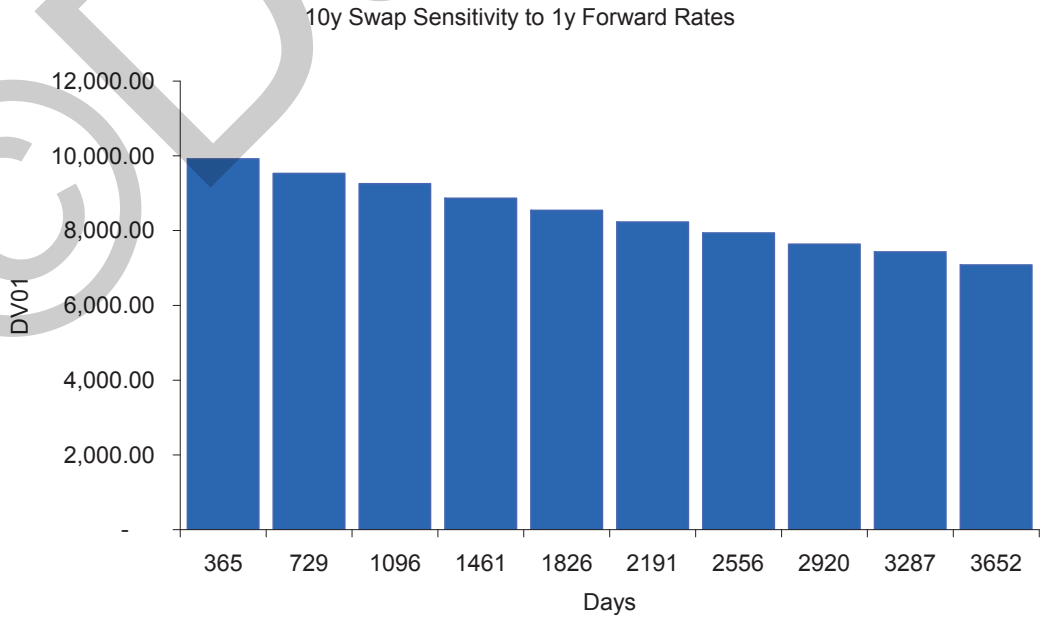
Dependent on sufficient liquid futures contracts

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Yield Curve Hedging/Trading

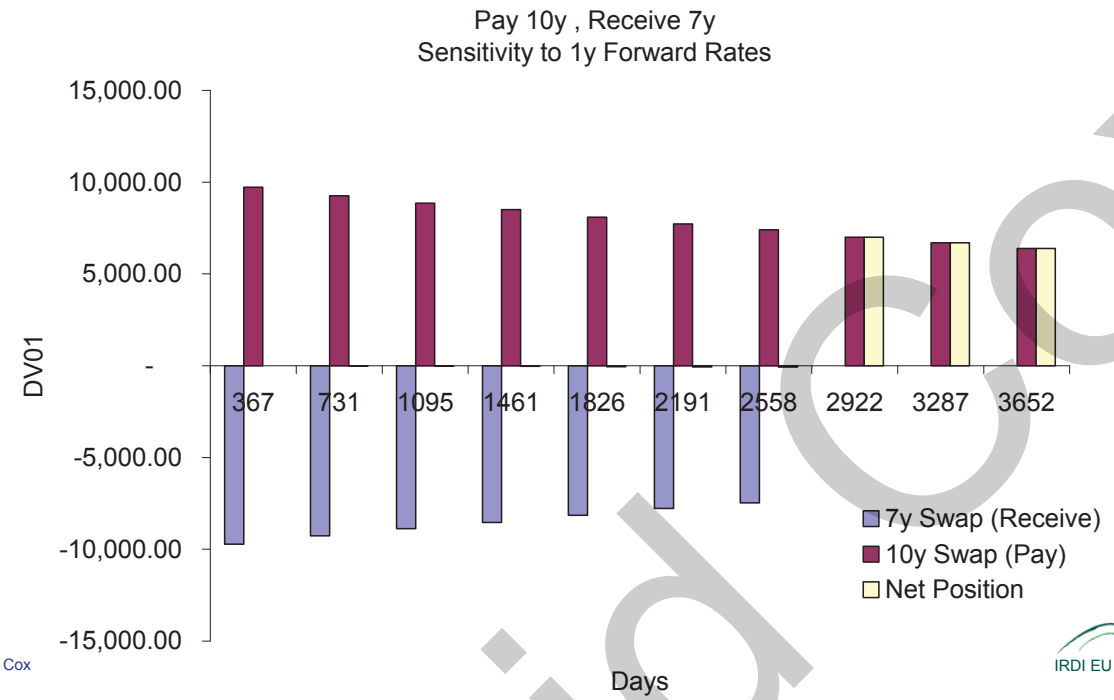
Single swap rate is sensitive to a weighted average of forward rates



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Yield Curve Hedging/Trading

Two swaps equal notionals sensitive to forward rates between their maturities.

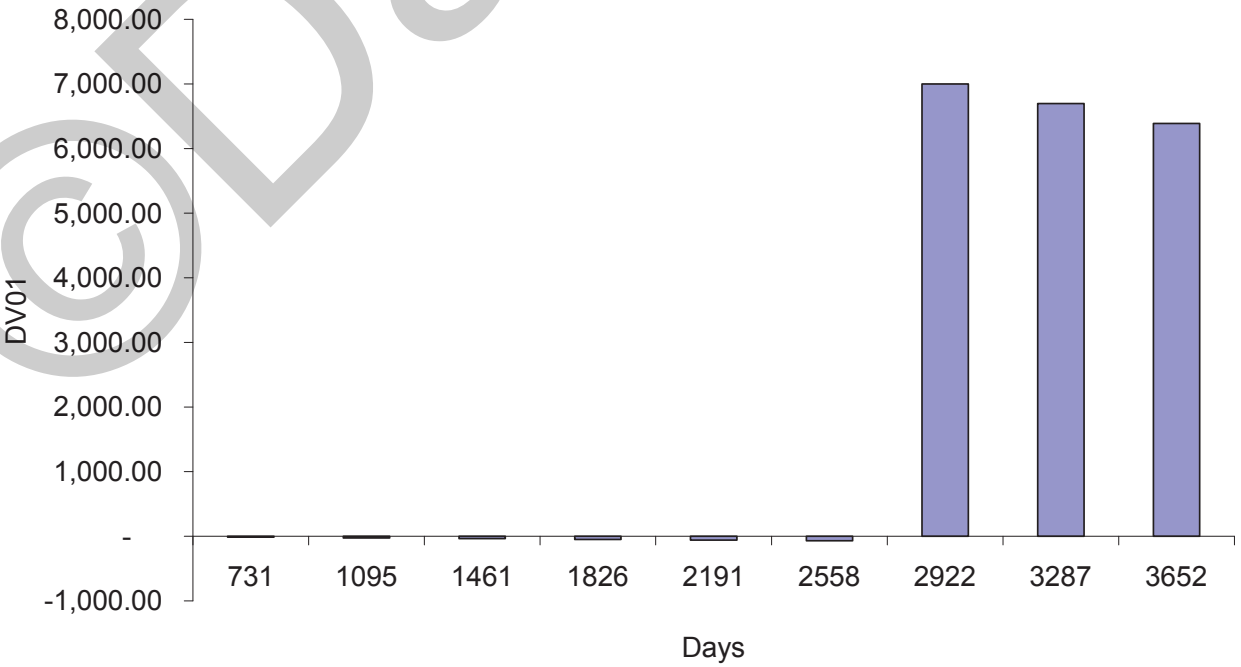


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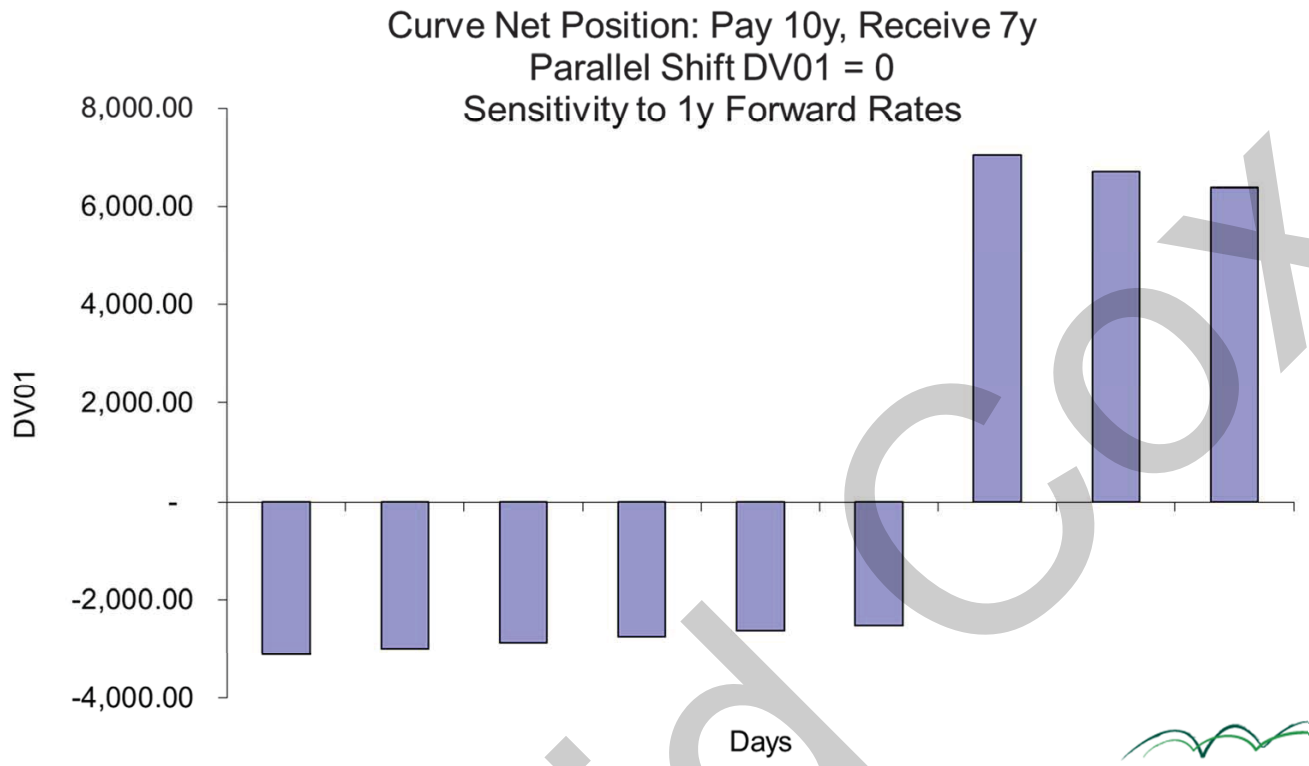
Yield Curve Hedging/Trading

Net position equivalent to a forward starting swap



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Yield Curve Hedging/Trading

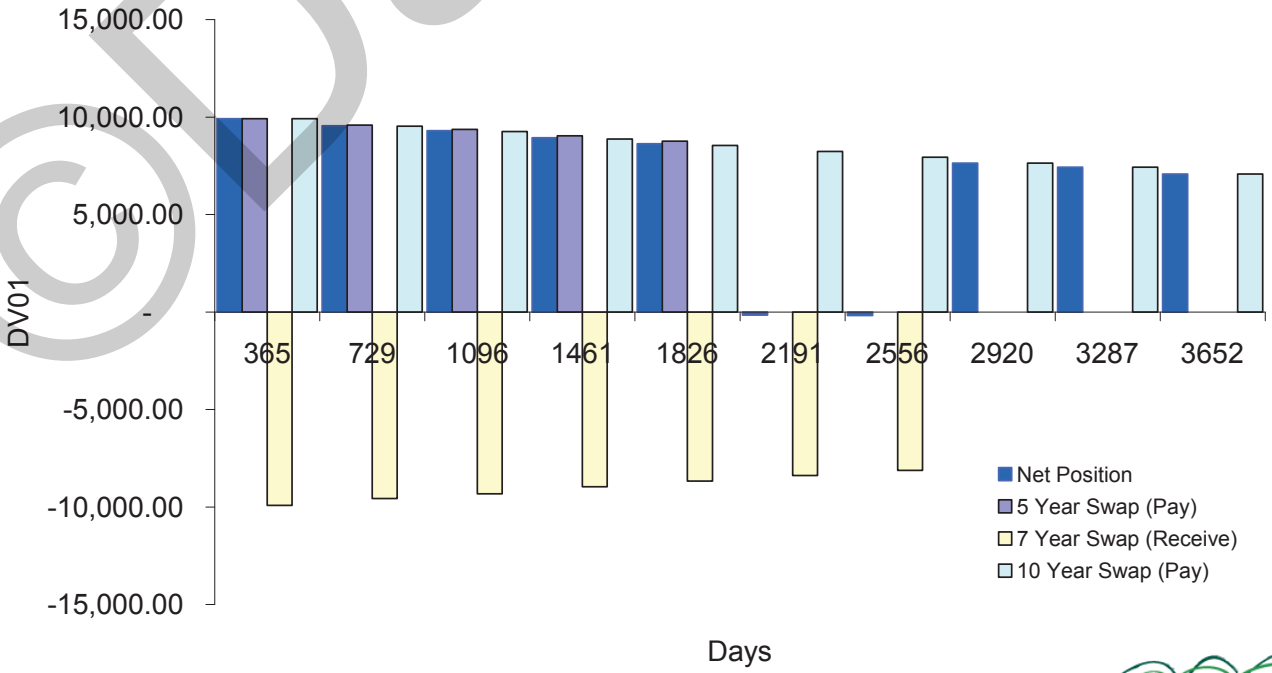


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Yield Curve Hedging/Trading

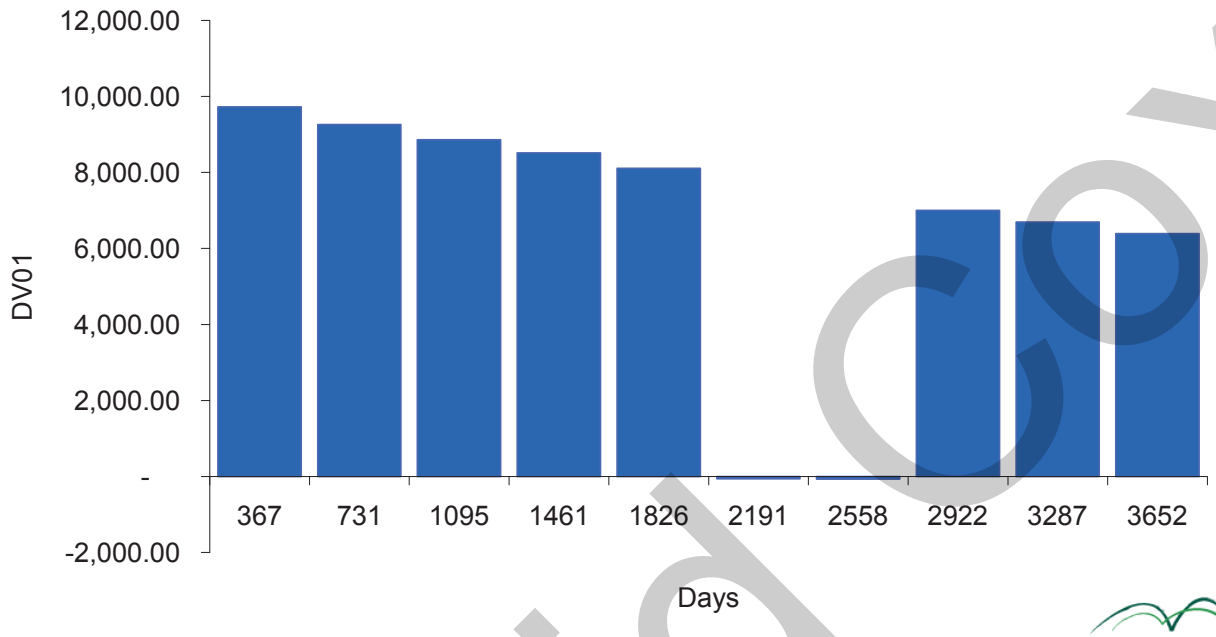
3 Swaps with equal notionals: Pay 5y, Receive 7y, Pay 10y



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Yield Curve Hedging/Trading

Pay 5y, Receive 7y, Pay 10y, equal notionals
Sensitivity to 1y forward rates

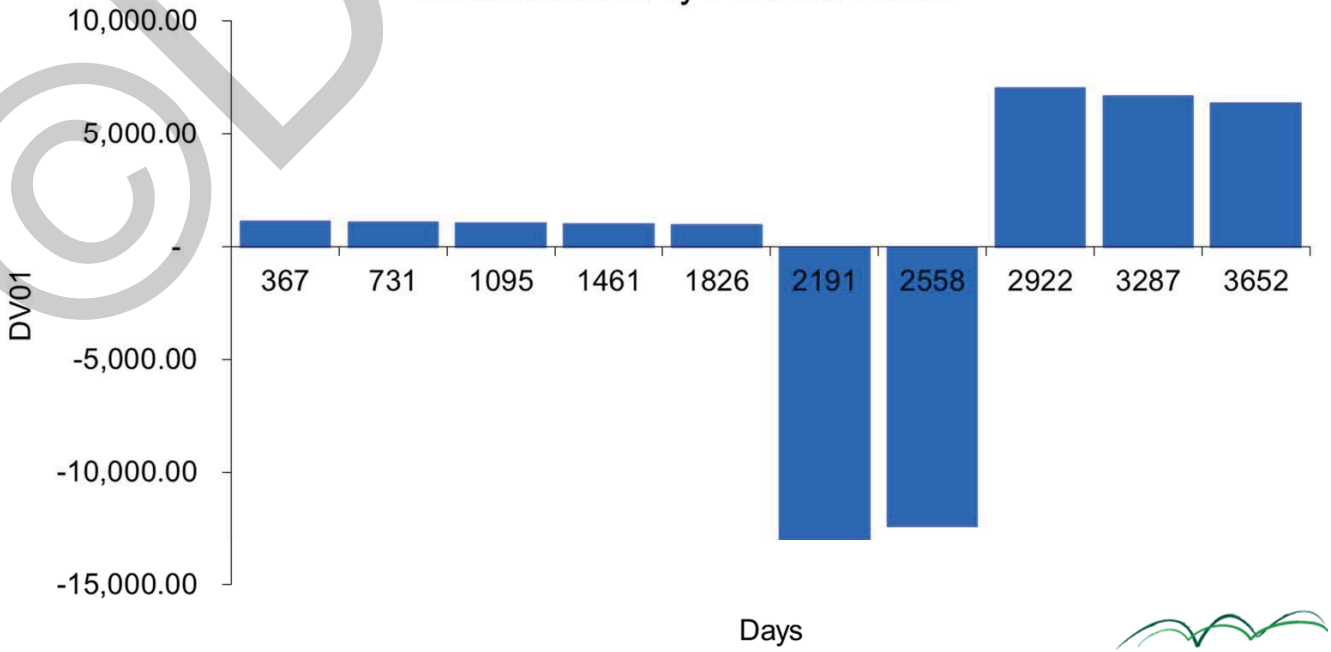


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Yield Curve Hedging/Trading

"Fly" Net Position: Pay 5y and 10y, Receive 7y.
Total Parallel Shift DV01 = 0
Sensitivities to 1y Forward Rates



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Day Three

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Goldman Sachs University

LondonFinancialStudies
capital markets learning

www.londonfs.com

Day Three

8.30 – 10.30

Interest Rate Swaps Continued

- Factors affecting market quotation
- CSA and counterparty credit risk
- Introduction to CVA, FVA and other valuation adjustments
- Netting agreements
- Swap clearing, SEF

Cross Currency and Asset Swaps

- Background to cross currency and FX swaps
- The market today
- Cross currency basis swaps, Libor-OIS spreads and market forces
- Using IRS and basis swaps to give fixed-floating currency swaps
- Multi-currency swap collateral
- Asset swaps and their purpose
- Par and proceeds asset swaps
- The impact of credit
- Z spread, I spread and OAS

10.30 – 10.45 Break

10.45 – 13.00 **Exercise: Using and pricing Asset Swaps**

13.00 – 14.00 Lunch

14.00 – 15.30 **Option Essentials for the Practitioner**

- Time value and its causes
- The basics of volatility and simple rules of thumb
- Standard assumptions about how asset prices move and their limitations (simple Monte Carlo illustration)
- Trees and distributions
- Option payoffs
- Valuing standard options
- Hedging and an introduction to risk neutrality
- Using trees for pricing
- Early exercise: American and Bermudan structures
- Sensitivities (“The Greeks”) and how they work
- The impact of volatility on hedging
- Hedging costs v time decay (the relationship between theta and gamma)
- Volatility skew and smile

15.30 – 15.45 Break

15.45 – 17.30 **Exercise: Volatility and the “Greeks”**

17.30 – 18.00 **Guest speaker if available**

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Quoting the Swap

What factors affect quotation?

Counterparty credit risk if OTC

- If CSA (credit support annex of ISDA agreement) is used then credit charge ("CVA") is subject to quality of collateral
- Other "VA" adjustments due to funding, options etc.

Volatility

- Ability to hedge the swap in the market
- In a volatile market the bid-ask spread will widen

Size of principal

- Too small and it is not worth doing the deal without a wide spread
- Too large and it will not be possible to hedge the position without moving the market against you so the spread widens

Trader's existing position and market view

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Swap Counterparty Risk

Hedging with futures or bonds offsets market risk on the basis of no default

- Default on a swap contract leaves the hedge in place

Probability of default or credit downgrade

- Credit rating
- Internal assessment

Recovery

- Derivatives normally rank "pari-pasu" (equally) with senior debt

Cost of default

- Mark to market represents replacement of transaction's cashflows
- Asymmetric risk as you loose if your MtM is positive and still have to pay if it is negative
- Liquidity risk includes transaction costs and the impact of market dislocation

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Swap Counterparty Risk

Risk Mitigation

Only trade with good credits
- Easier said than done
- Give up swap to a clearing house (limited vanilla swaps only)

Trade with many counterparties
- Diversification works, but not against systemic failure

Collateralise
- Hold securities or cash against default

Include termination clauses
- Allow for contracts to be ended early on a credit downgrade

Net cashflows
- Enter netting contracts to offset cashflows from all transactions with the same counterparty

Hedge credit exposure
- Trade CDS to offset credit exposure with counterparty

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Swap Counterparty Risk

Collateral

Post cash or securities against default by party with negative mark to market to party with positive MtM
- Reduce credit exposure to within lines enabling more business
- Enable trades with risky counterparty
- Reduce capital requirements
- Produce more competitive pricing

Normally bilateral agreement for posting either way depending on mark to market

ISDA CSA is standard documentation

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Swap Counterparty Risk

Collateral Agreement

Collateral delivered by party with negative MtM subject to minimum transfer amounts and threshold

- Regular mark to market (daily or intraday)
- Threshold level below which collateral is not required
- Minimum transfer amount and rounding keeps workload manageable
- Independent amount (initial margin) often required at outset
- Deliverable securities (often incentives for cash)
- Currency
- Haircut (discount from market price for risky securities)
- Rate payable on cash (normally overnight rate)
- Substitution and Rehypothecation

Collateral posting based upon net exposure

Valuation agent

- Calculate exposure
- Market value of existing collateral
- Posting or return amounts

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Swap Counterparty Risk

Netting

Close-out

- Right to terminate on insolvency
- Cancellation and immediate payment
- Creates debt to solvent party resulting from a negative MtM trade
- Crystallises exposure
- Solvent party can replace transaction and claim loss

Offset

- Payment netting: combine all payments to be made in one day
- Close-out netting: combine MtM of all contracts on a default to reduce the impact of asymmetric risk

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Swap Counterparty Risk

Netting

Swaps between A and B	A's Mark to Market
A pays fixed for 10 years on 500m	+5.0m
A pays fixed for 8 years on 100m	-1.5m
A pays fixed for 13 years on 250m	+3.0m
A receives fixed for 7 years: on 150m	-0.5m
A receives fixed for 15 years: on 100m	-5.0m

Exposure of A to B:
Without netting: 8m
With netting: 1m

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Swap Counterparty Risk

Netting

Advantages:

- Reduction of exposure
- Increases stability due to lack of need to suddenly cease trading with troubled counterparty

Disadvantages

- Exposure more volatile due to incremental impact of new business
- Incentive to unwind illiquid transaction with original counterparty causing uncompetitive quotes
- Incentive to concentrate new trades with one counterparty

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Swap Clearing

Market reforms prompted by the 2008/9 crisis:

- Electronic execution, transparent pricing, data publication and central clearing

Used widely in the interdealer market since 2007.

- Mandated for US institutions by CFTC from 2013.
- EMIR for European entities mandates from mid 2016-19 depending on category
- Execution through SEF

Mainly through LCH Clearnet (Swapclear) & CME

- As with futures the clearing house becomes counterparty to both sides of the trade
- Margin is paid to/from the clearing house by the clearing member

To check if clearing is mandatory:

http://www.swapclear.com/what/mandatory_clearing_checker.html

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Swap Clearing

ISDA agreement between counterparties replaced

Agreement with the clearing house is similar to ISDA, but incorporates additional terms.

- Clearing house calculates all payment amounts
- Payments are netted
- Daily NPV of swap calculated with CH's yield curve
- Initial and variation margin operates as with futures positions

Variation margin

- Daily cash settlement of NPV changes in cash

Initial margin

- Paid in cash or securities when a swap is submitted for clearing

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Clearing on LCH

Vanilla IRS, ZC and basis swaps (not all mandatory):

- USD, EUR and GBP out to 51 years
- CAD and JPY out to 41 years
- AUD, CAD, CHF, DKK, ZAR and SEK out to 31 years
- NZD HUF, SGD and MXN out to 21 years
- NOK and PLN, out to 16 years
- CZK, HKD and ILS out to 11 years
- USD, EUR CHF GBP Overnight Index Swaps

Inflation swaps

- French CPI, US CPIU out to 30y:
- HCPI, UK RPI out to 50y:

Overnight Index Swaps:

- AUD, CAD, CHF, EUR, GBP, JPY, NZD, SGD, USD

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Clearing on LCH

Overnight v Ibor basis swaps:

- AUD, CAD, EUR, GBP, JPY, NZD, SGD, USD

Variable Notional Swaps are cleared as long as the notional is known at the time of execution

FRAs are cleared

Not cleared:

- Variable notional OIS Swaps
- Swaps with an embedded option (CME Clears some Swaptions)
- Dual-currency swaps
- Swaps with undetermined future notionals

Current list at:

https://www.lch.com/services/swapclear/what-we-clear

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Swap Counterparty Risk

Pricing

Theoretical models treated with caution
- Focus on specific instruments
- Generally ignore collateral
- Use risk neutral valuation (only OK if hedging is possible)

Practical situation is complex
- Multiple trades with the same counterparty
- Netting agreements
- Collateral agreements
- Liquidity effects

Cost of counterparty risk calculated firmwide
- Trading decision made with reference to this cost
- Cost passed on to counterparty if appropriate
- Trader's P/L adjusted by the cost of counterparty risk

CVA calculated with reference to cost and probability of default

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Swap Counterparty Risk

Credit Value Adjustment

Simplified unilateral CVA with no correlation between default probability and recovery values

$$CVA = (1-\delta) \times \sum_{t=1}^{T} \left(D_t \times EE_t \times q_t \right)$$

- δ = recovery rate
- D_t = discount factor
- EE_t= expected exposure for relevant future dates
- q_t = relevant marginal default probability

With netting the incremental CVA is calculated
- This can be negative due to beneficial effects of netting

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Funding Value Adjustment (FVA)

Cash variation margin (collateral)

- Required for cleared and interdealer swaps
- Cash can be reposted to support offsetting transaction
- Cash earns overnight rate
- Future cashflows discounted at overnight rate

Non cash collateral

- Cash for reposting can be borrowed against this at a higher rate
- Depends on rehypothecation
- Swap cashflows discounted at the higher rate
- Rate depends on assets and swap dealer

No collateral

- Cash borrowed at cost of dealer's unsecured funding
- This is the highest rate used for discounting
- Different FVAs from different dealers

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Other VAs

OIS discounting for perfect collateralisation

- VAs make adjustments to this

FVA Funding Value Adjustment

- Cost of funding uncollateralised position

KVA: Capital Value Adjustment

- Cost of funding regulatory capital

MVA: Margin Value Adjustment

- Cost of funding initial margin (independent amount)

ColVA: Value of optionality in CSA

- Substitution of securities
- Ability to choose to post collateral in different currencies
- Cheapest to deliver option

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Currency Swaps

Background

Growth of multinational companies

Expansion of overseas earnings and investment

Need to raise foreign funds

Need to manage FX exposure

Exchange controls

Short term FX market provided some answers

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FX Swap

Spot		90 Days	
→	Sell USD 1	Buy USD 1	←
⟸	Buy CHF 1.5	Sell CHF 1.5±Pts	⟹
→	Invest USD 1 @ 5%	Receive USD 1.0125	←
⟸	Borrow CHF 1.5 @ 1%	Repay CHF 1.5*1.0025	⟹

$$\text{Forward} = \text{Spot} \times \frac{1 + R_{CHF} \times \frac{90}{360}}{1 + R_{USD} \times \frac{90}{360}} = \text{Spot} \times \frac{D_{USD}}{D_{CHF}}$$

$$\text{CHF } 1.5000 \underset{-}{+} \text{Pts.} = \text{CHF } 1.5000 \times \frac{1.0025}{1.0125} = 1.4852 = \text{Forward Rate}$$

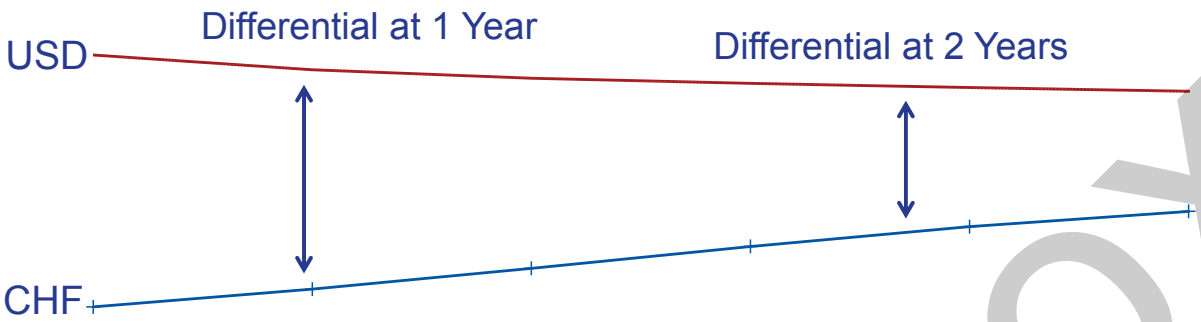
Forward - Spot = -0.0148 = -148 Points

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USD and CHF Yield Curves

Forward points are not a simple function of time



Points can be both positive and negative

Converting multiple cashflows over time using FX forwards is possible but messy and expensive

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IBM v World Bank (1981)

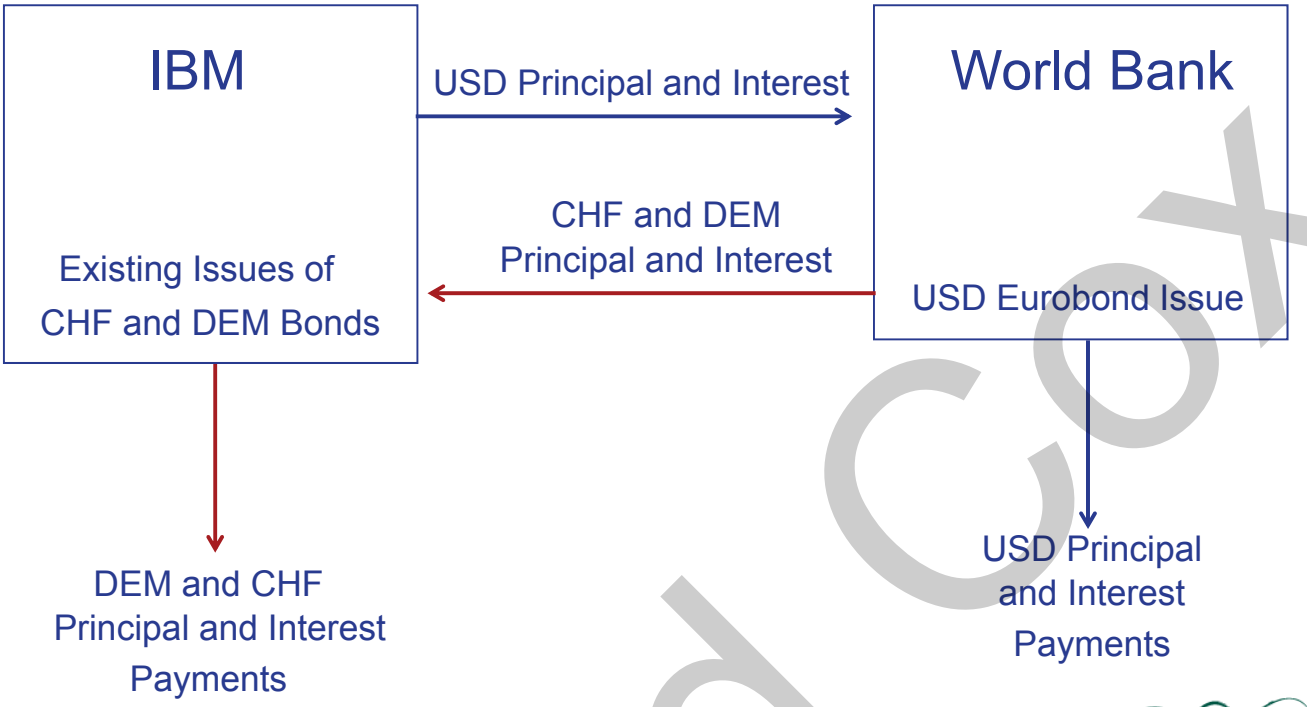
IBM	World Bank
Needed large amounts of funds	Policy of raising funds in low interest rate currencies (DEM & JPY)
World wide borrowing program in DEM and CHF	Demand for fixed rate funds greater than market supply
Proceeds converted to USD	Issue Dollar Eurobond
USD risen against CHF to give large unrealised FX gain	
Agree to pay principal and interest on World Bank Eurobond in USD	⇔ Agree to pay principal and interest on IBM CHF & DEM debt

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IBM v World Bank (1981)

Fixed to Fixed Currency Swap Payments



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Currency Swaps Today

An exchange of sets of cash flows in different currencies

Based on:

- A principal sum
- Fixed rates and/or
- Floating rates

The principal is normally exchanged

Initial exchange can be through FX market at spot

Often related to bond structures

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Currency Swaps

Today's Market

Similar to IRS market for standard structures

"Plain Vanilla": $ LIBOR v fixed currency

Quotes based upon IRS and Floating v Floating (basis swap)

Standard quotes: 2,3,4,5,7 & 10 - 40 years

Minimum market amount: $50 million principal

Standard ISDA documentation

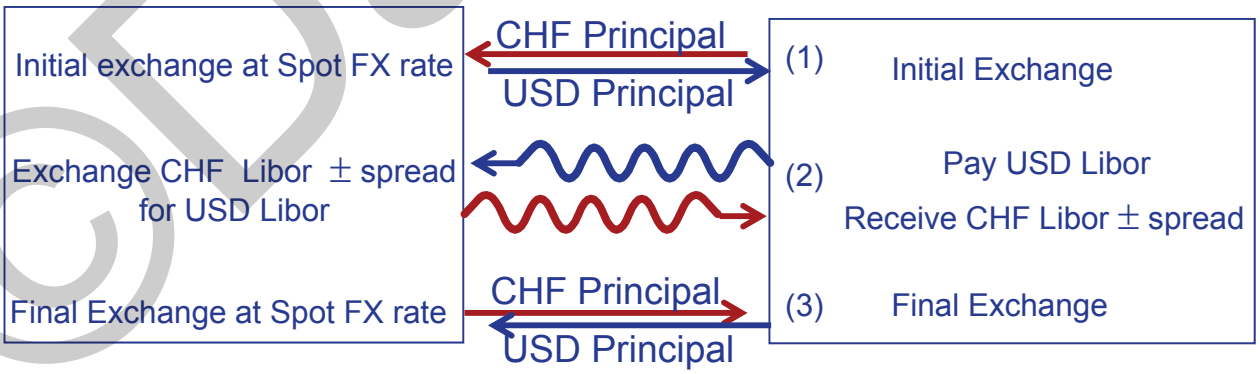
Much greater credit risk than IRS

Longer term swaps are arranged by banks

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Basis Swap

Floating - Floating



Exchange of principal and Libor payments
Equivalent to exchange of FRNs paying Libor
Theoretically, for zero NPV both sides should be at Libor flat
Not so in practice (we need a dynamic framework to understand why)
Currencies trade at Libor ± spread v $ Libor

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Basis Swap Rates

07:44 16APR03 GARBAN-INTERCAPITAL UK04138 ICAB1
Basis Swaps - All currencies vs. 3m USD LIBOR - Also see <ICAB2>

	REC/PAY EUR	REC/PAY JPY	REC/PAY GBP	REC/PAY CHF	REC/PAY SEK
1 Yr	+2.500/+0.500	-02.50/-05.50	+01.00/-03.00	+3.25/+0.25	-2.50/-5.50
2 Yr	+2.125/+0.125	-02.50/-05.50	+01.00/-03.00	+2.75/-0.25	-2.00/-5.00
3 Yr	+1.875/-0.125	-03.00/-06.00	+01.00/-03.00	+1.75/-1.25	-0.75/-3.75
4 Yr	+1.750/-0.250	-03.25/-06.25	+00.50/-03.50	+1.25/-1.75	+0.25/-2.75
5 Yr	+1.500/-0.500	-03.50/-06.50	+00.25/-03.75	+1.00/-2.00	+0.25/-3.00
7 Yr	+1.125/-0.875	-04.50/-07.50	-01.00/-05.00	+0.75/-2.25	+0.75/-2.25
10Yr	+1.000/-1.000	-07.00/-10.00	-02.75/-06.75	+0.75/-2.25	+0.75/-2.25
15Yr	+2.000/-2.000	-11.75/-14.75	-03.00/-07.00	+1.50/-2.50	+2.50/-2.50
20Yr	+1.750/-2.250	-16.00/-19.00	-02.75/-06.75	+1.50/-2.50	+3.00/-2.00
30Yr	+1.625/-2.375	-21.25/-24.25	-02.50/-06.50		

	DKK	NOK	CAD	CZK	PLN
1 Yr	-2.25/-6.25	-3.50/-7.50	+09.50/+05.50	+02.00/-07.00	+01.00/-09.00
2 Yr	-2.75/-5.75	-3.50/-7.50	+10.50/+06.50	+01.50/-06.50	+00.50/-08.50
3 Yr	-2.50/-5.50	-3.50/-7.50	+12.00/+08.00	+01.50/-06.50	+00.50/-08.50
4 Yr	-2.50/-5.50	-3.50/-7.50	+12.75/+08.75	+01.50/-06.50	+00.50/-08.50
5 Yr	-2.50/-5.50	-3.50/-7.50	+13.50/+09.50	+01.50/-06.50	+00.50/-08.50
7 Yr	-2.25/-5.25	-3.50/-7.50	+13.75/+09.75	+01.50/-06.50	+00.50/-08.50
10Yr	-2.00/-5.00	-3.50/-7.50	+14.00/+10.00	+01.50/-06.50	+00.50/-08.50

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Basis Swap Rates

08:28 21FEB06 ICAP UK69580 ICAB1
Basis Swaps - All currencies vs. 3m USD LIBOR - Also see <ICAB2>

	REC/PAY EUR	REC/PAY JPY	REC/PAY GBP	REC/PAY CHF	REC/PAY SEK
1 Yr	+2.875/+0.875	-00.50/-3.50	+01.00/-3.00	+01.50/-1.50	-04.50/-7.50
2 Yr	+3.125/+1.250	-00.50/-3.50	+01.50/-2.50	+01.00/-2.00	-03.25/-6.00
3 Yr	+3.500/+1.500	-00.50/-3.50	+02.25/-1.75	00.00/-2.75	-02.25/-5.00
4 Yr	+3.750/+1.750	-00.75/-3.75	+02.50/-1.50	-00.50/-3.25	-01.00/-4.00
5 Yr	+3.875/+1.875	-01.00/-4.25	+02.50/-1.50	-00.75/-3.75	-00.75/-3.75
7 Yr	+3.875/+1.875	-02.00/-5.00	+02.00/-2.00	-01.25/-4.00	-00.75/-3.50
10Yr	+3.875/+1.875	-04.25/-7.25	+01.25/-2.75	-01.50/-4.25	-00.75/-3.50
15Yr	+5.625/+1.625	-07.75/-10.75	-00.25/-4.25	-01.00/-5.00	+00.75/-4.00
20Yr	+5.875/+1.875	-10.00/-13.00	-00.75/-4.50	-01.00/-5.00	+01.00/-4.00
30Yr	+6.000/+2.000	-13.00/-16.00	-00.75/-4.50	-01.00/-5.00	+01.25/-3.75

	DKK	NOK	CAD
1 Yr	+00.75/-2.00	-05.00/-9.00	+07.00/+3.00
2 Yr	+01.00/-2.00	-04.25/-8.25	+07.25/+3.25
3 Yr	+01.50/-1.50	-04.00/-8.00	+07.50/+3.50
4 Yr	+01.50/-1.50	-03.75/-7.75	+07.75/+3.75
5 Yr	+01.75/-1.00	-03.75/-7.75	+08.00/+4.00
7 Yr	+02.50/-0.75	-03.75/-7.75	+08.25/+4.25
10Yr	+02.50/-0.50	-03.75/-7.75	+08.50/+4.50
15Yr	+04.00/-1.00	-03.25/-8.25	+10.50/+6.50
20Yr	+04.75/-0.50	-03.25/-8.25	+10.50/+6.50
30Yr	+04.75/-0.50	-03.25/-8.25	+11.00/+7.00

*FOR 3M V 6M EUR/EUR <ICAB4>**

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Basis Swap Rates

12:50 11JAN07 ICAP UK69580 ICAB1

Basis Swaps - All currencies vs. 3m USD LIBOR - Also see <ICAB2>

	REC/PAY EUR	REC/PAY JPY	REC/PAY GBP	REC/PAY CHF	REC/PAY SEK
1 Yr	+3.625/+1.625	+02.75/-0.25	+02.75/-1.25	+01.75/-1.25	-04.50/-7.50
2 Yr	+3.625/+1.625	+03.13/+0.13	+03.25/-0.75	+01.00/-1.75	-03.25/-6.00
3 Yr	+3.625/+1.750	+04.00/+1.00	+04.00/ 0.00	+00.75/-2.25	-01.75/-4.75
4 Yr	+3.750/+1.750	+04.25/+1.25	+04.25/+0.25	+00.25/-2.75	-00.75/-3.75
5 Yr	+3.750/+1.875	+04.25/+1.25	+04.25/+0.50	-00.25/-3.25	00.00/-3.00
7 Yr	+4.125/+2.250	+04.25/+1.25	+03.75/-0.25	-00.75/-3.50	+01.00/-2.00
10Yr	+4.625/+2.625	+04.00/+1.00	+03.00/-1.00	-01.00/-4.00	+02.00/-1.25
15Yr	+6.125/+2.250	+02.50/-0.50	+01.25/-2.75	-00.75/-4.75	+03.25/-0.75
20Yr	+7.000/+3.000	+01.75/-1.25	+00.50/-3.50	-01.00/-4.75	+03.50/-0.50
30Yr	+7.500/+3.500	+00.25/-2.75	+00.25/-3.75	-01.00/-5.00	+04.00/-1.00

	DKK	NOK	CAD
1 Yr	+01.00/-2.00	-03.75/-8.00	+04.75/+0.75
2 Yr	+01.00/-2.00	-03.50/-7.50	+05.25/+1.25
3 Yr	+01.25/-1.75	-03.00/-7.25	+05.50/+1.50
4 Yr	+02.00/-1.00	-03.00/-7.00	+06.00/+2.00
5 Yr	+02.75/-0.25	-02.75/-6.50	+06.25/+2.25
7 Yr	+03.75/+0.75	-02.75/-6.50	+06.50/+2.50
10Yr	+04.50/+1.50	-02.75/-6.50	+06.75/+2.75
15Yr	+06.50/+1.50	-02.75/-6.50	+09.00/+5.00
20Yr	+07.50/+2.50	-02.75/-6.50	+09.50/+5.50
30Yr	+07.75/+2.75	-02.75/-6.50	+10.00/+6.00

*FOR 3M V 6M EUR/EUR <ICAB4>**

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30-09-2008 Basis Swaps!

	EUR		JPY		GBP	
	Receive $	Pay $	Receive $	Pay $	Receive $	Pay $
1Y	-73.000	-77.000	-66.500	-67.500	-57.000	-61.000
2Y	-46.500	-50.500	-47.000	-48.000	-39.000	-43.000
3Y	-33.000	-37.000	-35.500	-36.500	-28.000	-32.000
4Y	-24.500	-28.500	-26.125	-27.125	-24.500	-28.500
5Y	-21.000	-25.000	-19.000	-20.000	-22.000	-26.000
7Y	-14.000	-18.000	-9.000	-10.000	-18.000	-22.000
10Y	-8.500	-12.500	-2.375	-3.375	-16.000	-20.000
15Y	-2.750	-6.750	-0.500	-1.500	-14.000	-18.000
20Y	-0.250	-4.250	0.875	-0.125	-13.000	-17.000
30Y	1.750	-2.250	1.875	0.875	-11.000	-15.000
40Y	2.125	-1.875	3.625	2.625	-10.500	-14.500
50Y	2.375	-1.625			-10.250	-14.250

Spreads apply to non Dollar currency

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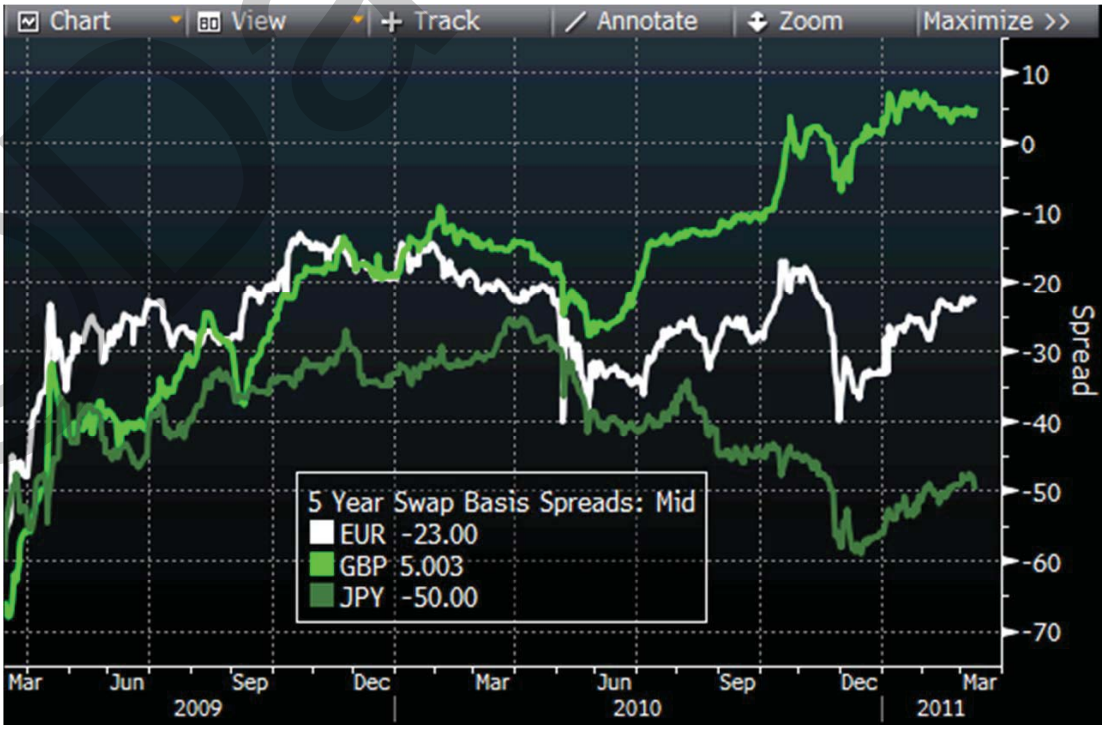
6 May 2009 Basis Swaps

05/06 16:30 GMT [REUTERS CAPITAL MARKETS SOURCE: ICAP] 05/06 1:04

	EUR/USD 3M LIBOR	GBP/USD 3M LIBOR	CHF/USD 3M LIBOR	CAD/USD 3M LIBOR
1YR	-30.00/ -20.00	-43.00/-33.00	-16.75/ -6.75	+14.50/+24.50
2YR	-30.00/ -25.00	-41.50/-36.50	-20.00/-14.00	+11.00/+21.00
3YR	-31.50/ -26.50	-42.00/-37.00	-25.00/-19.00	+9.00/+19.00
5YR	-32.00/ -27.00	-43.50/-38.50	-30.00/-24.00	+9.00/+19.00
7YR	-29.00/ -24.00	-38.00/-33.00	-29.00/-23.00	+7.00/+17.00
10YR	-23.00/ -18.00	-32.00/-27.00	-28.00/-22.00	+6.00/+16.00
15YR	-13.50/ -8.50	-27.00/-22.00	-21.50/-15.50	+0.50/+10.50
20YR	-8.50/ -3.50	-19.50/-14.50	-15.25/ -9.25	-2.50/ +7.50
30YR	-4.50/ +0.50	-14.00/ -9.00	-10.75/ -4.75	-7.50/ +2.50

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Basis Swaps 2009 - 2011



Source: Bloomberg

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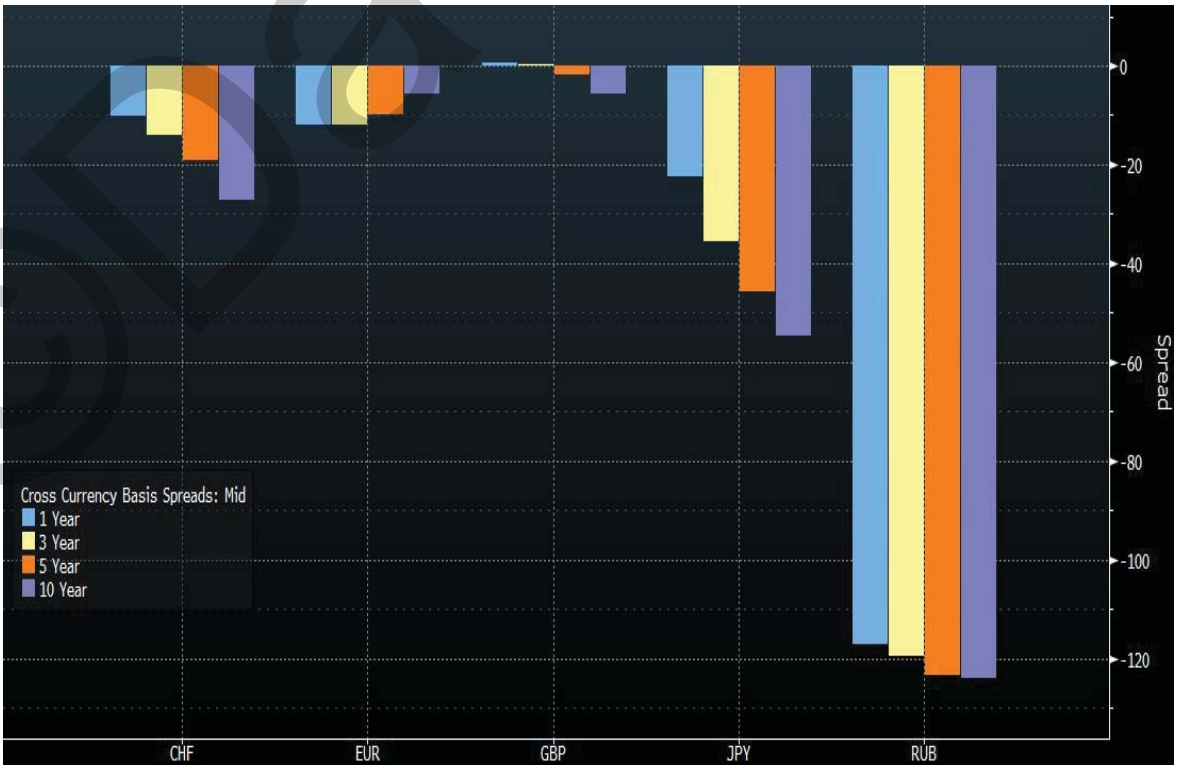
Basis Swaps 2014 - 2016



Source: Bloomberg XCCY

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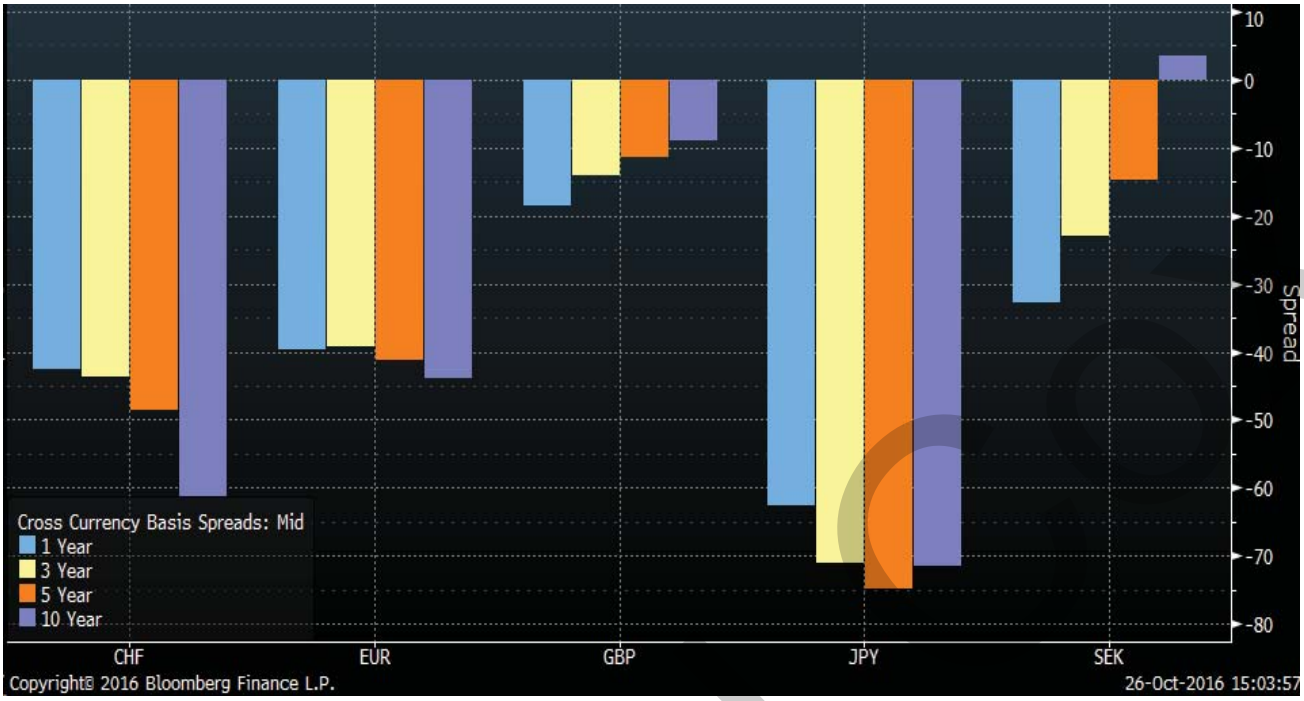
Basis Swaps 20 Aug 2014



Source: Bloomberg

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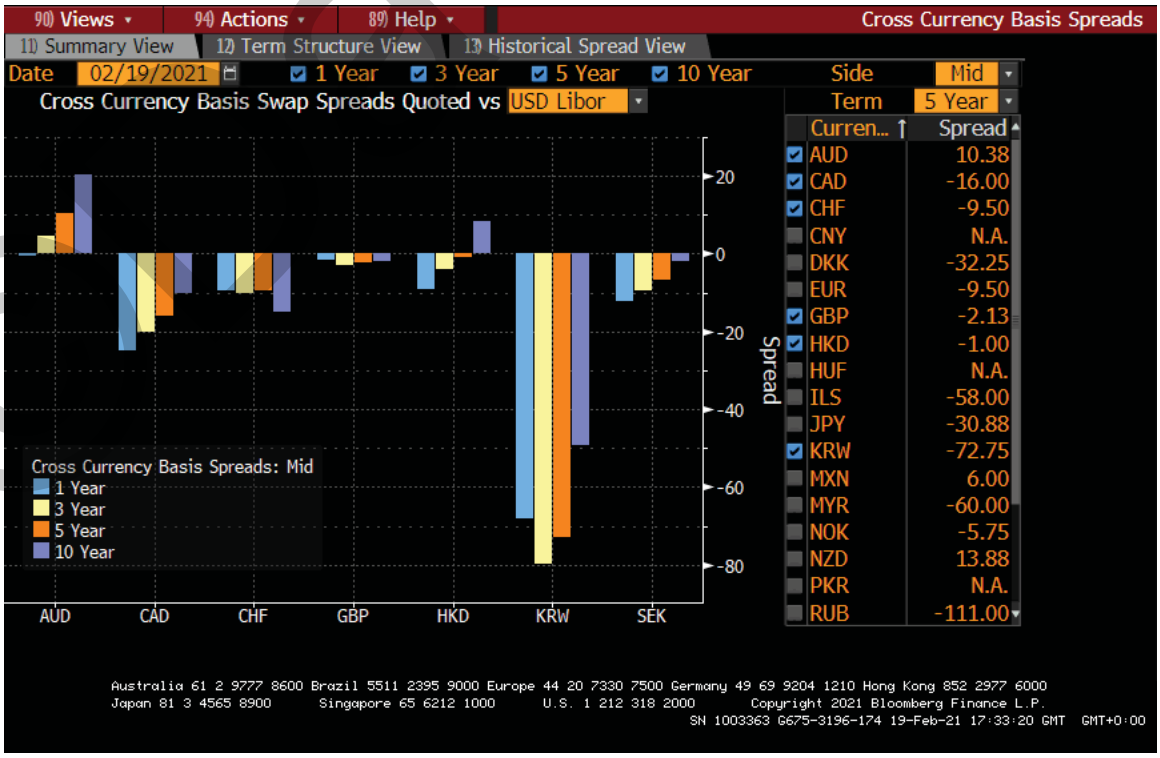
Basis Swaps 26 Oct 2016



Source: Bloomberg XCCY

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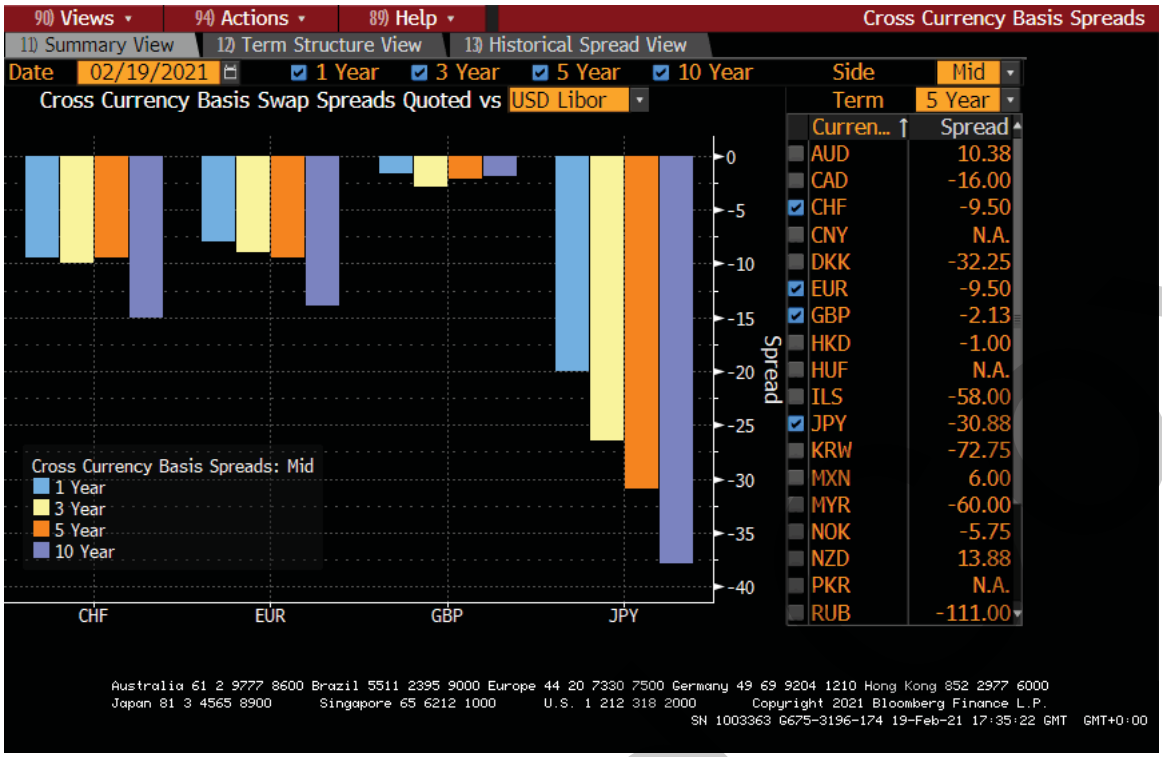
Basis Swaps 19 Feb 2021



Source: Bloomberg XCCY

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Basis Swaps 19 Feb 2021



Source: Bloomberg XCCY

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Basis Swap Spread

Libor discounting

Basis Swap spread is theoretically zero as basis swaps can be treated as floating rate notes in the two currencies valued at par and exchanged at spot

OIS discounting

Basis swap spread is not zero even if there are no other factors present as FRNs paying Libor will not be valued at par

Theoretical spread is close to the difference in the Libor v OIS spreads in the two currencies of the swap

Market Forces

The supply and demand for swaps in particular currencies will move the level of the Basis Swap spread from its theoretical value

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Basis Swap Adjusted YC

An adjusted OIS yield curve can be implied from the market level for basis swap spreads

- An adjusted Eonia yield curve consistent with €/$ basis swap market levels can be generated by adjusting the Eonia discount factors so that a €/$ basis swap for each tenor has zero NPV.

Adjusted Eonia discount factors will provide the value of Euro cashflows on a Fixed/Floating €/$ swap for conversion to Dollars at spot.

Adjusted Eonia discount factors can be used to determine the value of a Euro IRS when collateral is posted in Dollars.

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Fixed v Floating CC Swap Pricing

Start with basis swap

- Spread over or under currency Libor v $ Libor flat

Convert currency Libor ± spread to fixed with IRS

- Use the techniques of IRS pricing
- Add spread to forward rate when calculating implied cashflows on floating side
- Find fixed rate that gives zero NPV for total swap

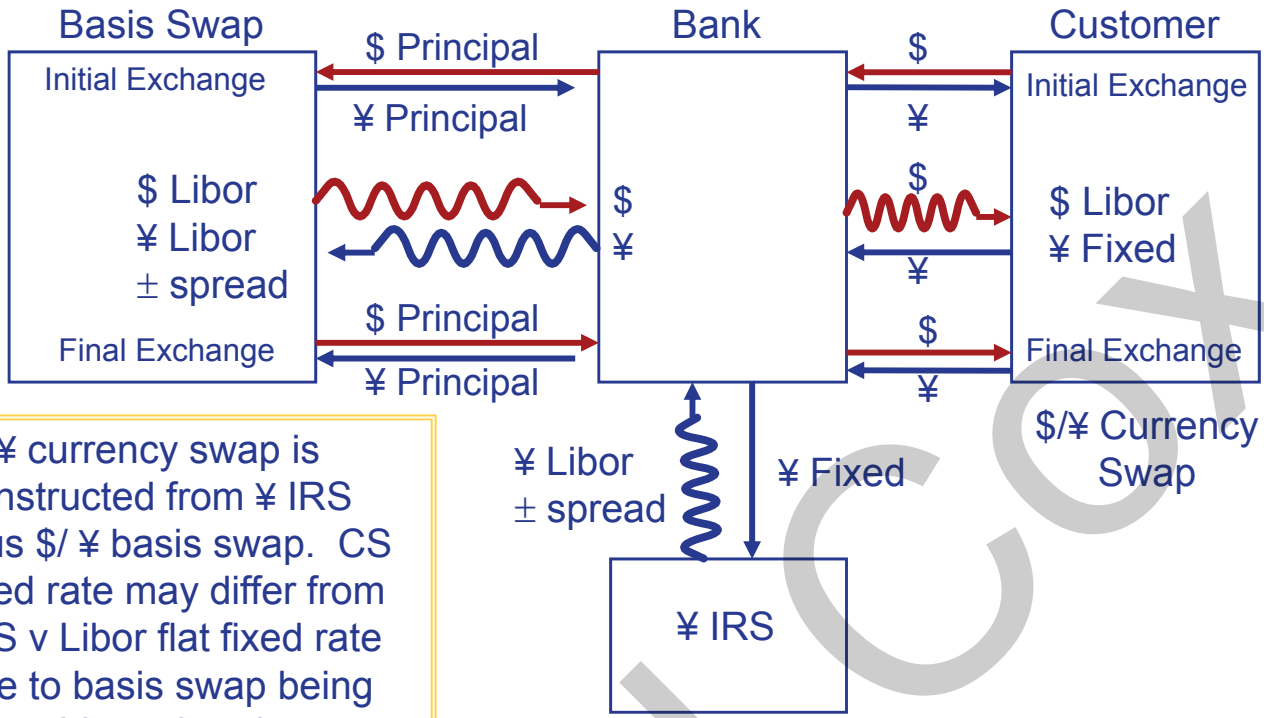
Spread above or below fixed on vanilla IRS (Libor flat) will not necessarily be the same as that over or under Libor on floating side

- Value of 1bp spread is different for different currencies
- Payment interval may be different

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Basis Swap + IRS = Currency Swap



$/ ¥ currency swap is constructed from ¥ IRS plus $/ ¥ basis swap. CS fixed rate may differ from IRS v Libor flat fixed rate due to basis swap being something other than zero

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Asset Swap

Asset swaps create instruments not available to investors

Combination of asset and IRS or Currency Swap

Higher yield

Different credit characteristics

Specific cash flows

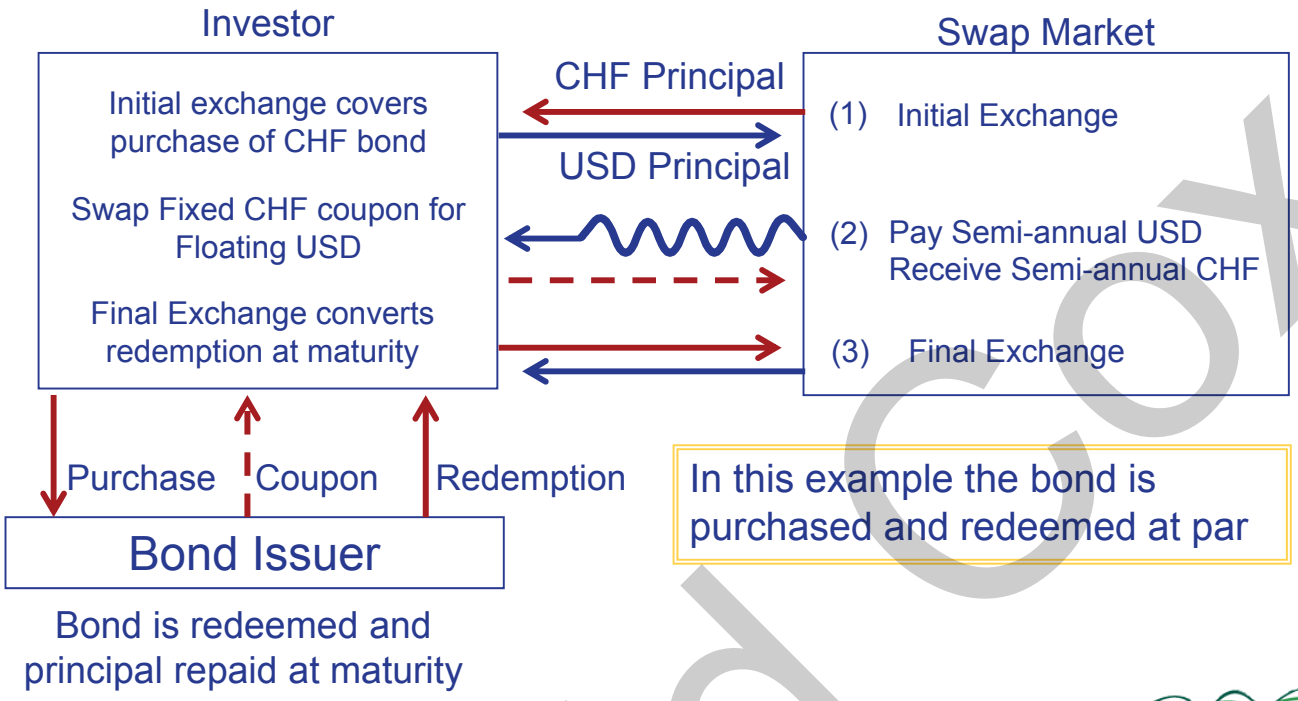
Specific currency or interest rate

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Cross Currency Asset Swap

Synthetic Dollar FRN



In this example the bond is purchased and redeemed at par

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Asset Swap

Bond price not necessarily par

Par asset swap

- Difference between bond price and par paid to or received from the swap market at start.
- Net effect is that investor buys an asset at par and receives Libor ± spread and par at maturity
- Spread reflects bond price in relation to par

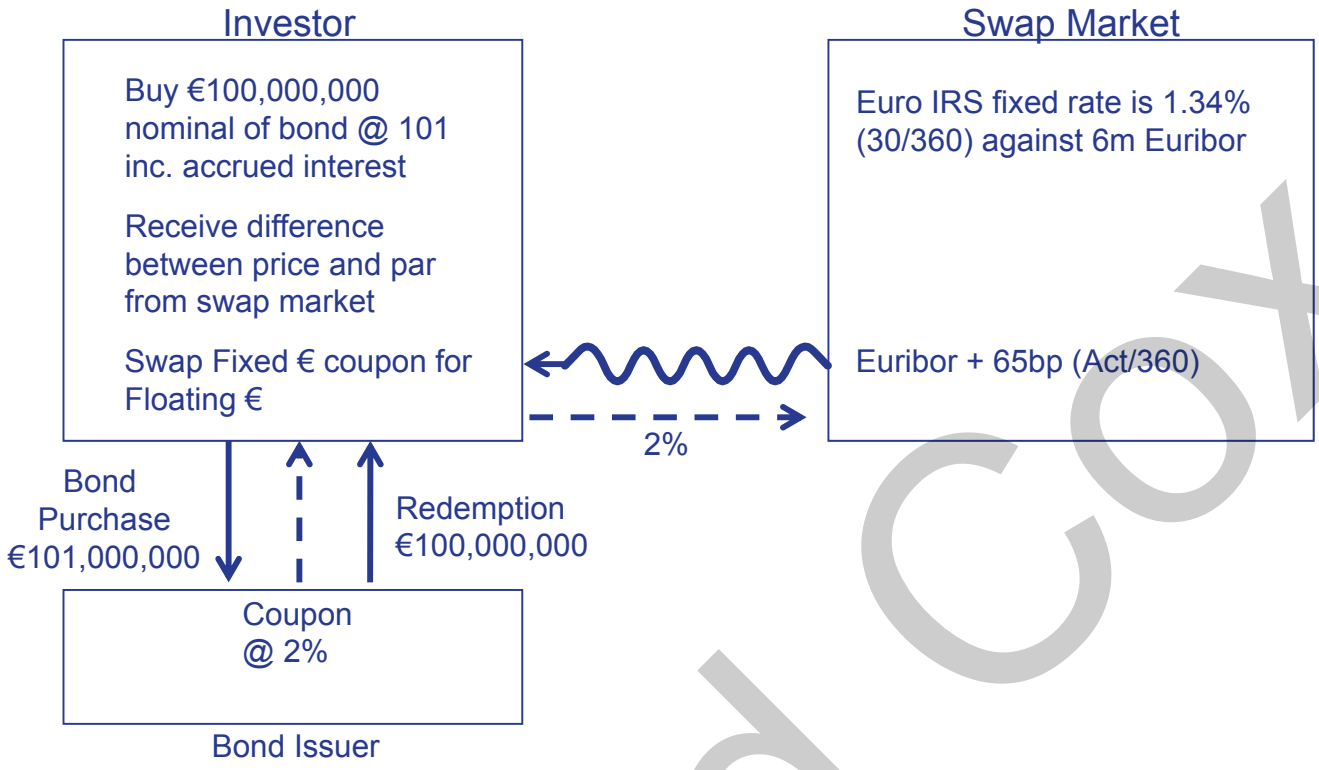
Proceeds asset swap

- Difference between bond price and par paid to or received from the swap market at end.
- Net effect is that investor buys an asset at initial price (plus accrued interest) and receives Libor ± spread and initial price at maturity
- Spread reflects bond price in relation to par

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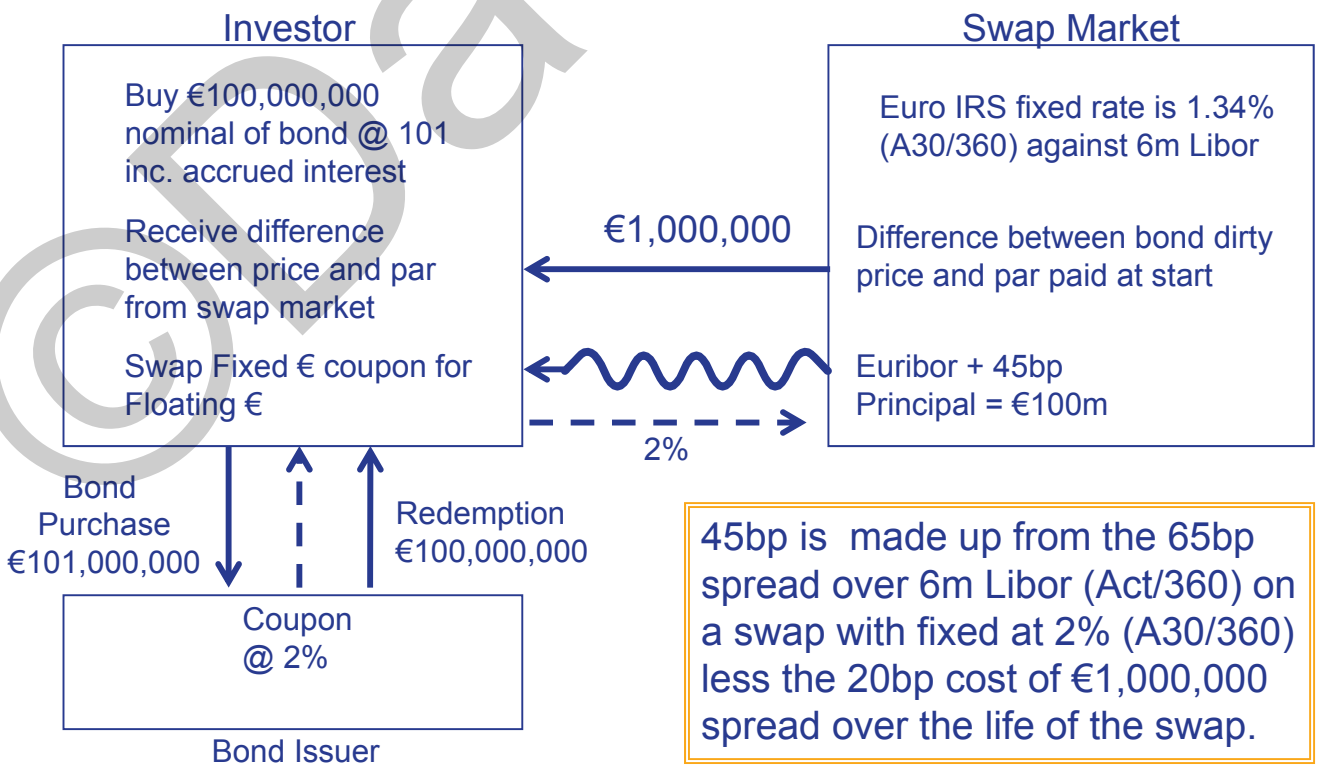
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Single Currency Coupon Swap



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Single Currency Par Asset Swap

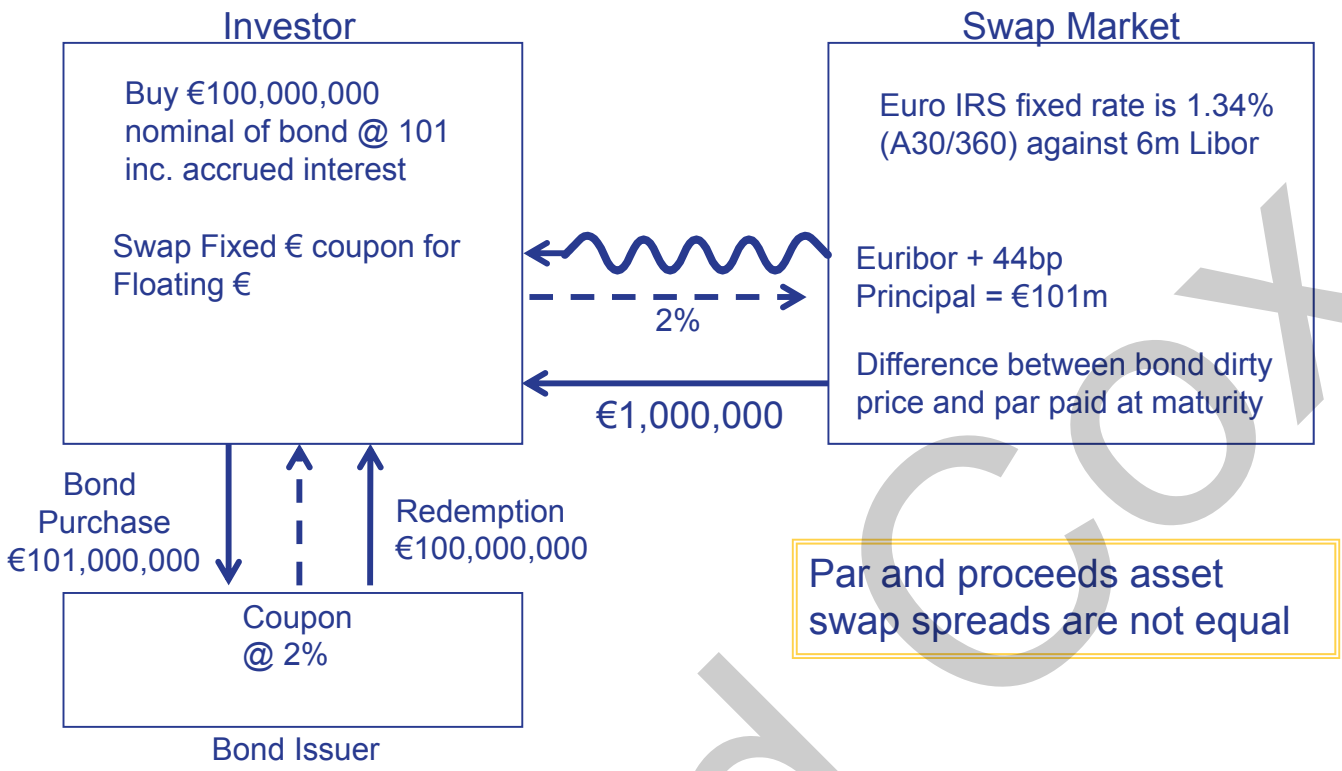


45bp is made up from the 65bp spread over 6m Libor (Act/360) on a swap with fixed at 2% (A30/360) less the 20bp cost of €1,000,000 spread over the life of the swap.

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Single Currency Proceeds Asset Swap



Par and proceeds asset swap spreads are not equal

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Asset Swap

Investor's View

	Par	Proceeds
Asset at Discount	Make up-front payment Interest on par amount	Make payment at maturity Interest on invested amount
Asset at Premium	Receive up-front payment Interest on par amount	Receive payment at maturity Interest on invested amount

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"Z" Spread

The spread added to the zero coupon rate (r_t) that gives discounted cashflows equal to bond price



$$103 = \frac{5}{(1+r_1+z)^1} + \frac{5}{(1+r_2+z)^2} + \frac{5}{(1+r_3+z)^3} + \frac{105}{(1+r_4+z)^4}$$

Z is a constant that represents a parallel shift in the yield curve

Does not make any assumptions about the yield curve being flat

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I (interpolated) Spread

Difference between a bond's yield to maturity and a swap rate for the same maturity as the bond.

For a flat YC this will be the same as the Z spread regardless of price and coupon

For a non flat curve the difference from the z spread will depend on the difference between the bond coupon and swap rate

Subject to all the inaccuracies of YTM and does not use information from the full term structure of rates

Not preferred by analysts for relative value comparison between bonds

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Spread Comparison

Flat YC @ 3%, 5y Bond, Annual Coupons

Coupon	Price	Z	Par AS	Proceeds	I
1%	110	-3.94%	-4.18%	-3.80%	-3.94%
1%	100	-2.00%	-2.00%	-2.00%	-2.00%
1%	90	0.20%	0.18%	0.20%	0.20%
3%	110	-2.06%	-2.18%	-1.99%	-2.06%
3%	100	0.00%	0.00%	0.00%	0.00%
3%	90	2.33%	2.18%	2.43%	2.33%
5%	110	-0.17%	-0.18%	-0.17%	-0.17%
5%	100	2.00%	2.00%	2.00%	2.00%
5%	90	4.47%	4.18%	4.65%	4.47%

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Spread Comparison

Z Spread

Parallel shift to Zero Coupon yield curve needed to give bond price

I Spread

Numerical difference between bond yield and swap rate of equal maturity to the bond

Par Asset Swap Spread

Shift to forward curve plus up-front payment needed to convert bond into par value FRN paying spread against Libor

Proceeds Asset Swap Spread

Shift to forward curve plus payment at end needed to convert bond into par value FRN paying spread against Libor with notional equal to bond price

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Option Adjusted Spread (OAS)

Like Z spread, but takes into account the optionality of a bond

To calculate, strip out the value of the bond option from the market price of the bond and shift the yield curve so that the PV of cashflows equals the bond price without option

- This requires the accurate valuation of the bond option
- Call option has negative price impact
- Put option has positive price impact

The difference between the Z spread and OAS of a callable bond reflects the value of the call

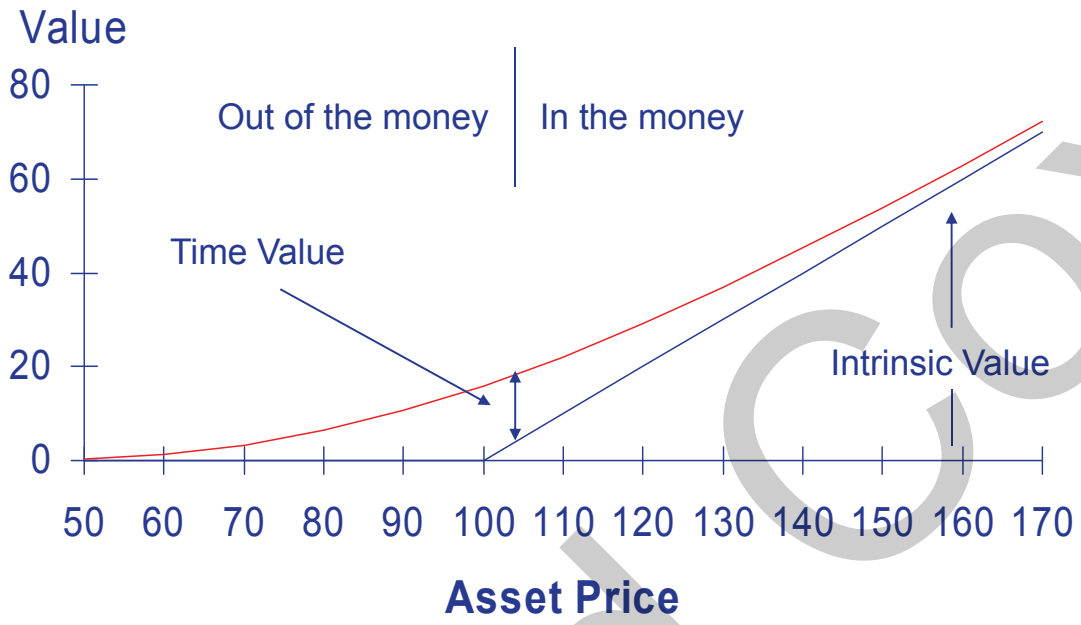
If a bond has no optionality the OAS and Z spread are the same

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Option Value

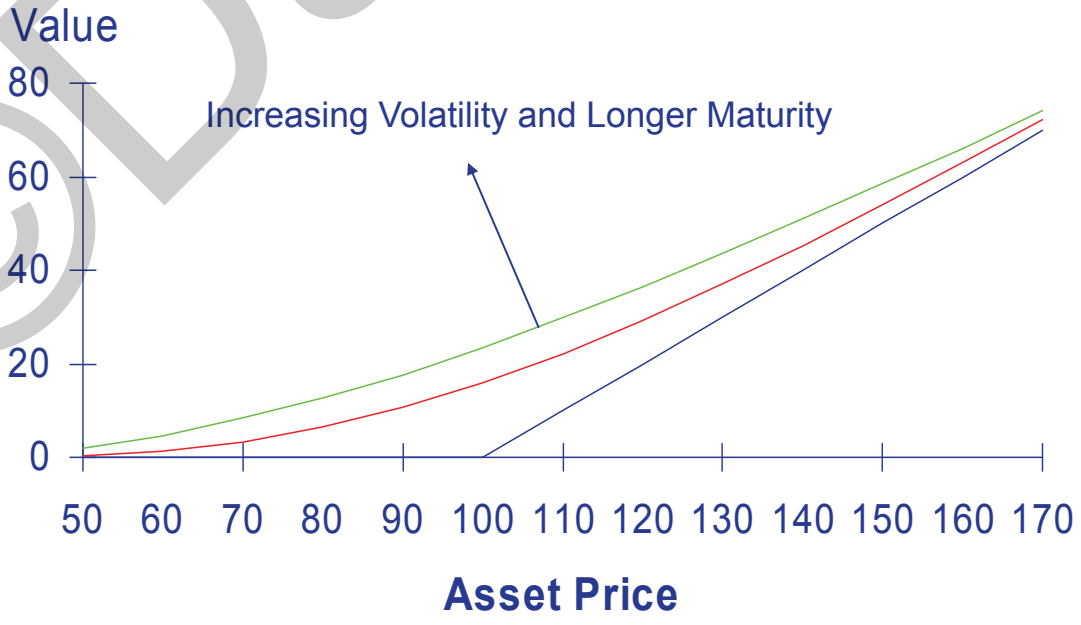
Call Option (Strike = 100)



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Option Value

Call Option (Strike = 100)



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Option Value

Key Factors

Volatility

Strike Price

Asset Price

Time to Maturity

Interest Rate

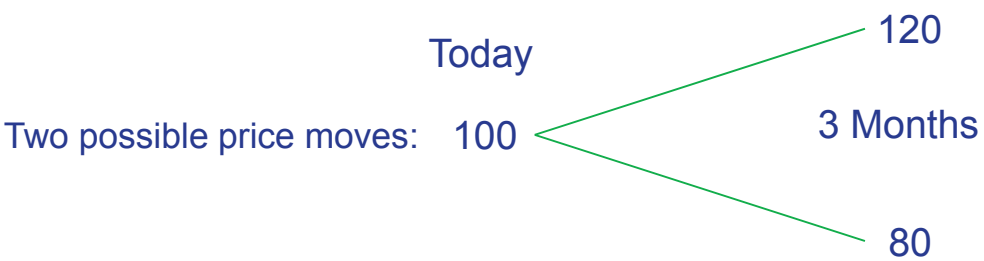
Dividends

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Option Value

European Call

Contract Size:	1000 Units
Maturity:	3 months
Strike:	100
Asset Price:	100
Interest Rate:	10%



No assumptions about the likelihood of price changes or investors attitude to risk

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Option Value

European Call

Today	3 Months	3 Months
Price = $100	Price = $80	Price = $120
Sell European Call on 1000 Units. Strike =100 @ Price = $c	Call not exercised	Call exercised Deliver 1000 units @ 100
Invest c @ 10%	Receive $1.025c	Receive $1.025c
Borrow $100H @ 10%	Repay $102.5H	Repay $102.5H
Buy H units of Asset @ 100	Sell H units @ $80	Buy 1000-H units @ $120

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Option Value

European Call

Price goes down to $80:

1.025c - 102.5H +80H = 0

Price goes up to $120:

1.025c - 102.5H + 100,000 - (1000-H) * 120 = 0

So:

c = 10,975.60

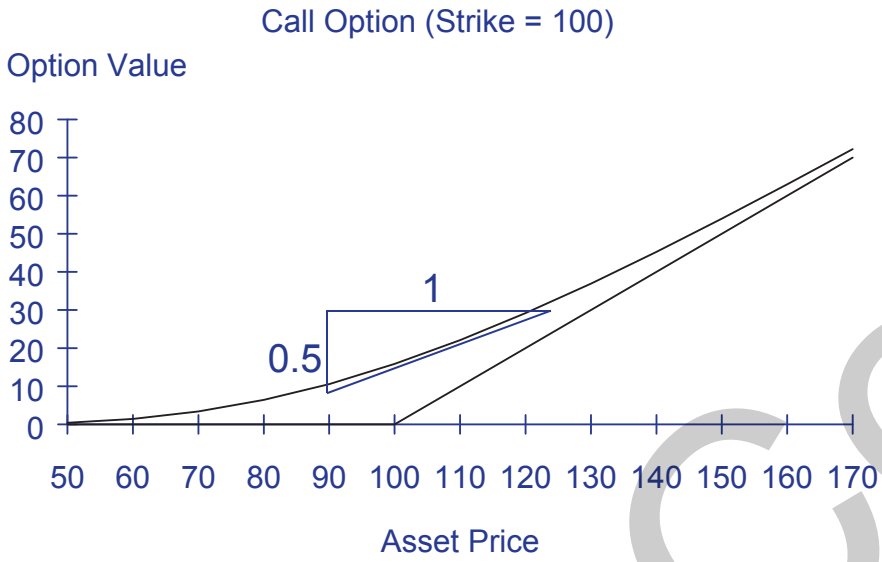
H = 500 (Delta =0.5)

Hedged option is instantaneously riskless

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Delta



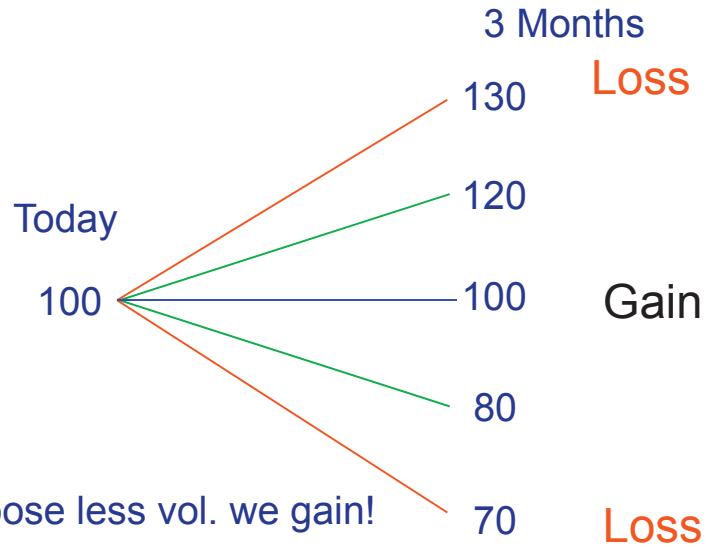
Delta changes with price
so the hedge needs to be adjusted to remain riskless

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Option Value

European Call

What happens if we get our estimate of future moves wrong?



More volatility we loose less vol. we gain!

Short option means short volatility

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Volatility

Annualized standard deviation of asset returns (or s.d. of log of asset price)

Implied volatility

- The volatility that makes the output of an option pricing model equal to the market value
- The most commonly quoted vols. (stock and FX options) are Black-Scholes volatilities, where assets are assumed to have a lognormal distribution
- Basis point volatilities (swaptions, caps floors) assume a normal distribution of rates
- Implied volatility can be loosely interpreted as a measure of "expected future volatility"

Historical volatility

- Annualized standard deviation of observed asset returns
- Calculation period and observation frequency affect results
- To measure this properly is not simple

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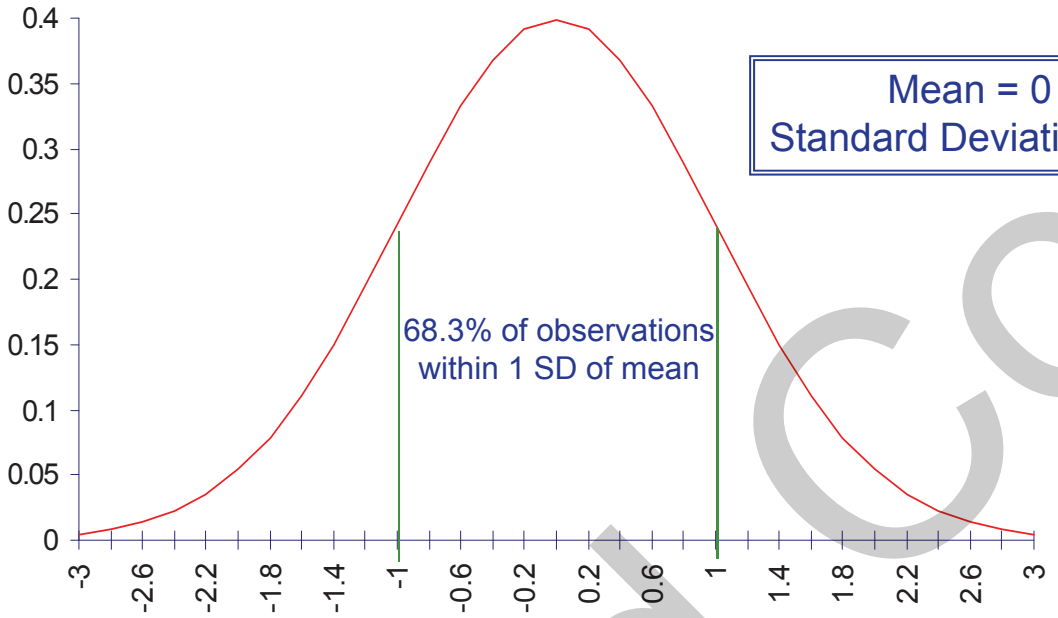
Historical v Implied Volatility

Implied Vol has generally been a leading directional indicator of future realized volatility, but not a highly accurate predictor.



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Normal Probability Distribution



Mean = 0
Standard Deviation = 1

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Mean and Variance

Mean is the arithmetic average: $S^* = \frac{\sum_{i=1}^{n} S_i}{n}$

Variance is the average of the square of the deviations from that mean

- S = Asset return for a given period t (in years)
- S* = mean of asset returns
- n = number of observations

Variance of asset returns: $\theta_s^2 = \frac{\sum_{i=1}^{n} (S_i - S^*)^2}{n}$

Sum of squared deviations from the mean

Annualised volatility: $\sigma_s = \frac{\theta_s}{\sqrt{t}}$

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Using Volatility

If an asset, currently priced at 100, has a volatility of 16% how much is it likely to move in one day?

If the one year return is normally distributed the asset has 68% probability of moving by 16% or less up or down.

Over one day the standard deviation will be:

$$\frac{16\%}{\sqrt{\text{Days in a year}}}$$

Use 256 trading days so daily vol. is 1%

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How Asset Prices Move

Geometric Brownian Motion

Return over Δt

Expected return

Stochastic Part. Mean = 0, standard deviation = $\sigma\sqrt{\Delta t}$

$$\frac{\Delta S}{S} = \mu\Delta t + \sigma\varepsilon\sqrt{\Delta t}$$

Volatility

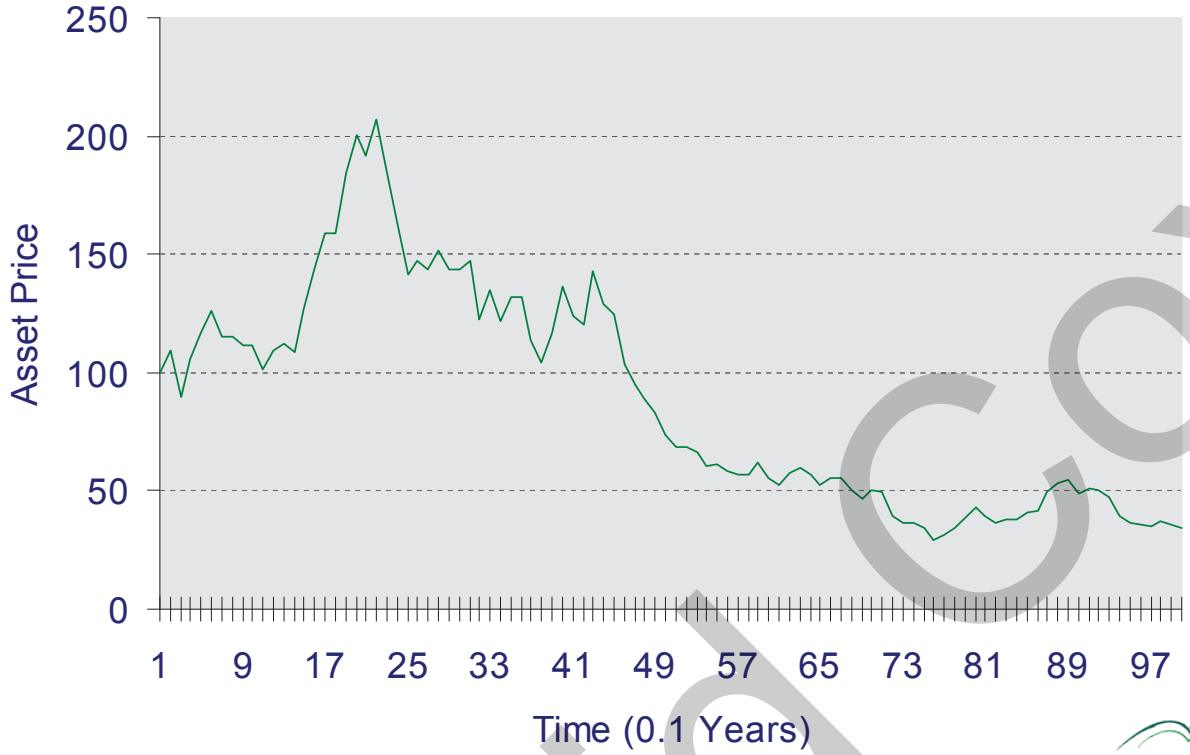
ε = Random number from a normal distribution with mean = 0 and standard deviation = 1

Overall the return of the asset over a small interval of time is normally distributed with mean $\mu\Delta t$ and standard deviation $\sigma\sqrt{\Delta t}$

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Asset Price Over Time
Geometric Brownian Motion
σ = 30% μ = 5%



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Asset Prices

Lognormal Probability Distribution



Returns normal
Price lognormal
Price above zero
Skewed to right

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Modelling Volatility

Using Trees

As with a Monte Carlo simulation of asset price moves, divide the period into small intervals of time

Allow the price to move in more than one possible direction from a given starting point in each interval

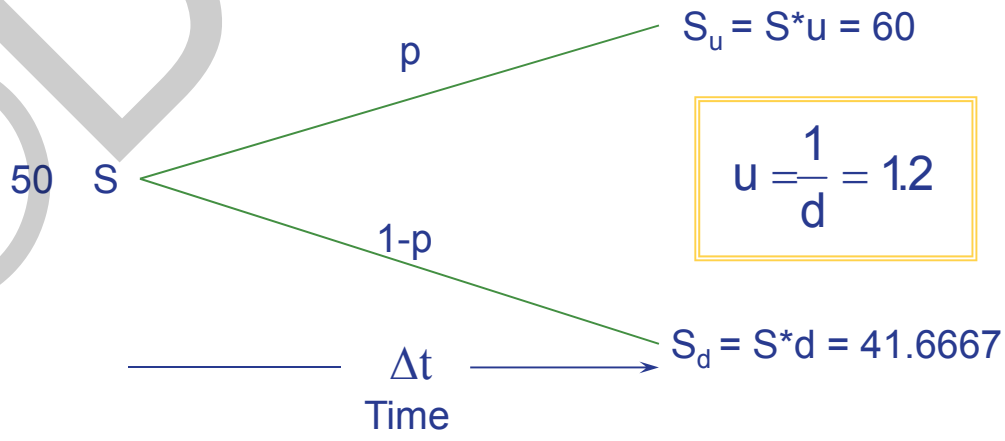
Let each possibility have an associated "probability"

Fit the parameters of the tree to the process for the asset price so that the expected return and the variance of returns are equivalent

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Binomial Trees

Single Period (Geometric)



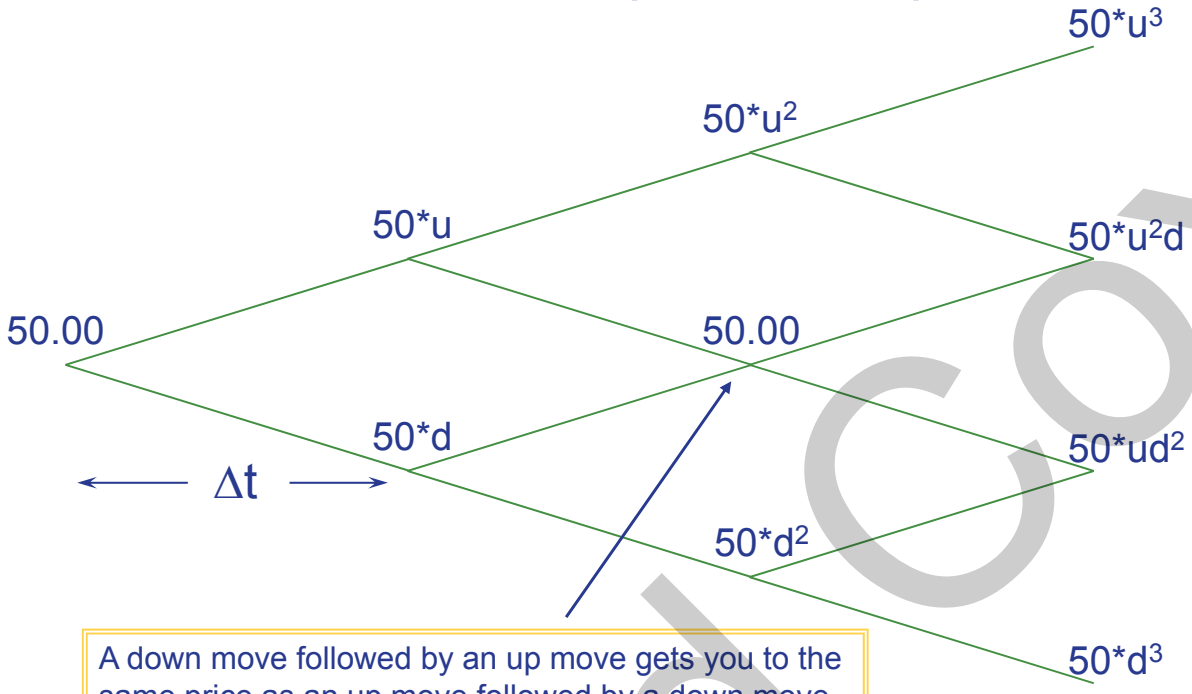
This can be used to represent Geometric Brownian Motion

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Binomial Trees

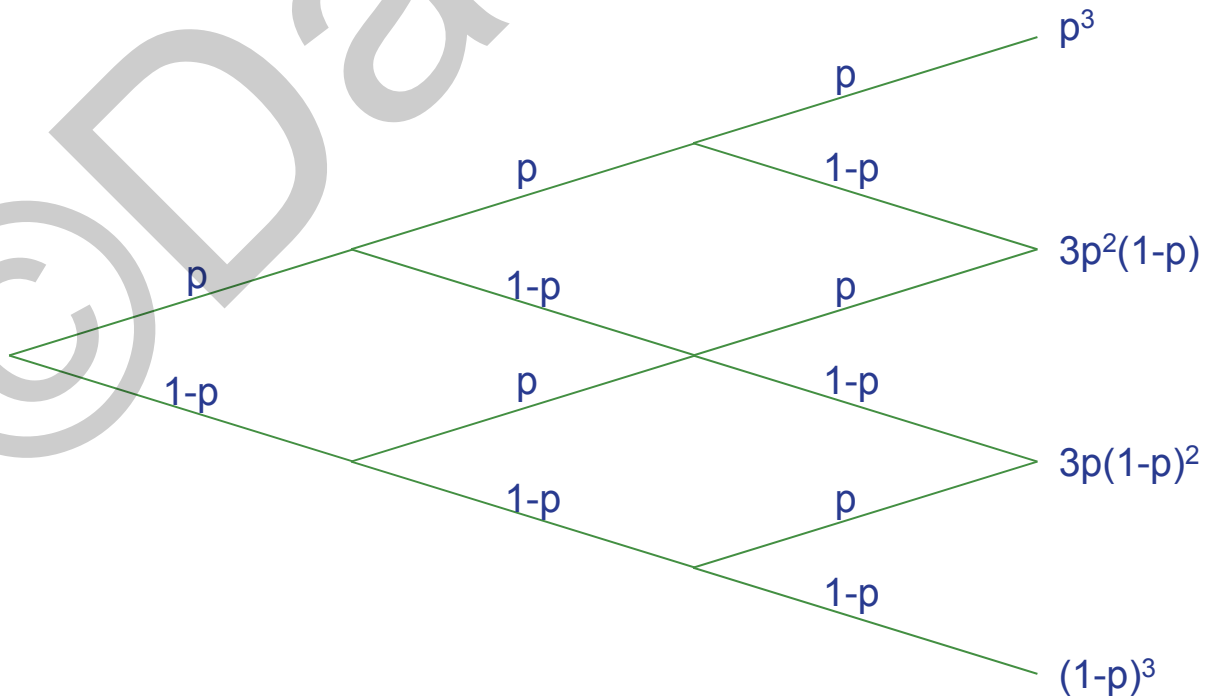
Multiperiod (Geometric)



A down move followed by an up move gets you to the same price as an up move followed by a down move. Past prices do not affect future prices

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Probability



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Probability

The probability of a asset being a particular node at the end of the period is:

$$\binom{l}{j} p^j (1-p)^{l-j}$$

l = number of periods in the tree

j = the number of upward moves in the asset price

AND

$\binom{l}{j}$ Is the number of ways of picking j items from l items (the jth binomial coefficient)

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Binomial Tree

The probability distribution produced by the binomial tree approximates to that of geometric Brownian motion if there is a large enough number of steps

- Using trees to price options gives results that will fluctuate around the results from analytic (Black – Scholes) methods
- Fluctuation will reduce as the number of steps in the tree is increased
- Trade off accuracy against calculation time

What do these probabilities really represent?

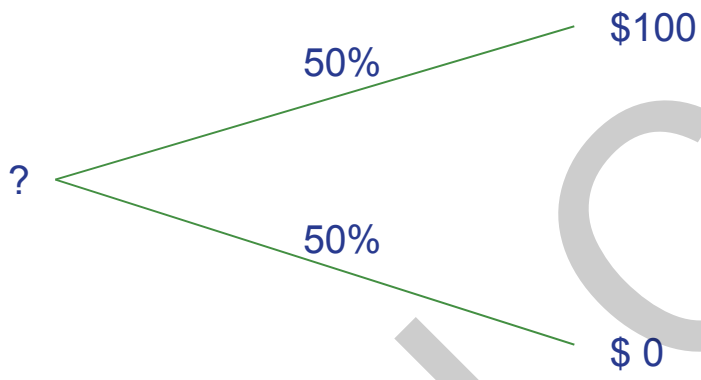
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Risk Preferences

How much would you pay for this bet?

A 50% chance of getting $100 and a 50% chance of getting nothing

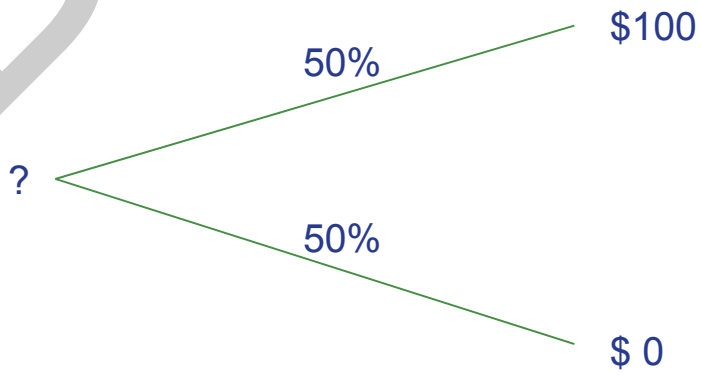


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Risk Preferences

How much would you pay?

The price you set expresses your preference for risk



> $50 = Risk Seeking < $50 = Risk Averse
$50 Exactly = Risk Neutral

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Risk Neutral Valuation

Essential Technique

It is possible to hedge many derivatives in such a way that the portfolio of hedge and derivative is riskless *at any instant in time*

If so their values do not depend on investor risk preferences

Therefore we are free to value these instruments assuming that all investors are risk neutral

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Risk Neutral Valuation

If the pricing of a securities does not involve the risk preferences of investors then any set of risk preferences can be used

We choose Risk Neutrality

A convenience not a statement about the way the world works!

In a risk neutral world the expected rate of return from all traded assets is the risk free rate of interest

In a risk neutral world present values are calculated by discounting the expected future payoff with the risk free rate of interest

So what is the “risk free” rate?

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Binomial Trees

Fitting Tree to Model

$$\frac{\Delta S}{S} = \mu\Delta t + \sigma\varepsilon\sqrt{\Delta t}$$

Risk Neutrality Implies:

Expected value of asset price at end of Δt = $Se^{r\Delta t}$

r is the continuously compounded risk free rate of interest

This is the same as $S\times(1+r_s\,\Delta t)$ if r_s is a simple interest rate

Variance of asset price over Δt = $S^2\sigma^2\Delta t$

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Binomial Trees

Fitting Tree to Model

Expected value of asset price at end of Δt

The two outcomes multiplied by their probabilities

$$pSu + (1-p)Sd = Se^{r\,\Delta t}$$

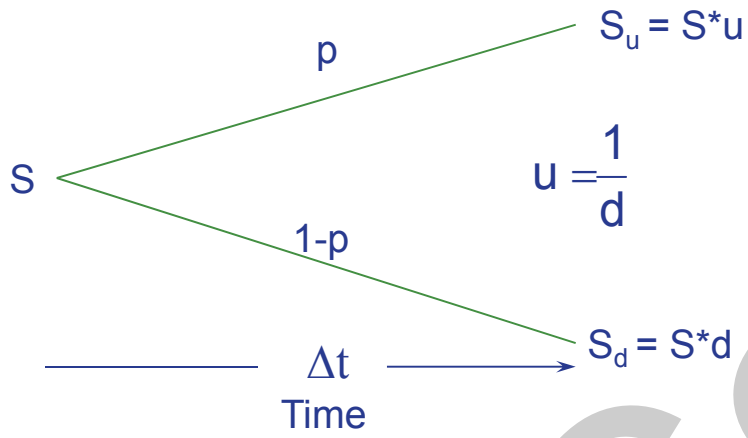
Variance of asset price over Δt

The expectation of the square of the outcomes less the square of the expectation

$$pS^2u^2 + (1-p)S^2d^2 - S^2\left[pu + (1-p)d\right]^2 = S^2\sigma^2\Delta t$$

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The Standard Cox-Ross Tree

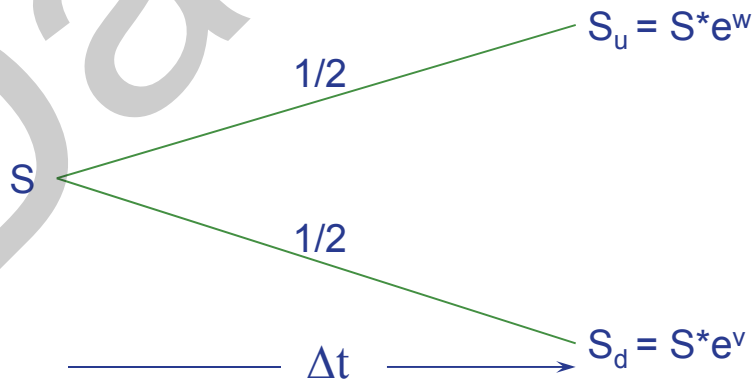


$$u = e^{\sigma\sqrt{\Delta t}} \qquad d = e^{-\sigma\sqrt{\Delta t}} \qquad a = e^{r\Delta t} \qquad p = \frac{a - d}{u - d}$$

p is a risk neutral or "pseudo" probability

© David Cox

The Jarrow-Rudd Tree



$$w = \left(r - \tfrac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} \qquad v = \left(r - \tfrac{1}{2}\sigma^2\right)\Delta t - \sigma\sqrt{\Delta t}$$

Mean and variance are the same as asset price process for all numbers of steps as well as when the number is large

This type of tree is better for calculating implied option volatilities and when volatilities are very low

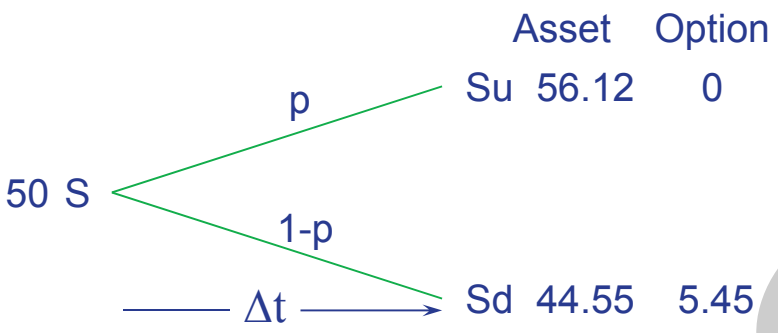
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Put Option

Strike (X) = 50
Interest Rate (r) = 10%

Volatility = 40%
Δt = 0.0833 years (1 month)



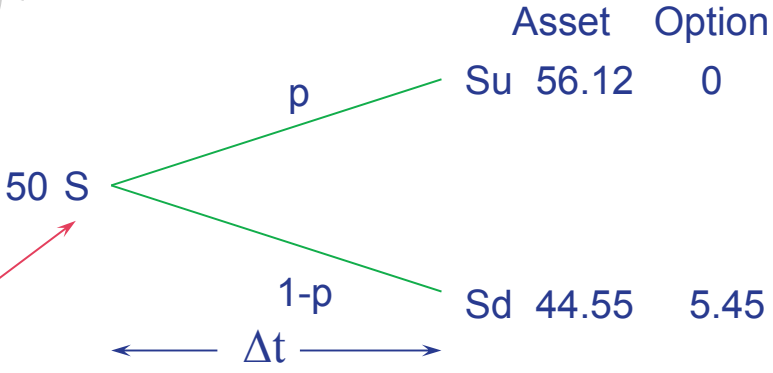
Payoff at Maturity:
Max(0, X-S_T)

$$u = e^{\sigma\sqrt{\Delta t}} \qquad d = e^{-\sigma\sqrt{\Delta t}} \qquad a = e^{r\Delta t} \qquad p = \frac{a - d}{u - d}$$

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Put Option

Strike = 50
Interest Rate (r) = 10%
Volatility = 40%
Δt = 0.0833 years
u = 1.1224
d = 0.8909
a = 1.0084
p = 0.5076



Option Price = $\left[p \times 0 + (1-p) \times 5.45\right] e^{-0.1 \times 0.0833} = 2.66$

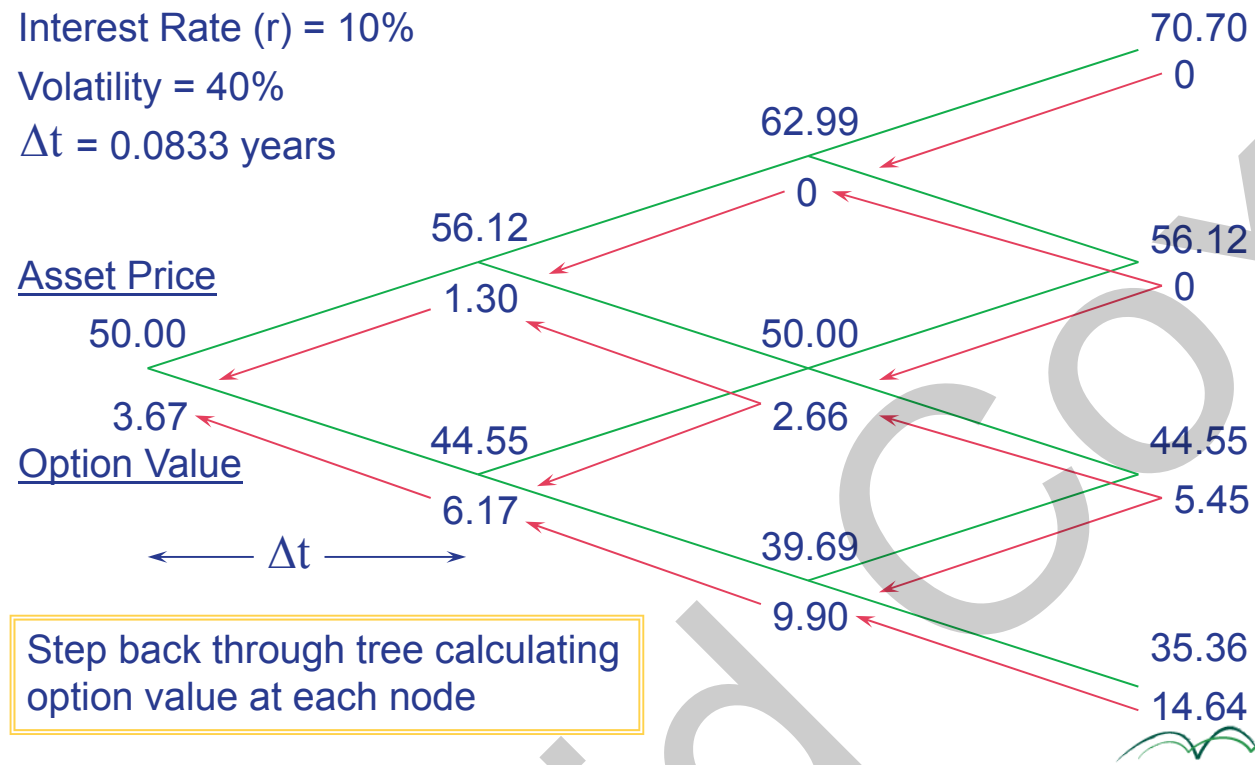
Option price is the expected future payoff discounted to today at the risk free rate of interest

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Three Month European Put

Strike = 50
Interest Rate (r) = 10%
Volatility = 40%
Δt = 0.0833 years

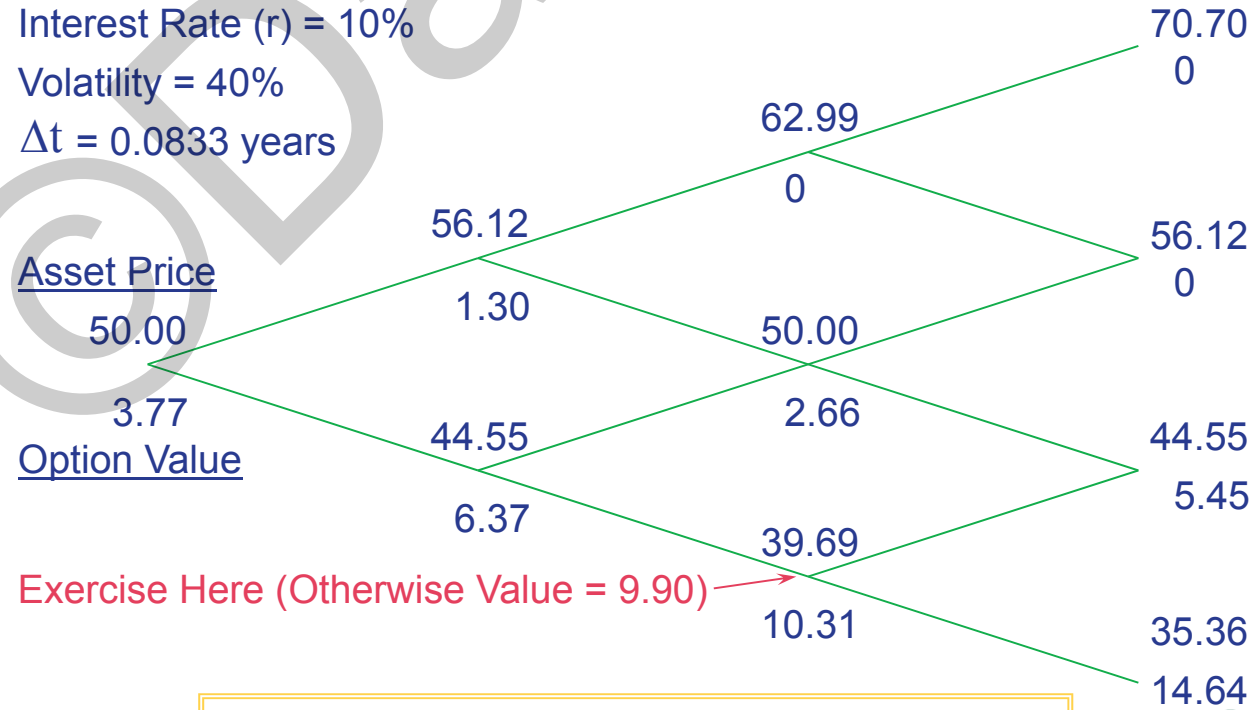


Step back through tree calculating option value at each node

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Three Month American Put

Strike = 50
Interest Rate (r) = 10%
Volatility = 40%
Δt = 0.0833 years



Exercise Here (Otherwise Value = 9.90)

American put is worth more that a European put

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Dividends

A continuous dividend stream can be incorporated into the expected return of the asset

- If the asset yields a rate of q then the return on the asset without dividend will be r-q in a risk neutral world (the total return will be r)

When building a tree set the parameters:

$$u = e^{\sigma\sqrt{\Delta t}} \qquad d = e^{-\sigma\sqrt{\Delta t}} \qquad a = e^{(r-q)\Delta t} \qquad p = \frac{a - d}{u - d}$$

- This is used for currency options where the domestic currency rate is r and the foreign rate q
- The option does not pay a dividend so its expected future payoff is discounted at r

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Options on Forward Contracts

In a risk neutral world a forward contract can be treated as an asset that pays a dividend equal to the risk free rate of interest

- This is because you pay nothing until you get the asset in the future
- The money that would otherwise have been used to buy the asset can be invested at the risk free rate

So set q = r and the tree parameters are:

$$u = e^{\sigma\sqrt{\Delta t}} \qquad d = e^{-\sigma\sqrt{\Delta t}} \qquad a = 1 \qquad p = \frac{a - d}{u - d}$$

Forward price does not grow

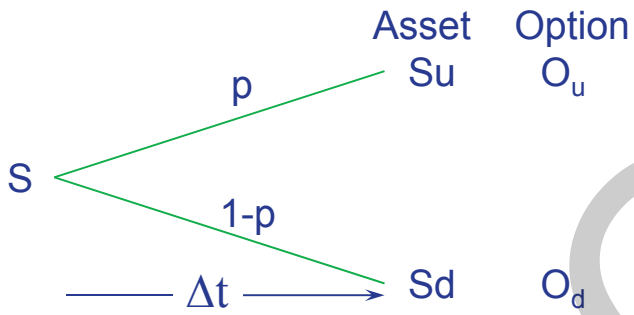
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Sensitivities

Delta

Rate of change of option price with respect to underlying security



$$\delta = \frac{O_u - O_d}{S_u - S_d}$$

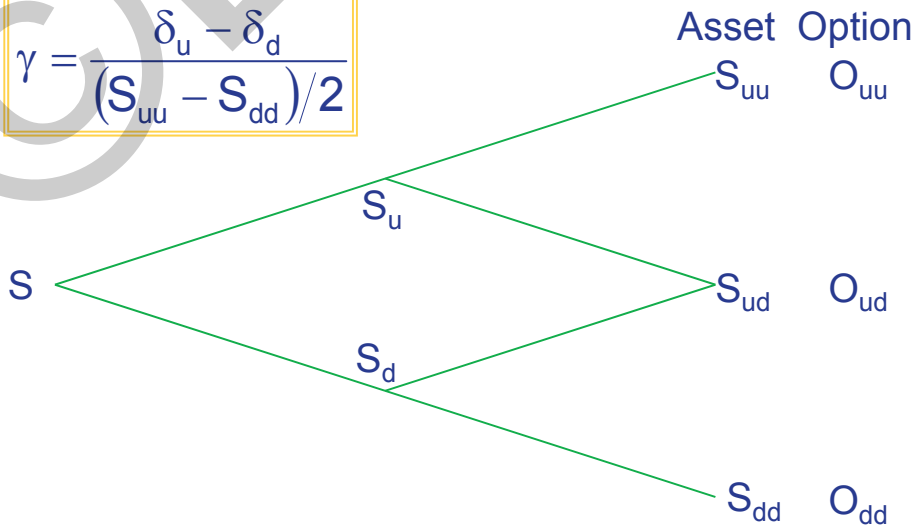
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Sensitivities

Gamma

Rate of change of Delta with respect to underlying security

$$\gamma = \frac{\delta_u - \delta_d}{(S_{uu} - S_{dd})/2}$$



$$\delta_u = \frac{O_{uu} - O_{ud}}{S_{uu} - S_{ud}}$$

$$\delta_d = \frac{O_{ud} - O_{dd}}{S_{ud} - S_{dd}}$$

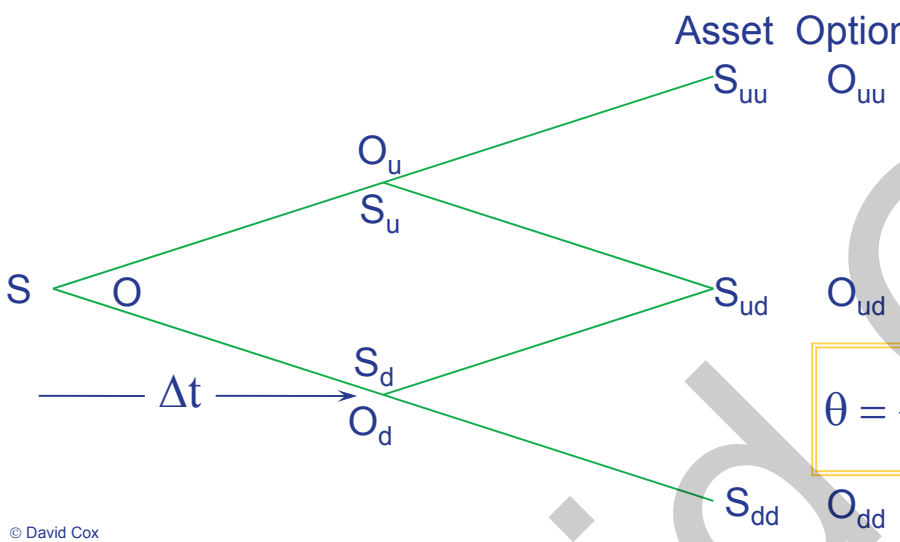
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Sensitivities

Theta

Rate of change of option price with respect to time, but not underlying security



If $S = S_{ud}$

$$\theta = \frac{O_{ud} - O}{2\Delta t}$$

If S not = S_{ud}

$$\theta = \frac{O_{ud} - \delta \times (S_{ud} - S) - O}{2\Delta t}$$

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Other "Greeks"

Have to be found by changing inputs and recalculating the tree

Vega

Chang in option value with respect to volatility (not really a Greek Letter)

Should have stochastic vol, but works well in B-S framework

$$V = \frac{\Delta O}{\Delta \sigma}$$

Rho

Chang in option value with respect to interest rate

Two Rhos: one for risk free rate and other for yield (fixed currency in FX quote)

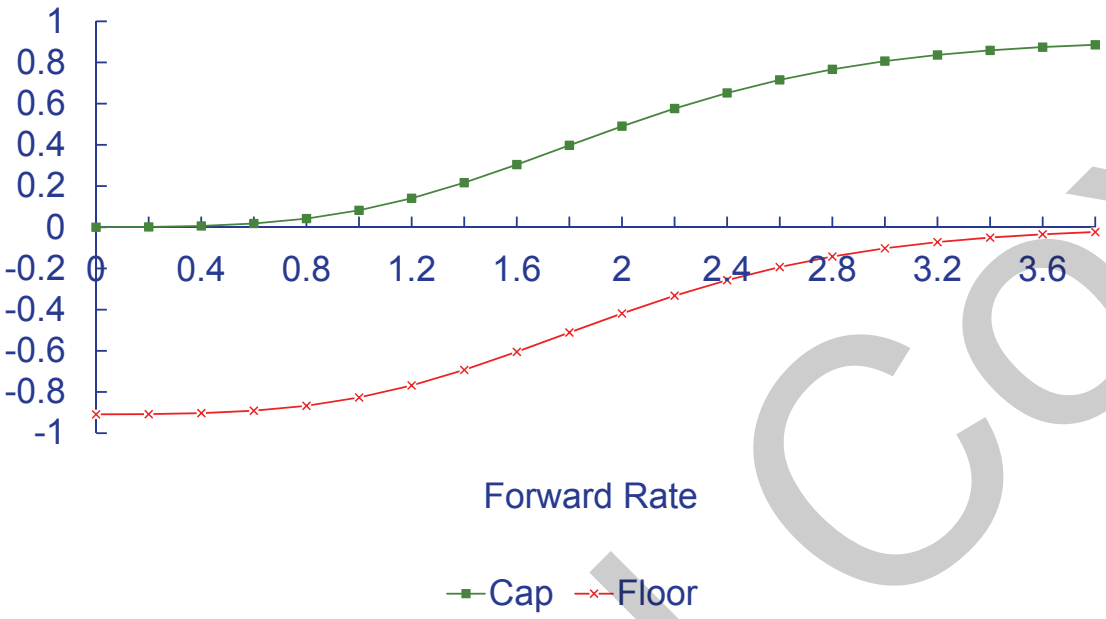
$$\rho = \frac{\Delta O}{\Delta r}$$

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Delta

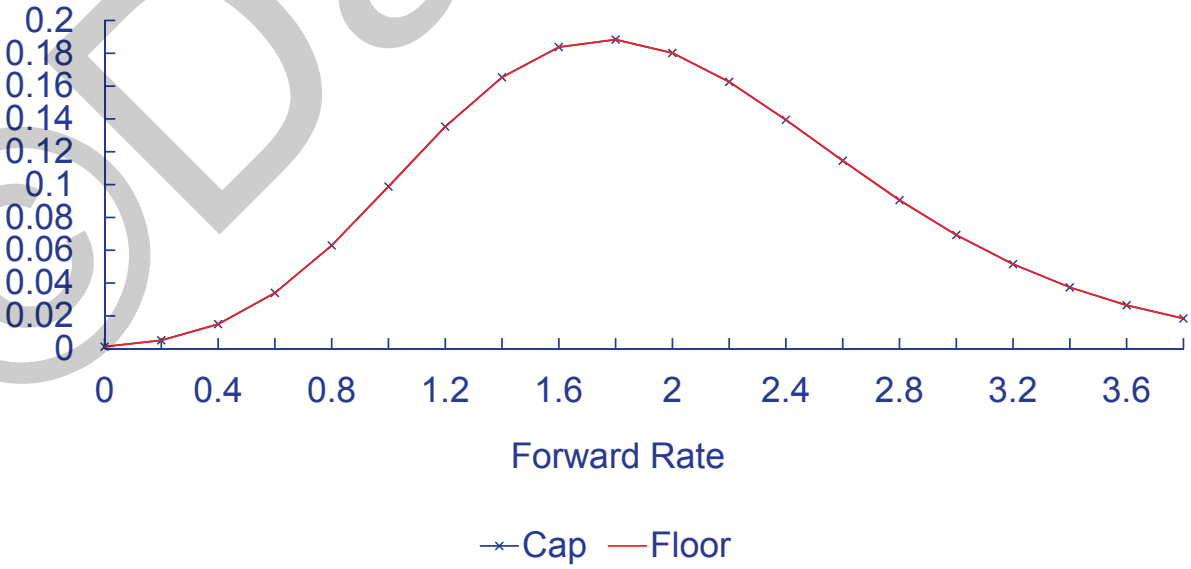
Single Period Cap and Floor Strike = 2%



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Gamma

Single Period Cap and Floor Strike = 2%

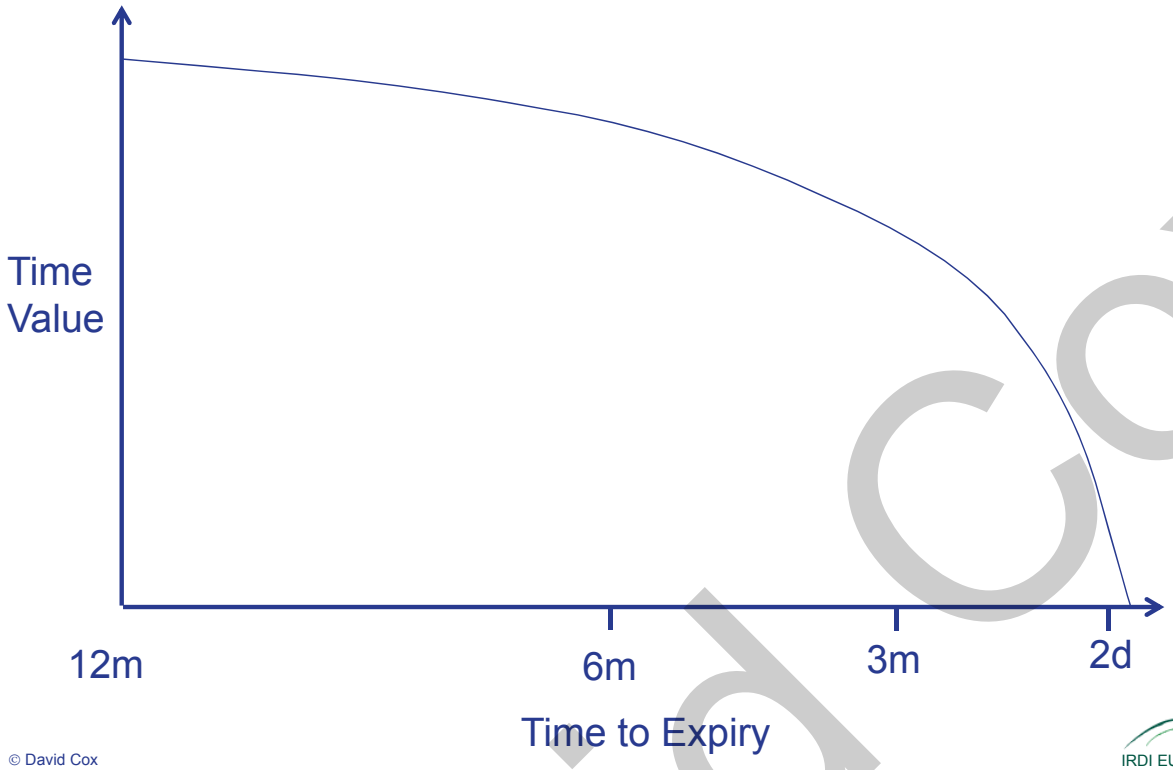


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Option Time Decay

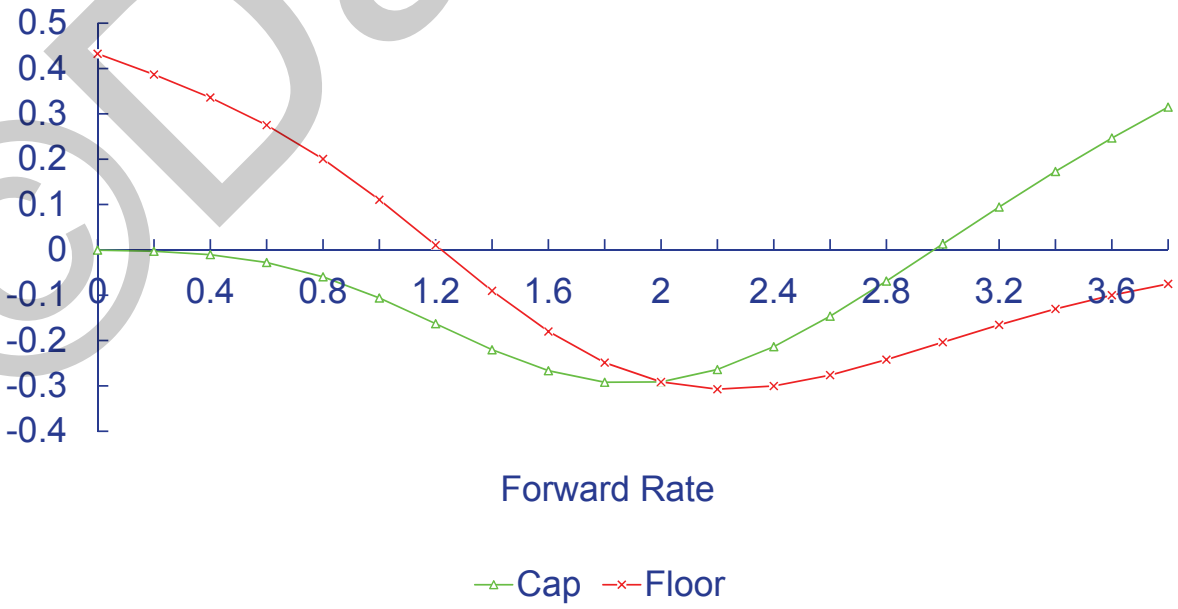
Single Period Cap at the Money



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Theta

Single Period Cap and Floor Strike = 10%



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Option Value Day by Day

Short European Call (Asset Liquid)

Contract Size:	1000 Assets	Strike:	100	Premium:	$11,000
Maturity:	3 months	Interest rate:	10%	Delta:	0.5

S	δ	Hedge	Realised Loss
100	0.5	+ 500 @ 100	
110	0.6	+ 100 @ 110	
120	0.7	+ 100 @ 120	
110	0.6	- 100 @ 110	10 × 100
100	0.5	- 100 @ 100	10 × 100
90	0.4	- 100 @ 90	
80	0.3	- 100 @ 80	
90	0.4	+ 100 @ 90	10 × 100
100	0.5	+ 100 @ 100	10 × 100

Loss due to rehedging (γ) is offset by gain on time decay (θ)

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Hedge Short European Call

Asset not Liquid or Size Large

Price moves against you when hedge is placed

S	δ	Hedge	Realised Loss
100	0.5	+ 500 @ 101	
110	0.6	+ 100 @ 111	
120	0.7	+ 100 @ 121	
110	0.6	- 100 @ 109	12 × 100
100	0.5	- 100 @ 99	12 × 100
90	0.4	- 100 @ 89	
80	0.3	- 100 @ 79	
90	0.4	+ 100 @ 91	12 × 100
100	0.5	+ 100 @ 101	12 × 100

More costly re-hedging requires higher option price

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The Impact of Delta Hedging

Delta hedging has an impact on asset price

As an ATM option moves closer to expiry delta hedging has to be more extreme.

If most of the delta hedgers are long options then they will sell when the asset goes up and buy when it goes down.

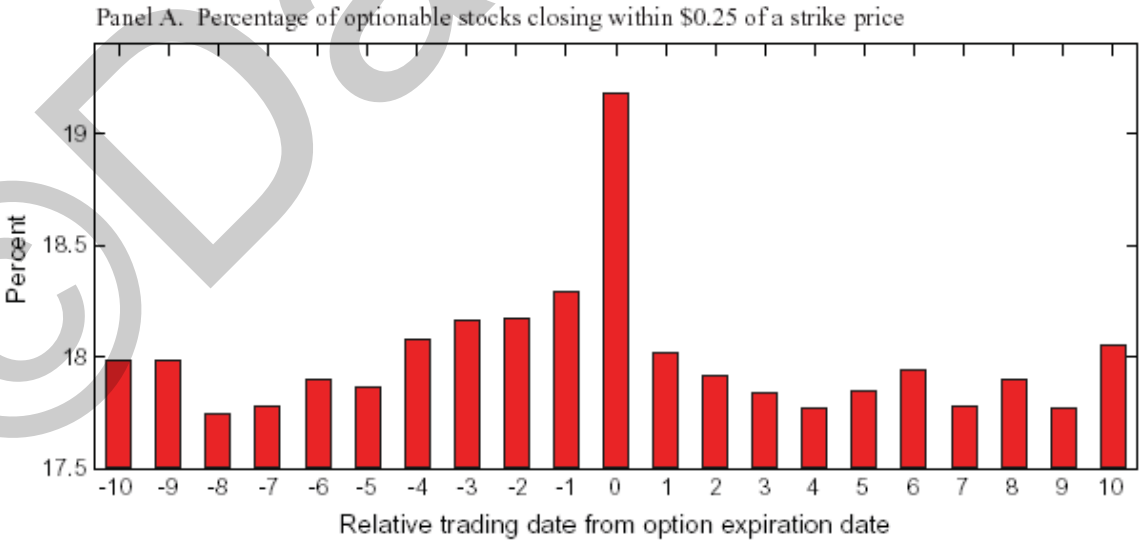
This has the effect of moving the asset price closer to the ATM option strike at expiry

If most delta hedgers are short options the reverse is true.

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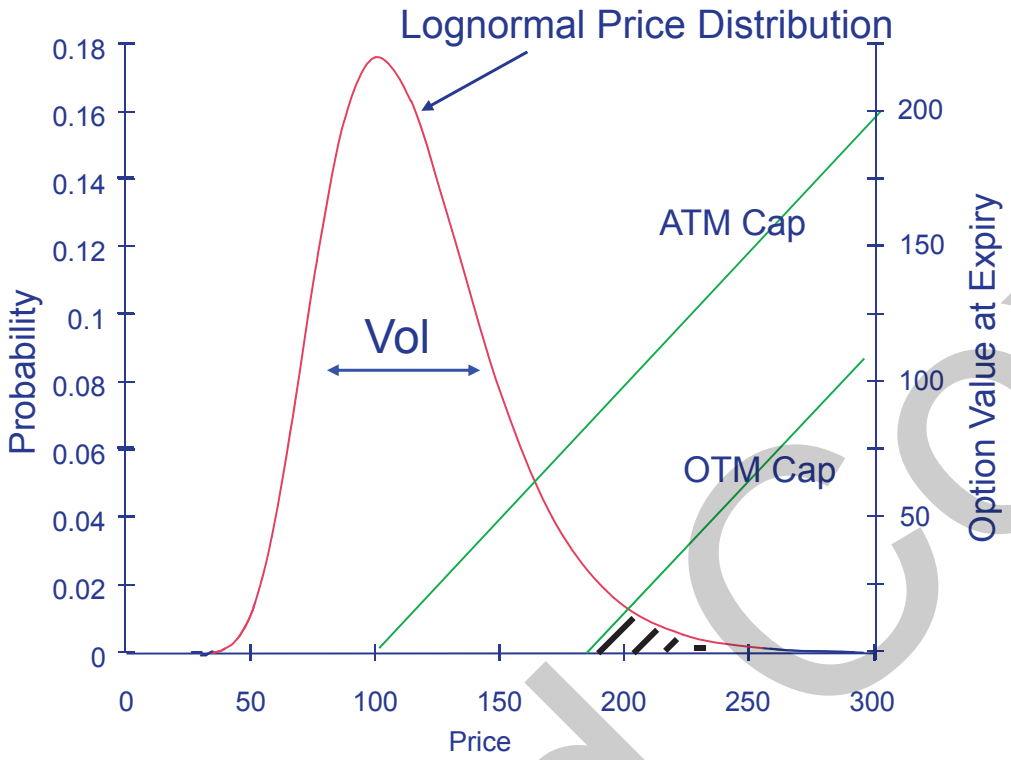
The Impact of Delta Hedging



Extract from S. Xiaoyan Ni et al. / Journal of Financial Economics 78 (2005) 49–87

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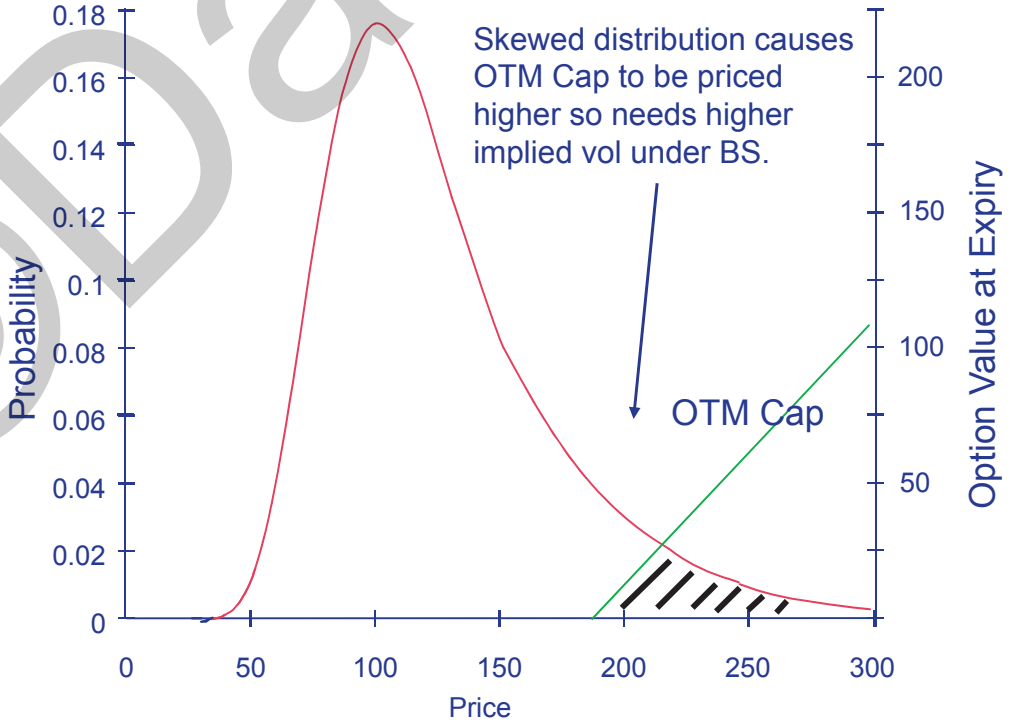
Call Option Value



Option Value = discounted product of payoff and probability over the whole range of prices

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Call Option Value

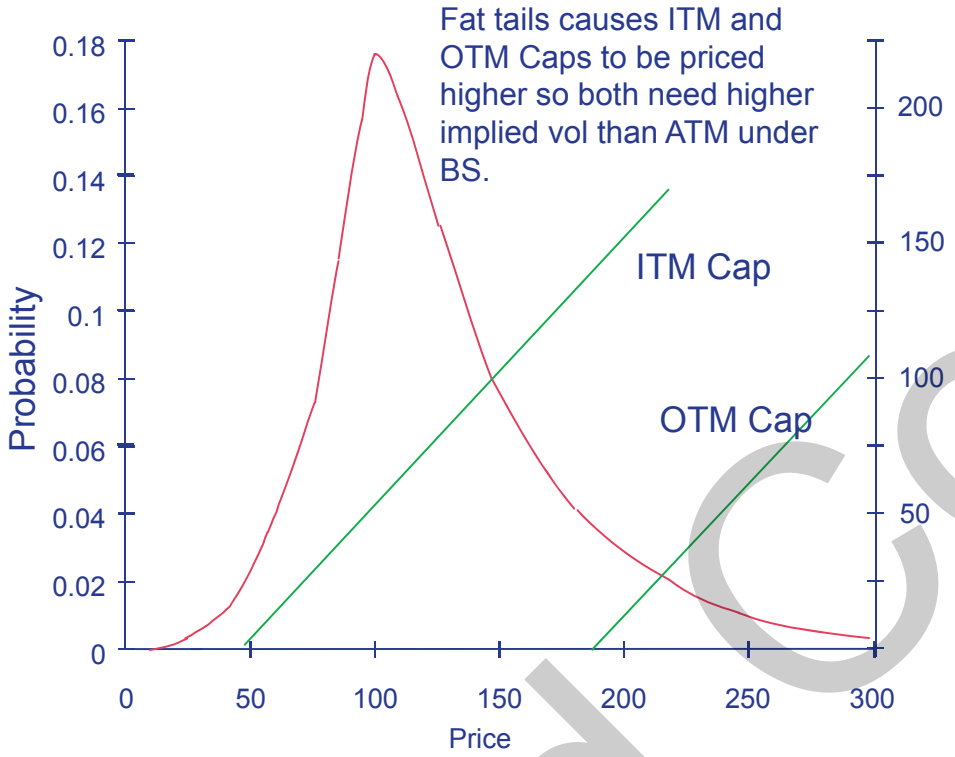


Increase vol. to reflect higher OTM option price or change shape of distribution to reflect higher likelihood of large move

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Call Option Value



Make tails fatter at both ends to reflect smile

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Day Four

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Day Four

8.30 – 10.30	**Interest Rate Options: Swaptions** • Swaption mechanics and how values are determined. • Cash settlement v physical delivery • Hedging swaptions (swaption "Greeks") • Swaptions and bond options • Using swaptions in ALM • Bermudan structures and how they are used • Constructing callable swaps • Valuing Bermudan v European style swaptions • Implementing market views • Single swaption breakeven analysis • Payer/receiver spreads under different conditions • Spreads and skew
10.30 – 10.45	Break
10.45 – 13.00	**Exercise: Swaption strategies**
13.00 – 14.00	Lunch
14.00 – 15.30	**Interest Rate Options: Caps, Floors and Collars (Ibor and CMS)** • Definitions and structure • Ibor caps and floors • Understanding market quotes • Cap-floor-swap parity • Caps and floors in operation • Hedging caps and floors • Structuring and using combinations of caps and floors • Zero NPV collars • Participating caps • Embedded caps
15.30 – 15.45	Break
15.45 – 16.45	**Exercise: Using caps, collars and related structures**
16.45 – 17.30	• Constant Maturity Swaps (CMS) • CMS caps and floors in relation to swaptions • CMS linked notes • Understanding interest rate volatility quotes • Basis point v yield (lognormal) vol. and its impact on skew • Pricing swaptions in today's market • SABR, stochastic volatility and why it matters • What SABR parameters do and how they link to volatility skew

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Swaption

An Option on an Interest Rate Swap

The right but not the obligation to enter into a specific interest rate swap at a given fixed rate.

Payer and Receiver Swaptions are options to pay or receive fixed under the Swap.

The swap starts when the option expires.

Swaptions can be settled for cash or physical delivery
- Cash is more standard in EUR, GBP
- Physical more common in USD

European style Swaptions are the most common.
- Bermudan style Swaptions also common and combine with swaps to give callable/puttable structures.
- Purely American structures are exotic

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Swaptions

Rights and obligations

	Buyer	Seller
Payer Swaption	Right to pay fixed and receive Libor	Obligation to receive fixed and pay Libor
Receiver Swaption	Right to receive fixed and pay Libor	Obligation to pay fixed and receive Libor



You can receive fixed by buying a receiver or selling a payer and vice-versa

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Swaptions

The Market

Mainly OTC at the moment

CME clears European style with a maximum option maturity of 2 years and maximum swap maturity of 30 years

Exists in all major currencies

Generally for standard structures:

European style, non-amortising

Minimum notional generally larger than for swaps

ISDA documentation for OTC

Bloomberg:

- Forward and spot swaps on FWCM
- Vols and premiums on VOLM, VCUB

Reuters:

- VCAP1 to VCAP6 for ATM vols.

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Swaption Valuation

Value Underlying Forward Starting Swap

- Yield Curve
- Conventions
- Reset dates
- Tenor
- Breakeven fixed rate
- DV01

Value Option

- Expiry
- Strike
- Volatility: lognormal (yield) or normal (bp)
- Skew / Smile

Option Model

- B-S, Local Vol. or SABR for European
- Term structure (HJM/BGM or 2 factor HW) for Bermudan

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1 into 5 year Payer Swaption

Buyer pays premium for right to pay fixed for five years in one year at Strike = 5%



Payoff at expiry of option: = Max(0,NPV of swap with fixed rate = 5%)

$$NPV = 100 - \left[5 \times \sum_{t=2}^{6} D_t + 100 \times D_6 \right]$$

FRN — Bond

Physical delivery: discount factors taken from yield curve

Cash settlement: swap rate fixing used for discount factors

What sort of bond option is this equivalent to?

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Swaption Payoff



The swaption cash value is the basis point value multiplied by the DV01 of the underlying swap

- DV01 is calculated using the yield curve for physical delivery
- DV01 is calculated using the underlying swap breakeven fixed rate (YTM) for cash settlement
- DV01 is not a constant. Swaps have convexity.

Payoff at expiry of option: = Max(0,S-X)*DV01

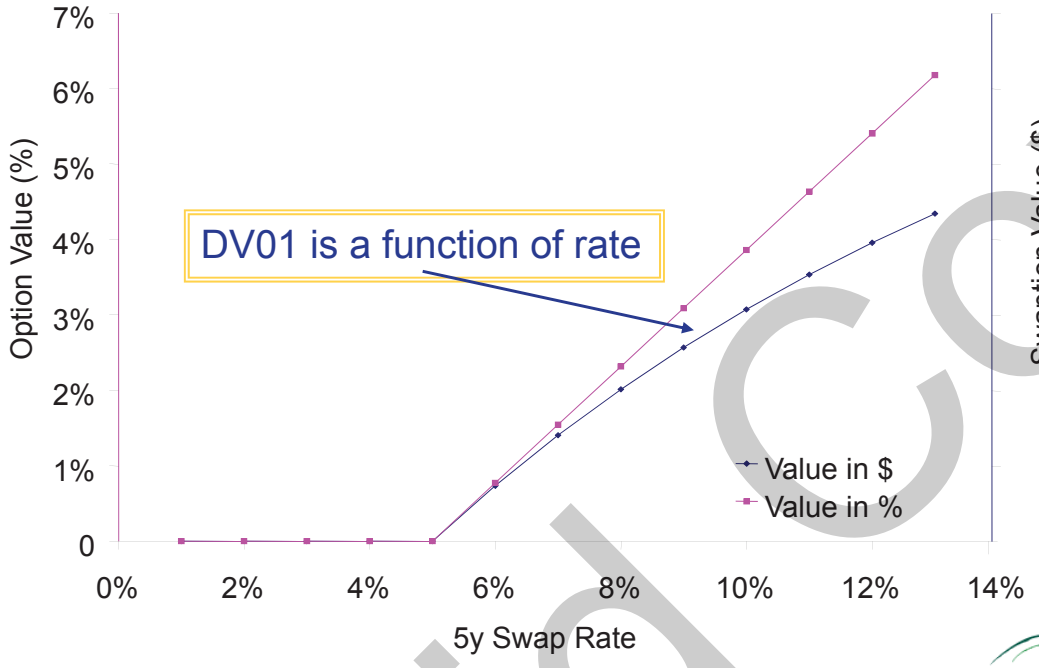
- S = swap fixed rate
- X = strike rate

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Swaption Payoff at Expiry

1 in to 5 year Payer Swaption



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Swaption Delta

Delta hedge is with the underlying forward starting swap

Swaption delta (δ) is the sensitivity of the swaption value in basis points to a small change in the underlying swap rate (in basis points)

Swaption DV01 is the cash value of a one basis point change in the swap rate = swap DV01 * δ

Notional of swap required to delta hedge is therefore: δ * notional of swap underlying swaption

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Swaption Gamma

Swaption DV01 = swap DV01 * option δ

So swaption gamma must take into account:

- The dependence of the swap DV01 on swap rate "S" (swap convexity)
- The dependence of the option delta on swap rate (option gamma)

$$\gamma_{\text{Swaption}} = \frac{\partial \text{DV01}}{\partial S} \times \delta + \frac{\partial \delta}{\partial S} \times \text{DV01}$$

Swap convexity

Option gamma

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Using Swaptions

Corporate Funding

Corporation

Sell 2, 3y Receiver Swaption

Issue 5y Bond
Callable after 2 years

Pay Coupon

Premium

Swaption Market

Buy 2, 3y Receiver Swaption

Cost of funds reduced if swaption premium greater than NPV of additional coupon over non-callable debt

	Rates Sell Off	Rates Rally
Swaption buyer	No action, swaption expires	Exercise swaption, receive fixed, pay floating (ibor or o/n)
Bond issuer	No action	Call bond, pay fixed refinance @ floating*

*For this to work refinancing must be available at the floating rate on the swap or lower

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Swaptions in ALM

Swaption can be used to offset changes in relative value of assets and liabilities due to rate moves

Shorter dated assets have lower duration than longer dated liabilities

This can be hedged with a forward starting swap

The hedge can also be done with a swaption

Assets:	Liabilities:*
Nominal 50m	Nominal 100m
Maturity: 7y	Maturity: 30y
Duration 5.9y	Duration 17y

*Remaining 50m in equities

Swap: Receive Fixed on 125m 7v23y Forward Start Swap

Find principal by matching DV01

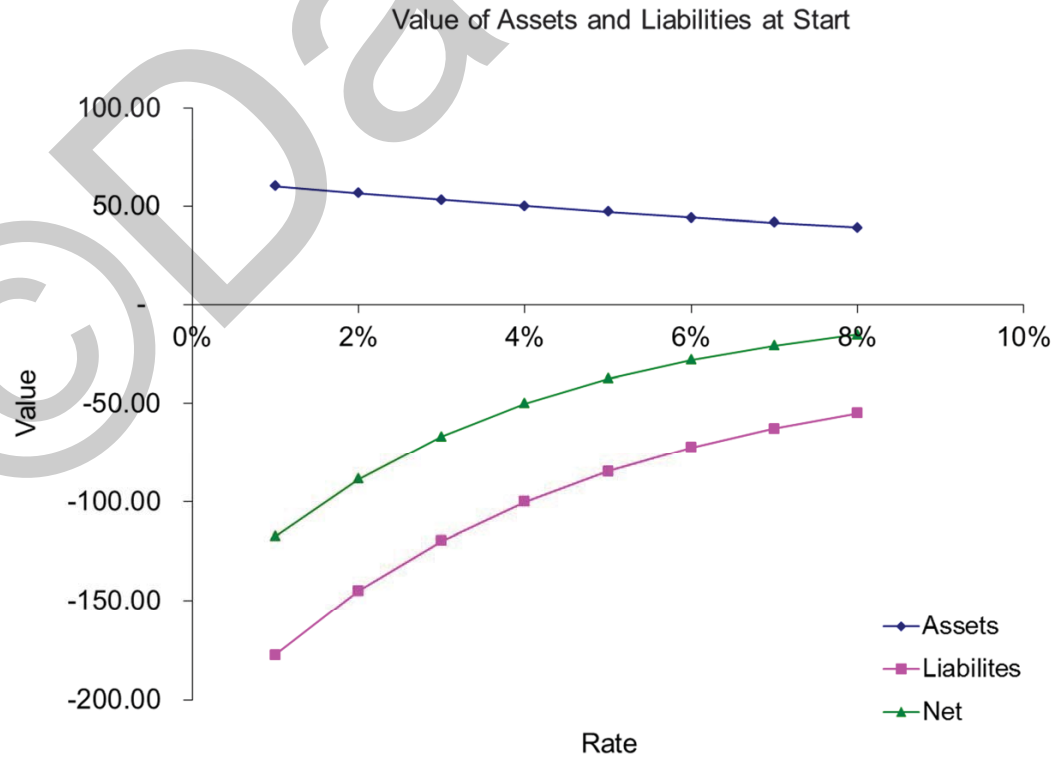
Swaption: Buy 296m 7v23y Receiver Swaption

DV01 matched. Premium to pay, but Gamma provides extra gain

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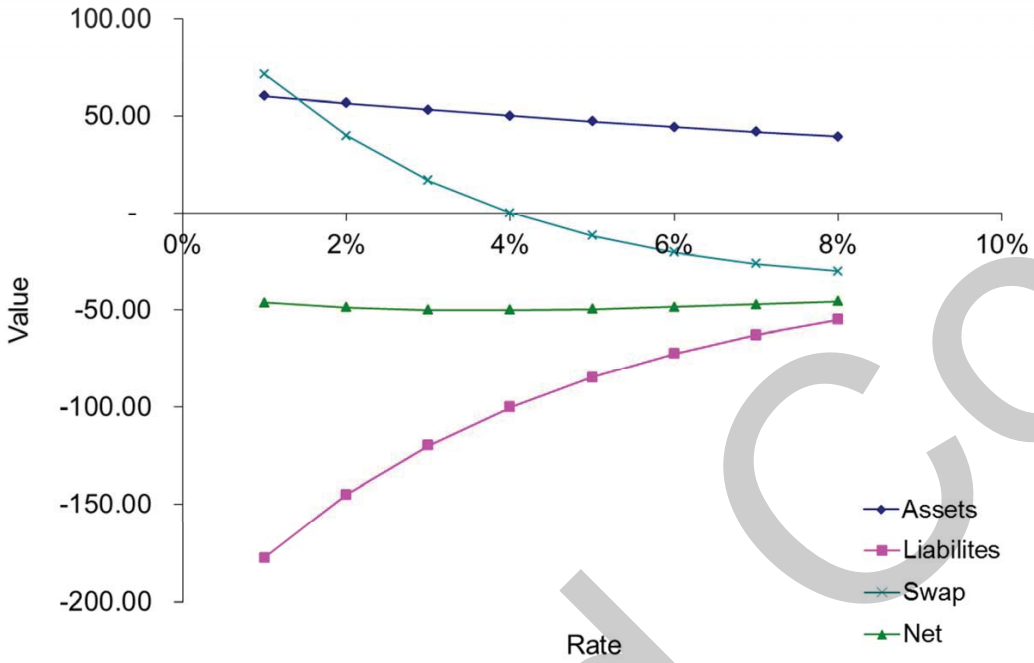
Swaptions in ALM



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Swaptions in ALM

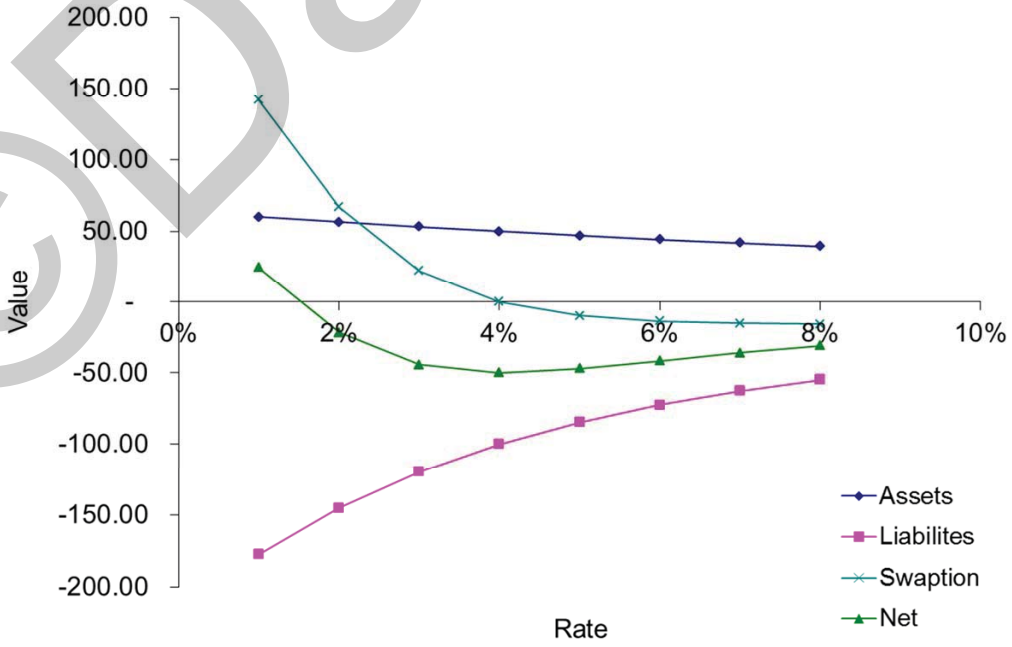
Value of Assets, Liabilities and Swap at Start



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Swaptions in ALM

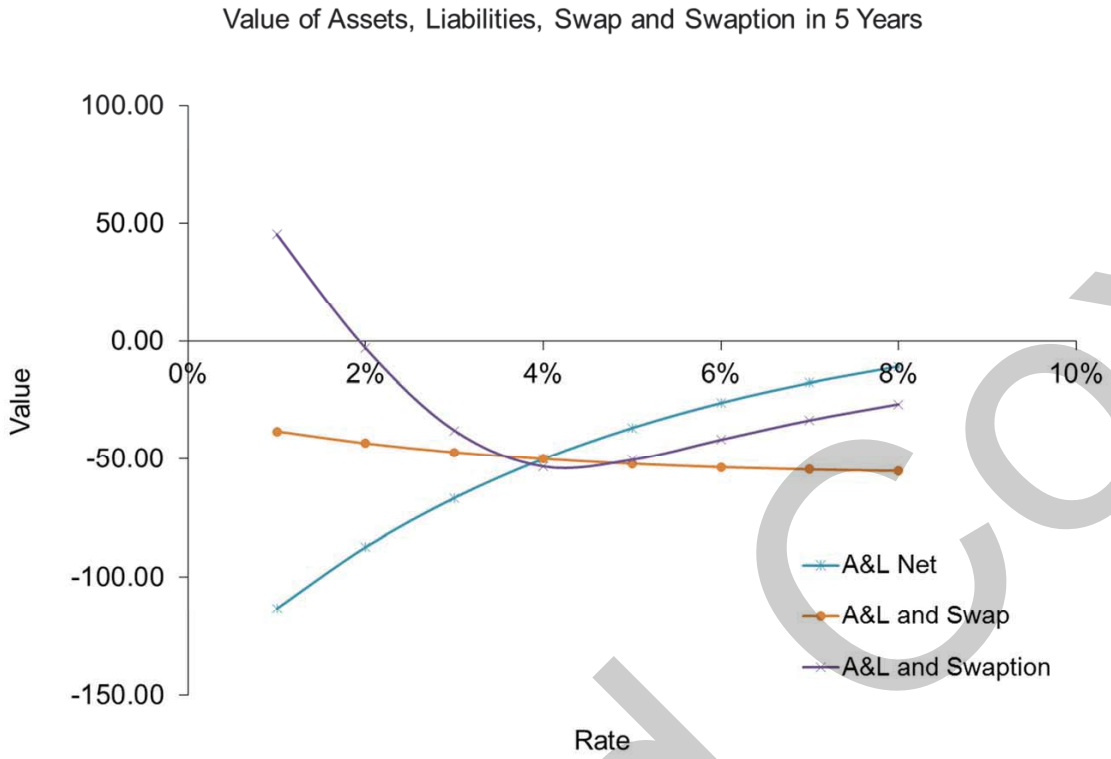
Value of Assets, Liabilities and Swaption at Start



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Swaptions in ALM

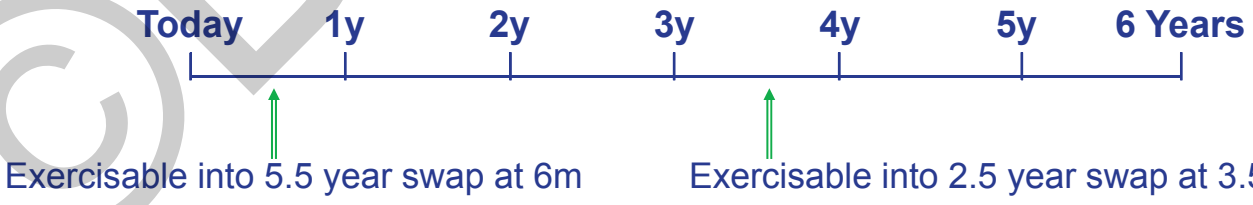


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Bermudan Swaptions

Exercisable at intervals equal to swap reset dates

E.g. Option exercisable at one of the six monthly intervals from 6m to 5.5y into a semi-semi swap starting at the exercise date and ending in six years.



Can also have swaptions exercisable into a swap of the same maturity on all dates (less common)

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Bermudan Swaptions

Valuation very different from European

- Don't be tempted to use a B-S model

Single option but exercisable every six or three months

- Exercisable into a swap of decreasing maturity

Whether option is worth more alive or exercised depends on value of swap that could be entered into now and the expected value of those in the future

- Normal method is MC to determine optimal early exercise path and valuation using this path
- Techniques are model dependent

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Bermudan Swaptions

Euro Payer Swaption Premia (bp)

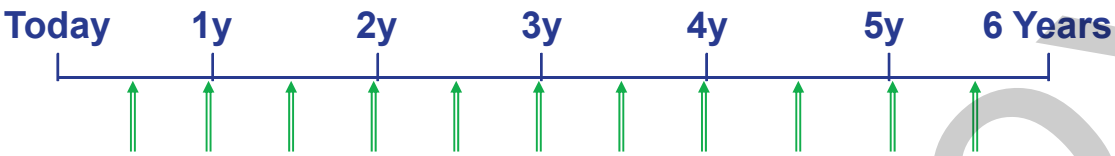
Strike:	2%	3%	4%	5%	Fwd Swap Rate
European 1y v 4y	390	124	26	6	3.0087
European 2y v 3y	383	177	64	26	3.3226
European 3y v 2y	299	158	69	32	3.5490
European 4y v1y	172	100	48	26	3.5490
5y Bermudan (annual exercise)	428	206	88	43	

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Multi-Callable Swap

Fixed rate payer has option to cancel swap on each reset date.

IRS plus Bermudan Receiver Swaption with strike equal to swap fixed rate.



Fixed rate payer has the right to cancel swap
Option is exercisable into swap receiving fixed with the remaining life of the IRS

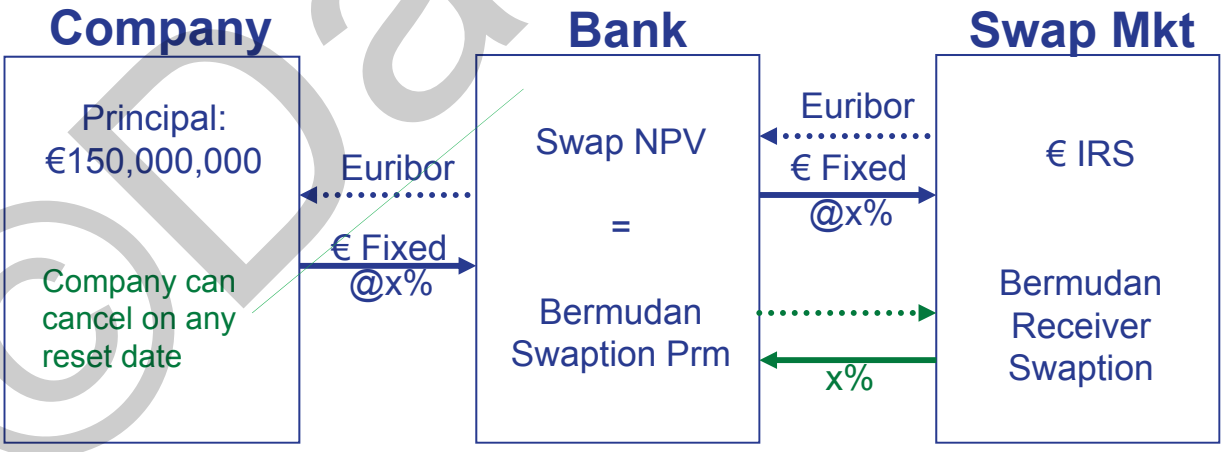
Parties have entered an IRS. Fixed rate payer has also bought a Bermudan style receiver swaption from the IRS counterparty.

Swaption expiry date is the last reset of the IRS.

Swaption is exercisable into a swap cancelling the IRS.

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Multi-Callable Swap



NPV of off market swap equals premium on Bermudan Receiver Swaption

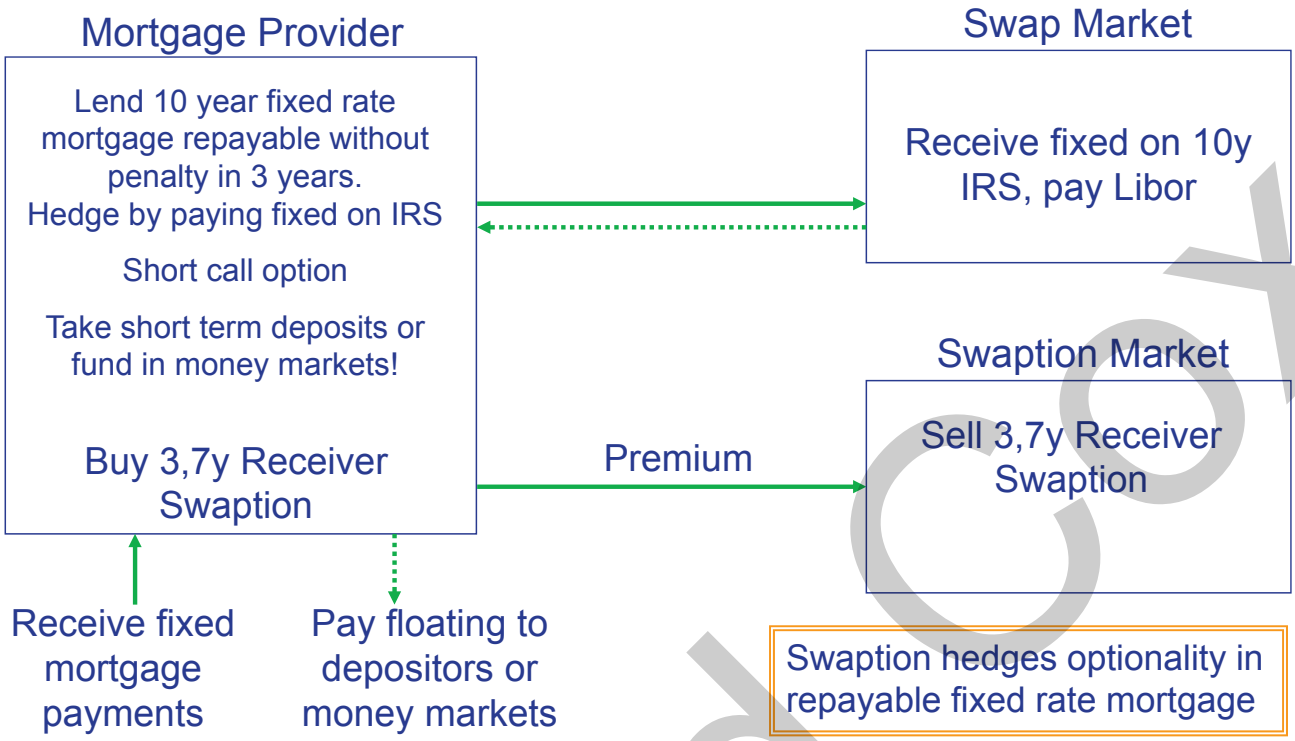
Swaption strike is equal to swap fixed rate

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Using Swaptions

Housing Finance



© David Cox

Swaption Breakeven

You buy a "1 in to 1 year ATM Payer Swaption"

Notional = $100

ATM Strike = 2% (you have the right to pay fixed at 2%)

ATM Prm = 35c

Vol = 90bp (annualised SD of changes in the swap rate)

1v1y forward swap DV01 = 0.98 (cents / bp)

Spot 1y swap rate = 1.30% (curve is quite steep)

What is your underlying view?

How far do rates have to move for the position to be in the money?

What is vol telling you?

What is the roll-down of the position?

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Swaption Breakeven

You buy a "1 in to 1 year ATM Payer Swaption"

Strike = 2% ATM
Prm = 35c
Vol = 90bp
Swap DV01 = 0.98
Spot 1y swap rate = 1.30%

How far do rates have to move for the position to be in the money?
35c premium is equivalent to a rate move of 35/0.98 in bp
= 35.7bp
Breakeven = 2.357% (exactly?)
Vol is annualised SD of change in Forward Swap rate
Trade is negative roll-down due to the shape of the YC
If nothing changes the trade loses money

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Swaption Breakeven

1 in to 1 year Payer Swaption

ATM Strike = 2%, Prm = 35c, Vol = 90bp, 1v1y Swap DV01 = 0.98



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Strategy: ?

View: US inflation will increase and long term rates will sell-off, but limited by the demand for treasuries from real money investors.

What could you do?

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Payer Spread

View: US inflation will increase and long term rates will sell off, but limited by the demand for treasuries from real money investors.

Strategy: buy 1v30 payer spread:

- Buy 1v30 ATM payer @ 2.10 strike
- Sell 1v30 OTM payer @ 3.10 strike
- Net premium = 495c
- 1v30 swap DV01 = 17.2 c/bp on $100 notional
- 495c / 17.2 = premium in bp = 28.8bp
- Breakeven if rates sell off to 2.388%

Max gain = (100- 28.8)bp ~ 2.5 × premium

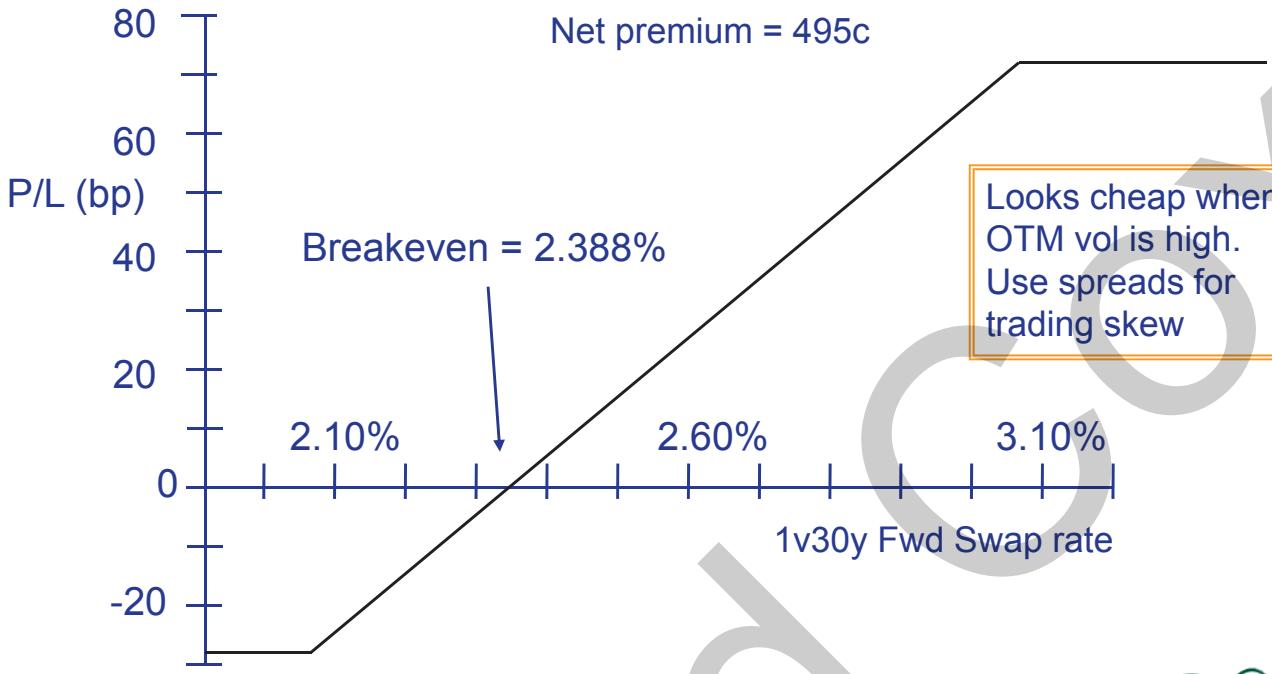
Vol skew position?

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Payer Spread

Buy 1v30 ATM payer @ 2.10 strike
Sell 1v30 OTM payer @ 3.10 strike
Net premium = 495c



© David Cox

Spreads

Payer Spread

- Bearish View
- Good when:
  - Potential sell-off over forwards of limited size
  - Risk of rally is high
- Profit in sell-off, maximum loss is premium (less than a single payer)

Receiver Spread

- Bullish view and/or roll down trade
- Good when:
  - Steep curve
  - Limited rally likely
  - Risk of sell off over forwards is high
  - Low implied vol
- Profit in rally and from roll-down, maximum loss is premium

Both used for trading skew

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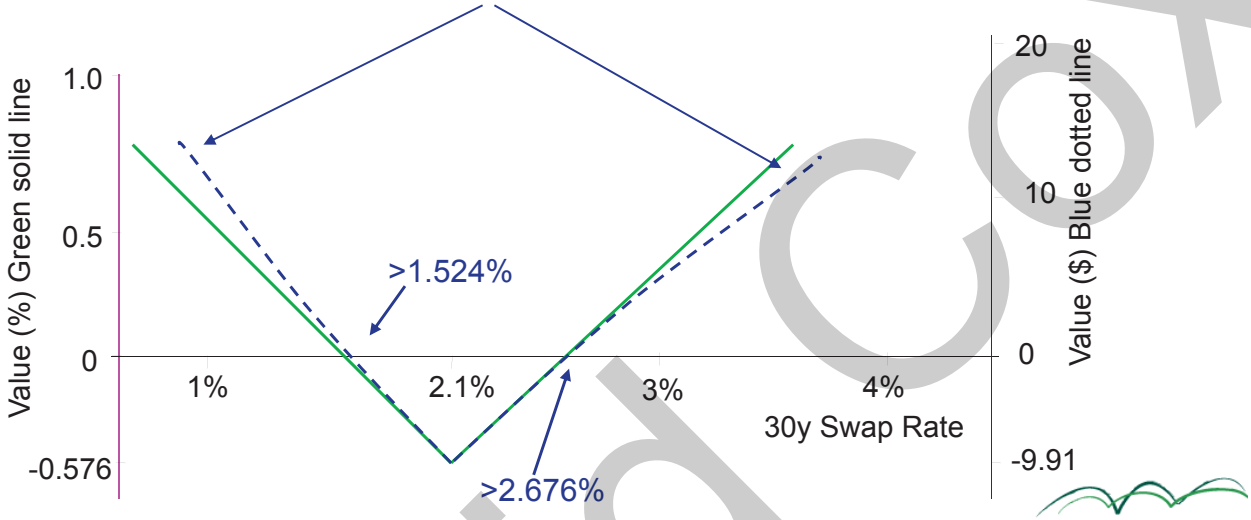
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Swaption Straddle

Buy 1 in to 30y Payer and Receiver Swaption

ATM Strike = 2.10%, Prm = 495c (28.8bp) , Vol = 90bp

Underlying swap DV01 reduces as payer becomes ITM and increases as receiver becomes ITM so this trade has a directional component. (Slightly long)

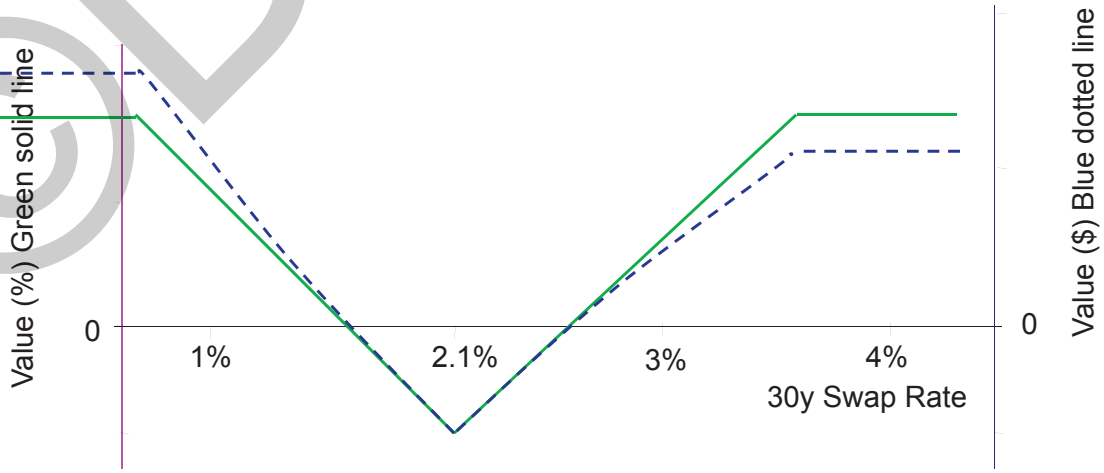


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Swaption Butterfly

Buy 1 in to 30y Payer and Receiver Spreads

Not symmetrical due to convexity of swap



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Caps and Floors

The buyer pays a premium

Receives the right to a payment based on:

- The difference between a variable rate (ibor or CMS) and a strike rate
- A notional principal
- A series of dates

Portfolio of short term options on a forward rate

Volatility for each forward period

Many structures possible

Normally first period is not included because there is no uncertainty about payout

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Swaptions and Caps

Swaption

- Single option
- Underlying instrument is a forward start interest rate swap
- Model of movement in swap fixed rate is used for valuation
- Swap can be viewed as an exchange of bond and FRN
- Swaptions have similar characteristics to bond options

Cap

- Multiple period instrument: a portfolio of options
- Underlying is forward Libor rates
- Model of Libor moves used for valuation
- Simple assumptions about Libor moves (lognormal) not consistent with same assumptions about swap rate moves

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18m Cap v 3m Ibor

Ibor (L_1) set for second period (3 months from today)

Cap payoff for second period: $P \times \frac{91}{360} \times \text{Max}(L_1 - R_s, 0)$



Payments at intervals of 3 months

No settlement at end of period 1

First period rate normally not used

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Caps and Floors

A Series of Options on Future Ibor



Strike Rate: 100bp

3m Ibor (bp)		90	110	150	120	80	
Cap Payoff (bp)			0	10	50	20	0
Floor Payoff (bp)			10	0	0	0	20
Cap Premium (bp)	5.09		8.09	10.2	11.83	13.27	

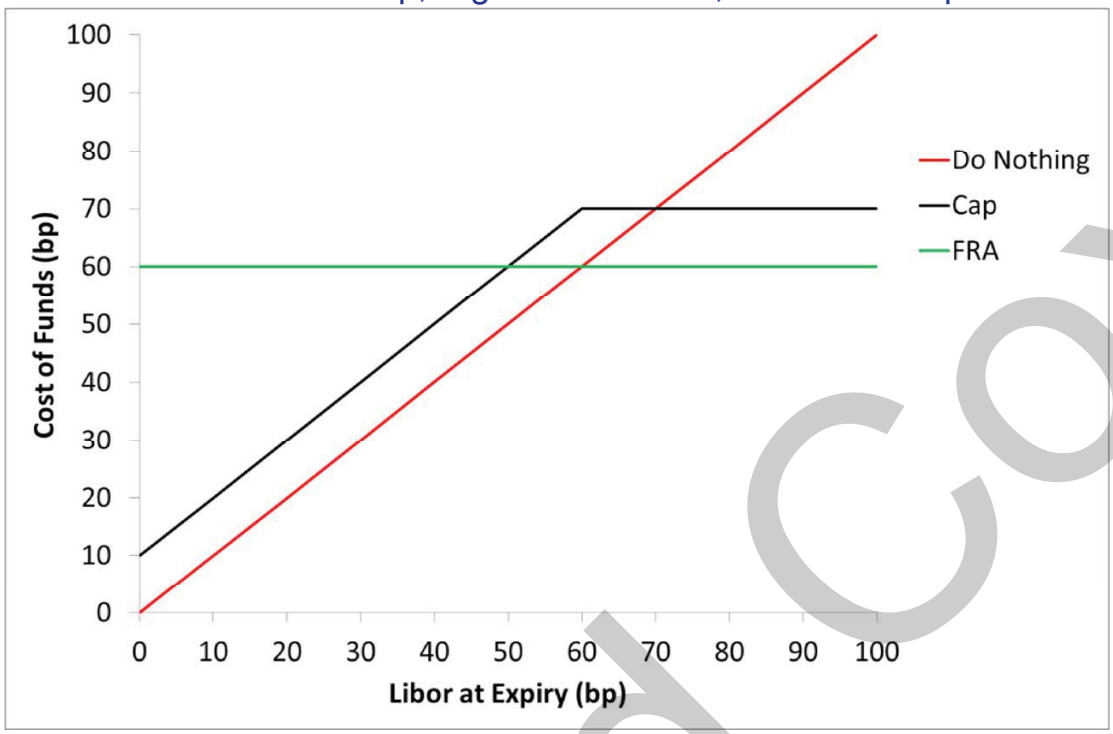
Total Cap Value = 48.48 bp

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Single Period Cap

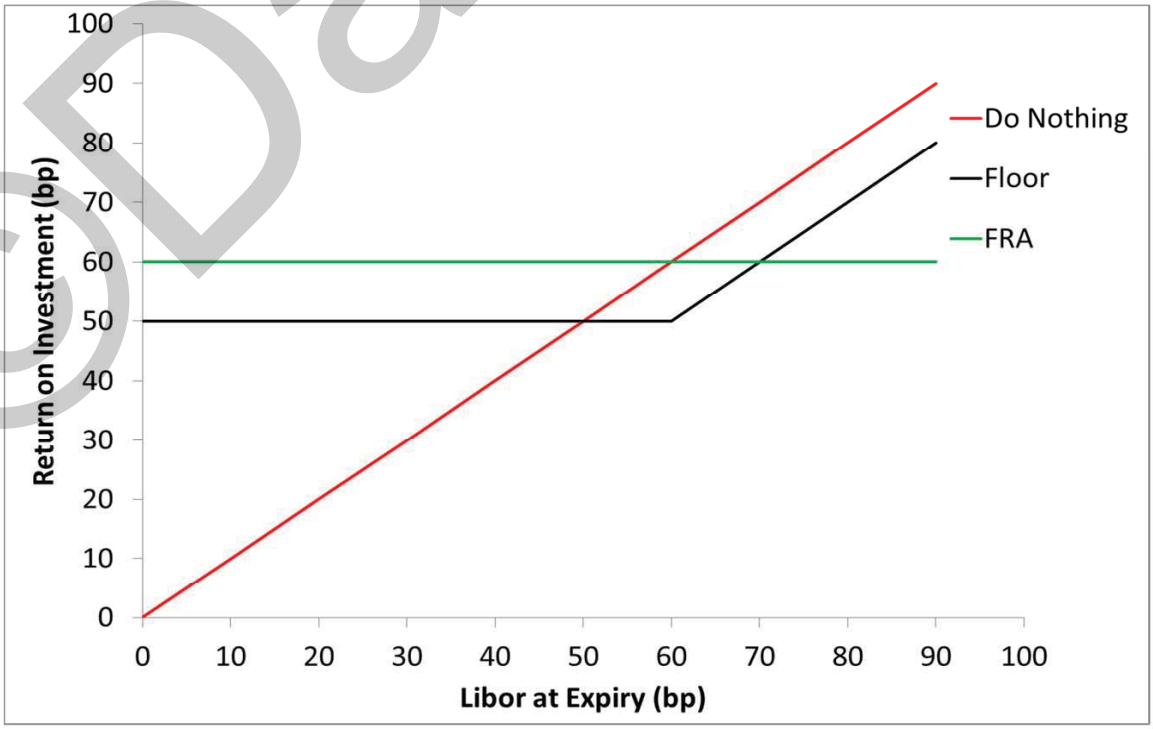
Strike Rate 60bp, Against 12m Ibor, Premium 10bp



© David Cox

Single Period Floor

Strike Rate 60bp, Against 12m Ibor, Premium 10bp

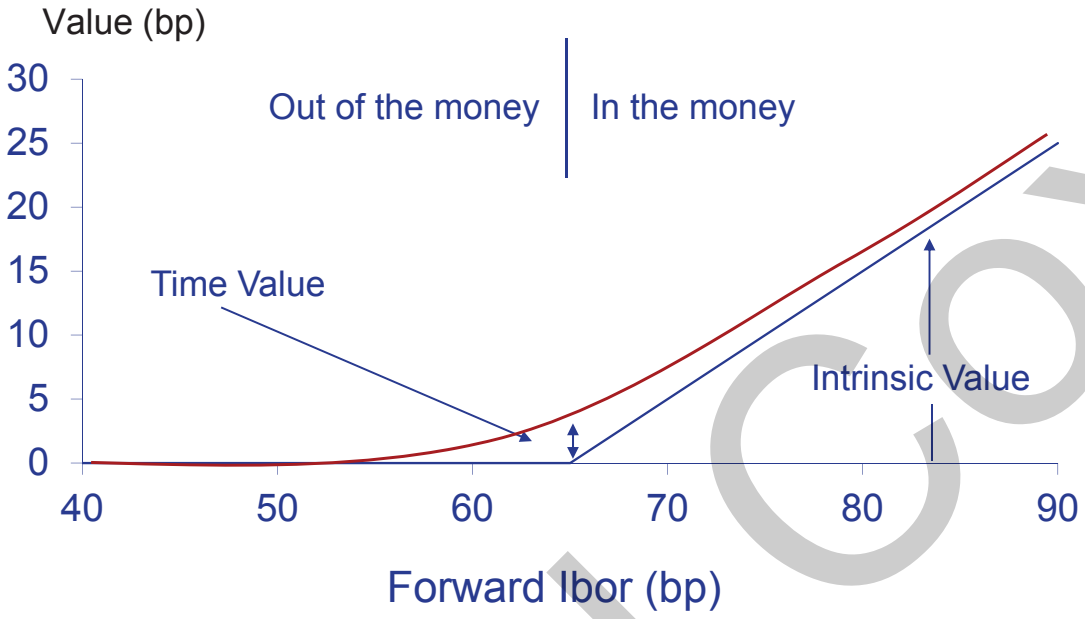


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Caplet Value

Single Period Cap, Strike = 65bp



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Cap Prices

```
09:15 16APR    GARBAN-INTERCAPITAL                    UK04138              VCAP
               EUR Caps - Premium Mids
               Please call +44 (0)20 7588 7588 for further details
     STK   ATM 1.75  2.0 2.25  2.5  3.0  3.5  4.0  4.5  5.0  6.0  7.0  8.0 10.0
 1Y|2.48|   12|    |    |    |  11|   2|    |    |    |    |    |    |    |
 2Y|2.76|   55|    |    |    |    |  39|  20|  10|   5|   3|   1|    |    |
 3Y|3.16|  105|    |    |    |    |    |  74|  43|  25|  15|   6|   3|   2|   1
 4Y|3.43|  172|    |    |    |    |    | 162| 104|  65|  40|  17|   8|   5|   3
 5Y|3.66|  242|    |    |    |    |    |    | 185| 121|  77|  34|  18|  10|   6
 6Y|3.85|  313|    |    |    |    |    |    | 281| 191| 126|  57|  30|  18|  10
 7Y|4.01|  384|    |    |    |    |    |    |    | 269| 182|  84|  45|  27|  14
 8Y|4.15|  454|    |    |    |    |    |    |    | 357| 246| 116|  64|  38|  20
 9Y|4.27|  522|    |    |    |    |    |    |    | 449| 315| 152|  85|  50|  25
10Y|4.37|  586|    |    |    |    |    |    |    | 540| 383| 188| 106|  63|  31
12Y|4.53|  712|    |    |    |    |    |    |    |    | 523| 265| 155|  95|  46
15Y|4.70|  868|    |    |    |    |    |    |    |    | 717| 370| 219| 138|  64
20Y|4.86| 1088|    |    |    |    |    |    |    |    | 996| 531| 326| 213| 104
```

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Floor Prices

09:15 16APR GARBAN-INTERCAPITAL UK04138 VCAP

EUR Floors - Premium Mids

Please call +44 (0)20 7588 7558 for further details

STK	ATM	1.75	2.0	2.25	2.5	3.0	3.5	4.0	4.5	5.0	6.0	7.0	8.0	10.0
1Y|2.48|	12|	1|	2|	6|	|	|	|	|	|	|	|	|	|	|
2Y|2.76|	55|	5|	10|	18|	31|	|	|	|	|	|	|	|	|	|
3Y|3.16|	105|	7|	13|	22|	36|	84|	|	|	|	|	|	|	|
4Y|3.43|	172|	11|	18|	29|	44|	100|	|	|	|	|	|	|	|
5Y|3.66|	242|	14|	23|	35|	53|	114|	206|	|	|	|	|	|	|
6Y|3.85|	313|	18|	28|	42|	61|	127|	226|	|	|	|	|	|	|
7Y|4.01|	384|	22|	33|	48|	70|	140|	244|	380|	|	|	|	|	|
8Y|4.15|	454|	26|	38|	55|	78|	152|	260|	404|	|	|	|	|	|
9Y|4.27|	522|	30|	43|	61|	86|	164|	276|	425|	|	|	|	|	|
10Y|4.37|	586|	34|	49|	68|	94|	175|	291|	446|	|	|	|	|	|
12Y|4.53|	712|	45|	61|	84|	112|	200|	323|	488|	698|	|	|	|	|
15Y|4.70|	868|	60|	80|	105|	138|	233|	364|	539|	764|	|	|	|	|
20Y|4.86|	1088|	92|	118|	149|	187|	296|	437|	627|	873|	|	|	|	|

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Cap Premium



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Cap/Floor Volatility

Contributor Pricing | Jump To Monitor Hotspot | Properties

ICAP | USD CAPS/FLOORS - IMPLIED VOLS | Page 1/4 | 12:47:19

ICAP | Zoom 100% | Show Navigation

ICAP Interest Rate Options Directory (VOLS) -> Current Monitor (GDCO 2840 7)

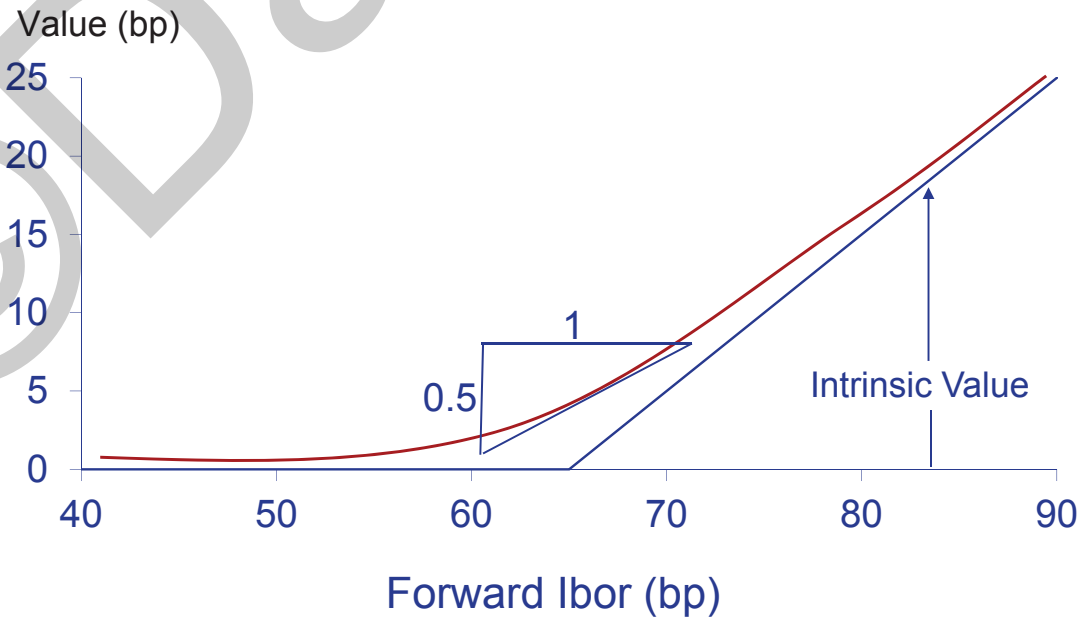
	Term	Strike	Time		Term	ATM	Time
1)	1 Year	0.34	03:41	14)	1 Year	56.66	07:45
2)	2 Year	0.43	07:12	15)	2 Year	72.49	07:45
3)	3 Year	0.57	04:28	16)	3 Year	74.07	07:46
4)	4 Year	0.78	04:52	17)	4 Year	71.99	07:46
5)	5 Year	1.03	04:51	18)	5 Year	63.15	07:46
6)	6 Year	1.29	07:42	19)	6 Year	55.42	07:46
7)	7 Year	1.52	07:31	20)	7 Year	49.97	07:45
8)	8 Year	1.73	07:30	21)	8 Year	46.05	07:46
9)	9 Year	1.91	07:30	22)	9 Year	42.93	07:46
10)	10 Year	2.07	05:37	23)	10 Year	40.37	07:46
11)	12 Year	2.33	07:46	24)	12 Year	36.62	07:46
12)	15 Year	2.59	07:46	25)	15 Year	33.06	07:46
13)	20 Year	2.82	07:42	26)	20 Year	29.97	07:46

ICAP

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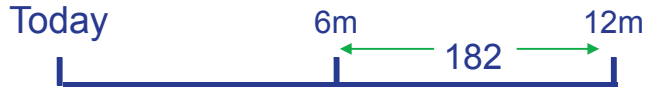
Caplet Delta

Single Period Cap, Strike = 65bp



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Hedging Cap with FRA



Each caplet has its own delta and will need its own hedge

Sensitivity of caplet to 1bp change in forward rate:

$$DV01_{Cap} = \text{Caplet Principal} \times \delta \times [0.01\% \times 182/360]$$

Sensitivity of FRA to 1bp change in forward rate = DV01

$$DV01_{FRA} = \text{FRA Principal} \times [0.01\% \times 182/360] \times D_{12}$$

Hedge to match DV01s:

$$\text{FRA Principal} = \frac{\text{Caplet Principal} \times \delta}{D_{12}}$$

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Implied Volatility from Caps

Vol. for each caplet period can be different

Term structure can be estimated from cap prices

Method similar to swap bootstrapping

- Take 1y, 18m and 2y caps v 6m Ibor
- Implied vol. for 6-12 period is derived from price of one year cap
- 18m cap price is derived from 6-12 and 12-18 vols
- Use known 6-12 vol. to imply 12-18 vol. from price of 18m cap
- Use known 6-12 and 12-18 vols to derive 18-24 vol. from 2y cap

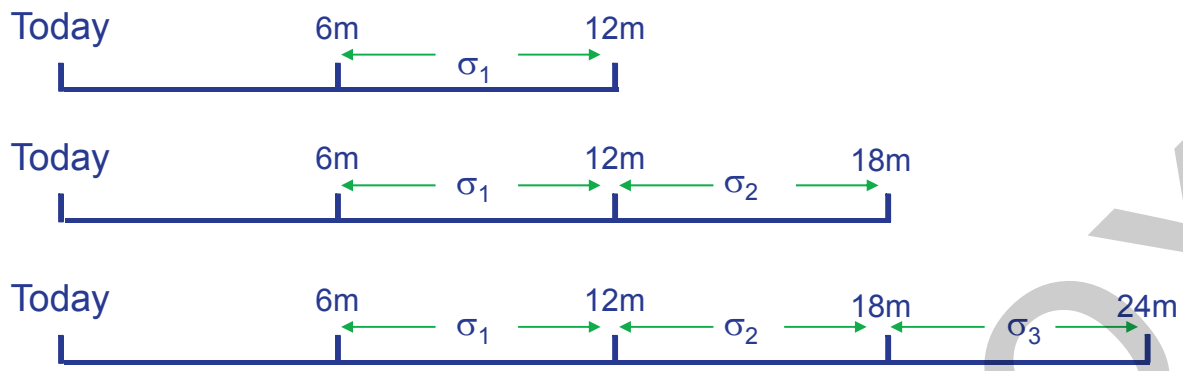
As long as there are enough cap prices available a term structure of B-S vol. can be derived.

This provides a partial picture, but takes no account of smile or skew

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Implied Volatility from Caps



Derive vols period by period starting with the shortest

Need to adjust for skew and smile effects as not all caplets will be at the money

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Cap-Floor-Swap Parity

Cap strike = Floor strike = Swap rate

Equal maturities

Same floating rate

Pay floating on swap

Cap premium - Floor premium = Swap NPV

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Cap-Floor-Swap Parity

Example

Cap strike = Floor strike = Swap rate = 50bp

Floating rate = 6 month Ibor

Maturity = 3 years

Cap @ 50bp	premium:	308,621.73
Floor @ 50bp	premium:	2,181,998.71
Swap fixed rate = 50bp	NPV:	1,873,376.98

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Zero Cost Collar

Cap premium = Floor premium

Cap strike > Floor strike

Borrower buys cap and sells floor

Upper and lower limits on cost of funds

Give away potential gain for "free protection"

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Zero Cost Collar

Cap premium = Floor premium = €308,621

- Cap strike = 60bp
- Floor strike = 41.32bp

Borrower buys cap and sells floor

- Upper limit on cost of funds: 60bp
- Lower limit on cost of funds: 41.32bp

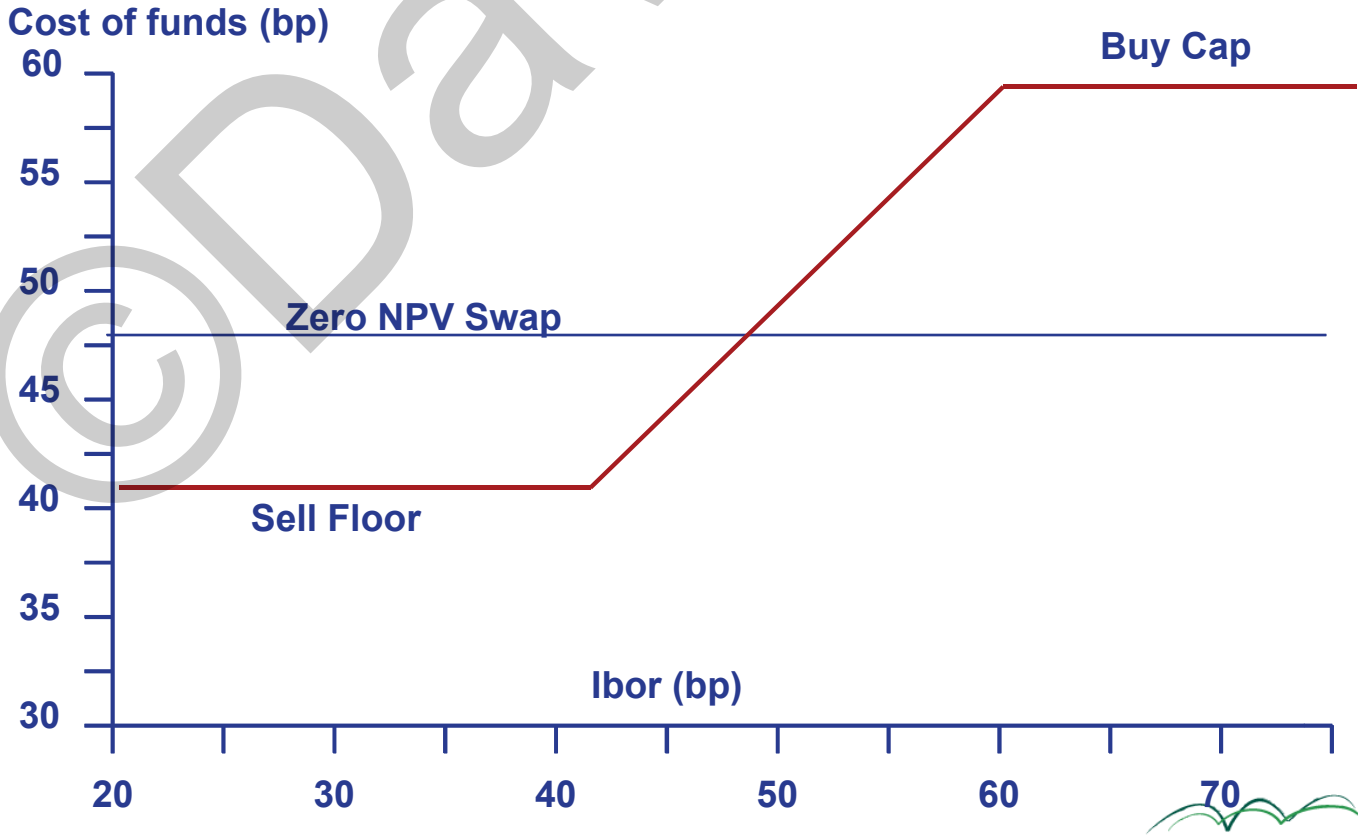
Give away potential gain for "free protection"

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Zero Cost Collar

Floor Strike: 41.32bp Cap: 60bp



© David Cox

Participating Cap

"PIRA"

Cap strike = Floor strike

Cap principal > Floor principal

Total Cap premium = total Floor premium

Borrower buys cap and sells floor

"Free protection" for reduction in potential gain

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Participating Cap

"PIRA"

Cap strike = Floor strike = 60bp

- Cap principal = € 50,000,000
- Floor principal = € 7,072,000 (~14% of 50m)

Cap premium = Floor premium = € 308,621

Borrower buys Cap and sells Floor

- Protection above 60bp
- For each 10bp Ibor is below 60bp floor pays out on 14% of €50m so funding cost decreases by 8.6bp.

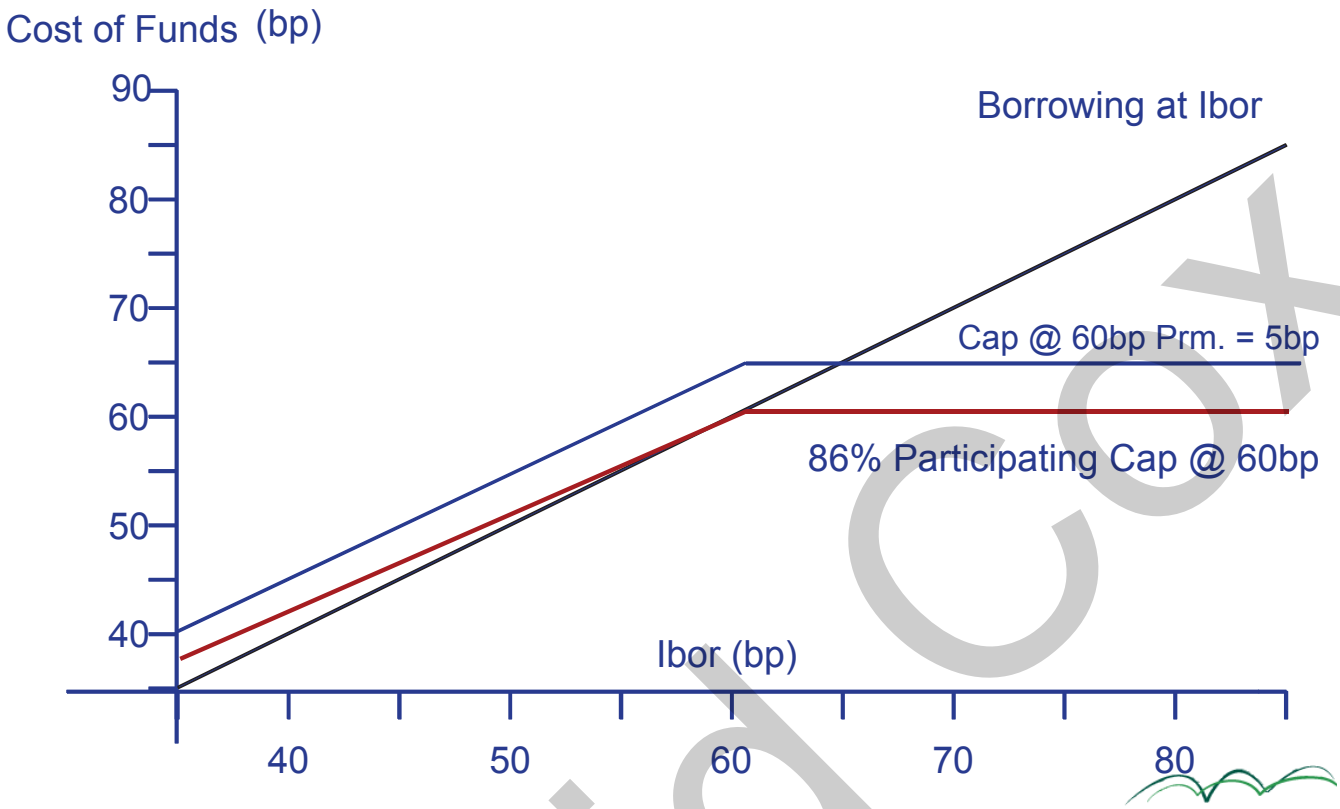
"Free protection" for reduction in potential gain

$$\text{Participation} = 1 - \frac{\text{Floor principal}}{\text{Cap principal}} = 86\%$$

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Participating Cap

Strike: 60bp 86% Participation



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Constant Maturity Swaps

Fixed v 10Y swap rate paid semiannually

- Payments every six months on the fixed side like a standard IRS
- Payments on the floating side every six months calculated with reference to the 10Y swap rate not Ibor

Six month Euribor v 20 year Euro swap rate

- Euribor side like a standard semi-semi IRS
- Payments on long term side every six months calculated with reference to the 20 year swap rate
- Swap rate adjusted by a spread or scaling factor for zero NPV
- Allows asset-liability matching

2y swap rate v 10y swap rate

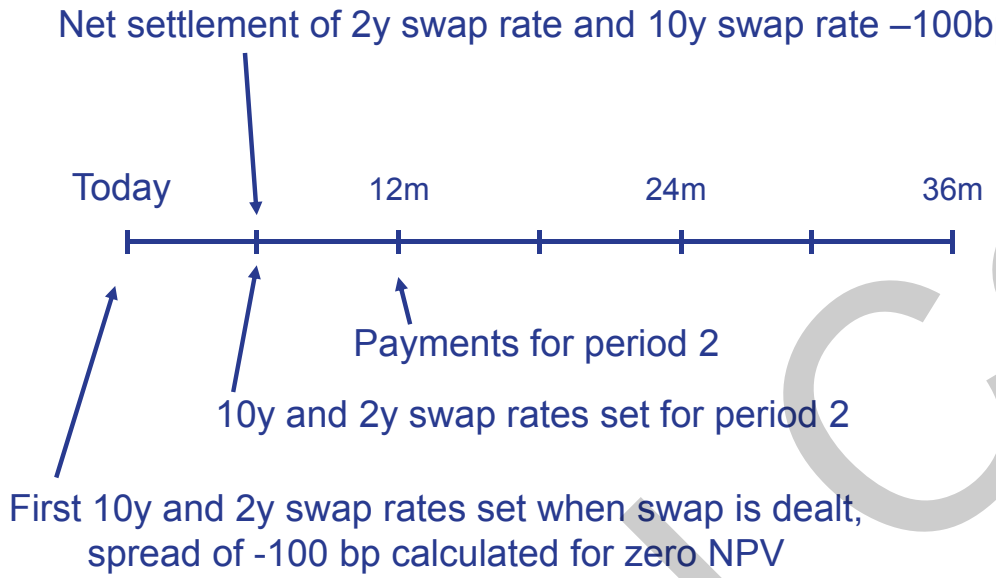
- Payments made every six months one period after the rates set
- Classic steepener or flattener trade depending on whether paying or receiving

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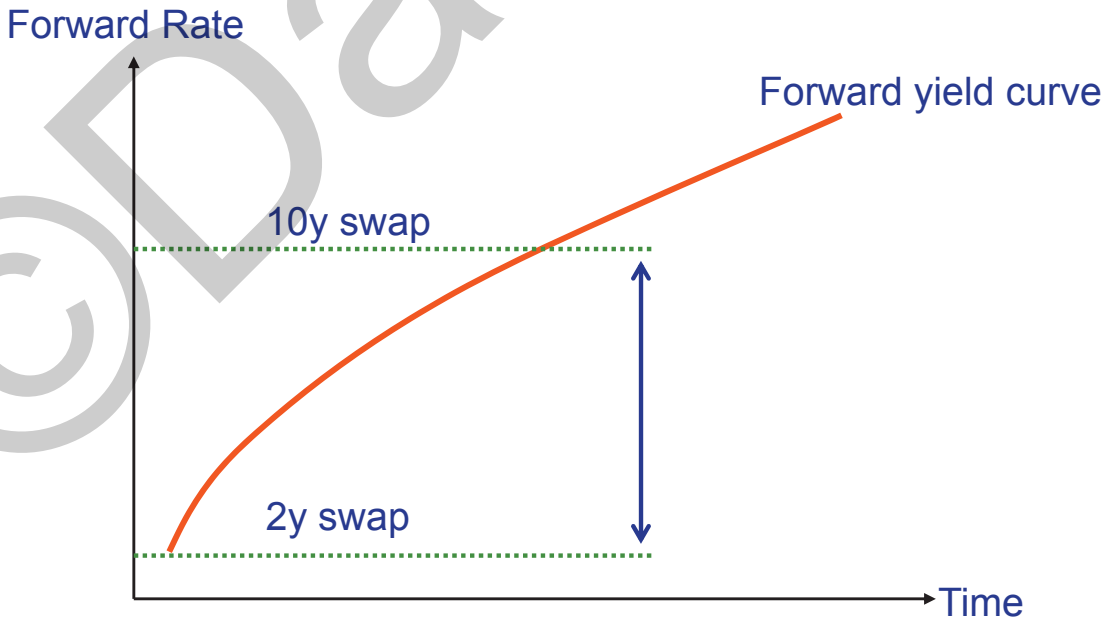
Constant Maturity Swap

10y swap rate –100bp against 2y swap rate



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CMS Yield Curve Trade



Unhedged payer of 10y swap rate -100bp and receiver of 2y swap rate takes the view that the curve will get flatter

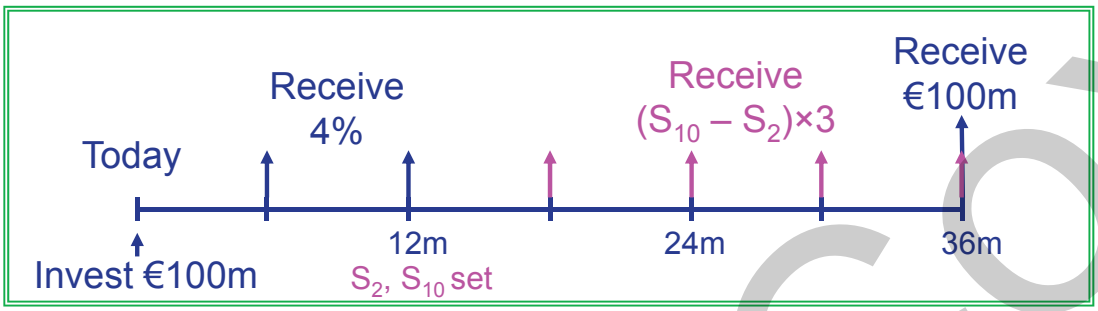
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3y Steepener CMS Note

Fund manager invests €100m

- Receive a fixed coupon of 4% paid semi-annually for one year followed by the 10y – 2y swap spread times 3 paid semi-annually for two years
- Also available as a swap where the client pays Euribor on €100m notional



In general:

- Receive z% for period one and $(S_{T2} - S_{T1})\times g$ for period two
- z = initial above market fixed coupon
- g = gearing factor (can be up to 50 or more in extreme cases)
- S_T = swap rate for tenor T

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3y Steepener CMS Note

Valuation depends on the slope of the YC

First attempt:

- Take the difference between the two and ten year forward swap rates starting on the four reset dates periods (12m, 18m, 24m, & 30m)
- Multiply differences by gearing, calculate implied cashflows using principal and day-count and take NPV
- Adjust first coupon rate so that the total NPV of cashflows including the final maturity payment =100

This is not completely accurate due to convexity, but provides a good idea of how the structure will perform when the curve changes

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Steepener CMS Note

Typical users are looking for enhanced yields:

- Hedge funds
- Fixed income fund managers
- High net worth account managers

People buy these when the curve is flat and they anticipate a steepening.

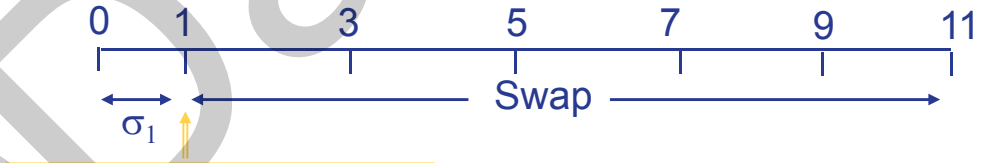
- Flat curve: high gearing is possible
- When the curve is steep gearing is lower

Notes can be callable and may have a knockout feature which triggers if the total coupon reaches a maximum

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CMS Cap v Payer Swaption

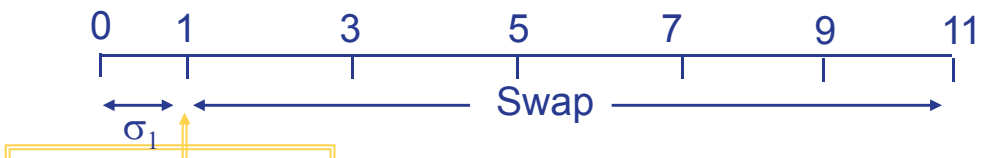
One into 10 year payer swaption (cash settlement)



Max(0, S-X)*Swap DV01

S = 10y swap rate, X= Strike rate, P = principal, τ = day count

One year single period CMS cap on 10y swap rate:



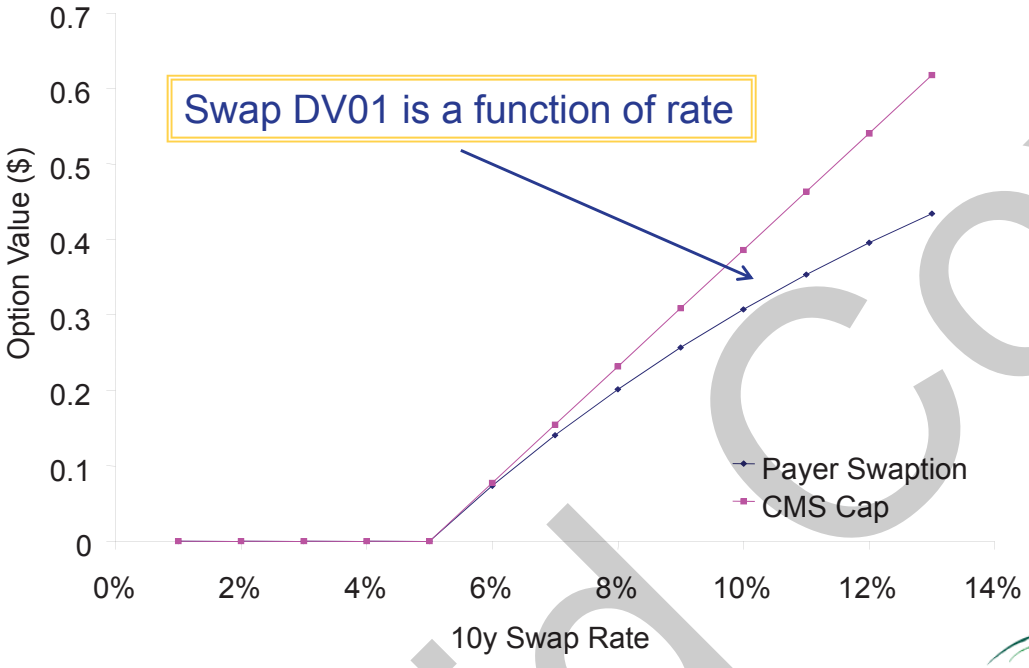
Max(0, S-X)$*\tau*$P

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CMS Cap v Payer Swaption

One into 10 year European payer swaption strike 5%
Single period cap on 10y CMS expiry in 1y, strike 5%

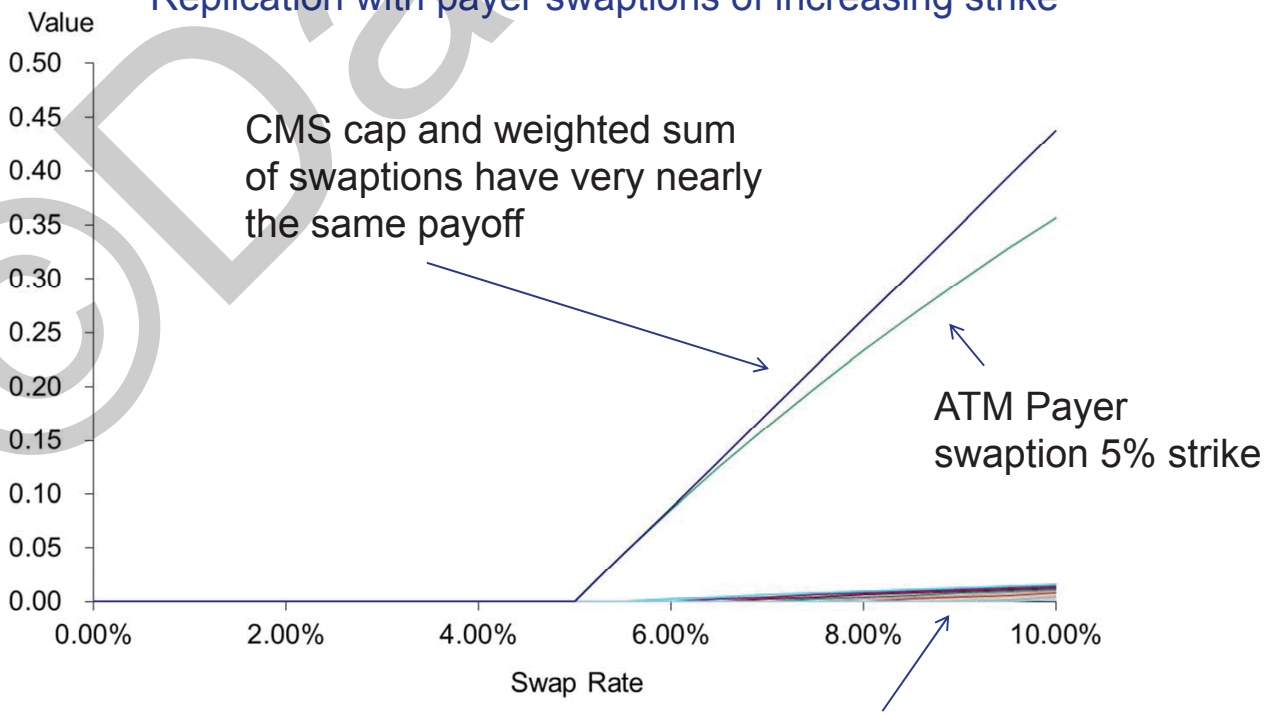


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CMS Cap v Payer Swaption

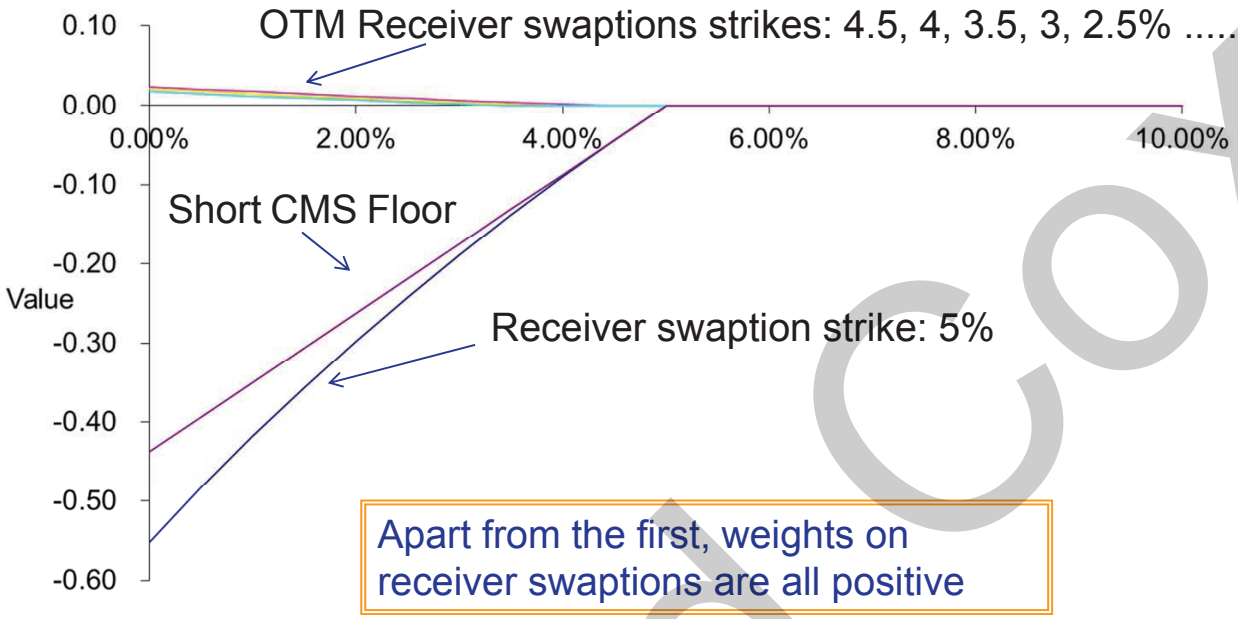
Replication with payer swaptions of increasing strike



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CMS Floor v Receiver Swaption

Short CMS Floor replicated with receiver swaptions (long positions apart from the one struck @ 5%)



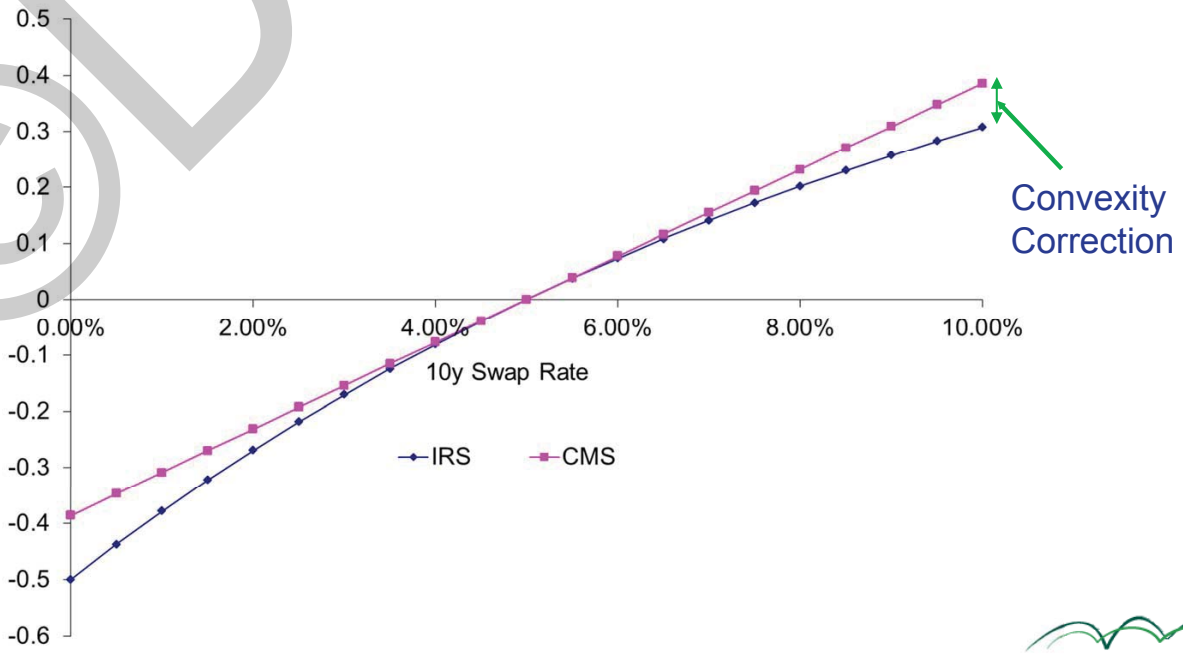
Apart from the first, weights on receiver swaptions are all positive

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IRS v CMS

Turn options into swaps by put-call-forward parity:

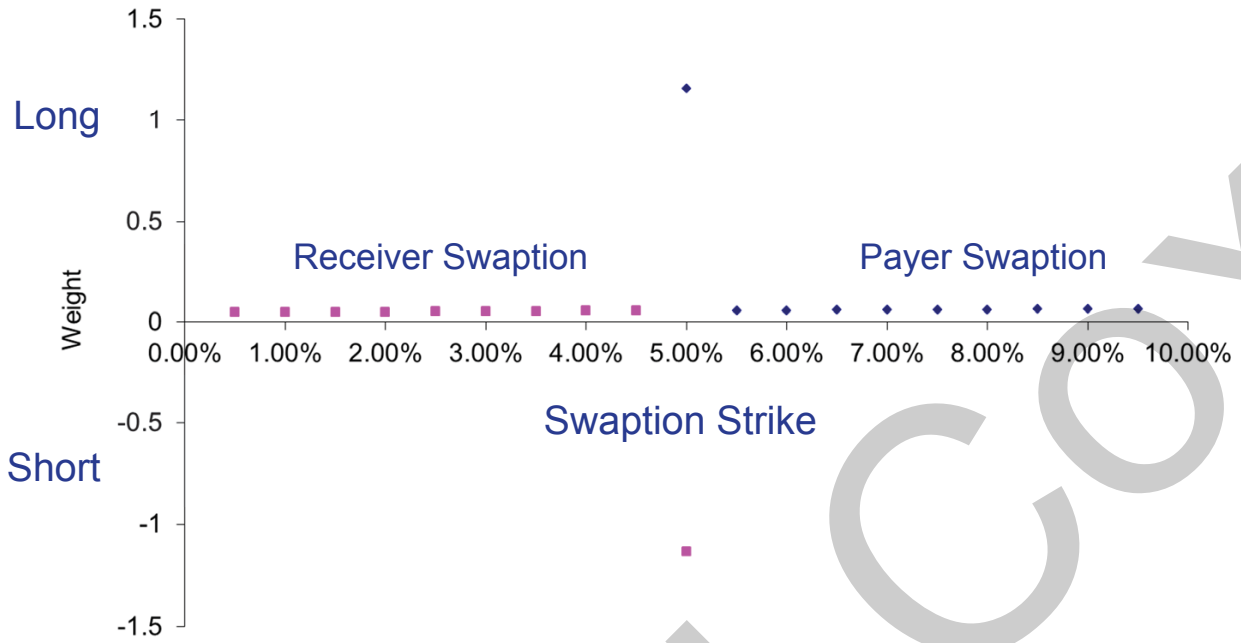
CMS swap is replicated by a portfolio of swaptions



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CMS Swap Hedging Weights



The value of the out of the money swaptions is the convexity correction
Smile is the key to accurate valuation

© David Cox

CMS Adjustments

Formulae can also give us the CMS convexity correction:

To generate the NPV of 10y CMS cashflows:

- From the yield curve generate the 10y forward swap rates out of each reset date on the swap
- Adjust the swap rates for convexity and correlation
- Use the adjusted forward rates to calculate implied cashflows
- Discount the implied cashflows and add them up.

The convexity correction reflects the value of the out of the money swaptions

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Volatility Quotes

Large 2y v 10y CMS Note trade will impact swaption volatilities quoted in the market

If the issuer is paying 10y-2y CMS spread what are the rate and vol. positions needed to hedge?

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Volatility Quotes

Large 2y v 10y CMS Note trade will impact swaption volatilities quoted in the market

If the issuer is paying 10y-2y CMS spread what are the rate and vol. positions needed to hedge?

Issuer needs to receive as 10y swap rate increases:

- Buy 10y CMS cap sell 10y CMS floor or:
- Buy 10y ATM payer, Sell 10y ATM receiver (forward IRS)
- Buy 10y OTM receivers, buy 10y OTM payers (long OTM vol in 10y)

Issuer needs to receive as 2y swap rate decreases:

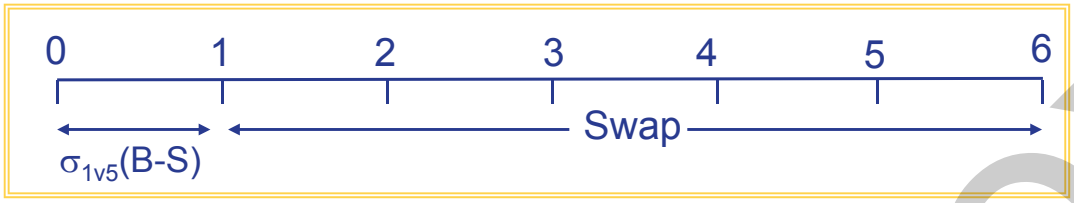
- Buy 2y CMS floor sell 2y CMS cap or
- Buy 2y ATM receiver, sell 2y ATM payer (forward IRS)
- Sell 2y OTM receivers, sell 2y OTM payers (short OTM vol in 2y)

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Volatility

European Swaptions

1 into 5 year Swaption:



Single European style option on IRS

If you use B-S (Black) to price distribution of swap rate is Lognormal

Volatility: Annualised SD of proportional changes in 1v5 year forward swap between today and option expiry

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Swaption Vol (Black)

Vol in %: lognormal distribution

92) Actions | 93) Settings | Interest Rate Volatility Cube

USD | USD BVOL Cube (Default) | Mid | Date 07/31/17

99) Analyze Cube | 98) Market Data

Swap Curve (23) US Dollar | Index Tenor 3M | Show Vol Black

View Strike ATM | Discounting OIS | Show Strikes

Table | Charts

Expiry	1Yr	2Yr	3Yr	4Yr	5Yr	7Yr	10Yr	12Yr	15Yr	20Yr	25Yr	30Yr
9Mo	22.71	27.37	29.29	30.54	31.04	30.10	28.51	27.43	26.19	24.85	24.49	24.35
1Yr	24.62	29.15	30.43	31.45	31.81	30.58	29.04	28.07	26.96	25.72	25.33	25.14
2Yr	30.91	32.89	32.95	32.46	32.07	30.97	29.65	28.80	27.77	26.62	26.33	26.19
3Yr	34.10	33.72	33.04	32.39	31.94	30.92	29.82	28.93	27.91	26.83	26.59	26.50
4Yr	34.21	33.32	32.52	31.71	31.09	30.35	29.39	28.60	27.58	26.49	26.30	26.29
5Yr	33.61	32.55	31.85	31.08	30.37	29.82	29.02	28.26	27.14	26.16	26.03	26.10
6Yr	31.96	31.18	30.66	30.10	29.40	29.03	28.25	27.50	26.45	25.49	25.44	25.57
7Yr	30.81	30.17	29.78	29.36	28.66	28.39	27.62	26.85	25.86	24.89	24.90	25.08
8Yr	29.59	29.10	28.76	28.40	28.02	27.60	26.94	26.18	25.27	24.34	24.40	24.61
9Yr	28.56	28.17	27.90	27.76	27.45	27.04	26.32	25.64	24.72	23.83	23.92	24.15
10Yr	27.65	27.37	27.40	27.16	26.92	26.53	25.75	25.15	24.22	23.35	23.48	23.71
12Yr	26.69	26.24	25.96	25.85	25.72	25.36	24.68	24.15	23.39	22.63	22.72	22.96
15Yr	24.95	24.44	24.25	23.98	23.75	23.86	23.10	22.75	22.22	21.60	21.64	21.89
20Yr	22.47	22.21	22.17	22.05	21.95	21.64	21.29	20.96	20.46	20.30	20.53	20.78
25Yr	21.26	21.25	21.24	21.15	21.12	21.59	21.14	20.97	20.72	20.72	21.06	22.65
30Yr	20.93	20.90	20.97	21.01	21.11	21.15	21.14	20.88	20.49	21.25	23.29	25.05

Quick Pricer For Swaption | 99) Swap Manager (SWPM)

Expiry 1YR Tenor 5YR Strike ATM 2.1111 Index Tenor 3M

Source: Bloomberg VCUB

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Swaption Vol (bp)

Vol in basis points: normal distribution

92) Actions 93) Settings Interest Rate Volatility Cube

USD | USD BVOL Cube (Default) | Mid | Date 07/31/17

95) Analyze Cube 96) Market Data

Swap Curve (23) US Dollar Index Tenor 3M Show Vol Normal

View Strike ATM Discounting OIS Show Strikes

Table Charts

Expiry	1Yr	2Yr	3Yr	4Yr	5Yr	7Yr	10Yr	12Yr	15Yr	20Yr	25Yr	30Yr
9Mo	38.16	48.91	55.23	60.32	63.89	66.31	67.34	66.80	65.73	64.07	63.73	63.51
1Yr	42.76	53.67	58.88	63.57	66.86	68.53	69.49	69.06	68.20	66.71	66.22	65.82
2Yr	59.96	66.69	69.54	71.05	72.47	73.58	74.12	73.43	72.23	70.41	69.89	69.36
3Yr	71.91	73.82	74.84	75.63	76.43	76.93	76.98	75.91	74.20	72.02	71.37	70.76
4Yr	77.25	77.77	78.09	77.91	77.88	78.34	77.85	76.56	74.57	71.90	71.15	70.57
5Yr	80.55	80.16	80.05	79.49	78.83	79.21	78.44	76.80	74.30	71.52	70.77	70.25
6Yr	80.48	79.89	79.80	79.37	78.47	78.52	77.31	75.56	72.92	69.99	69.27	68.83
7Yr	80.01	79.53	79.45	79.19	78.02	77.78	76.17	74.30	71.53	68.45	67.76	67.40
8Yr	78.90	78.47	78.36	77.95	77.04	76.45	74.69	72.78	69.97	66.89	66.23	65.92
9Yr	77.68	77.37	77.12	76.67	76.00	75.07	73.18	71.24	68.39	65.32	64.68	64.41
10Yr	76.49	76.11	75.86	75.33	74.96	73.68	71.66	69.70	66.83	63.78	63.17	62.88
12Yr	72.96	72.30	72.07	71.50	71.07	70.18	67.99	66.30	63.85	61.00	60.32	60.02
15Yr	67.82	66.70	66.38	65.87	65.36	65.07	62.63	61.36	59.52	56.98	56.19	55.88
20Yr	59.41	58.75	58.64	58.30	57.97	56.89	55.61	54.37	52.58	51.32	50.97	50.71
25Yr	54.65	54.57	54.42	54.08	53.86	54.36	52.67	51.90	50.78	49.84	49.72	49.99
30Yr	51.09	50.84	50.80	50.71	50.78	50.54	50.03	49.09	47.73	48.43	48.74	48.91

Quick Pricer For Swaption 99) Swap Manager (SWPM)

Expiry 1YR Tenor 5YR Strike ATM 2.1111 Index Tenor 3M

Source: Bloomberg VCUB

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Types of Volatility

Lognormal (B-S or yield) vol. (σ_l) is the annualised standard deviation of the proportional change in the forward swap rate (S):

$$\frac{\Delta S}{S} = \sigma_l \varepsilon \sqrt{\Delta t} \quad *$$

Normal (basis point) vol. (σ_n) is the annualised standard deviation of the absolute change in the forward swap rate:

$$\Delta S = \sigma_n \varepsilon \sqrt{\Delta t} \quad *$$

To convert from one to the other multiply by the forward swap rate:

$$\sigma_n = S \times \sigma_l$$

* ε is a normally distributed random number and Δt is a small time interval. Drift is zero for forward

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Daily Volatility

Daily vol gives an idea of the one standard deviation absolute move in the swap rate over one day.

This often gives you a better "feel for the market" than annual vol

Lognormal Vol. (annual) = 32%

Swap Rate = 2%

Normal Vol. (annual) = 2%×32% = 64bp

Normal Vol (daily) = 64/16 = 4bp per day

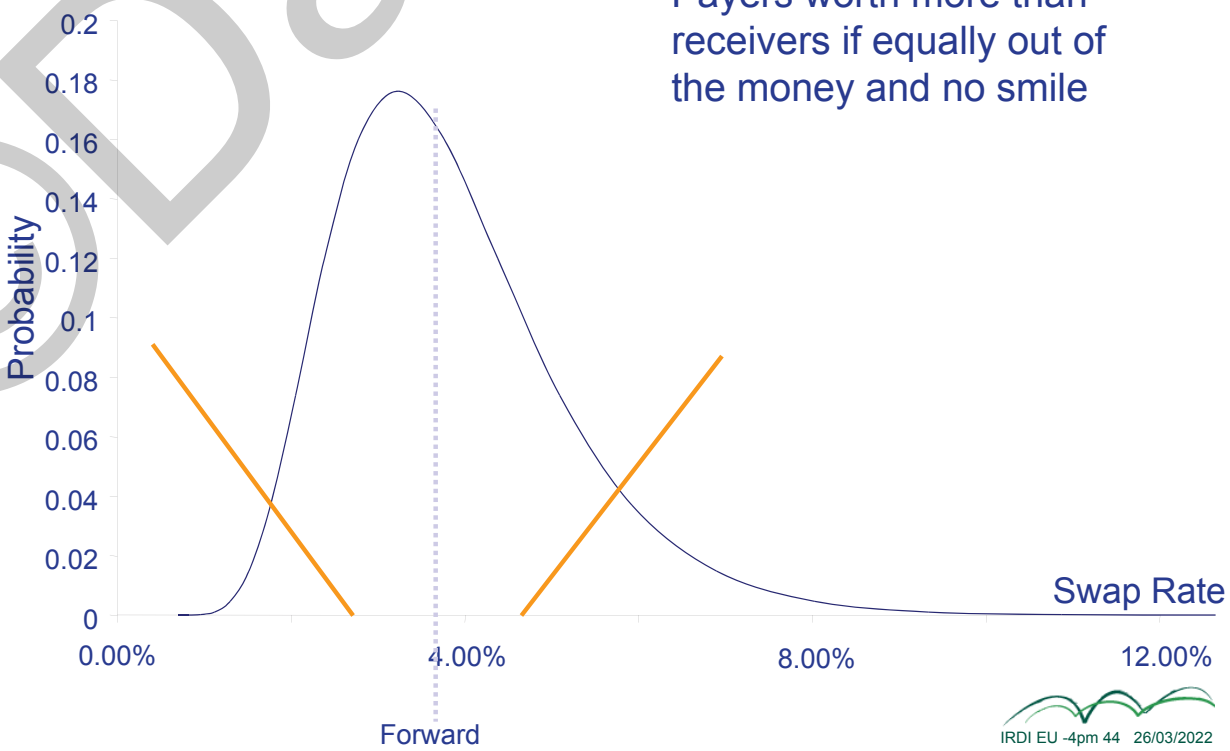
Normally 252 trading days in a year and $\sqrt{252} \cong 16$

(Scaling vol with the square root of time assumes independence between rate changes in each period)

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Lognormal Rate Distribution

Payers worth more than receivers if equally out of the money and no smile

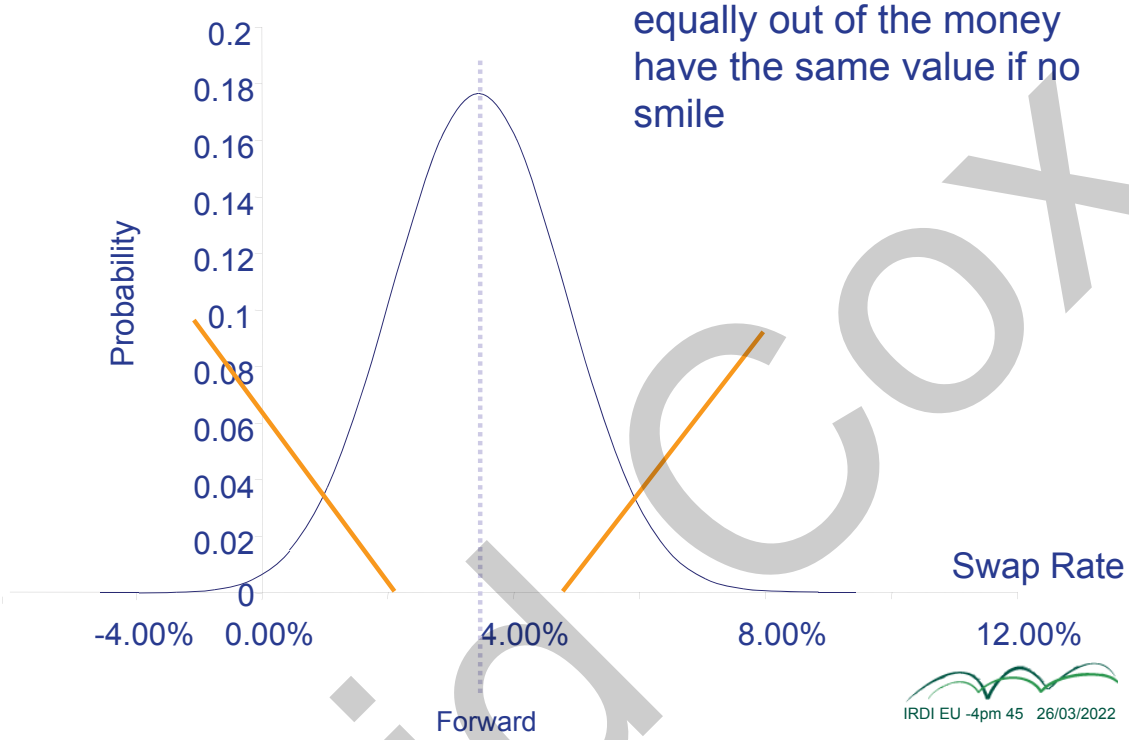


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Normal Rate Distribution

Payers and receivers equally out of the money have the same value if no smile



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Smile / Skew in Swaption Vol

Smile and skew result from market activity not being captured by models.

- Dealt with by altering the distribution of the swap rate or by keeping a Gaussian distribution and having vol. become a property of the swaption strike (model abuse!).

Lognormal process for rates:

- OTM puts less valuable than OTM calls with strikes the same distance from the forward if no smile or skew.

Normal process for rates:

- OTM puts and calls have same value with strikes the same distance from forward if no smile or skew

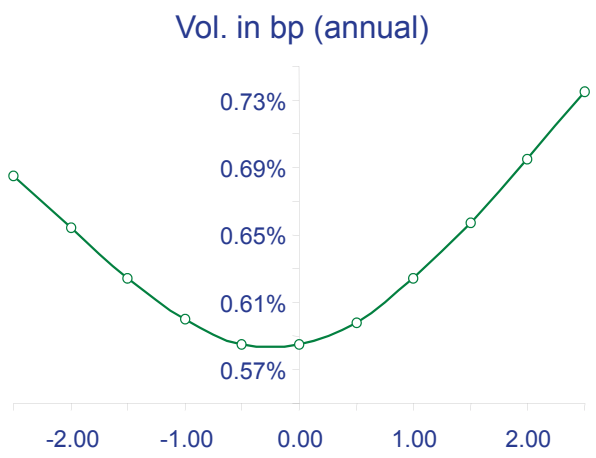
What smile or skew looks like depends upon assumptions about how rates move

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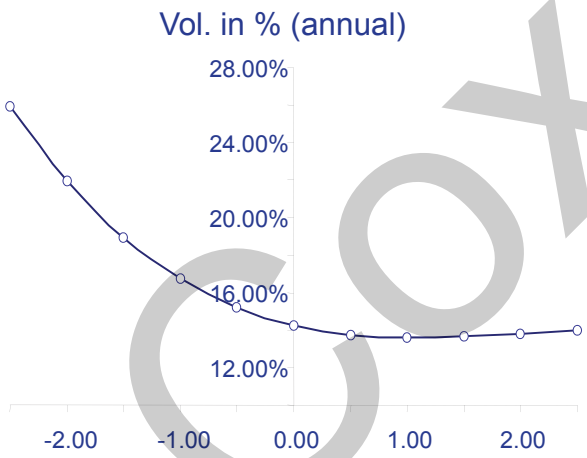
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Implied Swaption Volatilities Under Normal and Lognormal Distributions

Normal Rate Distribution



Lognormal Rate Distribution



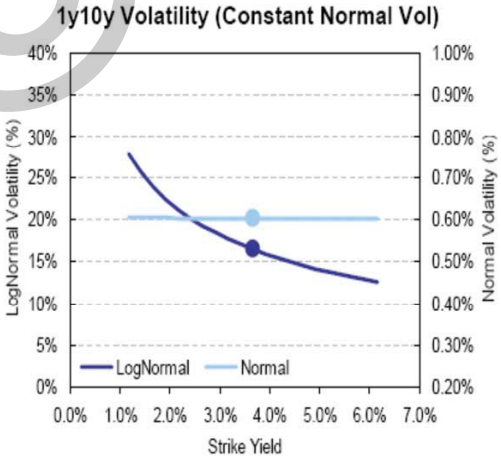
Payer Swaption: % in (+) or out (-) of the money

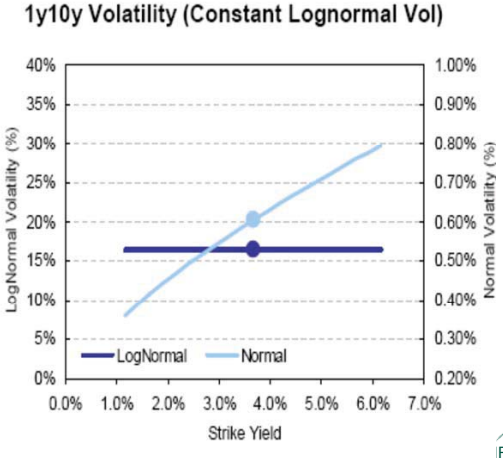
© David Cox

Lognormal (Yield) v Normal (bp) Vol.

Start with option price and imply volatility

- Skew or smile will look different depending on choice of normal or lognormal vol
- Some Greeks will also be different
- ATM Lognormal vol is affected by rate changes
- Normal vol is level independent so can often make a better cross market comparison





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Smile / Skew in Swaption Vol

If volatility is a function of swaption strike then:

Problems with aggregating risk in swaption books

- If different vols are used with different strikes then can deltas be offset across strike?
- Can vega risk be consolidated?
- Part of what looks like vega risk because of skew is actually delta risk (and can be hedged more easily and cheaply)

Problems with implied volatility

- Implied Vol will also change with the swap rate
- Which volatility should be the variable (proportional or parallel shifts)?

Problems with pricing exotics

- Which volatility to use, at strike, at barrier, at some other point or an average?

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Smile / Skew in Swaption Vol

Can make swaption volatility depend on time and swap rate (S,t). This is a "**Local Volatility** Model"

$$\Delta S = S\sigma(S,t)\varepsilon\sqrt{\Delta t}$$

Alter how vol changes with swap rate and time until all swaptions are priced correctly

- Deltas can be aggregated
- Vegas can be aggregated
- Quite a popular in the FX and equity markets

But the dynamics of the Smile/Skew are wrong

- Local vol predicts that the smile moves in the opposite direction to the forward which is not the case

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SABR

CEV model with stochastic volatility

Allows for volatility to be driven by a separate random variable with it's own volatility (vol of vol)

Captures smile well for swaptions and caps

No links between forward rates of different maturities

SABR is not a true term structure model (like BGM, HJM etc.)

In outline:

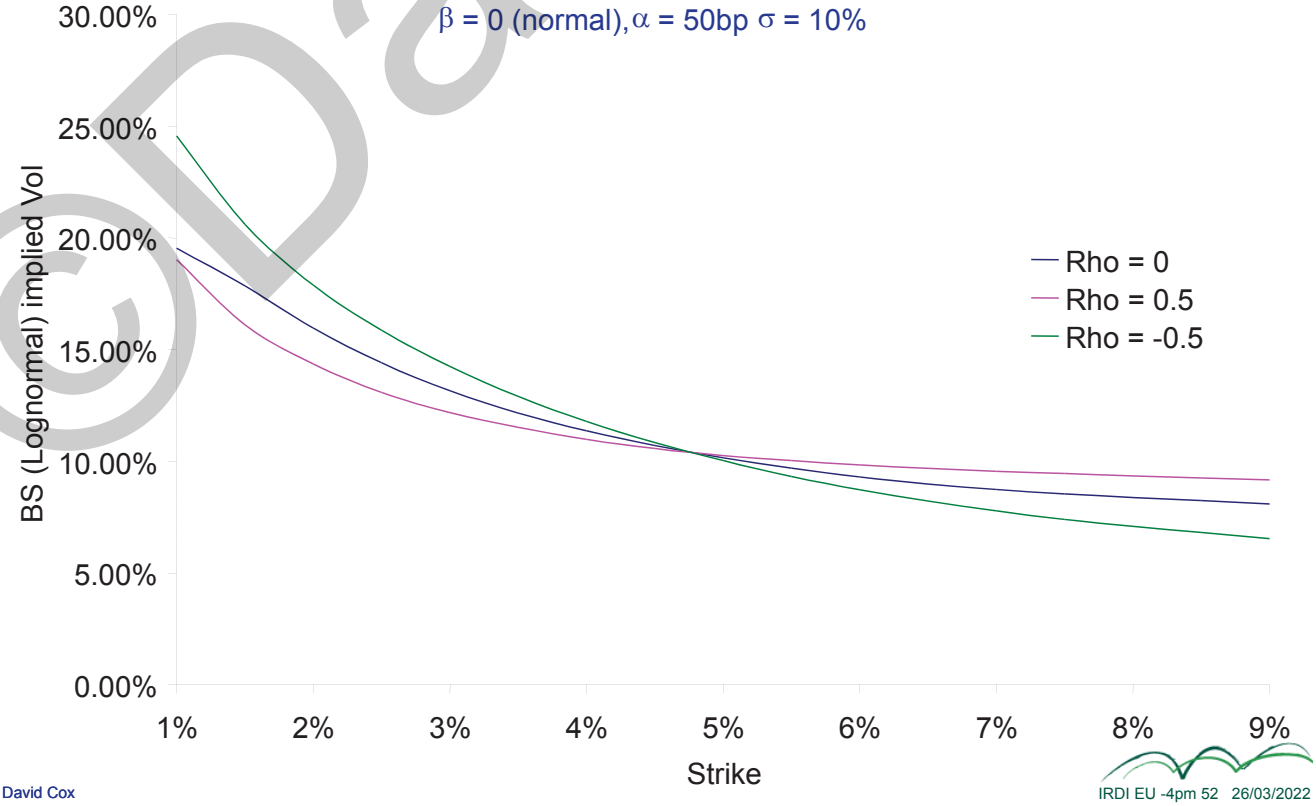
$$\Delta S = V S^{\beta} \varepsilon_s \sqrt{\Delta t}$$
$$\Delta V = V \sigma_v \varepsilon_v \sqrt{\Delta t}$$
$$Corr(\varepsilon_s, \varepsilon_v) = \rho$$
$$V_{t=0} = \alpha$$

S = forward swap fixed rate
V = swap rate volatility (stochastic)
σ_v = vol of swap rate vol
α = swap rate vol today
β = [1,0] determines underling normal or lognormal distribution
ρ = correlation between moves in swap rate and volatility
ε = random number from N(0,1)

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SABR

Implied Vols
β = 0 (normal), α = 50bp σ = 10%



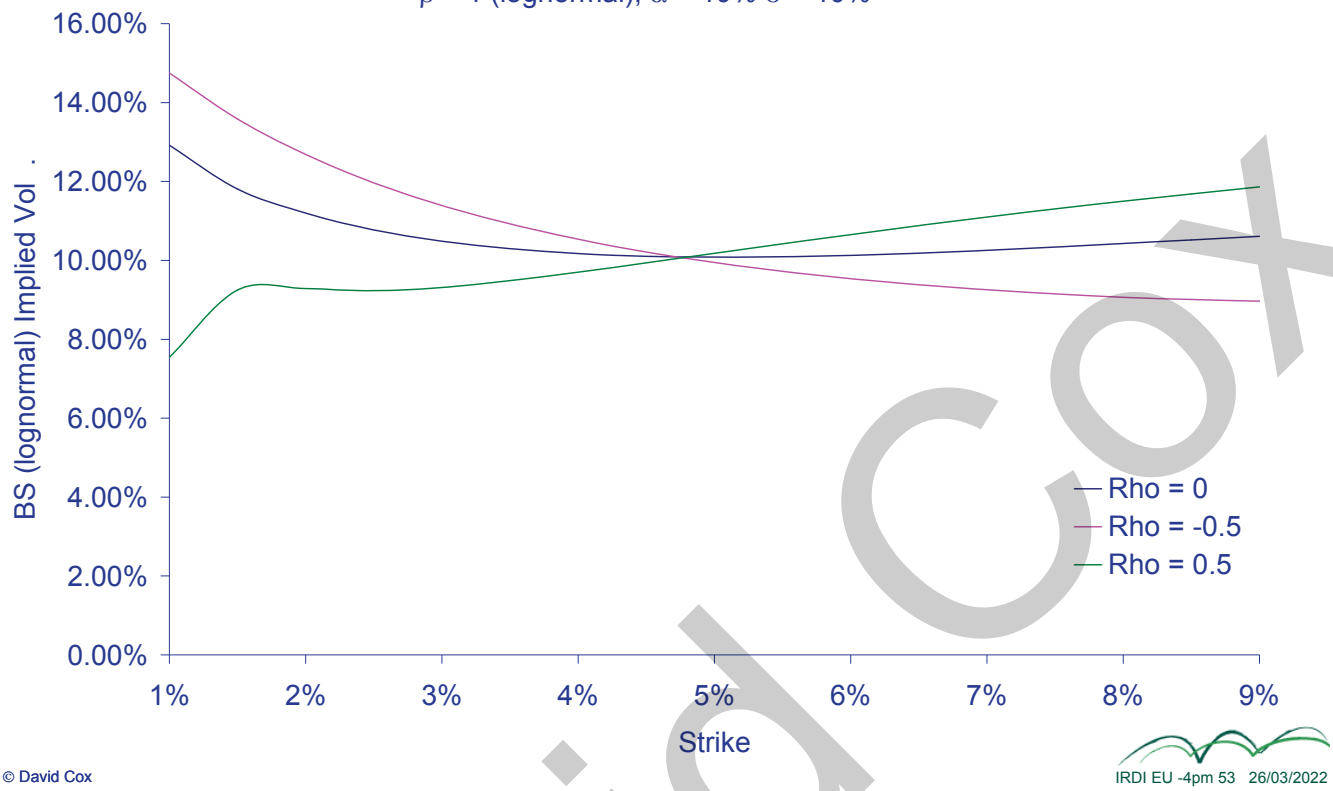
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SABR

Implied Vols

$\beta = 1$ (lognormal), $\alpha = 10\%$ $\sigma = 10\%$



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Further Reading and Useful Publications

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